

Laurent Martini-Bonnet

Mathematical Foundations of Diffusion-Based Generative Modeling

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Advisor: Prof. Yuansi Chen
Co-Advisor: Dr. Francesco Pedrotti

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Chapter 1

Introduction

1.1 Scope and objectives

Generative modeling has emerged as one of the most active research domains in machine learning. In contrast to classical discriminative methods, which aim to predict a label given input data, generative modeling seeks to learn the underlying probability distribution of the data itself, in order to be able to generate new data that looks "similar" in some sense to the input data.

Among modern generative methods, diffusion models have achieved particularly strong empirical results, as noted in [Son+21]. Their basic principle is to progressively perturb the data through a forward stochastic process ("adding noise") until an easy to sample from distribution is reached, and then to generate new samples by approximately reversing this process ("denoising"). Although this description is intuitive, its mathematical formulation relies on several non-trivial ingredients: stochastic differential equations, time reversal, approximations by neural networks, and numerical discretization. Diffusion models can also be interpreted through deterministic probability-flow ordinary differential equations or through Schrödinger bridges, providing complementary viewpoints on the same generative problem.

A recurring difficulty is that the ideal mathematical model and the algorithm implemented in practice are distinct objects. The reverse-time diffusion (see [And82a]) depends on the score of the intermediate population distributions, whereas an implemented model only has access to finitely many observations and replaces this score by a learned approximation. The resulting equation is then solved through a numerical scheme. Consequently, a complete analysis must distinguish the population distribution from the empirical measure, the exact score from its estimator, and the continuous-time process from its discretization. Additional difficulties arise when the data possess structure that is not adequately represented by an unconstrained Euclidean space, as is the case for time series and protein configurations.

The central objective of this thesis is to present, connect, and critically examine the mathematical foundations underlying diffusion-based generative modeling. The thesis addresses the following guiding questions:

1. How can diffusion generative models be derived from stochastic analysis?
2. How are these models related to optimal transport and Schrödinger bridges?
3. Which assumptions separate population-level theoretical results from finite-sample and discretized implementations?
4. How must the framework be adapted when the generated data satisfy temporal or geometric constraints?

The thesis is primarily expository, but its methodology is also critical. Several proofs are reconstructed in detail in order to identify their precise hypotheses and conclusions. When an argument from the literature cannot be applied directly in the stated setting, the obstruction is identified and, when possible, a corrected formulation is proposed.

The first part of the thesis develops the mathematical background required for this analysis. It introduces the relevant tools from stochastic analysis, optimal transport, Schrödinger bridge theory, and differential geometry. The sections about stochastic analysis and differential geometry can be seen as recipes of definitions and results that will be used throughout the thesis, whereas the sections about optimal transport and Schrödinger bridge consist of more descriptive parts and provide more intuition. These topics are presented with the later applications in mind: Girsanov's theorem is used to relate relative entropy to stochastic control, the Schrödinger bridge problem provides a path-space formulation of generative modeling, and differential geometry supplies the appropriate framework for diffusion processes on Riemannian manifolds.

The second part studies the principal mathematical formulations of diffusion generative modeling. Starting from Langevin dynamics and score matching, it then introduces diffusion models and some of their variants. It then discusses the different sources of approximation error appearing in convergence results, under different hypotheses. Finally, Schrödinger bridges are presented as an alternative formulation in which generation is interpreted as the optimal modification of a reference path measure, following the approach of [Bor+23].

The final part investigates two structured applications, financial time series and protein backbones, highlighting temporal constraints and geometrical constraints respectively. For time-series generation, the main paper studied is [Alo+25]. In this setting, the Schrödinger bridge is constrained by the complete joint distribution of the process on an observation grid. The thesis explains why this constraint yields an explicit path-space minimizer and reconstructs its SDE representation. In particular, we provide an alternative and more detailed proof of the original work main result. Then, the thesis examines the practical approximation via a Nadaraya-Watson estimator for the population drift. This leads to a discussion of the practical challenges faced by the method and its fidelity to the generative modeling framework. For protein-backbone generation, the 3D structure of proteins naturally leads to generative models on manifolds. The starting point of the analysis is the paper [Yim+23]. The thesis presents the mathematical derivation of a diffusion generative model on the considered manifolds, which involves representing proteins on some product group, and getting global rotation invariance for the output distribution. We show that one of the structural claims of the original paper does not hold: the centered configuration space it works with is not a subgroup of the product group considered, so those arguments do not apply as stated. We therefore reformulate the setting geometrically, as a Riemannian manifold on which global rotations act by isometries, and prove the invariance result again in that framework. The conclusion of the source paper survives; the route to it does not.

The scope of the thesis is deliberately mathematical. It does not aim to provide an exhaustive survey of generative architectures or perform a large-scale empirical comparison. Moreover, several results concern ideal population quantities and take place in a theoretical setting; practical challenges are discussed but not implemented. These limitations are stated explicitly, since distinguishing what is proved from what is assumed is itself one of the main objectives of the work.

1.2 Contributions and provenance

All the details about the experiments can be found at: <https://github.com/Laurentmb/Mathematical-Foundations-of-Diffusion-Based-Generative-Modeling/tree/main>.

The purpose of Table 1.1 is to distinguish results reproduced or adapted from the literature from the numerical, methodological, and mathematical contributions specific to this thesis.

Table 1.1: Summary of the contributions and intellectual provenance of the thesis.

Location or item	Closest source	Contribution of this thesis	Classification and scope
Chapter 3, Figures 3.7, 3.2, 3.4, 3.5, 3.6.	Inspired from the github: https://mfkasim1.github.io/2022/07/04/sgm-2/	Design of the synthetic target distributions; implementation of the comparisons; generation, visualization, and interpretation of the numerical outputs.	Original numerical illustrations. Full protocols, seeds, raw outputs, and code are provided in the reproducibility archive.
Section 4.1.2, static SBTS derivation	The time-series Schrödinger bridge construction of [Alo+25].	Expanded derivation using the chain rule for relative entropy, including construction, grid constraint, entropy value, and uniqueness of the optimal path measure.	Expository reconstruction, not present in the original work.
Theorem 4.4:	The corresponding dynamic representation theorem in [Alo+25].	Expanded construction of the density process, regularity analysis of h_i , and identification of the explicit drift with the Föllmer drift.	Detailed and adapted a proof of an already existing result.
Section 4.1.3, financial application	The SBTS experiments in [Alo+25] and the cited time-series evaluation literature.	Analysis of overlapping windows, sample dependence, and metrics scores.	Original methodological critique and conjecture, but not an empirically validated finding.
Nonexistence of a translation-invariant probability measure	Standard measure theory.	An elementary proof explaining the obstruction relevant to the protein-frame construction.	Expository proof of a standard fact; no novelty claim.
Proposition 4.10	Proposition G.3 in [Yim+23].	A proof is supplied for a statement for which the cited source does not provide the corresponding argument in the main text.	Original proof of a cited result.
Centered $SE(3)^N$ closure counterexample	The centered-space construction of [Yim+23].	An explicit counterexample showing that the centered configuration space is not closed under the inherited semi-direct product group law, together with a diagnosis of the issue.	Original correction, to the best of our knowledge.
Lemma 4.15	Standard Riemannian differential geometry.	A self-contained proof of the transformation rule for the Riemannian gradient under an isometry.	Standard supporting lemma; no novelty claim.
Proposition 4.16	Proposition 3.6 and Appendix F of [Yim+23], together with standard symmetry results for martingale problems.	A generator-level formulation for isometric group actions on Riemannian manifolds, tailored to the centered protein-frame space.	Original result and proof, to our best knowledge.
Exact centered forward process	Standard manifold-diffusion theory and Proposition 4.16.	Detailed verification of the geometry, covariance, invariance, well-posedness, and nonexplosion of the centered forward process.	Original technical synthesis, not provided in the original article.

Continued on the next page

Table 1.1: Summary of the contributions and intellectual provenance of the thesis (continued).

Location or item	Closest source	Contribution of this thesis	Classification and scope
Approximate reverse process	Proposition 4.16 and the reverse-diffusion construction of [Yim+23].	Extension of the invariance argument to a learned equivariant score.	Conditional corollary, subject to a clean proof for well-posedness and nonexplosion of the martingale problem.
Figures 4.1 and 4.2,	The mathematical constructions summarized in Chapter 4.	Design of a conceptual diagram synthesizing the geometric and probabilistic structure of the protein diffusion model.	Original figures; editable source included in the reproducibility archive.

Chapter 2

Mathematical foundations

2.1 Stochastic analysis

This section will cover some classical notions and results from stochastic calculus, assuming a basic knowledge in this field. Most definitions are classical and come from the lectures "Brownian Motion and Stochastic Calculus" (by Prof. Martin Schweizer) and "Mathematical Finance" (by Prof. Dylan Possamaï) from ETH Zürich. Throughout all of it, we consider a filtered probability space $(\Omega, \mathbb{F}, (\mathcal{F}_t)_{t \geq 0}, P)$.

Definition 2.1. Let m be a strictly positive integer. An $\mathbb{R}^m$ -valued, $\mathbb{F}$ -adapted process $M := (M_t)_{t \geq 0}$ is said to be an $(\mathbb{F}, P)$ -martingale if M_t is P -integrable for any $t \geq 0$ and if, with P -probability 1:

$$\mathbb{E}^P[M_t | \mathcal{F}_s] = M_s, \quad \forall (s, t) \in [0, T]^2 \text{ with } s \leq t. \quad (2.1)$$

Definition 2.2 (Local Martingale). A right continuous left limit (RCLL) $\mathbb{F}$ -adapted process X is an $(\mathbb{F}, P)$ -local martingale if it is locally in the class of RCLL $(\mathbb{F}, P)$ -martingales.

More precisely, there exists a localising sequence of $\mathbb{F}$ -stopping times $(\tau_n)_{n \in \mathbb{N}}$, increasing to $+\infty$ with P -probability 1, such that for any $n \in \mathbb{N}$, the process $1_{\{\tau_n > 0\}} X^{\tau_n}$ is an $(\mathbb{F}, P)$ -martingale.

Definition 2.3 (Semimartingale). A RCLL process $X = (X_t)_{t \geq 0}$ is called an $(\mathbb{F}, P)$ -semi-martingale if there exists a RCLL $(\mathbb{F}, P)$ -local martingale M and a RCLL $\mathbb{F}$ -adapted process V with P -finite variation such that

$$X = M + V. \quad (2.2)$$

Equivalently, it can be written as $X = X_0 + M + A$ with $M \in \mathcal{M}_{0, \text{loc}}$ (the set of local martingales starting at 0) and A an adapted process starting at 0 with RCLL trajectories of finite variation.

Definition 2.4 (σ -martingale). An $(\mathbb{F}, P)$ -semi-martingale X is called an $(\mathbb{F}, P)$ - σ -martingale if there exists a non-decreasing sequence of $\mathbb{F}$ -predictable sets $(D_n)_{n \in \mathbb{N}}$, with $\bigcup_{n \in \mathbb{N}} D_n = \Omega \times \mathbb{R}_+$, and such that for any $n \in \mathbb{N}$ the process

$$\int_0^\cdot 1_{D_n}(s) dX_s \quad (2.3)$$

is a P -uniformly integrable $(\mathbb{F}, P)$ -martingale.

Let d be a strictly positive integer. An $\mathbb{R}^d$ -valued $(\mathbb{F}, P)$ -semi-martingale X is called an $(\mathbb{F}, P)$ - σ -martingale if X^i is an $(\mathbb{F}, P)$ - σ -martingale for any $i \in \{1, \dots, d\}$.

The following lemma involves the space $L^1(X, \mathbb{F}, P)$. If the reader is not familiar with general stochastic integration theory, it is advisable to just view it as the space of processes for which the integral with respect to X is defined.

Lemma 2.5. *Let d be a strictly positive integer, let X be a d -dimensional $(\mathbb{F}, P)$ - σ -martingale, and let $\xi \in L^1(X, \mathbb{F}, P)$. Then*

$$\int_0^\cdot \xi_s \cdot dX_s \quad (2.4)$$

is an $(\mathbb{F}, P)$ - σ -martingale.

Lemma 2.6. *[Ansel–Stricker] Let X be an $(\mathbb{F}, P)$ - σ -martingale which is either bounded from below or $(\mathbb{F}, P)$ -locally bounded. Then X is an $(\mathbb{F}, P)$ -local martingale. This holds true in particular if X is continuous.*

The following lemma tells us when a stochastic integral is continuous:

Lemma 2.7. *Let X be an $(\mathbb{F}, P)$ -semimartingale, let $\xi \in L^1(X, \mathbb{F}, P)$, and define $Y := \int_0^\cdot \xi_s dX_s$. Then*

- (i) $\Delta Y_t = \xi_t \Delta X_t$, for all $t \geq 0$;
- (ii) if X is continuous, then so is Y .

The following lemma will be useful to tell when a local martingale is actually a true martingale.

Lemma 2.8 (Condition to be a true martingale). *Let M be a local martingale null at 0 such that $\forall t \geq 0, \mathbb{E}^P[M_t^2] < \infty$, and $\mathbb{E}^P[[M]_\infty] < +\infty$. Then M is a martingale, and both M and M^2 are P -uniformly integrable. In particular, M_∞ exists P -a.s. and $\mathbb{E}^P[M_\infty^2] = \mathbb{E}^P[[M]_\infty]$.*

Definition 2.9 (Stochastic exponential). *For a continuous semimartingale X with $X_0 = 0$, the analogous stochastic differential equation (SDE)*

$$dZ_t = Z_t dX_t, \quad Z_0 = 1, \quad (2.5)$$

has the unique solution:

$$\mathcal{E}(X)_t := Z_t = \exp\left(X_t - \frac{1}{2}\langle X \rangle_t\right), \quad t \geq 0. \quad (2.6)$$

*The process $\mathcal{E}(X)$ is called the **stochastic exponential** of X .*

Definition 2.10 (Density process). *Fix a probability measure $Q \ll P$ and choose an RCLL version of the P -martingale*

$$Z_t := \mathbb{E}^P\left[\frac{dQ}{dP} \mid \mathcal{F}_t\right], \quad \text{for } 0 \leq t \leq \infty. \quad (2.7)$$

*Then Z is a P -martingale on the right-closed interval $[0, \infty]$, hence UI (uniformly integrable) under P , and $Z_t \geq 0$, P -a.s. for each $t \in [0, \infty]$ since $Q \ll P$. We call Z the **density process** (of Q with respect to P). If $Q \approx P$, we even have $Z_t > 0$, P -a.s. and Q -a.s. for each $t \in [0, \infty]$.*

Lemma 2.11. *Suppose that $Q \ll P$ with density process Z . Then:*

- 1) *For each stopping time τ , Z_τ is the density of Q with respect to P on $\mathcal{F}_\tau$, i.e., we have $Q[A] = \mathbb{E}^P[Z_\tau \mathbf{1}_A]$ for $A \in \mathcal{F}_\tau$. As a consequence, $Z_\tau > 0$ Q -a.s. (but not necessarily P -a.s.).*
- 2) *The stopping time $\varrho := \inf\{t \geq 0 : Z_t = 0 \text{ or } Z_{t-} = 0\}$ equals $+\infty$ Q -a.s. (but not necessarily P -a.s.).*

Definition 2.12. For $M \in \mathcal{M}_{0,loc}$, the space $L^2_{loc}(M)$ consists of all (equivalence classes of) predictable processes for which there is a sequence of stopping times $\tau_n \nearrow \infty$ $\mathbb{P}$ -a.s. such that $H\mathbf{1}_{[0,\tau_n]}$ is in $L^2(M)$ for each n , i.e.,

$$\mathbb{E} \left[\int_0^{\tau_n} H_s^2 d[M]_s \right] < \infty \quad \text{for each } n.$$

Definition 2.13 (Euler-Maruyama scheme). Consider the stochastic differential equation:

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t,$$

with initial condition $X_0 = x_0$, where W_t denotes a standard Brownian motion, and suppose that we wish to solve this SDE on some interval of time $[0, T]$. Then the Euler–Maruyama approximation to the true solution X is the Markov chain Y defined as follows:

- Partition the interval $[0, T]$ into N equal subintervals of width $\Delta t > 0$:

$$0 = \tau_0 < \tau_1 < \cdots < \tau_N = T \quad \text{and} \quad \Delta t = T/N;$$

- Set $Y_0 = x_0$;
- Recursively define Y_n for $0 \leq n \leq N-1$ by

$$Y_{n+1} = Y_n + a(Y_n, \tau_n)\Delta t + b(Y_n, \tau_n)\Delta W_n,$$

where

$$\Delta W_n = W_{\tau_{n+1}} - W_{\tau_n}.$$

Theorem 2.14. [Girsanov theorem for Brownian motion] Suppose that W is a P -Brownian motion and $Q \stackrel{loc}{\approx} P$ has a density process of the form $Z = \mathcal{E}(\int_0^\cdot b_s dW_s)$ for some process $b \in L^2_{loc}(W)$. Then W is, under Q , a Brownian motion with drift b , i.e.,

$$W_t = \tilde{W}_t + \int_0^t b_s ds, \tag{2.8}$$

for a Q -Brownian motion $\tilde{W}$.

Definition 2.15 (Stochastic Kernel). A stochastic kernel from a measurable space $(\Omega, \mathcal{G})$ to a measurable space $(U, \mathcal{U})$ is a mapping $K : \Omega \times \mathcal{U} \rightarrow [0, 1]$ that satisfies the following two properties:

1. For each fixed set $B \in \mathcal{U}$, the mapping $\omega \mapsto K(\omega, B)$ is $\mathcal{G}$ -measurable.
2. For each fixed $\omega \in \Omega$, the mapping $B \mapsto K(\omega, B)$ is a probability measure on $(U, \mathcal{U})$.

Theorem 2.16 (Itô's isometry). (i) Let X be a RCLL $(\mathcal{F}, \mathbb{P})$ -local martingale and ξ be $\mathcal{F}$ -predictable and such that:

$$\mathbb{E}^\mathbb{P} \left[\int_0^\infty \xi_s^2 d[X]_s \right] < +\infty.$$

Then, $\xi \in L^1(X, \mathcal{F}, \mathbb{P})$, $\int_0^\cdot \xi_s dX_s$ is an $(\mathcal{F}, \mathbb{P})$ -martingale which is also $\mathbb{P}$ -square-integrable and for any $\mathcal{F}$ -stopping time τ taking values in $[0, +\infty]$:

$$\mathbb{E}^\mathbb{P} \left[\left(\int_0^\tau \xi_s dX_s \right)^2 \right] = \mathbb{E}^\mathbb{P} \left[\int_0^\tau \xi_s^2 d[X]_s \right].$$

Theorem 2.17 (Lévy's characterisation of Brownian motion). Suppose that $X = (X_t)_{t \geq 0}$ is a $\mathbb{R}^d$ -valued continuous $(\mathbb{F}, \mathcal{P})$ local martingale null at 0, with $\langle X_k, X_\ell \rangle_t = \delta_{k\ell}t$. Then X is a Brownian motion in $\mathbb{R}^d$ with respect to $\mathcal{P}$ and $\mathbb{F}$.

2.2 Optimal Transport

In this section, we describe the main foundations of Optimal Transport (OT), which will be useful as some generative models studied later are based on this theory. This section relies heavily on the book [CNR25].

2.2.1 Different formulations

Let us assume we have two probability distributions on $\mathbb{R}^d$, μ and ν . Optimal Transport can be described as transporting the mass from μ to ν , while minimizing the cost of doing so. If such a transport is deterministic, the problem takes the following form (which is known as the Monge formulation):

$$\inf_{T: T_{\#}\mu = \nu} \int_{\mathbb{R}^d} c(x, T(x)) \mu(dx)$$

The function c is called a cost function, and the function T is called a transport map. The notation $T_{\#}\mu = \nu$ indicates that if $X \sim \mu$, then $T(X) \sim \nu$. This approach has a major flaw: we do not have any guarantees for the existence of a solution. For instance, any μ which is a Dirac mass at one point can't be deterministically transported to any ν which is not a Dirac mass at one point.

To remedy these issues, Kantorovich introduced a relaxed version of the problem:

$$\inf_{\gamma \in \Gamma(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) \gamma(dx, dy)$$

where γ is a coupling between μ and ν , that is a measure on $\mathbb{R}^d \times \mathbb{R}^d$ which admits μ as first marginal on $\mathbb{R}^d$ and ν as second marginal on $\mathbb{R}^d$.

Because any coupling γ can be interpreted as the joint probability distribution of a random vector (X, Y) , integrating the cost function $c(x, y)$ with respect to this measure is equivalent to computing its mathematical expectation. Consequently, the objective can be rewritten in the following, much more intuitive form:

$$\inf_{(X, Y): X \sim \mu, Y \sim \nu} \mathbb{E}[c(X, Y)]$$

From this probabilistic perspective, the Kantorovich problem essentially asks: "Among all possible ways to correlate two random variables X and Y while strictly preserving their respective marginal distributions, which joint distribution minimizes the expected cost $\mathbb{E}[c(X, Y)]$?"

Moreover, by the disintegration of measures theorem, any coupling γ can be factored as $\gamma(dx, dy) = \mu(dx)\gamma(dy|x)$. Therefore, the Kantorovich formulation equivalently seeks the optimal transition kernel $\gamma(dy|x)$ that couples the source distribution μ to the target distribution ν . In contrast, the Monge formulation restricts this kernel to be entirely deterministic, forcing it to take the form of a Dirac measure $\gamma(dy|x) = \delta_{T(x)}(dy)$ concentrated on a single point.

The advantage of the Kantorovich relaxation is that the set of couplings is non-empty, and that under some assumptions, there exists a solution to the problem.

Indeed, we have this following result: (Proposition 1.1 in [CNR25])

Theorem 2.18. *Let μ, ν be two probability measures on $\mathbb{R}^d$. The set $\Gamma(\mu, \nu)$ of couplings between μ and ν is non-empty, convex, and compact with respect to the topology of weak convergence.*

Before stating the result of existence, we need the following definition:

Definition 2.19. *A function $f : X \rightarrow \mathbb{R}$ defined on a metric space (X, d) is said to be **lower semi-continuous** at a point $x_0 \in X$ if for every sequence $(x_n)_{n \in \mathbb{N}}$ in X such that $x_n \rightarrow x_0$ as $n \rightarrow \infty$, we have:*

$$\liminf_{n \rightarrow \infty} f(x_n) \geq f(x_0)$$

The function f is lower semi-continuous on X if it is lower semi-continuous at every $x_0 \in X$.

Theorem 2.20. *Let c be a lower semi-continuous and bounded from below function. The infimum in*

$$\inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) \gamma(dx, dy) \right) \quad (2.9)$$

is attained. Equivalently, the Kantorovich relaxation with cost function $c(x, y)$ admits a solution.

Proof: First define:

$$I(\gamma) := \int c(x, y) \gamma(dx, dy)$$

Let $m = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) \gamma(dx, dy) \right)$. By definition of the infimum, there exists a sequence $\gamma_n \in \Gamma_{\mu, \nu}$ such that $I(\gamma_n) \rightarrow m$. Moreover, as in metric spaces, the compactness is equivalent to sequential compactness, we can use the previous result and get that there exists a subsequence γ_{n_k} which converges weakly to some $\bar{\gamma} \in \Gamma_{\mu, \nu}$. Then, as c is lower semi-continuous and bounded from below, by Portemanteau theorem, we get:

$$\begin{aligned} I(\bar{\gamma}) &= \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) \bar{\gamma}(dx, dy) \\ &\leq \liminf_{n \rightarrow \infty} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) \gamma_n(dx, dy) \\ &= \liminf_{n \rightarrow \infty} I(\gamma_n) \\ &= m \end{aligned}$$

□

A particularly important choice of cost function is obtained by taking the power of a distance, which leads to the definition of the Wasserstein distance.

2.2.2 Wasserstein distance

For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$ let us consider the quantity

$$\inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p \gamma(dx, dy) \right)^{1/p}, \quad (2.10)$$

where $\mathcal{P}_p(\mathbb{R}^d)$ is the set of probability measures on $\mathbb{R}^d$ which admit finite p^{th} moment, for some $p \geq 1$.

We can then give the following definition (definition 1.2 in [CNR25]) which will be central for some results about convergence of measures:

Definition 2.21 (Wasserstein distance). *The p -Wasserstein distance between two probability measures $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$ is defined by:*

$$W_p(\mu, \nu) := \min_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p \gamma(dx, dy) \right)^{1/p}$$

We now state this result, confirming that the Wasserstein distance is indeed a distance:

Proposition 2.22. *The Wasserstein distance defines a metric (or distance) over $\mathcal{P}_p(\mathbb{R}^d)$, that is for every $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, it holds:*

1. $W_p(\mu, \nu) \geq 0$
2. $W_p(\mu, \nu) = W_p(\nu, \mu)$
3. $W_p(\mu, \nu) = 0 \iff \mu = \nu$
4. $W_p(\mu, \nu) \leq W_p(\mu, \rho) + W_p(\rho, \nu)$ for any $\rho \in \mathcal{P}_p(\mathbb{R}^d)$.

Proof: For the first and second points, it follows from the non-negativity and the symmetry of the function $(x, y) \rightarrow \|x - y\|^p$.

Now, we prove the third point. If $\mu = \nu$, then the measure $\gamma(dx, dy) = \mu(dx)\delta_x(dy)$ is a valid coupling. Indeed: To verify that $\gamma \in \Gamma_{\mu, \nu}$, we check its marginals. For any Borel sets $A, B \subset \mathbb{R}^d$, we have:

$$\begin{aligned} \gamma(A \times \mathbb{R}^d) &= \int_A \left(\int_{\mathbb{R}^d} \delta_x(dy) \right) \mu(dx) = \int_A \mu(dx) = \mu(A), \\ \gamma(\mathbb{R}^d \times B) &= \int_{\mathbb{R}^d} \left(\int_B \delta_x(dy) \right) \mu(dx) = \int_{\mathbb{R}^d} \mathbf{1}_B(x) \mu(dx) = \mu(B). \end{aligned}$$

Since $\mu = \nu$, the second marginal is $\nu(B)$, confirming that γ is a valid coupling. Concretely, γ is the law of (X, X) for $X \sim \mu$. Therefore,

$$0 \leq W_p^p(\mu, \mu) \leq \int \|x - y\|^p \gamma(dx, dy) = \int \|x - x\|^p \mu(dx) = 0.$$

To show the other implication, observe that if $W_p(\mu, \nu) = 0$, there exists an optimal coupling $\bar{\gamma} \in \Gamma_{\mu, \nu}$ such that for $(X, Y) \sim \bar{\gamma}$, we have $\mathbb{E}[\|X - Y\|^p] = 0$, which implies that $X = Y$ almost surely. In particular, they must have the same distribution, hence $\mu = \nu$.

To complete the proof, we check the triangle inequality 4. To that end, we employ the Gluing Lemma (Lemma B.5 in [CNR24]), which ensures that there exist random variables X, Y, Z such that $X \sim \mu$, $Y \sim \nu$, and $Z \sim \rho$, and such that the pairs (X, Z) and (Z, Y) are optimally coupled. Then,

$$\begin{aligned} W_p(\mu, \nu) &\leq (\mathbb{E}[\|X - Y\|^p])^{1/p} \\ &= (\mathbb{E}[\|X - Z + Z - Y\|^p])^{1/p} \\ &\leq (\mathbb{E}[\|X - Z\|^p])^{1/p} + (\mathbb{E}[\|Z - Y\|^p])^{1/p} \\ &= W_p(\mu, \rho) + W_p(\rho, \nu), \end{aligned}$$

where in the first line we used the suboptimality of the coupling (X, Y) , in the third line we used Minkowski's inequality, and in the last line, we used the optimality of the couplings (X, Z) and (Z, Y) . $\square$

For the choice $p=2$, the p -Wasserstein distance has a deep connection with convex analysis, that will allow for dual formulation of the OT problem.

2.2.3 Duality

Assume $p=2$. Then, the problem: $\min_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \gamma(dx, dy) \right)$ admits a dual representation:

$$\sup_{\substack{f \in \mathcal{L}^1(\mu), g \in \mathcal{L}^1(\nu) \\ f(x) + g(y) \leq \|x - y\|^2}} \left(\int_{\mathcal{X}} f(x) d\mu(x) + \int_{\mathcal{Y}} g(y) d\nu(y) \right) \quad (2.11)$$

This dual representation allows for a deep result, by Brenier (Theorem 1.14 in [CNR25]). We first need the following definition (definition 1.4 in [CNR25]).

Definition 2.23 (Convex conjugate). *For any function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$, we define its convex conjugate f^* via*

$$f^*(y) := \sup_{x \in \mathbb{R}^d} \{ \langle x, y \rangle - f(x) \}, \quad y \in \mathbb{R}^d. \quad (2.12)$$

The convex conjugate is also called the Fenchel-Legendre transform.

Theorem 2.24 (Fundamental theorem of optimal transport). *Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ be two probability measures such that μ has a density with respect to Lebesgue measure, and let $X \sim \mu$. For a given coupling $\bar{\gamma} \in \Gamma(\mu, \nu)$, the following are equivalent:*

(i) *The coupling $\bar{\gamma}$ is optimal, in the sense that:*

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \bar{\gamma}(dx, dy) = W_2^2(\mu, \nu). \quad (2.13)$$

(ii) *There exists a proper convex function ϕ such that $(X, \nabla \phi(X)) \sim \bar{\gamma}$. Moreover, $\nabla \phi$ is unique in the sense that if there exists a convex function ψ such that $\nabla \psi(X) \sim \nu$, then $\nabla \psi(X) = \nabla \phi(X)$ almost surely.*

(iii) *The coupling $\bar{\gamma}$ saturates the dual problem, meaning its cost is exactly equal to the supremum of the dual representation:*

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \bar{\gamma}(dx, dy) = \sup_{\substack{f \in \mathcal{L}^1(\mu), g \in \mathcal{L}^1(\nu) \\ f(x) + g(y) \leq \|x - y\|^2}} \left(\int_{\mathcal{X}} f d\mu + \int_{\mathcal{Y}} g d\nu \right). \quad (2.14)$$

Moreover, the above supremum is achieved for

$$\bar{f}(x) := \|x\|^2 - 2\phi(x) \quad \text{and} \quad \bar{g}(y) := \|y\|^2 - 2\phi^*(y), \quad (2.15)$$

where ϕ^* is the convex conjugate of ϕ . The functions $\bar{f}$ and $\bar{g}$ are then called Kantorovich potentials.

The fact that (i) $\implies$ (ii) crucially tells us that in the case where μ has a density with respect to the Lebesgue measure, the optimal transport is actually deterministic and does not split the mass. Hence in this case, the Monge formulation admits a solution which is the same as the one for Kantorovich relaxation.

The fact that (ii) $\implies$ (i) tells us that any valid coupling $\gamma \in \Gamma(\mu, \nu)$ of the form $(X, \nabla \psi(X)) \sim \gamma$ for some convex function ψ , must be the unique optimal coupling between μ and ν .

The fact that $(i) \implies (iii)$ means that if a coupling is optimal, its transport cost evaluates exactly to the supremum of the dual problem.

The fact that $(iii) \implies (i)$ means that if a coupling has a transport cost equal to the supremum on the right side, then it has to be an optimal coupling.

Remark: This theorem contains the statement of strong duality $((i) \implies (iii))$, given the existence of an optimal coupling). However, in most literature, strong duality is actually proven before stating this result and is used as a prerequisite in its proof.

2.2.4 Entropic regularization

The entropic regularization relies on a penalty that we bring to the objective function. Historically, researchers started to add an entropic term in optimal transport when they noticed that real life phenomena were often more diffuse than the optimal transport solutions, that tend to be very sparse (see section 4.1 in [PC20]). The main advantage of the Entropic regularization is computational. Indeed, we will see that the regularized minimization problem can be solved via an alternating minimization scheme. Crucially, these iterative updates reduce to elementary matrix-vector multiplications, rendering the algorithm highly efficient for GPU acceleration. Mathematically, the entropic regularization also brings some new advantages: thanks to the Kantorovich formulation, we got the existence of a solution to the Optimal Transport problem. We will see that the entropic regularization gives the uniqueness of the solution.

For probability measures μ and ν on $\mathbb{R}^d$ admitting densities with respect to the Lebesgue measure, the entropic regularized problem was first defined as:

$$\inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \gamma(dx, dy) - \varepsilon \text{Ent}(\gamma) \right), \quad (2.16)$$

where $\text{Ent}(\gamma)$ denotes the differential entropy $\int \gamma(x, y) \log \frac{1}{\gamma(x, y)} dx dy$ for an absolutely continuous (with respect to the Lebesgue measure) probability measure γ .

We note that $\Gamma(\mu, \nu)$ is convex (any convex combination of probability measures is a probability measure, and the marginal property is stable by convex combinations) and $\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \gamma(dx, dy)$ is linear in γ . Moreover, $-\text{Ent}(\gamma)$ is strictly convex in γ as $t \rightarrow t \log(t)$ is strictly convex. Hence, we minimize a strictly convex function over a convex set. Hence, if the minimization problem admits a solution, it is unique.

From an information-theoretic perspective, the entropic regularization of optimal transport can be elegantly interpreted through Jaynes' Maximum Entropy Principle (1957). In this context, the entropy of a distribution γ quantifies the uncertainty inherent in the distribution γ . Jaynes' principle asserts that the most unbiased probability distribution representing our current state of knowledge is the one that maximizes this entropy, subject to known prior constraints (here, the marginals μ and ν , and the expected transport cost). By penalizing distributions with low entropy, the regularized problem naturally avoids the almost deterministic and highly sparse solutions (such as Monge's permutation matrices) that essentially "overfit" the data. Instead, it favors more diffuse solutions where a single source distributes its mass across multiple nearby targets.

Another formulation of the regularized problem is more widely used, for which we need the following definition:

Definition 2.25. The **Kullback-Leibler divergence** (or relative entropy) between a probability measure μ and a σ -finite positive measure ν is defined as:

$$KL(\mu\|\nu) = \begin{cases} \int_{\mathbb{R}^d} \log\left(\frac{d\mu}{d\nu}(x)\right) d\mu(x) & \text{if } \mu \ll \nu \\ +\infty & \text{otherwise} \end{cases}$$

where $\frac{d\mu}{d\nu}$ denotes the **Radon-Nikodym derivative** of μ with respect to ν .

For a rigorous definition of the relative entropy when ν is unbounded, we refer to [Léo13], Appendix A.1-A.6.

Another formulation of the regularized Optimal Transport problem is then:

$$\inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \gamma(dx, dy) + \varepsilon KL(\gamma\|\mu \otimes \nu) \right) \quad (2.17)$$

Proposition 2.26. Assume that γ , μ , and ν admit densities with respect to the Lebesgue measure, and in addition that μ and ν have finite entropies. Then, the objectives of the two formulations of the regularized problem only differ by an additive constant depending exclusively on the fixed marginals constraints.

Proof: Let $\gamma \in \Gamma(\mu, \nu)$. As γ , μ , and ν admit densities with respect to the Lebesgue measure, the Radon-Nikodym derivative of γ with respect to the product measure $\mu \otimes \nu$ is given by the ratio of their densities:

$$\frac{d\gamma}{d(\mu \otimes \nu)}(x, y) = \frac{\gamma(x, y)}{\mu(x)\nu(y)}.$$

Hence, as the entropies of μ and ν are assumed to be finite, we can use Fubini's theorem to get:

$$\begin{aligned} KL(\gamma\|\mu \otimes \nu) &= \int_{\mathbb{R}^d \times \mathbb{R}^d} \log\left(\frac{d\gamma}{d(\mu \otimes \nu)}\right) d\gamma \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} \gamma(x, y) \log\left(\frac{\gamma(x, y)}{\mu(x)\nu(y)}\right) dx dy \\ &= \int \gamma \log \gamma dx dy - \int \gamma \log \mu dx dy - \int \gamma \log \nu dx dy \end{aligned}$$

By definition, the first term is $-\text{Ent}(\gamma)$. For the remaining terms, applying the marginal constraints of the valid coupling γ (i.e., $\int \gamma(x, y) dy = \mu(x)$ and $\int \gamma(x, y) dx = \nu(y)$) simplifies the integrals:

$$\begin{aligned} KL(\gamma\|\mu \otimes \nu) &= -\text{Ent}(\gamma) - \int_{\mathbb{R}^d} \mu(x) \log \mu(x) dx - \int_{\mathbb{R}^d} \nu(y) \log \nu(y) dy \\ &= -\text{Ent}(\gamma) + \text{Ent}(\mu) + \text{Ent}(\nu). \end{aligned}$$

Since the marginals μ and ν are fixed in the infimum over $\gamma \in \Gamma(\mu, \nu)$, the entropies $\text{Ent}(\mu)$ and $\text{Ent}(\nu)$ are constants with respect to γ . Therefore, the penalty $\varepsilon KL(\gamma\|\mu \otimes \nu)$ is equivalent for the argmin to $-\varepsilon \text{Ent}(\gamma)$ in the optimization problem. $\square$

This formulation admits a dual problem. From now on, we will work on a compact set Ω in $\mathbb{R}^n$ and we denote by $C_b(\Omega)$ the set of continuous functions on Ω (hence bounded as Ω is compact).

The dual problem is:

$$\sup_{f,g \in C_b(\Omega)} \left(\int_{\Omega} f(x) d\mu(x) + \int_{\Omega} g(y) d\nu(y) - \iota_{\epsilon}(f, g) \right) \quad (2.18)$$

where:

$$\iota_{\epsilon}(f, g) = \epsilon \int_{\Omega \times \Omega} \left(e^{\frac{f(x) + g(y) - \|x - y\|^2}{\epsilon}} - 1 \right) \mu(dx) \nu(dy) \quad (2.19)$$

We then have this useful result (Proposition 4.3 in [CNR25]) which links both formulations and gives simple conditions for a coupling to be optimal:

Theorem 2.27. *If $f^*, g^* \in C_b(\Omega)$ are such that*

$$\omega(dx, dy) = \exp \left(\frac{f^*(x) + g^*(y) - \|x - y\|^2}{\epsilon} \right) \mu(dx) \nu(dy) \quad (2.20)$$

is a valid coupling between μ and ν , then ω is an optimal coupling for the primal problem and f^, g^* are optimal potentials for the dual problem. Moreover, solving the primal problem (2.17) is equivalent to solving the dual problem (2.18).*

Equivalently, f^ and g^* are optimal for the dual problem if and only if the coupling ω satisfies the marginal constraints, which can be explicitly written as the following set of equations, known as the **Schrödinger system**:*

$$\left\{ \begin{array}{l} \int_{\Omega} \exp \left(\frac{f^*(x) + g^*(y) - \|x - y\|^2}{\epsilon} \right) \nu(dy) = 1 \quad \mu\text{-a.e.} \\ \int_{\Omega} \exp \left(\frac{f^*(x) + g^*(y) - \|x - y\|^2}{\epsilon} \right) \mu(dx) = 1 \quad \nu\text{-a.e.} \end{array} \right\}. \quad (2.21)$$

Remark: The existence of f^* and g^* is not detailed here. However, it can be proven that if μ and ν are compactly supported, then f^* and g^* exist. On a side-note, if μ and ν have finite second moments, then Theorem 2.27 and the existence of optimal potentials still hold, with this time $f^* \in L^1(\mu), g^* \in L^1(\nu)$.

Thanks to this theorem, we see that to solve the regularized Optimal Transport problem, it suffices to find f and g that satisfy the Schrödinger system. This idea is at the core of the so-called Sinkhorn's algorithm:

Definition 2.28 (Sinkhorn's algorithm). *Sinkhorn's algorithm is defined by: $f_0 \equiv 0$, and for $n \geq 1$:*

$$\left\{ \begin{array}{l} g_n(y) = -\epsilon \log \int_{\Omega} \exp \left(\frac{f_{n-1}(x) - \|x - y\|^2}{\epsilon} \right) \mu(dx), \\ f_n(x) = -\epsilon \log \int_{\Omega} \exp \left(\frac{g_n(y) - \|x - y\|^2}{\epsilon} \right) \nu(dy). \end{array} \right. \quad (2.22)$$

This is the alternative minimization scheme introduced at the beginning of this section. To understand how it reduces to fast matrix-vector operations and accelerates the computation of optimal transport, we now introduce the discrete formulation.

Discrete Optimal Transport Let us consider discrete probability measures $\mu = \sum_{i=1}^n a_i \delta_{x_i}$ and $\nu = \sum_{j=1}^m b_j \delta_{y_j}$, where $a_i, b_j > 0$ for $1 \leq i, j \leq n$. Any coupling between μ and ν takes the form of a matrix $P \in \mathbb{R}_+^{n \times m}$. The coefficient $P_{i,j}$ describes the amount of mass flowing from x_i to y_j . Hence the set of feasible solutions is:

$$U(a, b) := \left\{ P \in \mathbb{R}_+^{n \times m} : P \mathbf{1}_m = a, P^T \mathbf{1}_n = b \right\},$$

The Optimal Transport problem is now:

$$L_C(a, b) := \min_{P \in U(a, b)} \langle C, P \rangle := \min_{P \in U(a, b)} \sum_{i,j} C_{i,j} P_{i,j}.$$

where $C \in \mathbb{R}^{n \times m}$ is the cost matrix, by analogy to the cost function in the continuous setting.

Let $\epsilon > 0$. The entropic regularized formulation is then:

$$L_C^\epsilon(a, b) \stackrel{\text{def.}}{=} \min_{P \in U(a, b)} \langle P, C \rangle - \epsilon H(P) \quad \text{where } H(P) \text{ is } H(P) = - \sum_{i,j} P_{i,j} (\log(P_{i,j}) - 1)$$

Observe that we easily get existence and uniqueness of a solution. Indeed, $U(a, b)$ is bounded and closed, hence compact, and $P \rightarrow \langle P, C \rangle - \epsilon H(P)$ is continuous, hence we get the existence of a solution. Moreover, it is also strictly convex, and as $U(a, b)$ is convex and closed, we get uniqueness.

The computational advantage of the regularized formulation over the classical one comes from the following result:

Theorem 2.29. *Let $\mathbb{R}_{+,*}^n$ be the set of elements in $\mathbb{R}^n$ which have positive coordinates, and $\mathbb{R}_+^n$ the set of elements in $\mathbb{R}^n$ which have non-negative coordinates. Let P^* denote the unique minimizer of the discrete regularized problem. Then:*

$$\exists (u, v) \in \mathbb{R}_+^n \times \mathbb{R}_+^m, \quad K \in \mathbb{R}_{+,*}^{n \times m}, \quad \text{such that:}$$

$$\forall (i, j) \in \{1, \dots, n\} \times \{1, \dots, m\}, \quad P_{i,j}^* = u_i K_{i,j} v_j$$

Proof: The Lagrangian associated to the regularized problem reads for $f \in \mathbb{R}^n, g \in \mathbb{R}^m$:

$$\mathcal{L}(P, f, g) = \langle P, C \rangle - \epsilon H(P) - \langle f, P \mathbf{1}_m - a \rangle - \langle g, P^T \mathbf{1}_n - b \rangle.$$

We know that since P^* is a minimizer of the discrete regularized problem:

$$\frac{\partial \mathcal{L}(P^*, f, g)}{\partial P_{i,j}^*} = C_{i,j} + \epsilon \log(P_{i,j}^*) - f_i - g_j = 0,$$

$$\implies \log(P_{i,j}^*) = \frac{f_i + g_j - C_{i,j}}{\epsilon},$$

$$\implies P_{i,j}^* = e^{f_i/\epsilon} e^{-C_{i,j}/\epsilon} e^{g_j/\epsilon},$$

$$:= u_i K_{i,j} v_j \quad \text{with} \quad K_{i,j} = e^{-C_{i,j}/\epsilon}.$$

□

Although we know that these vectors u and v exist, we don't have any explicit expression for them. Our goal is to find a way to approximate them.

From this theorem we derive directly:

$$P^* = \text{diag}(u)K\text{diag}(v)$$

Hence the variables u and v must satisfy the marginal constraints, namely:

$$\text{diag}(u)K\text{diag}(v)\mathbf{1}_m = a \quad \text{and} \quad \text{diag}(v)K^T\text{diag}(u)\mathbf{1}_n = b$$

which is equivalent to the formulation:

$$u \odot (Kv) = a,$$

$$v \odot (K^T u) = b,$$

where $\odot$ is the entry-wise multiplication of vectors.

The principle of the discrete Sinkhorn's algorithm is to iterate u and v , by modifying first u so that it satisfies $u \odot (Kv) = a$ and then v to satisfy $v \odot (K^T u) = b$. This gives us:

$$\begin{aligned} v^{(0)} &= \mathbf{1}_m \\ u^{(k+1)} &\stackrel{\text{def.}}{=} \frac{a}{Kv^{(k)}}, \\ v^{(k+1)} &\stackrel{\text{def.}}{=} \frac{b}{K^T u^{(k+1)}}, \end{aligned}$$

where $\frac{a}{Kv^{(k)}}$ and $\frac{b}{K^T u^{(k+1)}}$ denote element-wise division. □

As noted in [Cut13], the computational cost of solving the unregularized discrete problem scales poorly, typically $O(N^3 \log N)$, becoming prohibitive for large support sizes N . By contrast, a single iteration of the discrete Sinkhorn's algorithm consists solely of matrix-vector multiplications, taking $O(N^2)$ operations. The total computational complexity is therefore $O(N_{\text{iter}}N^2)$, where N_{iter} is the number of iterations required to reach a given stopping tolerance.

However, we need to distinguish the entropically regularized cost computed by this algorithm from the exact unregularized OT cost: using Sinkhorn's algorithm to approximate the unregularized cost requires setting the regularization parameter ϵ to a very small value, which drastically increases the number of iterations K needed to converge.

2.3 Schrödinger Bridge Problem

In this section, we will cover the foundations of the Schrödinger Bridge Problem. It is linked to optimal transport as well as generative modeling, and we will see that it suits particularly well the setting for time series generation.

2.3.1 The main variations

In plain words, the Schrödinger Bridge Problem (SBP) consists of the following. Given a reference diffusion (defined as any stochastic process which is solution to a SDE with non-zero diffusion part) with finite time horizon T ; and two probability distributions on $\mathbb{R}^d$, μ_0 and μ_T , we aim at finding the closest diffusion, in terms of KL divergence (see Definition 2.25), to the reference; which admits μ_0 as marginal at time 0, and μ_T as marginal at time T .

In the following, for a given measurable space Ω , $\mathcal{M}_+(\Omega)$ will denote the set of positive measures on Ω , and $\mathcal{P}(\Omega)$ the set of probability measures on Ω .

The first formulation describes mathematically what is written above, generalizing it to reference measures which are not necessarily probability measures (Definition 2.1 in [Léo13]). It might not be very clear now why we need measures more general than probability measures. A motivating example is provided in subsection 2.3.2.

Definition 2.30 (Dynamic SBP in continuous time). *Let $\mathcal{R}$ be a σ -finite non-zero element of $\mathcal{M}_+(\mathcal{C}([0, T], \mathbb{R}^d))$, which we call the reference measure, and μ_0 and μ_T be probability measures on $\mathbb{R}^d$. We want to solve:*

$$\Pi^\star = \operatorname{argmin}\{KL(\Pi|\mathcal{R}) : \Pi \in \mathcal{P}(\mathcal{C}([0, T], \mathbb{R}^d)), \Pi_0 = \mu_0, \Pi_T = \mu_T\},$$

where Π_0 and Π_T are the marginals of Π at time 0 and T respectively.

This is called the dynamic formulation since we look for a measure on the path space $\mathcal{C}([0, T], \mathbb{R}^d)$.

Remark: We restrict ourselves to $\mathcal{C}([0, T], \mathbb{R}^d)$ instead of the whole space $(\mathbb{R}^d)^{[0, T]}$ for the following reasons:

1. **Finiteness of the KL Divergence:** To minimize $KL(\Pi|\mathcal{R})$, the measure Π must be absolutely continuous with respect to the reference measure $\mathcal{R}$. Because $\mathcal{R}$ (typically the law of a diffusion process like Brownian motion) is in general supported almost surely on continuous paths, any measure Π assigning mass to discontinuous trajectories would instantly result in $KL(\Pi|\mathcal{R}) = +\infty$.

2. **Topological setting.** Endowed with the supremum norm, $\mathcal{C}([0, T], \mathbb{R}^d)$ is a Polish space (separable and completely metrizable). This yields a well-behaved Borel σ -algebra and the validity of the tools used later, in particular the disintegration theorem for conditional probabilities.

Another formulation of the problem is the following (definition 2.2 in [Léo13]).

Definition 2.31 (Static SBP). *Let $\mathcal{R}_{0,T}$ be a σ -finite non-zero element of $\mathcal{M}_+((\mathbb{R}^d)^2)$, μ_0, μ_T be probability measures on $\mathbb{R}^d$. We want to solve:*

$$\Pi_S^\star = \operatorname{argmin}\{KL(\Pi_S|\mathcal{R}_{0,T}) : \Pi_S \in \mathcal{P}((\mathbb{R}^d)^2), \Pi_{S_0} = \mu_0, \Pi_{S_T} = \mu_T\},$$

where Π_{S_0} and Π_{S_T} are the marginals of Π_S at time 0 and T respectively.

This is called the static formulation as now we are just trying to find a coupling, and not a measure on the path space.

The two formulations are actually equivalent in the following way. Consider the dynamic formulation of the SBP with reference measure $\mathcal{R}$ and marginals μ_0 and μ_T . We want to solve:

$$\operatorname{argmin}\{KL(\Pi|\mathcal{R}) : \Pi \in \mathcal{P}(\mathcal{C}([0, T], \mathbb{R}^d)), \Pi_0 = \mu_0, \Pi_T = \mu_T\} \quad (*)$$

The Kullback-Leiber divergence between two paths measure can actually be split into two different terms, thanks to the following theorem (A8 from [Léo13]).

Theorem 2.32 (Disintegration and Chain Rule for Relative Entropy). *Let Y and Z be two Polish spaces equipped with their Borel σ -fields. For any measurable function $\varphi : Y \rightarrow Z$ and any positive measure $q \in \mathcal{M}_+(Y)$, we have the disintegration formula:*

$$q(dy) = \int_Z q(dy \mid \varphi = z) \varphi_{\#} q(dz),$$

where the mapping $z \in Z \mapsto q(\cdot \mid \varphi = z) \in \mathcal{P}(Y)$ is measurable.

Furthermore, the following additive property holds:

$$KL(p \mid r) = KL(\varphi_{\#} p \mid \varphi_{\#} r) + \int_Z KL(p(\cdot \mid \varphi = z) \mid r(\cdot \mid \varphi = z)) \varphi_{\#} p(dz),$$

which is valid for any probability measure $p \in \mathcal{P}(Y)$ and any σ -finite measure $r \in \mathcal{M}_+(Y)$.

Using this theorem with $p = \Pi$, $r = \mathcal{R}$, $\varphi(f) = (f(0), f(T))$ for $f \in (\mathcal{C}([0, T], \mathbb{R}^d))$ yields:

$$KL(\Pi \mid \mathcal{R}) = KL(\varphi_{\#} \Pi \mid \varphi_{\#} \mathcal{R}) + \int_{\mathbb{R}^d \times \mathbb{R}^d} KL(\Pi(\cdot \mid \varphi = (x, y)) \mid \mathcal{R}(\cdot \mid \varphi = (x, y))) \varphi_{\#} \Pi(dx, dy)$$

But now recall that for any Borel measurable set of endpoints $B \in \mathcal{B}(\mathbb{R}^d \times \mathbb{R}^d)$, the push-forward measure $\varphi_{\#} \Pi$, is defined as:

$$\varphi_{\#} \Pi(B) = \Pi(\varphi^{-1}(B)) = \Pi(\{\omega \in \mathcal{C}([0, T], \mathbb{R}^d) \mid (\omega(0), \omega(T)) \in B\}),$$

which is exactly the definition of the endpoints marginal of Π .

By the disintegration theorem with respect to the endpoint mapping $\varphi(X) = (X_0, X_T)$, we can decompose the measures Π and $\mathcal{R}$. Letting $\Pi_{0,T} := \varphi_{\#} \Pi$ and $\mathcal{R}_{0,T} := \varphi_{\#} \mathcal{R}$ denote the marginal distributions on the endpoints, there exist probability kernels $(x, y) \mapsto \Pi_{\cdot \mid x, y}$ and $(x, y) \mapsto \mathcal{R}_{\cdot \mid x, y}$ such that for $\Pi_{0,T}$ -almost every (x, y) ,

$$\Pi_{\cdot \mid x, y}, \mathcal{R}_{\cdot \mid x, y} \in \mathcal{P}(C([0, T], \mathbb{R}^d)).$$

Hence, for $\Pi_{0,T}$ -almost every (x, y) ,

$$\begin{aligned} KL(\Pi_{\cdot \mid x, y} \mid \mathcal{R}_{\cdot \mid x, y}) &= \int \frac{d\Pi_{\cdot \mid x, y}}{d\mathcal{R}_{\cdot \mid x, y}} \log \left(\frac{d\Pi_{\cdot \mid x, y}}{d\mathcal{R}_{\cdot \mid x, y}} \right) d\mathcal{R}_{\cdot \mid x, y} \\ &\geq \left(\int \frac{d\Pi_{\cdot \mid x, y}}{d\mathcal{R}_{\cdot \mid x, y}} d\mathcal{R}_{\cdot \mid x, y} \right) \log \left(\int \frac{d\Pi_{\cdot \mid x, y}}{d\mathcal{R}_{\cdot \mid x, y}} d\mathcal{R}_{\cdot \mid x, y} \right) \\ &= \Pi_{\cdot \mid x, y}(C([0, T], \mathbb{R}^d)) \log \Pi_{\cdot \mid x, y}(C([0, T], \mathbb{R}^d)) = 0, \end{aligned}$$

where the inequality follows from Jensen's inequality applied to the convex function $t \mapsto t \log t$, and the last equality uses that $\Pi_{\cdot \mid x, y}$ is a probability measure. It follows that

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} KL(\Pi_{\cdot \mid x, y} \mid \mathcal{R}_{\cdot \mid x, y}) \Pi_{0,T}(dx dy) \geq 0.$$

Moreover, this term vanishes when choosing $\Pi_{\cdot \mid x, y} = \mathcal{R}_{\cdot \mid x, y}$, since then

$$KL(\Pi_{\cdot \mid x, y} \mid \mathcal{R}_{\cdot \mid x, y}) = 0$$

for $\Pi_{0,T}$ -almost every (x, y) . As this choice does not depend on the choice of $\Pi_{0,T}$, solving (*) just amounts to solve:

$$\operatorname{argmin}\{KL(\pi|\mathcal{R}_{0,T}) : \pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d), \pi_0 = \mu_0, \pi_T = \mu_T\}$$

and then to choose Π such that $\Pi_{\cdot|x,y} = \mathcal{R}_{\cdot|x,y}$.

Consequently, solving the dynamic problem reduces to solving its static counterpart. Since the conditional measures (bridges) of the optimal solution are necessarily identical to those of the reference, knowing the reference bridges allows us to directly construct the optimal path measure.

In practice, we deal with discretizations of stochastic processes, hence we need the following version of the Schrödinger Bridge Problem (equation (9) in [Bor21]).

Definition 2.33 (SBP in discrete time). *Let $\{t_0 = 0, \dots, t_N = T\}$ be a discretization of $[0, T]$. Let $\mathcal{R}_{disc} = (X_{t_0}, \dots, X_{t_N})_{\#} \mathcal{R}$ be the finite-dimensional distribution of the reference measure on the grid, and μ_0, μ_T be probability measures on $\mathbb{R}^d$. We want to solve:*

$$\Pi^* = \operatorname{argmin}\{KL(\Pi|\mathcal{R}_{disc}) : \Pi \in \mathcal{P}((\mathbb{R}^d)^{N+1}), \Pi_0 = \mu_0, \Pi_N = \mu_T\}$$

2.3.2 Links between Schrödinger Bridge Problem and Optimal Transport

Now, let us consider the following random system:

$$X_t = x + \sqrt{\frac{\varepsilon}{2}} W_t, \text{ for } t \in [0, 1]$$

The first point x is distributed according to an improper uniform prior on $\mathbb{R}^d$. That is, if $\mathcal{R}$ is the corresponding path measure on $\mathcal{C}([0, 1], \mathbb{R}^d)$, then $R_0(dx) = dx$ is the Lebesgue measure on $\mathbb{R}^d$.

In the static SBP, we focus on $\mathcal{R}_{0,1}$. Its Radon-Nikodym derivative $r_{0,1}$ with respect to the Lebesgue measure on $\mathbb{R}^d \times \mathbb{R}^d$ is then:

$$r_{0,1}(x, y) = \frac{d\mathcal{R}_{0,1}}{dx dy}(x, y) \implies r_{0,1}(x, y) = r_0(x) p_{0,1}(y | x) \implies r_{0,1}(x, y) \propto 1 \times \frac{1}{(\pi\varepsilon)^{d/2}} \exp\left(-\frac{\|x - y\|^2}{\varepsilon}\right)$$

where r_0 is the Radon-Nikodym derivative of R_0 with respect to the Lebesgue measure on $\mathbb{R}^d$ and $p_{0,1}(y | x)$ is the (this time) probability density function defining the likelihood of arriving at state y at time $t = 1$ given the initial state x at time $t = 0$. In our case:

$$p_{0,1}(y | x) = \frac{1}{(\pi\varepsilon)^{d/2}} \exp\left(-\frac{\|x - y\|^2}{\varepsilon}\right)$$

because $X_1|x \sim N(x, \frac{\varepsilon}{2}\mathcal{I}_d)$.

The static Schrödinger Bridge Problem is formulated as:

$$\min_{\pi \in \Gamma(\mu_0, \mu_1)} \text{KL}(\pi \| \mathcal{R}_{0,1})$$

By definition of the Kullback-Leibler divergence and separating the logarithmic terms, we obtain:

$$\begin{aligned} \text{KL}(\pi \| \mathcal{R}_{0,1}) &= \int \pi(x, y) \log \pi(x, y) dx dy - \int \pi(x, y) \log r_{0,1}(x, y) dx dy \\ &= -\text{Ent}(\pi) - \int \pi(x, y) \left(\log C - \frac{\|x - y\|^2}{\varepsilon} \right) dx dy \end{aligned}$$

Since π is a probability density, $\int \pi(x, y) dx dy = 1$. The expression simplifies to:

$$\text{KL}(\pi \| \mathcal{R}_{0,1}) = -\text{Ent}(\pi) + \frac{1}{\varepsilon} \int \|x - y\|^2 \pi(dx, dy) - \log C$$

Dropping the additive constant $\log C$ which is independent of π and multiplying the objective function by ε , the minimization reduces strictly to the standard Entropic Optimal Transport problem:

$$\min_{\pi \in \Gamma(\mu_0, \mu_1)} \left[\int \|x - y\|^2 \pi(dx, dy) - \varepsilon \text{Ent}(\pi) \right]$$

Remark: Choosing the Radon-Nikodym derivative: $r_{0,1}(x, y) \propto \exp\left(-\frac{c(x, y)}{\varepsilon}\right)$ would yield the same connection with Entropic Optimal Transport as before but with the cost function $c(., .)$.

Moreover:

It is customary in the literature to define the reference measure $\mathcal{R}$ using the improper Lebesgue prior $R_0(x) = 1$. However, this specific choice is not strictly necessary to establish the equivalence between the static SBP and Entropic Optimal Transport. If we had chosen any proper prior density $R_0(x)$ with respect to the Lebesgue measure, the term $\int \pi(x, y) \log C dx dy$ would have become:

$$\int \pi(x, y) \log R_0(x) dx dy = \int \mu_0(x) \log R_0(x) dx.$$

Since the optimization is performed over couplings $\pi \in \Gamma(\mu_0, \mu_1)$ with fixed marginals, this integral is a constant with respect to π . It therefore drops out of the argmin, consistently recovering the standard Regularized Optimal Transport problem. The primary motivation for using the Lebesgue prior is thus to maintain mathematical symmetry in the reference measure $\mathcal{R}_{0,1}(x, y) = \mathcal{R}_{0,1}(y, x)$, reflecting the spatial symmetry of the Brownian transition kernel.

Indeed, we know that:

$$\mathcal{R}_{0,1}(x, y) = R_0(x) p_{0,1}(y | x)$$

but:

$$p_{0,1}(y | x) = p_{0,1}(x | y) \propto \exp\left(-\frac{\|x - y\|^2}{\varepsilon}\right)$$

Hence, in order to have $\mathcal{R}_{0,1}(x, y) = \mathcal{R}_{0,1}(y, x)$, we would need that $R_0(x) = R_0(y) \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d$, which means that R_0 has to be constant.

2.4 Time reversal of SDEs and Langevin dynamics

In this section, we will cover theoretical knowledge about the time reversal of SDEs and Langevin dynamics, which are essential to the score-based approach of generative modeling. First, we need a few definitions.

2.4.1 Markov processes and transition semigroups

Different definitions of Markov processes can be found in the literature. We choose here a quite simple definition, that is still general enough for our purpose.

Definition 2.34. *Given a process $X = \{X_t\}_{t \geq 0}$ on $\mathbb{R}^n$, it is Markov if for any $t, s \geq 0$ and any bounded measurable function $g : \mathbb{R}^n \rightarrow \mathbb{R}$,*

$$\mathbb{E}[g(X_{t+s}) \mid \mathcal{F}_t] = \mathbb{E}[g(X_{t+s}) \mid X_t] \quad \mathbb{P} - a.s.$$

A Markov process induces a so-called transition semigroup:

Definition 2.35. *For a Markov process X , we define the transition semigroup $(P_{s,t})_{0 \leq s \leq t}$ associated to X as $P_{s,t}g(x) = \mathbb{E}[g(X_t) \mid X_s = x]$, where g is any bounded measurable function.*

One can easily check using the properties of conditional expectations that $\forall t \geq 0 : P_{t,t} = I_d$.

Remark:

If we consider a time-homogeneous setting, meaning that $s - s' = t - t' \implies P_{s,s'} = P_{t,t'}$, we can then define the transition semigroup family as $P_t = P_{0,t}, t \geq 0$. Then we can prove the Chapman-Kolmogorov equation:

$$\begin{aligned} P_{s+t}f(x) &= \mathbb{E}[f(X_{s+t}) \mid X_0 = x] \\ &= \mathbb{E}[\mathbb{E}[f(X_{s+t}) \mid \mathcal{F}_s] \mid X_0 = x] \\ &= \mathbb{E}[\mathbb{E}[f(X_{s+t}) \mid X_s] \mid X_0 = x] \quad (\text{Markov property}) \end{aligned}$$

As the process is time-homogeneous, we get:

$$\mathbb{E}[f(X_{s+t}) \mid X_s] = (P_t f)(X_s)$$

Hence:

$$\begin{aligned} P_{s+t}f(x) &= \mathbb{E}[(P_t f)(X_s) \mid X_0 = x] \\ &= P_s(P_t f)(x) \quad (\text{by definition of } P_s \text{ applied to the function } P_t f) \\ &= (P_s \circ P_t)f(x) \end{aligned}$$

As it is true for all f and for all x , we deduce:

$$P_{s+t} = P_s P_t$$

Remark: This in particular implies that the composition $P_s P_t$ stays in the set $\{P_t, t \geq 0\}$ and that it is associative, as $(P_s P_t)P_v = P_{s+t+v} = P_s(P_t P_v)$. This means that the set $\{P_t, t \geq 0\}$ with the composition law is a semigroup (in the algebraic sense of the term).

However, if we consider a time-inhomogeneous situation, the composition $P_{s,t}P_{u,v}$ does not

belong in the set $\{P_{s,t}, t \geq s \geq 0\}$ as soon as $t \neq u$. Hence, although we still call it a transition semigroup, it is not a semigroup in the algebraic sense of the term.

But we still have the Chapman-Kolmogorov equation, which now takes the form:

$$P_{s,s'}P_{s',t} = P_{s,t} \quad \forall s \leq s' \leq t$$

Working on $\mathbb{R}^n$ allows us to define the following quantity (definition 1.2.3 in [Che26]).

Definition 2.36. For $0 \leq s \leq t$, we define the infinitesimal generator L_t as $L_t g(x) = \lim_{\delta \rightarrow 0} \frac{P_{t,t+\delta}g(x) - g(x)}{\delta}$, for all functions g such that the limit is well defined.

These quantities will be of the most importance when dealing with the time reversal of SDE.

2.4.2 Langevin dynamics

As it is the starting point of score-based models, we will focus on one important SDE, the Langevin dynamics:

$dX_t = -\nabla U(X_t)dt + \sqrt{2}dB_t$, where $U : \mathbb{R}^n \rightarrow \mathbb{R}$ is a twice differentiable function and B is a standard Brownian motion on $\mathbb{R}^n$.

Our goal will be to describe the evolution of the law of X_t with respect to time. For this, we compute the infinitesimal generator of the Langevin diffusion, for functions $g \in C_c^\infty$.

Fix $x \in \mathbb{R}^n$ and write $\mathbb{E}_x[\cdot] := \mathbb{E}[\cdot \mid X_0 = x]$. Set

$$h := \langle \nabla g, -\nabla U \rangle + \Delta g.$$

Since $g \in C_c^\infty(\mathbb{R}^n)$ and ∇U is continuous, both ∇g and h are continuous with compact support, hence: $C := \|\nabla g\|_\infty < \infty$ and $\|h\|_\infty < \infty$.

Step 1 (Itô's formula on the finite horizon $[0, \delta]$). As the diffusion matrix is $\sigma\sigma^\top = 2I_d$, Itô's formula applied to $g(X)$ over $[0, \delta]$ gives $\frac{1}{2} \text{tr}(\sigma\sigma^\top D^2 g) = \Delta g$, so

$$g(X_\delta) - g(x) = \int_0^\delta h(X_s) ds + \sqrt{2} M_\delta, \quad M_t := \int_0^t \langle \nabla g(X_s), dB_s \rangle.$$

Step 2 (the martingale term vanishes in mean, finite horizon). The quadratic variation of M satisfies, on $[0, \delta]$,

$$\mathbb{E}_x[\langle M \rangle_\delta] = \mathbb{E}_x \left[\int_0^\delta |\nabla g(X_s)|^2 ds \right] \leq C^2 \delta < \infty.$$

Hence M is a square-integrable (true) martingale on $[0, \delta]$ and $\mathbb{E}_x[M_\delta] = M_0 = 0$.

Step 3 (take expectation, divide by δ , and apply Fubini). Taking $\mathbb{E}_x$ in Step 1 and using Step 2,

$$\frac{P_\delta g(x) - g(x)}{\delta} = \frac{1}{\delta} \mathbb{E}_x \left[\int_0^\delta h(X_s) ds \right].$$

Since $\mathbb{E}_x \int_0^\delta |h(X_s)| ds \leq \|h\|_\infty \delta < \infty$, we can apply Fubini's theorem to get:

$$\frac{P_\delta g(x) - g(x)}{\delta} = \frac{1}{\delta} \int_0^\delta \phi(s) ds, \quad \phi(s) := \mathbb{E}_x[h(X_s)].$$

Step 4 (continuity of ϕ at 0). By path continuity, $X_s \rightarrow X_0 = x$ a.s. as $s \rightarrow 0^+$; since h is continuous, $h(X_s) \rightarrow h(x)$ a.s., and $|h(X_s)| \leq \|h\|_\infty$. Dominated convergence then yields

$$\phi(s) = \mathbb{E}_x[h(X_s)] \xrightarrow{s \rightarrow 0^+} h(x),$$

i.e. ϕ is continuous at 0 with $\phi(0) = h(x)$.

Step 5 (pass to the limit, outside the expectation). The time-average of a function continuous at 0 converges to its value at 0 by the fundamental theorem of calculus. Hence, we get:

$$Lg(x) = \lim_{\delta \rightarrow 0^+} \frac{1}{\delta} \int_0^\delta \phi(s) ds = \phi(0) = h(x) = \langle \nabla g(x), -\nabla U(x) \rangle + \Delta g(x).$$

Remark (Beyond compactly supported g). If $g \in C^2$ has only locally bounded derivatives, or ∇U is unbounded, replace $[0, \delta]$ by $[0, \delta \wedge \tau_n]$ with $\tau_n := \inf\{t \geq 0 : |X_t| \geq n\}$. Steps 1–2 hold with M stopped at τ_n , since it is a bounded local martingale, hence a true martingale. Steps 3–5 are then valid on the localized interval. Thus, letting $n \rightarrow \infty$ after $\delta \rightarrow 0$ recovers the same formula, since $\mathbb{P}(\tau_n > \delta) \rightarrow 1$ and h is continuous at x .

Note that this allows us to derive a PDE for $P_t g$. Indeed, we have (proposition 1.2.5 in [Che26]):

Theorem 2.37. *For all $t \geq 0$, it holds that*

$$\partial_t P_t f = \mathcal{L} P_t f = P_t \mathcal{L} f$$

In particular, $\mathcal{L}$ commutes with the semigroup $(P_t)_{t \geq 0}$.

Proof. Using $P_{t+h} = P_t P_h = P_h P_t$, we get:

$$\lim_{h \searrow 0} \frac{P_{t+h} f - P_t f}{h} = \lim_{h \searrow 0} \frac{P_h - \text{id}}{h} P_t f = \mathcal{L} P_t f.$$

Repeating the computation but factoring out P_t on the left yields the second equality.

Applying this result to Langevin dynamics and the expression of $\mathcal{L}$ with function $P_t g$, we get the *Kolmogorov backward equation*:

$$\partial_t P_t g = \langle \nabla P_t g, -\nabla U \rangle + \Delta P_t g$$

where the gradient and the Laplacian are taken with respect to the space variable x .

Recalling the definition of P_t , the Kolmogorov backward equation dictates how the conditional expectation evolves over time based purely on its spatial derivatives at the starting point. It gives us information about the law of X_t given a fixed future.

There is a natural dual formulation of this PDE, giving us information about the law of X_t given a fixed starting point.

Indeed, we can write $\mathbb{E}[g(X_t)]$ as $\int g(x) \rho_t(x) dx = \langle g, \rho_t \rangle$ where we use the classical inner product on $L^2(\mathbb{R}^n)$, and where ρ_t is the density of X_t .

However, we can also write it as:

$$\mathbb{E}[g(X_t)] = \mathbb{E}[\mathbb{E}[g(X_t) \mid X_0]]$$

And using that

$$\mathbb{E}[g(X_t) \mid X_0 = x] = P_t g(x)$$

we get:

$$\mathbb{E}[g(X_t)] = \mathbb{E}[P_t g(X_0)] = \int P_t g(x) \rho_0(x) dx = \langle P_t g, \rho_0 \rangle$$

Hence we have:

$$\langle g, \rho_t \rangle = \langle P_t g, \rho_0 \rangle$$

Differentiating both sides with respect to t , we obtain

$$\langle g, \partial_t \rho_t \rangle = \langle \mathcal{L} P_t g, \rho_0 \rangle$$

Then using $\langle f, \rho_t \rangle = \langle P_t f, \rho_0 \rangle$ with $f = \mathcal{L}g$, and the fact that $\mathcal{P}_t$ and $\mathcal{L}$ commute, we obtain:

$$\langle \mathcal{L}g, \rho_t \rangle = \langle P_t \mathcal{L}g, \rho_0 \rangle$$

Now if we define $\mathcal{L}^*$ by $\langle g, \mathcal{L}^* h \rangle = \langle \mathcal{L}g, h \rangle$ for all g, h , we get:

$$\langle g, \partial_t \rho_t - \mathcal{L}^* \rho_t \rangle = 0$$

As this is valid for all g , by the fundamental lemma of calculus of variations, we get that:

$$\partial_t \rho_t = \mathcal{L}^* \rho_t$$

which is known as the Fokker-Planck equation.

In the specific case of Langevin dynamics, we get:

$$\langle \mathcal{L}g, h \rangle = \int_{\mathbb{R}^d} (\langle \nabla g, -\nabla U \rangle + \Delta g) h dx = - \int_{\mathbb{R}^d} \langle \nabla g, h \nabla U \rangle dx + \int \Delta g h dx$$

But now we can use the identity (which just amounts to use the product rule):

$$\nabla \cdot (gh \nabla U) = \langle \nabla g, h \nabla U \rangle + g(\nabla \cdot h \nabla U)$$

Integrating over $\mathbb{R}^d$ yields:

$$\int_{\mathbb{R}^d} \nabla \cdot (gh \nabla U) dx = \int_{\mathbb{R}^d} \langle \nabla g, h \nabla U \rangle dx + \int_{\mathbb{R}^d} g(\nabla \cdot h \nabla U) dx$$

By the theorem of divergence, we have:

$$\int_{B_R} \nabla \cdot (gh \nabla U) dx = \int_{\partial B_R} \langle gh \nabla U, \mathbf{n} \rangle dS$$

where B_R is any ball of $\mathbb{R}^d$ of radius R .

But letting $R \rightarrow \infty$, as g and h are in $C_c^\infty(\mathbb{R}^n)$, the right-hand term vanishes. Hence we get:

$$\int_{\mathbb{R}^d} \nabla \cdot (gh \nabla U) dx = 0$$

And then:

$$\int_{\mathbb{R}^d} \langle \nabla g, h \nabla U \rangle dx = - \int_{\mathbb{R}^d} g(\nabla \cdot h \nabla U) dx$$

For the second part, applying the same identity and recalling that $\Delta g = \nabla \cdot \nabla g$ yields:

$$\int (\nabla \cdot \nabla g) h dx = - \int \langle \nabla g, \nabla h \rangle dx$$

Using the same identity once again exchanging the roles of g and h gives:

$$- \int \langle \nabla g, \nabla h \rangle dx = \int g \Delta h dx$$

So finally:

$$\langle \mathcal{L}_g, h \rangle = \langle g, \nabla \cdot h \nabla U + \Delta h \rangle$$

This gives us: $\mathcal{L}^* h = \nabla \cdot h \nabla U + \Delta h$

Recalling that $\partial_t \rho_t = \mathcal{L}^* \rho_t$, we can derive the so-called Fokker-Planck equation for the Langevin SDE:

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla U) + \Delta \rho_t$$

There is actually also a PDE for processes which have zero diffusion part, known as the continuity equation (Theorem 1.3.17 in [Che26]).

Theorem 2.38 (Continuity equation). *Let $(v_t)_{t \geq 0}$ be a family of vector fields and suppose that the process $(X_t)_{t \geq 0}$ evolves according to $\dot{X}_t = v_t(X_t)$, and that X_t admits density μ_t . Then, μ_t evolves according to the continuity equation*

$$\partial_t \mu_t + \operatorname{div}(\mu_t v_t) = 0. \tag{2.23}$$

Proof: Let $\varphi \in C_c^\infty(\mathbb{R}^d)$ be a test function. By the definition of the density μ_t , we have:

$$\frac{d}{dt} E[\varphi(X_t)] = \frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x) \mu_t(x) dx = \int_{\mathbb{R}^d} \varphi(x) \partial_t \mu_t(x) dx$$

On the other hand, using the Chain Rule and the dynamics $\dot{X}_t = v_t(X_t)$, we have:

$$\begin{aligned} \frac{d}{dt} \mathbb{E}[\varphi(X_t)] &= \mathbb{E} \left[\frac{d}{dt} \varphi(X_t) \right] \\ &= \mathbb{E} [\langle \nabla \varphi(X_t), \dot{X}_t \rangle] \\ &= \mathbb{E} [\langle \nabla \varphi(X_t), v_t(X_t) \rangle] \\ &= \int_{\mathbb{R}^d} \langle \nabla \varphi, v_t \rangle \mu_t dx \\ &= \int_{\mathbb{R}^d} \langle \nabla \varphi, \mu_t v_t \rangle dx. \end{aligned}$$

Now, consider the vector field $A = \varphi \mu_t v_t$. Applying the product rule, we have:

$$\nabla \cdot (\varphi \mu_t v_t) = \langle \nabla \varphi, \mu_t v_t \rangle + \varphi \nabla \cdot (\mu_t v_t)$$

Integrating this over $\mathbb{R}^d$ yields:

$$\int_{\mathbb{R}^d} \nabla \cdot (\varphi \mu_t v_t) dx = \int_{\mathbb{R}^d} \langle \nabla \varphi, \mu_t v_t \rangle dx + \int_{\mathbb{R}^d} \varphi \nabla \cdot (\mu_t v_t) dx$$

By the Divergence Theorem, for a ball B_R of radius R :

$$\int_{B_R} \nabla \cdot (\varphi \mu_t v_t) dx = \int_{\partial B_R} \langle \varphi \mu_t v_t, \mathbf{n} \rangle dS$$

Since $\varphi \in C_c^\infty(\mathbb{R}^d)$, it has compact support. Thus, for a sufficiently large R , φ vanishes on the boundary ∂B_R . Letting $R \rightarrow \infty$, the boundary integral vanishes:

$$\int_{\mathbb{R}^d} \nabla \cdot (\varphi \mu_t v_t) dx = 0$$

It follows that:

$$\int_{\mathbb{R}^d} \langle \nabla \varphi, \mu_t v_t \rangle dx = - \int_{\mathbb{R}^d} \varphi \nabla \cdot (\mu_t v_t) dx$$

Finally, comparing our two expressions for $\frac{d}{dt} E[\varphi(X_t)]$, we obtain:

$$\int_{\mathbb{R}^d} \varphi \partial_t \mu_t dx = - \int_{\mathbb{R}^d} \varphi \nabla \cdot (\mu_t v_t) dx$$

Rearranging the terms:

$$\int_{\mathbb{R}^d} \varphi (\partial_t \mu_t + \nabla \cdot (\mu_t v_t)) dx = 0$$

Since this holds for every test function $\varphi \in C_c^\infty(\mathbb{R}^d)$, we conclude that:

$$\partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0$$

which is equivalent to the statement in the theorem since by definition of the divergence, $\operatorname{div}(u) = \nabla \cdot u$ for any $u : \mathbb{R}^d \rightarrow \mathbb{R}$. $\square$

From the generative modeling point of view, we will see that continuity equations are useful to define Flow-based generative models.

There is actually a nice link between the Fokker-Planck equation and the continuity equation, which we will illustrate with any SDE with constant diffusion part.

We consider:

$$dX_t = b(X_t)dt + \sigma dW_t$$

The Fokker-Planck equation associated with the given SDE is:

$$\partial_t \rho_t = -\nabla \cdot (\rho_t b) + \frac{\sigma^2}{2} \Delta \rho_t = -\nabla \cdot (\rho_t b) + \nabla \cdot \left(\frac{\sigma^2}{2} \nabla \rho_t \right).$$

Indeed:

Let $f \in C_c^2(\mathbb{R}^d)$ be an arbitrary test function. Applying Itô's Lemma to $f(X)$ for the SDE:

$$dX_t = b(X_t)dt + \sigma dW_t$$

yields:

$$df(X_t) = \left(b(X_t) \cdot \nabla f(X_t) + \frac{\sigma^2}{2} \Delta f(X_t) \right) dt + \sigma \nabla f(X_t) \cdot dW_t. \quad (2.24)$$

Taking the expectation, using lemma 2.6 and lemma 2.8 on the σ -martingale $\int_0^\cdot f(X_s) dW_s$ yields that it is a true martingale, hence of constant expectation 0. Hence:

$$\frac{d}{dt} \mathbb{E}[f(X_t)] = \mathbb{E} \left[b(X_t) \cdot \nabla f(X_t) + \frac{\sigma^2}{2} \Delta f(X_t) \right]. \quad (2.25)$$

Expressing the expectations in terms of the probability density $\rho_t(x)$:

$$\int_{\mathbb{R}^d} f(x) \partial_t \rho_t(x) dx = \int_{\mathbb{R}^d} \left(b(x) \cdot \nabla f(x) + \frac{\sigma^2}{2} \Delta f(x) \right) \rho_t(x) dx. \quad (2.26)$$

Applying integration by parts in the same way that for the proof of Theorem 1.4.2, the boundary terms vanish and we get:

$$\int_{\mathbb{R}^d} (b(x) \rho_t(x)) \cdot \nabla f(x) dx = - \int_{\mathbb{R}^d} f(x) \nabla \cdot (b(x) \rho_t(x)) dx, \quad (2.27)$$

$$\int_{\mathbb{R}^d} \rho_t(x) \Delta f(x) dx = \int_{\mathbb{R}^d} f(x) \Delta \rho_t(x) dx. \quad (2.28)$$

Substituting these back yields:

$$\int_{\mathbb{R}^d} f(x) \left[\partial_t \rho_t(x) + \nabla \cdot (\rho_t(x) b(x)) - \frac{\sigma^2}{2} \Delta \rho_t(x) \right] dx = 0. \quad (2.29)$$

Since this identity holds for any $f \in C_c^2$, the bracketed term must be zero almost everywhere, which gives the Fokker-Planck equation:

$$\partial_t \rho_t = -\nabla \cdot (\rho_t b) + \frac{\sigma^2}{2} \Delta \rho_t = -\nabla \cdot (\rho_t b) + \nabla \cdot \left(\frac{\sigma^2}{2} \nabla \rho_t \right). \quad (2.30)$$

Now by the linearity of the divergence:

$$\partial_t \rho_t = -\nabla \cdot \left(\rho_t b - \frac{\sigma^2}{2} \nabla \rho_t \right).$$

Assuming $\rho_t(x) > 0$:

$$\partial_t \rho_t = -\nabla \cdot \left(\rho_t \left(b - \frac{\sigma^2}{2} \frac{\nabla \rho_t}{\rho_t} \right) \right) = -\nabla \cdot \left(\rho_t \left(b - \frac{\sigma^2}{2} \nabla \log \rho_t \right) \right).$$

By defining the velocity field as:

$$v_t(x) := b(x) - \frac{\sigma^2}{2} \nabla \log \rho_t(x),$$

we obtain the continuity equation:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0.$$

Stationary distribution and convergence To understand why is the Fokker-Planck equation useful, we need to define what are stationary and invariant measures (definition 1.2.10 in [Che26]).

Definition 2.39. For a stochastic process X , a measure π is said to be stationary if $X_0 \sim \pi \implies X_t \sim \pi$, $\forall t \geq 0$. In other words, $\partial_t \pi_t = 0$ where π_t is the law of X_t .

Definition 2.40. A measure μ is said to be invariant for a transition semigroup $(P_t)_{t \geq 0}$ if it satisfies

$$\int_{\mathbb{R}^n} P_t g(x) \mu(dx) = \int_{\mathbb{R}^n} g(x) \mu(dx)$$

for all bounded measurable functions $g : \mathbb{R}^n \rightarrow \mathbb{R}$ and all $t \geq 0$.

It is easy to check that if a measure μ is stationary, then it is also invariant. Indeed, we know that X_0 and X_t have the same law if and only if for any measurable and bounded function g , $\mathbb{E}[g(X_0)] = \mathbb{E}[g(X_t)]$. So let g be such a function. We have then:

$$\langle \mu, g \rangle = \mathbb{E}[g(X_0)] = \mathbb{E}[g(X_t)] = \langle \mu, P_t g \rangle$$

where the second equality comes from the fact that μ is stationary.

On the other hand, if we consider a Markov process $X = (X_t)_{t \geq 0}$ such that μ is invariant for the transition semigroup associated to X , then μ is stationary. Indeed:

$$\mathbb{E}[g(X_t)] = \langle \mu, P_t g \rangle = \langle \mu, g \rangle = \mathbb{E}[g(X_0)]$$

Our aim is now to find a stationary distribution for the Langevin dynamics. We are looking for a measure such that:

$$\partial_t \rho_t = \mathcal{L}^* \rho_t = \nabla \cdot (\rho_t \nabla U) + \Delta \rho_t = 0$$

One easily sees that $\rho_t = \frac{e^{-U}}{Z}$, with Z the finite normalization constant, satisfies the desired equation. Indeed:

$$\nabla \cdot \left(\frac{e^{-U}}{Z} \nabla U \right) = \sum_{i=1}^n \frac{\partial}{\partial u_i} \left(\frac{e^{-U}}{Z} \frac{\partial U}{\partial u_i} \right) = -\Delta \left(\frac{e^{-U}}{Z} \right)$$

Now that we found a stationary distribution for the Langevin process, we would like to assess and quantify the convergence of the distribution of the process at time t towards the stationary distribution.

The first result, from section 1.5.1 in [Che26] relies on the assumption that the potential U is α -strongly convex.

Theorem 2.41 (Convergence in Wasserstein distance for the Langevin SDE). *If U is α -strongly convex, then for any initial distribution μ_0 of X_0 that admits a second moment, the law μ_t of X_t satisfies*

$$W_2(\mu_t, \mu) \leq e^{-\alpha t} W_2(\mu_0, \mu).$$

Proof: Consider two processes $\{X_t\}_{t \geq 0}$ and $\{Y_t\}_{t \geq 0}$ defined by the SDEs:

$$\begin{aligned} dX_t &= -\nabla U(X_t)dt + \sqrt{2}dB_t, & X_0 &\sim \mu_0, \\ dY_t &= -\nabla U(Y_t)dt + \sqrt{2}dB_t, & Y_0 &\sim \mu. \end{aligned}$$

Subtracting the two SDEs, we have:

$$d(X_t - Y_t) = -(\nabla U(X_t) - \nabla U(Y_t))dt.$$

Now define $Z_t = X_t - Y_t$. using Itô's formula, we get:

$$d|Z_t|^2 = \langle 2Z_t, dZ_t \rangle$$

as Z_t is of finite variation. Hence:

$$d|X_t - Y_t|^2 = \langle 2(X_t - Y_t), d(X_t - Y_t) \rangle = \langle 2(X_t - Y_t), -(\nabla U(X_t) - \nabla U(Y_t))dt \rangle = -2dt \langle X_t - Y_t, \nabla U(X_t) - \nabla U(Y_t) \rangle$$

Hence:

$$\frac{d}{dt} |X_t - Y_t|^2 = -2 \langle X_t - Y_t, \nabla U(X_t) - \nabla U(Y_t) \rangle \leq -2\alpha |X_t - Y_t|^2. \mathbb{P} - a.s$$

where the last inequality comes from the following:

Let $x, y \in \mathbb{R}^n$ and $t \in [0, 1]$: By the Fundamental Theorem of Calculus:

$$\nabla U(y) - \nabla U(x) = \int_0^1 \nabla^2 U(x + t(y - x))(y - x)dt$$

Taking the inner product with $(y - x)$ on both sides yields:

$$\langle \nabla U(y) - \nabla U(x), y - x \rangle = \left\langle \int_0^1 \nabla^2 U(x + t(y - x))(y - x) dt, y - x \right\rangle$$

Then:

$$\langle \nabla U(y) - \nabla U(x), y - x \rangle = \int_0^1 \langle \nabla^2 U(x + t(y - x))(y - x), y - x \rangle dt$$

By the definition of α -strong convexity, the Hessian matrix minus αI_n is positive semi-definite, which implies that for any vector v and any point z , $\langle \nabla^2 U(z)v, v \rangle \geq \alpha \|v\|^2$. Applying this property with $v = y - x$, we obtain:

$$\langle \nabla U(y) - \nabla U(x), y - x \rangle \geq \alpha \|y - x\|^2$$

Applying Gronwall's inequality, we obtain:

$$|X_t - Y_t|^2 \leq e^{-2\alpha t} |X_0 - Y_0|^2 \quad \mathbb{P} - a.s$$

We can choose $X_0 \sim \mu_0$ and $Y_0 \sim \mu$ coupled such that:

$$\mathbb{E}|X_0 - Y_0|^2 = W_2(\mu_0, \mu)^2.$$

Taking the infimum over all couplings on the left hand side of the previous inequality, we obtain:

$$W_2(\mu_t, \mu) \leq e^{-\alpha t} W_2(\mu_0, \mu).$$

□

Remark: This theorem also ensures that the invariant measure μ is unique among measures which admit a second moment. Indeed, assume that there is such a measure ν which is different from μ . Then, as $W_2(\mu_t, \mu) \leq e^{-\alpha t} W_2(\mu_0, \mu)$ is true for any starting distribution μ_0 , if we take $\mu_0 = \nu$ then $\mu_t = \nu \quad \forall t$ and we get:

$$W_2(\nu, \mu) \leq e^{-\alpha t} W_2(\nu, \mu)$$

Now as μ is different from ν , $W_2(\nu, \mu)$ is different than 0 and we get:

$$1 \leq e^{-\alpha t}$$

which is absurd as both α and t are positive.

We now introduce another convergence result, which relies this time on the chi-squared divergence. As we will see in the next chapter, this result has a lot to do with the performance of generative models based on Langevin SDE.

We first need the two following definitions:

Definition 2.42 (Chi-squared divergence). For two probability measures μ and ν on $\mathbb{R}^n$, the chi-square divergence is defined as:

$$\chi^2(\mu \parallel \nu) = \int \frac{(d\mu - d\nu)^2}{d\nu} = \int \frac{(d\mu)^2}{d\nu} - 1$$

Definition 2.43 (Poincaré inequality). A probability measure μ on $\mathbb{R}^n$ is said to satisfy a Poincaré inequality with constant $C > 0$ if for all smooth functions $g : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\text{Var}_\mu(g) \leq C \int |\nabla g|^2 d\mu.$$

The smallest number C_P such that the above inequality holds is then called the Poincaré constant.

Remark: It can be proven that if U is α -strongly convex, then $\mu \propto e^{-U}$ satisfies the Poincaré inequality with a constant $C_P \leq \frac{1}{\alpha}$ by combining Theorem 1.2.30 and Theorem 1.2.31 in [Che26].

The Poincaré inequality states that a high global variance $\text{Var}_\mu(g)$ must be compensated for by steep gradients $|\nabla g|^2$ in regions where the probability measure μ is actually concentrated. This is what makes bottlenecks fatal. If μ has two wells of comparable mass separated by a region of negligible density, take $g \approx -1$ on one well, $g \approx +1$ on the other, and let it interpolate across the barrier: then $\text{Var}_\mu(g) \approx 1$ while $\int |\nabla g|^2 d\mu \approx 0$, forcing C_P to be large.

Strong convexity forbids exactly this configuration. If $\nabla^2 U \succeq \alpha \text{Id}$, then $\mu \propto e^{-U}$ is a single well with no low-density corridor separating regions of mass, and $C_P \leq 1/\alpha$.

Now we can state the following result, from section 1.5.1 in [Che26].

Theorem 2.44 (Convergence in Chi-squared divergence for the Langevin SDE). Let μ_t be the density of X_t with $X_0 \sim \mu_0$, and assume $\mu_t \ll \mu$ for all $t \geq 0$. Define $f_t := \frac{d\mu_t}{d\mu}$. Then

$$\frac{d}{dt} \chi^2(\mu_t \parallel \mu) = -2 \int |\nabla f_t|^2 d\mu \leq 0.$$

Additionally, if μ satisfies a Poincaré inequality with constant C_P , then

$$\chi^2(\mu_t \parallel \mu) \leq \exp\left(-\frac{2t}{C_P}\right) \chi^2(\mu_0 \parallel \mu).$$

Proof: We assume the first part and only prove the convergence result.

First, note that:

$$\text{Var}_\mu(f_t) = \int f_t^2 d\mu - \left(\int f_t d\mu\right)^2 = \int \frac{(d\mu_t)^2}{d\mu} - \left(\int d\mu_t\right)^2 = \int \frac{(d\mu_t)^2}{d\mu} - 1 = \chi^2(\mu_t \parallel \mu)$$

Then, as μ satisfies a Poincaré inequality with constant C_P , we get:

$$\chi^2(\mu_t \parallel \mu) \leq C_P \int |\nabla f_t|^2 d\mu$$

Using the first part yields:

$$\chi^2(\mu_t \parallel \mu) \leq -\frac{C_P}{2} \frac{d}{dt} \chi^2(\mu_t \parallel \mu) \iff \frac{d}{dt} \chi^2(\mu_t \parallel \mu) \leq -\frac{2}{C_P} \chi^2(\mu_t \parallel \mu)$$

hence we can apply Gronwall's inequality to get:

$$\chi^2(\mu_t \parallel \mu) \leq \exp\left(-\frac{2t}{C_P}\right) \chi^2(\mu_0 \parallel \mu).$$

□

2.4.3 Time-reversal of SDEs

As we will see in next chapter, score-based generative models rely on the fact that under some assumptions on the SDE considered, we can derive an analytical expression for the reverse-time SDE. This section derives this expression.

One approach, developed by Föllmer in [Föl85], uses nice stochastic calculus tools. However, it gives us the desired formula only for the case where the volatility is constant. We will then follow another approach, based on transition semigroups defined earlier.

Let $T > 0$. Let X be a real-valued diffusion process satisfying the SDE:

$$dX_t = a(X_t, t)dt + \sigma_t dB_t \quad X_0 \sim \mu_0$$

where $a : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ and $\sigma_t > 0$ are smooth functions, globally Lipschitz and with linear growth. Let ρ_t be the density of X_t , that exists for every $t > 0$ thanks to a famous result by Friedman ([Fri64]). Crucially, we assume that a and σ depend only on the current state of time, so that X is a (strong) Markov process by Theorem 7.1.2 in [Øks03].

For $g \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, we have by Itô's formula:

$$dg(X_t) = g'(X_t)dX_t + \frac{1}{2}g''(X_t)d\langle X_t \rangle = \left(a(X_t, t)g'(X_t) + \frac{1}{2}\sigma_t^2 g''(X_t)\right)dt + \sigma_t g'(X_t)dB_t$$

Now we compute the infinitesimal generator:

$$\begin{aligned} L_t g(x) &= \lim_{\delta \rightarrow 0} \frac{P_{t, t+\delta} g(x) - g(x)}{\delta} \\ &= \lim_{\delta \rightarrow 0} \frac{\mathbb{E}[g(X_{t+\delta}) - g(X_t) \mid X_t = x]}{\delta} \\ &= \lim_{\delta \rightarrow 0} \frac{1}{\delta} \mathbb{E} \left[\int_t^{t+\delta} dg(X_s) \mid X_t = x \right] \\ &= \lim_{\delta \rightarrow 0} \frac{1}{\delta} \mathbb{E} \left[\int_t^{t+\delta} \left(a(X_s, s)g'(X_s) + \frac{1}{2}\sigma_s^2 g''(X_s) \right) ds + \int_t^{t+\delta} \sigma_s g'(X_s) dB_s \mid X_t = x \right] \end{aligned}$$

As $g \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, the stochastic integral is a martingale (by Lemma 2.8) at 0 hence of constant expectation 0, and by continuity of a and σ combined with the fact that the support of the derivatives of g is compact, the integrand of the finite variation part is bounded. Then we can apply dominated convergence to get:

$$L_t g(x) = \mathbb{E} \left[\lim_{\delta \rightarrow 0} \frac{1}{\delta} \int_t^{t+\delta} \left(a(X_s, s) g'(X_s) + \frac{1}{2} \sigma_s^2 g''(X_s) \right) ds \mid X_t = x \right]$$

And then by the fundamental theorem of calculus we get:

$$L_t g(x) = a(x, t) g'(x) + \frac{1}{2} \sigma_t^2 g''(x)$$

But then by the Fokker-Planck equation derived earlier:

$$\partial_t \rho_t = \mathcal{L}_t^* \rho_t(x)$$

Using integration by parts we find:

$$\begin{aligned} \int_{-\infty}^{+\infty} (L_t g(x)) \rho_t(x) dx &= \int_{-\infty}^{+\infty} \left(a(x, t) g'(x) + \frac{1}{2} \sigma_t^2 g''(x) \right) \rho_t(x) dx \\ &= \int_{-\infty}^{+\infty} a(x, t) \rho_t(x) g'(x) dx + \int_{-\infty}^{+\infty} \frac{1}{2} \sigma_t^2 \rho_t(x) g''(x) dx \\ &= - \int_{-\infty}^{+\infty} \frac{\partial}{\partial x} (a(x, t) \rho_t(x)) g(x) dx - \int_{-\infty}^{+\infty} \frac{\partial}{\partial x} \left(\frac{1}{2} \sigma_t^2 \rho_t(x) \right) g'(x) dx \\ &= - \int_{-\infty}^{+\infty} \frac{\partial}{\partial x} (a(x, t) \rho_t(x)) g(x) dx + \int_{-\infty}^{+\infty} \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma_t^2 \rho_t(x)) g(x) dx \\ &= \int_{-\infty}^{+\infty} g(x) \left[- \frac{\partial}{\partial x} (a(x, t) \rho_t(x)) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma_t^2 \rho_t(x)) \right] dx \\ &= \int_{-\infty}^{+\infty} g(x) (L_t^* \rho_t(x)) dx \end{aligned}$$

And hence:

$$\partial_t \rho_t(x) = - \frac{\partial}{\partial x} (a(x, t) \rho_t(x)) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma_t^2 \rho_t(x))$$

Now define: $(\overleftarrow{X})_{t \in [0, T]} = (X_{T-\cdot})_{t \in [0, T]}$. If we prove that $\overleftarrow{X}$ is a Markov process, then we will be able to go back to its SDE starting from its infinitesimal generator, doing the procedure we did earlier but backwards. We hence need to prove that $\overleftarrow{X}$ admits a transition semigroup satisfying the Chapman-Kolmogorov equation. We first need to define transitions kernels, as it will be easier to deal with them than to work directly with transition semigroups.

Definition 2.45. *Let X be a real-valued Markov process with associated transition semigroup $P_{s,t}$ for $t \geq s \geq 0$. We assume that the transition probabilities are absolutely continuous with respect to the Lebesgue measure, admitting transition densities $p_{s,t}(x, y)$. The transition operator $P_{s,t}$ acts on a function g as:*

$$P_{s,t} g(x) = \int_{\mathbb{R}} g(y) p_{s,t}(x, y) dy.$$

In this case, $p_{s,t}(x, \cdot)$ is the probability density function of X_t conditioned on $X_s = x$.

We will derive the transition kernel of $\overleftarrow{X}$. For any times $0 \leq s < t \leq T$ and any Borel sets $A, B \in \mathcal{B}(\mathbb{R})$, we consider the joint probability of the time-reversed process. Using the regular conditional probability of the forward process, the joint probability can be written as an integral over the sets A and B :

$$\begin{aligned} \mathbb{P}(\overleftarrow{X}_t \in A, \overleftarrow{X}_s \in B) &= \mathbb{P}(X_{T-t} \in A, X_{T-s} \in B) \\ &= \int_{y \in A} \mathbb{P}(X_{T-s} \in B \mid X_{T-t} = y) \rho_{T-t}(y) dy \\ &= \int_{y \in A} \left(\int_{x \in B} p_{T-t, T-s}(y, x) dx \right) \rho_{T-t}(y) dy \\ &= \int_A \int_B p_{T-t, T-s}(y, x) \rho_{T-t}(y) dx dy. \end{aligned}$$

Additionally, the marginal probability of the reversed process at time s is simply:

$$\mathbb{P}(X_s^{\leftarrow} \in B) = \mathbb{P}(X_{T-s} \in B) = \int_B \rho_{T-s}(x) dx.$$

Now applying the Bayes rule, the reverse transition kernel from state x at time s to a set A at time t is given by the ratio of the joint density to the marginal density of the starting point x . Hence, for $\rho_{T-s}(x) > 0$, we define the reverse transition probability:

$$\mathbb{P}(\overleftarrow{X}_t \in A \mid \overleftarrow{X}_s = x) = \int_A p_{T-t, T-s}(y, x) \frac{\rho_{T-t}(y)}{\rho_{T-s}(x)} dy.$$

We can therefore identify the reverse transition density $p_{s,t}^{\leftarrow}(x, y)$ as:

$$\overleftarrow{p}_{s,t}(x, y) = p_{T-t, T-s}(y, x) \frac{\rho_{T-t}(y)}{\rho_{T-s}(x)}.$$

Now we need to check that this is indeed a valid transition kernel. If so, it will induce a valid transition semigroup and hence the process $\overleftarrow{X}$ will be a Markov process.

We check the Chapman-Kolmogorov equation:

$$\begin{aligned} \int_{\mathbb{R}} \overleftarrow{p}_{s,t}(x, y) \overleftarrow{p}_{t,u}(y, z) dy &= \int_{\mathbb{R}} \left(p_{T-t, T-s}(y, x) \frac{\rho_{T-t}(y)}{\rho_{T-s}(x)} \right) \left(p_{T-u, T-t}(z, y) \frac{\rho_{T-u}(z)}{\rho_{T-t}(y)} \right) dy \\ &= \int_{\mathbb{R}} p_{T-t, T-s}(y, x) p_{T-u, T-t}(z, y) \frac{\rho_{T-u}(z)}{\rho_{T-s}(x)} dy \\ &= \left(\int_{\mathbb{R}} p_{T-u, T-t}(z, y) p_{T-t, T-s}(y, x) dy \right) \frac{\rho_{T-u}(z)}{\rho_{T-s}(x)} \\ &= p_{T-u, T-s}(z, x) \frac{\rho_{T-u}(z)}{\rho_{T-s}(x)} \\ &= \overleftarrow{p}_{s,u}(x, z) \end{aligned}$$

Moreover, the reverse transition kernel obviously integrates to 1. Hence it is a valid transition kernel and it induces the following valid transition semigroup:

$$\overleftarrow{P}_{s,t} g(x) = \int_{\mathbb{R}} \overleftarrow{p}_{s,t}(x, y) g(y) dy.$$

We will now try to find some relation between the transition semigroup of X and the one of $\overleftarrow{X}$, in order to derive the infinitesimal generator of $\overleftarrow{X}$ and hence recover the SDE it satisfies. Using the definition of the reverse transition kernel we just derived yields:

$$\begin{aligned} \int_{\mathbb{R}} g(x) \overleftarrow{P}_{s,t} h(x) \rho_{T-s}(x) dx &= \int_{\mathbb{R}} g(x) \left(\int_{\mathbb{R}} \overleftarrow{p}_{s,t}(x, y) h(y) dy \right) \rho_{T-s}(x) dx \\ &= \iint_{\mathbb{R}^2} g(x) p_{T-t, T-s}(y, x) \rho_{T-t}(y) h(y) dy dx \\ &= \int_{\mathbb{R}} h(y) \left(\int_{\mathbb{R}} p_{T-t, T-s}(y, x) g(x) dx \right) \rho_{T-t}(y) dy \\ &= \int_{\mathbb{R}} h(y) P_{T-t, T-s} g(y) \rho_{T-t}(y) dy. \end{aligned}$$

Now to make the infinitesimal generator appear, we just need to set: $t = s + \delta$, divide by δ , let δ goes to 0 and set $u = T-s$.

$$\begin{aligned} \int g(x) \overleftarrow{\mathcal{L}}_s h(x) \rho_u(x) dx &= \int h(y) \mathcal{L}_u g(y) \rho_u(y) dy + \int h(y) g(y) \partial_t \rho_u(y) dy \\ &= \int h(y) \mathcal{L}_u g(y) \rho_u(y) dy - \int h(y) g(y) \mathcal{L}_u^* \rho_u(y) dy \end{aligned}$$

Where we used Fokker-Planck equation. Now we recall:

$$\begin{aligned} \mathcal{L}_t g(x) &= a(x, t) g'(x) + \frac{1}{2} \sigma_t^2 g''(x) \\ \mathcal{L}_t^* \rho_t(x) &= -(a(x, t) \rho_t(x))' + \frac{1}{2} (\sigma_t^2 \rho_t(x))'' \end{aligned}$$

And then using integration by part, and once again the fundamental lemma of calculus of variations, we finally get:

$$\begin{aligned} \overleftarrow{\mathcal{L}}_u h(x) &= -a(x, u) h'(x) + \sigma_u^2 \frac{\frac{1}{2} h'' \rho_u + h' \rho_u'}{\rho_u(x)} \\ &= \left[-a(x, u) + \sigma_u^2 \frac{\rho_u'(x)}{\rho_u(x)} \right] h'(x) + \frac{1}{2} \sigma_u^2 h''(x). \end{aligned}$$

That is,

$$\overleftarrow{\mathcal{L}}_{T-s} h(x) = \left[-a(x, T-s) + \sigma_{T-s}^2 \frac{\rho'_{T-s}(x)}{\rho_{T-s}(x)} \right] h'(x) + \frac{1}{2} \sigma_{T-s}^2 h''(x).$$

As it is a Markovian diffusion, we know that the reverse process defined by $\overleftarrow{X}_s = X_{T-s}$ satisfies a SDE of the form:

$$d\overleftarrow{X}_s = \alpha(\overleftarrow{X}_s, s) ds + \beta(\overleftarrow{X}_s, s) d\bar{W}_s$$

Our goal will be to determine α and β from the expression we derived for the infinitesimal generator.

We have on the one hand:

$$\overleftarrow{\mathcal{L}}_s g(x) = \lim_{\delta \rightarrow 0} \frac{1}{\delta} \mathbb{E} \left[\int_s^{s+\delta} dg(\overleftarrow{X}_u) \mid \overleftarrow{X}_s = x \right]$$

Now by Itô's formula:

$$dg(\overleftarrow{X}_s) = g'(\overleftarrow{X}_s) \alpha(\overleftarrow{X}_s, s) ds + \frac{1}{2} g''(\overleftarrow{X}_s) \beta^2(\overleftarrow{X}_s, s) ds + g'(\overleftarrow{X}_s) \beta(\overleftarrow{X}_s, s) d\bar{W}_s$$

Hence:

$$\overleftarrow{\mathcal{L}}_s g(x) = g'(x) \alpha(x, s) + \frac{1}{2} g''(x) \beta^2(x, s)$$

But on the other hand, we know that the generator of the reverse process at time s corresponds to the reverse generator evaluated at the forward time $t = T - s$:

$$\overleftarrow{\mathcal{L}}_s g(x) = \left[-a(x, T-s) + \sigma_{T-s}^2 \frac{\rho'_{T-s}(x)}{\rho_{T-s}(x)} \right] g'(x) + \frac{1}{2} \sigma_{T-s}^2 g''(x).$$

As both of the expressions for $\overleftarrow{\mathcal{L}}_s$ are true for any bounded and twice differentiable function g , by taking some g such that $g'(x) = 1$ and $g''(x) = 0$, we identify:

$$\alpha(x, s) = -a(x, T-s) + \sigma_{T-s}^2 \frac{\rho'_{T-s}(x)}{\rho_{T-s}(x)}$$

Now taking some g such that $g''(x) = 1$, we identify:

$$\beta^2(x, s) = \sigma_{T-s}^2 \implies \beta(x, s) = \sigma_{T-s}$$

Hence the SDE satisfied by the reverse process is:

$$d\overleftarrow{X}_s = \left[-a(\overleftarrow{X}_s, T-s) + \sigma_{T-s}^2 \frac{\rho'_{T-s}(\overleftarrow{X}_s)}{\rho_{T-s}(\overleftarrow{X}_s)} \right] ds + \sigma_{T-s} d\bar{W}_s$$

Derivation with the continuity equation

We still consider the SDE:

$$dX_t = a(X_t, t)dt + \sigma_t dB_t$$

with the same hypotheses as in the previous paragraph.

There is actually a way to find processes which have the same marginals as X hence the same marginals as $\overleftarrow{X}$, without having to derive the closed form expression for the SDE satisfied by $\overleftarrow{X}$.

The evolution of the marginal probability densities p_t is governed by the Fokker-Planck equation (FPE):

$$\frac{\partial p_t(x)}{\partial t} = -\nabla \cdot (a(x, t)p_t(x)) + \frac{1}{2}\sigma_t^2 \Delta p_t(x). \quad (2.31)$$

By adapting what we did just after theorem 2.38 to non-constant diffusion parts, we can rewrite the Fokker-Planck equation as a continuity equation:

$$\frac{\partial p_t(x)}{\partial t} + \nabla \cdot (v_t(x)p_t(x)) = 0, \quad (2.32)$$

where the effective velocity field $v_t(x)$ transporting the probability fluid is defined as:

$$v_t(x) = a(x, t) - \frac{1}{2}\sigma_t^2 \nabla \log p_t(x). \quad (2.33)$$

Time Reversal via Mass Conservation To reverse the process, we define the backward time variable $\tau = T - t$ for $\tau \in [0, T]$, and the time-reversed density $\tilde{p}_\tau(x) = p_{T-\tau}(x)$. Differentiating $\tilde{p}_\tau(x)$ with respect to τ yields $\frac{\partial \tilde{p}_\tau}{\partial \tau} = -\frac{\partial p_{T-\tau}}{\partial t}$. Substituting this into Eq. (2.32) gives the reversed continuity equation:

$$\frac{\partial \tilde{p}_\tau(x)}{\partial \tau} + \nabla \cdot (\tilde{v}_\tau^{\text{fluid}}(x)\tilde{p}_\tau(x)) = 0, \quad \text{where} \quad \tilde{v}_\tau^{\text{fluid}}(x) = -v_{T-\tau}(x). \quad (2.34)$$

Hence, we get:

$$\tilde{v}_\tau^{\text{fluid}}(x) = -a(x, T - \tau) + \frac{1}{2}\sigma_{T-\tau}^2 \nabla \log \tilde{p}_\tau(x). \quad (2.35)$$

The Infinite Family of Matching Processes We now consider a candidate reverse-time SDE driven by a backward Brownian motion $\bar{B}_\tau$ with an arbitrary diffusion coefficient γ_τ :

$$d\tilde{X}_\tau = \tilde{a}(\tilde{X}_\tau, \tau)d\tau + \gamma_\tau d\bar{B}_\tau. \quad (2.36)$$

By applying the same continuity formulation, the effective velocity field $\tilde{v}_\tau(x)$ induced by this candidate process is:

$$\tilde{v}_\tau(x) = \tilde{a}(x, \tau) - \frac{1}{2}\gamma_\tau^2 \nabla \log \tilde{p}_\tau(x). \quad (2.37)$$

To guarantee that this generalized SDE has the target marginal distributions (i.e., the time-reversed marginals of the original forward process), its velocity field must match the field from Eq. (2.35) (i.e., $\tilde{v}_\tau(x) = \tilde{v}_\tau^{\text{fluid}}(x)$):

$$\tilde{a}(x, \tau) - \frac{1}{2}\gamma_\tau^2 \nabla \log \tilde{p}_\tau(x) = -a(x, T - \tau) + \frac{1}{2}\sigma_{T-\tau}^2 \nabla \log \tilde{p}_\tau(x). \quad (2.38)$$

Solving for the generalized reverse drift $\tilde{a}(x, \tau)$ yields:

$$\tilde{a}(x, \tau) = -a(x, T - \tau) + \frac{1}{2}(\sigma_{T-\tau}^2 + \gamma_\tau^2) \nabla \log \tilde{p}_\tau(x). \quad (2.39)$$

Eq. (2.39) defines an infinite family of stochastic processes, indexed by the arbitrary noise γ_τ , all sharing identical marginal probability densities (but not sharing law on the path space). In the next chapter, we will focus on these two particular cases:

- **Standard Reverse SDE** ($\gamma_\tau = \sigma_{T-\tau}$):

$$\tilde{a}(x, \tau) = -a(x, T - \tau) + \sigma_{T-\tau}^2 \nabla \log \tilde{p}_\tau(x). \quad (2.40)$$

• **Probability Flow ODE** ($\gamma_\tau = 0$):

$$d\tilde{X}_\tau = \left[-a(\tilde{X}_\tau, T - \tau) + \frac{1}{2}\sigma_{T-\tau}^2 \nabla \log \tilde{p}_\tau(\tilde{X}_\tau) \right] d\tau. \quad (2.41)$$

Finally, we also give the definition of the reverse SDE in higher dimensions, as it will be used in the next chapters.

Proposition 2.46. *We consider the forward SDE:*

$$dx = f(x, t)dt + G(x, t)dw,$$

where $f(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $G(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$.

We define the reverse-time process as $(y_t)_{0 \leq t \leq T} = (x_{T-t})_{0 \leq t \leq T}$. According to Anderson (1982), Y satisfies the SDE:

$$dy = \left\{ -f(y, T-t) + \nabla \cdot [G(y, T-t)G(y, T-t)^T] + G(y, T-t)G(y, T-t)^T \nabla_y \log p_{T-t}(y) \right\} dt + G(y, T-t)d\bar{w} \quad (2.42)$$

where we define $\nabla \cdot F(y) := (\nabla \cdot f_1(y), \nabla \cdot f_2(y), \dots, \nabla \cdot f_d(y))^T$ for a matrix-valued function $F(y) := (f_1(y), f_2(y), \dots, f_d(y))^T$, and $\bar{w}$ is a standard Brownian motion.

Remark: Note that this formulation allows for diffusion parts depending on state in addition of time. The proof is not given here but can be found in [And82b].

2.5 Differential geometry

This section introduces the notions of differential geometry that will be used in this work. Even though an in-depth understanding of this part is not necessary, it is advised that the reader takes some time to become familiar with the most basic notions.

2.5.1 Smooth and Riemannian manifolds

Definition 2.47 (Manifold). *An m -dimensional (topological) manifold M is a Hausdorff topological space with a countable basis of the topology, with the property that for every point $p \in M$ there exists a homeomorphism $\varphi : U \rightarrow \varphi(U)$ from an open neighborhood $U \subset M$ of p onto an open set $\varphi(U) \subset \mathbb{R}^m$.*

Definition 2.48 (Differentiable Manifold). *For $1 \leq r \leq \infty$, a differentiable structure of class C^r on a topological manifold is a maximal C^r atlas. A differentiable manifold of class C^r is a topological manifold equipped with a C^r structure.*

Definition 2.49 (Tangent Space). *Let M be an m -dimensional C^r manifold. On the set of all pairs (φ, ξ) , where φ is a chart of M around p and $\xi \in \mathbb{R}^m$, an equivalence relation is defined such that $(\varphi, \xi) \sim_p (\psi, \eta)$ if and only if $d(\psi \circ \varphi^{-1})_{\varphi(p)}(\xi) = \eta$. The tangent space TM_p of M at p is the set of all such equivalence classes.*

Definition 2.50 (Differential). *For a C^1 map $F : M \rightarrow N$, the differential of F at $p \in M$ is the unique linear map $dF_p : TM_p \rightarrow TN_{F(p)}$ such that for every local representation $H := \psi \circ F \circ \varphi^{-1}$ of F around p , the chain rule $dF_p = (d\psi_{F(p)})^{-1} \circ dH_{\varphi(p)} \circ d\varphi_p$ holds.*

Definition 2.51 (Diffeomorphism). *Let M, N be two C^r manifolds. A map $F : M \rightarrow N$ is a C^r diffeomorphism if F is bijective and both F and F^{-1} are C^r .*

Definition 2.52 (Submanifold). Let N be an n -dimensional C^∞ manifold. A subset $M \subset N$ is an m -dimensional submanifold of N if for every point $p \in M$ there is a chart $\psi : V \rightarrow \psi(V) \subset \mathbb{R}^n = \mathbb{R}^m \times \mathbb{R}^{n-m}$ of N such that $p \in V$ and $\psi(M \cap V) = \psi(V) \cap (\mathbb{R}^m \times \{0\})$.

Definition 2.53 (Riemannian Manifold). A Riemannian manifold is a smooth differentiable manifold M equipped with a Riemannian metric g . This metric assigns a positive-definite inner product g_p to the tangent space $T_p M$ at each point $p \in M$.

Definition 2.54 (Riemannian volume measure). Let (M, g) be an m -dimensional Riemannian manifold. In a chart (U, x) set

$$\mu_g(A) := \int_{x(A)} \sqrt{\det(g_{ij})} dx^1 \cdots dx^m, \quad A \subset U \text{ Borel.}$$

Under a change of chart, $\sqrt{\det g}$ picks up exactly the Jacobian factor $|\det J|$, so the change-of-variables formula makes this independent of the chart; a partition of unity then extends μ_g to a Borel measure on M , the Riemannian volume measure. It is finite on compact sets and positive on nonempty open sets.

Definition 2.55 (Gradient, divergence, Laplace–Beltrami operator). For $f \in C^1(M)$ the Riemannian gradient ∇f is the vector field characterized by $g_x(\nabla f(x), v) = df_x(v)$ for all $v \in T_x M$.

For a smooth vector field Z , the divergence is the unique continuous function such that

$$\int_M \langle Z, \nabla f \rangle d\mu_g = - \int_M f \operatorname{div} Z d\mu_g, \quad f \in C_c^\infty(M),$$

uniqueness holding because μ_g has full support.

The Laplace–Beltrami operator is $\Delta_M f := \operatorname{div}(\nabla f)$; in a chart, $\Delta_M f = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} g^{ij} \partial_j f)$.

Definition 2.56 (Isometry). A diffeomorphism $F : (M, g) \rightarrow (M, g)$ is an isometry if

$$g_{F(x)}(dF_x v, dF_x w) = g_x(v, w)$$

for all $x \in M$ and $v, w \in T_x M$.

Proposition 2.57 (Naturality under isometries). If F is an isometry, then

$$\nabla(f \circ F) = dF^{-1}((\nabla f) \circ F), \quad \Delta_M(f \circ F) = (\Delta_M f) \circ F,$$

and $F_\# \operatorname{vol}_g = \operatorname{vol}_g$.

2.5.2 Lie groups, $SO(3)$, and $SE(3)$

Definition 2.58 (Lie Group). A Lie group $(G, \cdot)$ is a group with the structure of a C^∞ manifold such that the map $G \times G \rightarrow G$, $(g, h) \mapsto gh^{-1}$, is C^∞ .

Important examples of Lie groups for Chapter 4 will be:

Definition 2.59 (The rotation group).

$$SO(3) := \{R \in \mathbb{R}^{3 \times 3} : R^\top R = I_3, \det R = 1\}.$$

Axis-angle representation: Let $(a, b, c) \in \mathbb{R}^3$ such that $a^2 + b^2 + c^2 = 1$, i.e. $\omega = (a, b, c) \in \mathbb{S}^2$ and $\theta \in \mathbb{R}_+$, then any element of $SO(3)$ is given by $\exp(Y)$ with $Y = \theta K$ and $K = aY_1 + bY_2 + cY_3$. The parameterization of $SO(3)$ using (ω, θ) is called the axis-angle parameterization. Using that $K^3 = -K$ we have

$$\exp[\theta K] = \operatorname{Id} + \sin(\theta)K + (1 - \cos(\theta))K^2.$$

Definition 2.60 (Oriented orthonormal frame). *An oriented orthonormal frame of $\mathbb{R}^3$ is an ordered orthonormal basis (u_1, u_2, u_3) satisfying $\det[u_1 \ u_2 \ u_3] = 1$. Hence $R \in SO(3)$ represents the frame formed by the columns of R . The determinant condition excludes reflections and fixes the right-handed orientation.*

Definition 2.61 (The special Euclidean group). *As a manifold, $SE(3) = SO(3) \times \mathbb{R}^3$. Its group law and inverse are*

$$(R, x)(S, y) = (RS, x + Ry), \quad (R, x)^{-1} = (R^\top, -R^\top x).$$

Thus $SE(3) = SO(3) \ltimes \mathbb{R}^3$ is a semidirect product, not the direct product group.

Definition 2.62 (Haar measure). *A left Haar measure on a locally compact group G is a nonzero measure μ satisfying*

$$(L_g)_\# \mu = \mu, \quad g \in G.$$

It is unique up to a positive multiplicative constant. On a compact group it can be normalized to a probability measure. For $SO(3)$ under the usual metric, normalized Riemannian volume equals normalized Haar measure, denoted by $\mathcal{U}(SO(3))$.

Lemma 2.63 (Naturality of the Riemannian volume and divergence). *Let (M, g) be a Riemannian manifold without boundary, and let*

$$\varphi : M \longrightarrow M$$

be a Riemannian isometry. Then φ preserves the Riemannian volume measure:

$$\varphi_\# \text{vol}_g = \text{vol}_g.$$

Moreover, for every smooth vector field Y on M ,

$$\boxed{\text{div}(\varphi_* Y) \circ \varphi = \text{div } Y,}$$

where the pushforward vector field $\varphi_ Y$ is defined by*

$$(\varphi_* Y)(\varphi(x)) = d\varphi_x(Y(x)).$$

Proof: We first prove that φ preserves the Riemannian volume measure. Fix $x \in M$ and let

$$(e_1, \dots, e_d)$$

be an orthonormal basis of $T_x M$. Since φ is an isometry,

$$g_{\varphi(x)}(d\varphi_x e_i, d\varphi_x e_j) = g_x(e_i, e_j) = \delta_{ij}.$$

Therefore,

$$(d\varphi_x e_1, \dots, d\varphi_x e_d)$$

is an orthonormal basis of $T_{\varphi(x)} M$. Hence the matrix of $d\varphi_x$ in orthonormal bases is orthogonal, and its Riemannian Jacobian satisfies

$$J_\varphi(x) = |\det(d\varphi_x)| = 1.$$

The change-of-variables formula therefore gives, for every nonnegative measurable function $\psi : M \rightarrow \mathbb{R}$,

$$\begin{aligned} \int_M \psi(y) \varphi_\# \text{vol}_g(dy) &= \int_M \psi(\varphi(x)) \text{vol}_g(dx) \\ &= \int_M \psi(y) J_{\varphi^{-1}}(y) \text{vol}_g(dy) \\ &= \int_M \psi(y) \text{vol}_g(dy), \end{aligned}$$

because $J_{\varphi^{-1}} = 1$. Since this equality holds for every nonnegative measurable function ψ , we conclude that

$$\varphi_{\#} \text{vol}_g = \text{vol}_g.$$

We now prove the transformation formula for the divergence. Set

$$Z := \varphi_* Y.$$

By definition,

$$Z(\varphi(x)) = d\varphi_x(Y(x)).$$

Let $\psi \in C_c^\infty(M)$. The integration-by-parts characterization of the divergence gives

$$\int_M \psi(y) \operatorname{div} Z(y) \operatorname{vol}_g(dy) = - \int_M d\psi_y(Z(y)) \operatorname{vol}_g(dy).$$

Performing the change of variables $y = \varphi(x)$ and using the invariance of the Riemannian volume measure, we obtain

$$\begin{aligned} \int_M \psi(y) \operatorname{div} Z(y) \operatorname{vol}_g(dy) &= - \int_M d\psi_{\varphi(x)}(Z(\varphi(x))) \operatorname{vol}_g(dx) \\ &= - \int_M d\psi_{\varphi(x)}(d\varphi_x(Y(x))) \operatorname{vol}_g(dx). \end{aligned}$$

By the chain rule,

$$d\psi_{\varphi(x)}(d\varphi_x(Y(x))) = d(\psi \circ \varphi)_x(Y(x)).$$

Therefore,

$$\begin{aligned} \int_M \psi(y) \operatorname{div} Z(y) \operatorname{vol}_g(dy) &= - \int_M d(\psi \circ \varphi)_x(Y(x)) \operatorname{vol}_g(dx) \\ &= - \int_M (\psi \circ \varphi)(x) \operatorname{div} Y(x) \operatorname{vol}_g(dx), \end{aligned}$$

where we used the integration-by-parts characterization of the divergence once again.

Applying the change of variables $y = \varphi(x)$ a second time gives

$$\int_M \psi(y) \operatorname{div} Z(y) \operatorname{vol}_g(dy) = \int_M \psi(y) \operatorname{div} Y(\varphi^{-1}(y)) \operatorname{vol}_g(dy).$$

Since this equality holds for every $\psi \in C_c^\infty(M)$, we have

$$\operatorname{div} Z(y) = \operatorname{div} Y(\varphi^{-1}(y))$$

for vol_g -almost every $y \in M$. Both sides are smooth, so the equality holds everywhere. Recalling that $Z = \varphi_* Y$, we obtain

$$\operatorname{div}(\varphi_* Y) = (\operatorname{div} Y) \circ \varphi^{-1}.$$

Composing both sides with φ yields

$$\boxed{\operatorname{div}(\varphi_* Y) \circ \varphi = \operatorname{div} Y.}$$

□

2.5.3 Group actions and equivariance

Definition 2.64 (Action of a Lie Group on a Manifold). *A smooth action of a Lie group H on a smooth manifold M is a smooth map $\theta : H \times M \rightarrow M$, typically written as $\theta(h, p) = h \cdot p$, such that $e \cdot p = p$ for the identity element $e \in H$, and $h_1 \cdot (h_2 \cdot p) = (h_1 h_2) \cdot p$ for all $h_1, h_2 \in H$ and $p \in M$.*

Definition 2.65 (Induced action on the tangent bundle). *Let H be a Lie group acting smoothly on a smooth manifold M through the map*

$$\alpha : H \times M \longrightarrow M, \quad (h, x) \longmapsto h \cdot x.$$

For every $h \in H$, define

$$\Phi_h : M \longrightarrow M, \quad \Phi_h(x) := h \cdot x.$$

Since α is a group action, one has

$$\Phi_e = \text{Id}_M, \quad \Phi_{gh} = \Phi_g \circ \Phi_h, \quad \Phi_h^{-1} = \Phi_{h^{-1}}.$$

In particular, each Φ_h is a diffeomorphism.

Recall that the tangent bundle of M is

$$TM = \bigsqcup_{x \in M} T_x M = \{(x, v) : x \in M, v \in T_x M\}.$$

The action of H on M induces an action on TM , called the tangent lift of α , defined by

$$\tilde{\alpha} : H \times TM \longrightarrow TM, \quad \tilde{\alpha}(h, (x, v)) := (h \cdot x, d\Phi_h|_x(v)).$$

Equivalently,

$$h \cdot (x, v) = (h \cdot x, d\Phi_h|_x(v)).$$

Here,

$$d\Phi_h|_x : T_x M \longrightarrow T_{h \cdot x} M$$

is the differential of Φ_h at x . Since Φ_h is a diffeomorphism, this differential is a linear isomorphism, with inverse

$$d\Phi_{h^{-1}}|_{h \cdot x} : T_{h \cdot x} M \longrightarrow T_x M.$$

Definition 2.66 (Equivariant map). *Suppose H acts on M and N . A map $F : M \rightarrow N$ is H -equivariant if*

$$F(h \cdot x) = h \cdot F(x) \quad \text{for all } h \in H, x \in M.$$

For a vector field b on M , equivariance means

$$b(\Phi_h(x)) = d\Phi_h|_x b(x).$$

Definition 2.67 (Pushforward measure). *If $F : M \rightarrow N$ is measurable and μ is a measure on M , its pushforward is*

$$F_{\#}\mu(A) := \mu(F^{-1}(A)).$$

Equivalently, for every nonnegative measurable or integrable f ,

$$\int_N f(y) F_{\#}\mu(dy) = \int_M f(F(x)) \mu(dx).$$

Definition 2.68 (Riemannian score). *If p_t is a positive smooth density with respect to vol_g , its score is the vector field*

$$s_t(x) := \nabla_x \log p_t(x) \in T_x M.$$

For a conditional density $p_{t|0}(x_t | x_0)$, the conditional score is

$$\nabla_{x_t} \log p_{t|0}(x_t | x_0).$$

The gradient is always taken with respect to the variable indicated by the subscript.

Proposition 2.69 (Invariant density implies equivariant score). *Let H act isometrically on M . If $p(\Phi_h(x)) = p(x)$, then*

$$\nabla \log p(\Phi_h(x)) = d\Phi_h|_x \nabla \log p(x).$$

Thus the score of an invariant density is equivariant.

2.5.4 Martingale Problem, Brownian motion, heat kernels

Definition 2.70. *Let M be a differentiable manifold and $(\Omega, (\mathcal{F}_t)_{t \geq 0}, \mathcal{F}, \mathbb{P})$ a filtered probability space. Let τ be an $\mathcal{F}$ -stopping time. A continuous, M -valued process X defined on $[0, \tau)$ is called an M -valued semimartingale if $f(X)$ is a real-valued semimartingale on $[0, \tau)$ for all $f \in C^\infty(M)$.*

Definition 2.71 (Martingale problem). *Let M be a smooth manifold and let $(\mathcal{L}_t)_{t \geq 0}$ be a family of second-order differential operators on M without zeroth-order term, with measurable dependence on t . Write $M_\Delta = M \cup \{\Delta\}$ for the one-point compactification and extend every $f \in C_c^\infty(M)$ by $f(\Delta) = 0$.*

A solution of the martingale problem for $(\mathcal{L}_t)$ with initial distribution μ_0 is a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ carrying an $(\mathcal{F}_t)$ -adapted process X with continuous paths in M_Δ , absorbed at Δ , such that $\text{Law}(X_0) = \mu_0$ and, for every $f \in C_c^\infty(M)$,

$$M_t^f := f(X_t) - f(X_0) - \int_0^t (\mathcal{L}_s f)(X_s) ds$$

is an $(\mathcal{F}_t)$ -martingale. The explosion time is $\zeta := \inf\{t \geq 0 : X_t = \Delta\}$, and the solution is non-explosive if $\mathbb{P}(\zeta = \infty) = 1$, in which case X has paths in $C([0, \infty), M)$. $(\mathcal{L}_s)_{s \geq 0}$ is then called the generator of X .

The martingale problem is well posed if for every initial distribution μ_0 a solution exists and any two solutions induce the same law on $C([0, \infty), M_\Delta)$.

Definition 2.72 (Brownian motion by the martingale problem). *Let (M, g) be a Riemannian manifold. A continuous process (B_t) is a Brownian motion on M if, for every $f \in C_c^\infty(M)$,*

$$f(B_t) - f(B_0) - \frac{1}{2} \int_0^t \Delta_M f(B_s) ds$$

is a local martingale, up to the explosion time. Equivalently, its generator is $\frac{1}{2} \Delta_M$.

Proposition 2.73. *Let M be a compact smooth manifold and let $(\mathcal{L}_t)_{t \geq 0}$ be a family of second-order differential operators on M without zeroth-order term. Then every solution of the martingale problem for $(\mathcal{L}_t)$ is non-explosive. That is, the explosive time ζ of any solution X satisfies $\mathbb{P}[\zeta = \infty] = 1$.*

Proof: Since M is compact, $\mathbf{1} \in C_c^\infty(M)$, and its extension to M_Δ by $\mathbf{1}(\Delta) = 0$ equals $\mathbf{1}_M$. Since $\mathcal{L}$ does not contain any zero-order term, $\mathcal{L}_t \mathbf{1} = 0$. Then, since X solves the martingale problem for $\mathcal{L}$, $1_{X_t \in M} - 1_{X_0 \in M}$ is a martingale. Hence: $\mathbb{E}[1_{X_t \in M}] = \mathbb{E}[1_{X_0 \in M}] = 1$ as X starts in M . But note that $1_{X_t \in M} = 1_{\zeta > t}$. Consequently: $\mathbb{P}(\zeta > t) = 1$ for every $t \geq 0$. We can then write: $\zeta = \infty = \bigcap_{n \in \mathbb{N}} \{\zeta > n\}$. Finally, as a countable intersection of events of probability 1, $\zeta = \infty$ has probability 1. $\square$

Definition 2.74 (Heat kernel). *When the transition law of Brownian motion admits a density with respect to vol_g , the heat kernel $k_t(x, y)$ is defined by*

$$\Pr(B_t \in dy \mid B_0 = x) = k_t(x, y) \text{vol}_g(dy).$$

It solves

$$\partial_t k_t(x, \cdot) = \frac{1}{2} \Delta_M k_t(x, \cdot)$$

with initial condition δ_x in the distributional sense.

Theorem 2.75 (Long-time limit on a compact connected manifold). *If M is compact and connected, then*

$$k_t(x, \cdot) \text{vol}_g \xrightarrow{t \rightarrow \infty} \frac{\text{vol}_g}{\text{vol}_g(M)}$$

Equivalently, if the density is taken with respect to normalized volume, then $k_t(x, y) \rightarrow 1$.

Definition 2.76 (Covariance symbol and ellipticity). *Let M be a smooth m -dimensional manifold and let L be a second-order differential operator without zeroth-order term, i.e. $L\mathbf{1} = 0$. Suppose that, in every local chart $(U, x^1, \dots, x^m)$, it can be written as*

$$Lf = \frac{1}{2} \sum_{i,j=1}^m a^{ij}(x) \partial_i \partial_j f + \sum_{i=1}^m b^i(x) \partial_i f,$$

where $a = (a^{ij})$ is symmetric and positive semidefinite.

*The covariance symbol of L is the symmetric bilinear form a_L on T^*M intrinsically defined by*

$$a_L(df, dg) := L(fg) - fLg - gLf, \quad f, g \in C^\infty(M).$$

In local coordinates, it is given by

$$a_L(x; \xi, \eta) = \sum_{i,j=1}^m a^{ij}(x) \xi_i \eta_j, \quad \xi, \eta \in T_x^*M.$$

In particular,

$$a_L(x; \xi, \xi) = \sum_{i,j=1}^m a^{ij}(x) \xi_i \xi_j.$$

Equivalently, the carré du champ operator associated with L is

$$\Gamma_L(f, g) := \frac{1}{2} (L(fg) - fLg - gLf),$$

so that

$$a_L(df, dg) = 2\Gamma_L(f, g).$$

The operator L is said to be elliptic at x if

$$a_L(x; \xi, \xi) > 0 \quad \text{for every } \xi \in T_x^*M \setminus \{0\},$$

and elliptic if it is elliptic at every point of M .

Lemma 2.77 (Covariance symbol of a divergence-form generator). *Let (M, g) be a Riemannian manifold and let*

$$Lf = \langle b, \nabla f \rangle_g + \frac{1}{2} \operatorname{div}(\Sigma \nabla f),$$

where b is a smooth vector field and Σ is a smooth field of g -symmetric, positive-semidefinite endomorphisms of TM . Define the covariance symbol of L by

$$a_L(df, dg) := L(fg) - fLg - gLf.$$

Then

$$a_L(df, dg) = \langle \Sigma \nabla f, \nabla g \rangle_g.$$

*In particular, for every $x \in M$ and $\xi \in T_x^*M$,*

$$a_L(x, \xi) = \left\langle \Sigma(x) \xi^\sharp, \xi^\sharp \right\rangle_g.$$

Consequently, L is elliptic at x if and only if $\Sigma(x)$ is positive definite.

Lemma 2.78 (Smooth Urysohn / bump function). *Let M be a smooth manifold, $A \subset M$ compact and $U \subseteq M$ open with $A \subset U$. Then there exists $\chi \in C_c^\infty(M)$ with*

$$0 \leq \chi \leq 1, \quad \chi \equiv 1 \text{ on } A, \quad \operatorname{supp} \chi \subset U.$$

Proof: By local compactness, choose an open set V with $A \subset V \subset \overline{V} \subset U$ and $\overline{V}$ compact. Since A is compact, hence closed, $\{V, M \setminus A\}$ is an open cover of M ; let $\{\psi_1, \psi_2\}$ be a smooth partition of unity subordinate to it, so that $\psi_1 + \psi_2 = 1$, $\operatorname{supp} \psi_1 \subset V$ and $\operatorname{supp} \psi_2 \subset M \setminus A$. Set $\chi := \psi_1$. On A we have $\psi_2 = 0$, hence $\chi \equiv 1$; moreover $\operatorname{supp} \chi$ is closed and contained in the compact set $\overline{V}$, hence compact, and $\operatorname{supp} \chi \subset V \subset U$. $\square$

Chapter 3

Diffusion Generative Modeling

In this section, we present the mathematical foundations of generative modeling, with a specific focus on the diffusion framework. We will formalize different theoretical settings, provide a comparative analysis, and study convergence results.

3.1 Basics on generative modeling and first approaches

3.1.1 First models

Let us consider an unknown probability measure μ . It is unknown in the sense that we only have access to μ through iid samples $X_1, \dots, X_N$. A generative model is a model designed to produce new data "similar" to the samples $X_1, \dots, X_N$; that is producing samples from a distribution as close (in some sense) as possible from the distribution induced by μ .

We will cover here some of the models that were introduced before diffusion based models.

Likelihood based models: In these models, the unknown density function is learned via Maximum Likelihood Estimation. Some examples include Energy-Based Models. However, these models are inherently limited. Indeed, one has to assume that the data distribution has a density with respect to Lebesgue measure, and model the density as:

$$p_\theta(x) = \frac{e^{-f_\theta(x)}}{Z_\theta}$$

One then wants to maximize the log-likelihood:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \log p_\theta(x_i) = -\frac{1}{N} \sum_{i=1}^N f_\theta(x_i) - \log Z_\theta$$

via computing the gradient:

$$\nabla_\theta \mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \nabla_\theta f_\theta(x_i) - \nabla_\theta \log Z_\theta$$

However:

$$\nabla_\theta \log Z_\theta = \int \frac{1}{Z_\theta} e^{-f_\theta(x)} (-\nabla_\theta f_\theta(x)) dx = -\mathbb{E}_{x \sim p_\theta} [\nabla_\theta f_\theta(x)]$$

Hence, to be able to compute this gradient, one must be able to approximate the expectation with respect to the current model distribution p_θ via Monte Carlo, which implies having to sample from p_θ . However, as Z_θ is unknown, to sample from p_θ , one needs to use expensive Markov Chains Monte Carlo techniques. This has to be done at each optimization step, which quickly becomes intractable.

Implicit models. In these models, the distribution is represented implicitly through a model of its sampling process: instead of learning the density, we learn how to generate samples. A standard example is Generative Adversarial Networks (GANs). Two networks are trained jointly, a generator $G : \mathcal{Z} \rightarrow \mathcal{X}$ and a discriminator $D : \mathcal{X} \rightarrow [0, 1]$. The generator aims to produce samples close to the data, while the discriminator aims to distinguish real data from generated data. The discriminator is a probabilistic binary classifier: $D(x) = \alpha$ with $0 \leq \alpha \leq 1$ means that x has probability α of being real.

Training solves the min–max problem

$$\min_G \max_D V(D, G)$$

where:

$$V(D, G) := \mathbb{E}_{x \sim \mu} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))] ,$$

The maximization drives D toward 1 on real samples (first term of the sum) and toward 0 on generated ones (second term of the sum), while the minimization drives $D(G(z))$ towards 1. The expectation over μ is approximated by Monte Carlo, i.e. computed with respect to the empirical data distribution $\hat{\mu}$.

Here p_z is a fixed latent distribution on $\mathcal{Z}$ —typically $\mathcal{N}(0, I_d)$ with $d \ll \dim \mathcal{X}$ —and $G : \mathcal{Z} \rightarrow \mathcal{X}$ is deterministic, so that all randomness enters through z . The induced distribution is the pushforward $p_g := G_{\#} p_z$. This is what makes the model implicit: one can sample from p_g without having any expression for its density—indeed, when $d < \dim \mathcal{X}$ the measure p_g is carried by a d -dimensional submanifold and has no density at all.

Before stating a convergence result for GANs, we introduce the following definition.

Definition 3.1. *The Jensen–Shannon divergence between two probability distributions P and Q is defined as*

$$\text{JS}(P \parallel Q) = \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q),$$

where $\text{KL}(\cdot \parallel \cdot)$ denotes the Kullback–Leibler divergence.

Theorem 3.2. *Assume that G and D have enough capacity, that at each step the discriminator is trained to reach its optimum given G , and that the induced distribution p_g (the law of $G(z)$ with $z \sim p_z$) is then updated so as to improve*

$$\mathbb{E}_{x \sim \mu} [\log D_G(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D_G(G(z)))] .$$

Then $p_g \rightarrow \mu$ in Jensen–Shannon divergence, where μ is the data distribution.

For simplicity we assume throughout this paragraph that μ and p_g admit densities, as in [Goo+14]; this is in fact never the case when $d < \dim \mathcal{X}$, and the resulting singularity is one source of the instability discussed below. The paper [Goo+14], from which this theorem is taken, provides an algorithm satisfying its hypotheses. Note, however, that convergence is established

for updates of the induced distribution p_g ranging over all of $\mathcal{P}(\mathcal{X})$: the criterion $\sup_D V(D, G)$ is a supremum of functions that are linear in p_g . Indeed, the original paper [Goo+14] rewrites the objective as:

$$V(D, G) = \int [\mu(x) \log D(x) + p_g(x) \log(1 - D(x))] dx$$

This functional is hence convex, and gradient descent on a convex functional converges. In practice one optimizes the weights θ_g instead. The accessible set $\{(G_\theta)_\# p_z\}$ need not contain μ , is not convex in $\mathcal{P}(\mathcal{X})$, and $\theta \mapsto p_\theta$ is nonlinear; hence convexity is lost. Moreover, the adversarial training is unstable and may fail to converge: if the discriminator becomes too strong, then $\nabla_G \log(1 - D_G(G(z))) \approx 0$ (the vanishing-gradients problem) and the generator receives no usable signal, so it stops improving.

In order to address the inherent issues of Likelihood-based models and Implicit models, another method have been developed, relying on Langevin dynamics.

3.1.2 An approach via Langevin diffusion and Score Matching

Recall the Langevin diffusion:

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}dB_t, \quad X_0 \sim \mu_0$$

where $U : \mathbb{R}^d \rightarrow \mathbb{R}$ is a twice differentiable potential function.

The stationary distribution of this process is given by the Gibbs measure $p \propto e^{-U}$. Consequently, taking the gradient of the logarithm yields $\nabla \log p(x) = -\nabla U(x)$, allowing us to rewrite the underlying SDE purely in terms of the stationary distribution:

$$dX_t = \nabla \log p(X_t)dt + \sqrt{2}dB_t.$$

In this setting, from a generative modeling perspective, we assume that the target data distribution is p (denoted as p_{data} in the following) and our goal is to sample from it. However, in practice, we do not have access to the analytical form of p_{data} , nor to its score $\nabla \log p_{\text{data}}(\cdot)$. To circumvent this issue, we rely on a technique known as *score matching*.

To approximate the score, we use a neural network s_θ , parameterized by $\theta \in \mathbb{R}^D$ and called the *score network*. It is trained by *score matching*, which aims at minimizing

$$J_{\text{ESM}}(\theta) = \frac{1}{2} \mathbb{E}_{p_{\text{data}}} [\|s_\theta(x) - \nabla_x \log p_{\text{data}}(x)\|_2^2],$$

where p_{data} is the data distribution, to which we do not have access. Using integration by parts [Hyv05], one shows that, under the assumptions

- (1) p_{data} and s_θ are differentiable;
- (2) $\mathbb{E}_{p_{\text{data}}} [\|\nabla_x \log p_{\text{data}}(x)\|^2] < \infty$ and $\mathbb{E}_{p_{\text{data}}} [\|s_\theta(x)\|^2] < \infty$;
- (3) $\lim_{\|x\| \rightarrow \infty} p_{\text{data}}(x) s_\theta(x) = 0$ (componentwise),

the objective equals, up to an additive real constant c independent of θ ,

$$J_{\text{ESM}}(\theta) = J_{\text{ISM}}(\theta) + c, \quad J_{\text{ISM}}(\theta) := \mathbb{E}_{p_{\text{data}}} \left[\text{tr}(\nabla_x s_\theta(x)) + \frac{1}{2} \|s_\theta(x)\|_2^2 \right].$$

The objective J_{ISM} no longer involves p_{data} , as long as the expectation can be approximated by data samples. Note, however, that this empirical average estimates the expectation under

the *empirical* distribution, so we approximate the score of the empirical distribution rather than that of the true distribution. While efficient in low dimension, this method does not scale to high-dimensional data because of the term $\text{tr}(\nabla_x s_\theta(x))$: computing the d diagonal entries $\partial_i(s_\theta)_i$ requires d separate backpropagation passes, where d is the data dimension.

A method used to circumvent this issue is *Denoising Score Matching* (DSM), introduced in [Vin11]. The idea is to corrupt the data with noise: for each sample $x^{(j)}$ we set

$$\tilde{x}^{(j)} = x^{(j)} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I).$$

Let $p_\sigma(x, \tilde{x}) = p_{\text{data}}(x) p_\sigma(\tilde{x} | x)$ denote the joint law of clean and corrupted samples, and $p_\sigma(\tilde{x}) = \int p_\sigma(x, \tilde{x}) dx$ the marginal of the corrupted samples. The score-matching objective on the corrupted data reads

$$J_{\text{ESM}}^{p_\sigma}(\theta) = \frac{1}{2} \mathbb{E}_{p_\sigma(\tilde{x})} [\|s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})\|_2^2].$$

We still cannot use $\nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$ directly. At the population level the reason is that the marginal $p_\sigma(\tilde{x}) = \int p_{\text{data}}(x) \mathcal{N}(\tilde{x}; x, \sigma^2 I) dx$ is unknown, since p_{data} is. If one replaces p_{data} by the empirical measure, p_σ becomes an explicit mixture of n Gaussians centred at the samples, and its score

$$\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}) = \sum_{j=1}^n w_j(\tilde{x}) \frac{x^{(j)} - \tilde{x}}{\sigma^2}, \quad w_j(\tilde{x}) = \frac{\mathcal{N}(\tilde{x}; x^{(j)}, \sigma^2 I)}{\sum_k \mathcal{N}(\tilde{x}; x^{(k)}, \sigma^2 I)},$$

can be written down—but each evaluation costs $O(nd)$, which is prohibitive at the sample sizes used in practice.

We do, however, have access to the conditional density

$$p_\sigma(\tilde{x} | x) = \frac{1}{(2\pi)^{d/2} \sigma^d} e^{-\frac{1}{2\sigma^2} \|\tilde{x} - x\|^2}, \quad \text{whose score is} \quad \nabla_{\tilde{x}} \log p_\sigma(\tilde{x} | x) = \frac{x - \tilde{x}}{\sigma^2}.$$

The latter points from the corrupted sample $\tilde{x}$ back to the clean one x , which is precisely why the method is called *denoising*. DSM uses this to define the objective

$$J_{\text{DSM}}^{p_\sigma}(\theta) = \frac{1}{2} \mathbb{E}_{p_\sigma(x, \tilde{x})} [\|s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log p_\sigma(\tilde{x} | x)\|_2^2].$$

It is proven in [Vin11] that, as long as $\log p_\sigma(\tilde{x} | x)$ is differentiable in $\tilde{x}$, minimizing $J_{\text{DSM}}^{p_\sigma}$ is equivalent to minimizing $J_{\text{ESM}}^{p_\sigma}$, in the sense that they differ by an additive constant independent of θ :

$$J_{\text{ESM}}^{p_\sigma}(\theta) = J_{\text{DSM}}^{p_\sigma}(\theta) + C, \quad C \text{ independent of } \theta.$$

This method avoids the second derivative of the log-density, making it faster to implement, and it holds under much weaker requirements on the data distribution.

That said, DSM targets the score of a *smoothed* distribution, namely $p_\sigma = \hat{p}_0 * \mathcal{N}(0, \sigma^2 I)$, where $\hat{p}_0$ is the empirical data distribution. One may ask whether this introduces a bias. As $\sigma \rightarrow 0$, we have $p_\sigma \rightarrow \hat{p}_0$. However, because the empirical law $\hat{p}_0$ is atomic, it does not admit a proper defined score, and the DSM loss $J_{\text{DSM}}^{p_\sigma}$ diverges. Nevertheless, if we consider the objective after subtracting the σ -dependent (but θ -independent) constant C , it formally tends to the implicit objective evaluated on the empirical distribution:

$$J_{\text{ISM}}^{\hat{p}_0}(\theta) := \frac{1}{n} \sum_{j=1}^n \left(\frac{1}{2} \|s_\theta(x^{(j)})\|^2 + \sum_{i=1}^d \frac{\partial(s_\theta(x^{(j)}))_i}{\partial x_i} \right).$$

Since the samples $x^{(j)}$ are assumed i.i.d. from p_{data} , the empirical average $J_{\text{ISM}}^{\hat{p}_0}$ is an unbiased Monte-Carlo estimator of J_{ISM} under p_{data} . In contrast, $J_{\text{DSM}}^{p_\sigma}$ is biased for $\sigma > 0$, since the corrupted samples $\tilde{x}^{(j)}$ do not follow p_{data} . However, [Vin11] notes that the noise typically yields an estimator with lower variance, and that the best results are obtained for non-negligible values of σ .

3.1.3 Challenges

The idea of generative modeling via Langevin diffusion is to approximate the score within the Langevin SDE via score matching, and then to run the diffusion until some time T large enough so that X_T is almost distributed like the stationary distribution of X , which is the data distribution by assumption. In practice, this is done via a grid discretization of time. Before moving to this setting, we first highlight some challenges that already arise from the continuous Langevin SDE.

As discussed in the last chapter, the Langevin diffusion suffers from the fact that if the distribution has a large Poincaré constant, for instance when it is multi-modal with widely separated modes, the convergence towards the stationary distribution will be much slower.

Furthermore, the score matching objective is inherently flawed when the distribution is multi-modal, with the modes far from each other. To state that rigorously, we will need the following definition, from [Hyn96].

Definition 3.3 (Highest Density Region). *The $100(1 - \alpha)\%$ Highest Density Region (HDR) of a probability density function p defined on $\mathbb{R}^d$ is:*

$$R_\alpha = \{x \in \mathbb{R}^d \mid p(x) \geq c_\alpha\}$$

where c_α is chosen such that :

$$\int_{R_\alpha} p(x) dx = 1 - \alpha$$

Now, assume that $p_{\text{data}} = \pi p_1 + (1 - \pi)p_2$, where the intersection of the $100(1 - \alpha)\%$ HDR of p_1 and p_2 is empty for some α close to 0.

Assume also that $(1 - \pi)p_2(x) \ll \pi p_1(x)$ on $\text{HDR}_\alpha(p_1)$ and $\pi p_1(x) \ll (1 - \pi)p_2(x)$ on $\text{HDR}_\alpha(p_2)$.

Then, the score takes the following form:

$$\begin{aligned} \nabla_x \log p_{\text{data}}(x) &\approx \frac{\pi \nabla_x p_1(x) + (1 - \pi) \nabla_x p_2(x)}{\pi p_1(x) + (1 - \pi) p_2(x)} \\ &= \nabla_x \log p_1(x) \cdot 1_{x \in \text{HDR}_\alpha(p_1)} \\ &\quad + \nabla_x \log p_2(x) \cdot 1_{x \in \text{HDR}_\alpha(p_2)} \\ &\quad + \frac{\pi \nabla_x p_1(x) + (1 - \pi) \nabla_x p_2(x)}{\pi p_1(x) + (1 - \pi) p_2(x)} \cdot 1_{x \in \mathbb{R}^d \setminus (\text{HDR}_\alpha(p_1) \cup \text{HDR}_\alpha(p_2))} \end{aligned}$$

where $\text{HDR}_\alpha(p_i)$ is the $100(1 - \alpha)\%$ HDR of p_i for $i \in 1, 2$.

This decomposition highlights a critical issue: inside the High Density Region of p_{data} , the terms containing π vanish, rendering the score virtually independent of the mixture weights. The information regarding π is almost entirely concentrated in $\mathbb{R}^d \setminus (\text{HDR}_\alpha(p_1) \cup \text{HDR}_\alpha(p_2))$, which is a region of low density by construction (as $p_{\text{data}}(x) \leq \max(p_1(x), p_2(x))$ and both $p_1(x)$ and $p_2(x)$ are low in $\mathbb{R}^d \setminus (\text{HDR}_\alpha(p_1) \cup \text{HDR}_\alpha(p_2))$).

This local decoupling yields a profound problem: the score operator is not injective on disconnected supports, rendering the mixture parameter π unidentifiable. To see why the inverse mapping from the score field to the density is not unique, assume the support is partitioned into strictly disjoint regions: $\Omega = \Omega_1 \cup \Omega_2$ with $\Omega_1 \cap \Omega_2 = \emptyset$. Since no path connects Ω_1 and Ω_2 within

the support, the score $\nabla_x \log p(x)$ cannot be integrated globally. We must therefore integrate the score independently on each connected component. Choosing an arbitrary reference point $x_i^* \in \Omega_i$ for each region, the local density is recovered via the integral:

$$p(x) = p(x_i^*) \exp \left(\int_{x_i^*}^x \langle \nabla \log p(u), du \rangle \right) \equiv C_i \tilde{p}_i(x), \quad \forall x \in \Omega_i, \quad i \in \{1, 2\}, \quad (3.1)$$

where $\tilde{p}_i(x) = \exp \left(\int_{x_i^*}^x \langle \nabla \log p(u), du \rangle \right)$ and $C_i = p(x_i^*) > 0$ are two unknown local constants of integration.

Enforcing the constraint $\int_{\Omega} p(x) dx = 1$ yields:

$$C_1 Z_1 + C_2 Z_2 = 1, \quad (3.2)$$

where $Z_i = \int_{\Omega_i} \tilde{p}_i(x) dx > 0$ are known constants computable from the score. Crucially, the mixture weight π allocated to the first component is defined by its total probability mass:

$$\pi = \int_{\Omega_1} p(x) dx = C_1 Z_1. \quad (3.3)$$

Because any pair (C_1, C_2) satisfying $C_2 = (1 - C_1 Z_1)/Z_2$ for $C_1 \in (0, 1/Z_1)$ is a valid solution to (3.2), the recovered weight $\pi = C_1 Z_1$ can take any arbitrary value in the open interval $(0, 1)$. Therefore, the relative weights of the mixture components cannot be uniquely identified from the score field alone.

For the mixtures we consider in practice (e.g. Gaussian components), p_{data} has support $\mathbb{R}^d$ which is connected, so the strictly-disjoint argument above does not apply strictly speaking. Still, disjoint HDRs actually reduce the problem to the strictly disjoint scenario from an algorithmic perspective. Indeed, recall that the score matching objective function is weighted by the data density $p_{\text{data}}(x)$. Because the region $\mathbb{R}^d \setminus (HDR_{\alpha}(p_1) \cup HDR_{\alpha}(p_2))$, which contains the information regarding π , the network receives almost no training signal there. Consequently, the model fails to learn the true score that bridges the modes.

Lastly, the continuous Langevin framework is fundamentally ill-defined under the *manifold hypothesis*. Assuming the data distribution p_{data} concentrates on a lower-dimensional submanifold $\mathcal{M} \subset \mathbb{R}^d$ with $k = \dim(\mathcal{M}) \ll d$, the score is not properly defined. Indeed, a k -dimensional submanifold with $k < d$ has Lebesgue measure zero in $\mathbb{R}^d$. Hence p_{data} , being concentrated on $\mathcal{M}$, is singular with respect to the Lebesgue measure and, by the Radon–Nikodym theorem, admits no density on $\mathbb{R}^d$. Hence, in practice the score for Langevin is itself learned by denoising score matching at a small fixed noise level σ , whose minimiser is the score of the smoothed density $p_{\text{data}} * \mathcal{N}(0, \sigma^2 I)$, which is then well defined. However, the choice of σ is a dilemma: small σ preserves fidelity to the data distribution but leaves the score hardly learnable off the data manifold, which is a problem as the first sample has typically a high probability of being off data manifold, and then the whole algorithm is derailed. Large σ fixes this but leads to samples from a distribution further to p_{data} .

3.1.4 Discretization of the Langevin diffusion

As discussed earlier, in practice, we discretize the time interval $[0, T]$, which yields a discretization of the Langevin diffusion. We present here the most common one. The ULA (Unadjusted Langevin Algorithm) is defined through the Euler-Maruyama approximation (see Definition 2.13) of the Langevin diffusion.

$$x_t = x_{t-1} + \frac{\epsilon}{2} \nabla_x \log p(x_{t-1}) + \sqrt{\epsilon} z_t, \quad z_t \sim N(0, 1)$$

An equivalent formulation is the Langevin Monte Carlo (LMC). To derive it from the ULA, we apply three simple substitutions:

1. **Potential representation:** We relate the target distribution $p(x)$ to the potential $V(x)$ via $p(x) \propto \exp(-V(x))$, which yields $\nabla_x \log p(x) = -\nabla V(x)$.
2. **Step-size rescaling:** We define the physical time step as $h = \epsilon/2$, mapping the scaling factor to h and the noise variance to $\sqrt{2h}$.
3. **Brownian increments:** We map the discrete indices to a time grid, identifying $x_{t-1} \equiv \hat{X}_{nh}$ and $x_t \equiv \hat{X}_{(n+1)h}$. Since $\sqrt{h}z_t \sim \mathcal{N}(0, hI)$, it is equal in distribution to the standard Brownian increment $(B_{(n+1)h} - B_{nh})$.

Injecting these into the original sequence yields the LMC scheme:

$$\hat{X}_{(n+1)h} = \hat{X}_{nh} - h\nabla V(\hat{X}_{nh}) + \sqrt{2}(B_{(n+1)h} - B_{nh}). \quad (3.4)$$

Convergence of LMC: There are multiple results for the convergence of the LMC algorithm, depending on the hypothesis about the potential V . We state and prove here one result (Theorem 4.1.2 in [Che26]) which assumes that V is α -strongly convex.

Theorem 3.4 (Coupling analysis of LMC). *For $n \in \mathbb{N}$, let $\hat{\mu}_{nh}$ denote the law of the n -th iterate of LMC with step size $h > 0$. Assume that the target $\pi \propto \exp(-V)$ satisfies $0 \prec \alpha I_d \preceq \nabla^2 V \preceq \beta I_d$, write $\kappa := \beta/\alpha$, and assume the initialization is warm in the sense that $\mathbb{E}\|\hat{X}_0 - x^*\|^2 \lesssim d/\alpha$, where x^* is the minimizer of V . Then, provided $h \lesssim \frac{1}{\beta\kappa}$, for all $N \in \mathbb{N}$,*

$$W_2(\hat{\mu}_{Nh}, \pi) \leq \exp\left(-\frac{\alpha Nh}{2}\right) W_2(\mu_0, \pi) + \mathcal{O}\left(\frac{\beta d^{1/2} h^{1/2}}{\alpha}\right). \quad (3.5)$$

If we set $h \asymp \frac{\epsilon^2}{\beta\kappa d}$, then for any $\epsilon \in (0, \sqrt{d}]$ we obtain $\sqrt{\alpha} W_2(\hat{\mu}_{Nh}, \pi) \leq \epsilon$ after

$$N = \mathcal{O}\left(\frac{\kappa^2 d}{\epsilon^2} \log\left(\frac{\alpha W_2^2(\mu_0, \pi)}{\epsilon^2}\right)\right)$$

iterations.

The proof relies on the following moment bound along the LMC trajectory, which is where the step-size condition $h \lesssim 1/(\beta\kappa)$ is used.

Lemma 3.5 (Moments of LMC and local discretization error). *Under the assumptions of Theorem 3.4, if $h \leq 1/\beta$ then*

$$\sup_{n \in \mathbb{N}} \mathbb{E}[\|\hat{X}_{nh} - x^*\|^2] \leq \max\left\{\mathbb{E}[\|\hat{X}_0 - x^*\|^2], \frac{2d}{\alpha}\right\} \lesssim \frac{d}{\alpha}, \quad \text{hence} \quad \sup_{n \in \mathbb{N}} \mathbb{E}[\|\nabla V(\hat{X}_{nh})\|^2] \lesssim \beta\kappa d.$$

Consequently, for the continuous-time interpolation $(\hat{X}_t)_{t \geq 0}$ and $t \in [nh, (n+1)h)$, if moreover $h \lesssim 1/(\beta\kappa)$,

$$\mathbb{E}[\|\hat{X}_t - \hat{X}_{nh}\|^2] \lesssim hd. \quad (3.6)$$

Proof: Write $X_n := \hat{X}_{nh}$. One LMC step reads $X_{n+1} = X_n - h\nabla V(X_n) + \sqrt{2h}\xi_n$ with $\xi_n \sim \mathcal{N}(0, I_d)$ independent of X_n , so that

$$\mathbb{E}\|X_{n+1} - x^*\|^2 = \mathbb{E}\|X_n - x^*\|^2 - 2h\mathbb{E}\langle \nabla V(X_n), X_n - x^* \rangle + h^2\mathbb{E}\|\nabla V(X_n)\|^2 + 2hd.$$

By co-coercivity of the β -smooth convex V (and $\nabla V(x^*) = 0$), $\langle \nabla V(x), x - x^* \rangle \geq \frac{1}{\beta} \|\nabla V(x)\|^2$, so for $h \leq 1/\beta$ the term $h^2 \|\nabla V\|^2$ is absorbed by half of the inner-product term; by α -strong convexity, $\langle \nabla V(x), x - x^* \rangle \geq \alpha \|x - x^*\|^2$ absorbs the other half. Hence $\mathbb{E} \|X_{n+1} - x^*\|^2 \leq (1 - \alpha h) \mathbb{E} \|X_n - x^*\|^2 + 2hd$, and the first claim follows by induction. The gradient bound follows from $\|\nabla V(x)\| \leq \beta \|x - x^*\|$ and $\beta^2 d/\alpha = \beta \kappa d$.

For the local error, on $t \in [nh, (n+1)h)$ the interpolation gives explicitly

$$\hat{X}_t - \hat{X}_{nh} = -(t - nh) \nabla V(\hat{X}_{nh}) + \sqrt{2} (B_t - B_{nh}).$$

Since $B_t - B_{nh}$ is independent of $\hat{X}_{nh}$ and centered, the cross term vanishes in expectation and

$$\mathbb{E} \|\hat{X}_t - \hat{X}_{nh}\|^2 = (t - nh)^2 \mathbb{E} \|\nabla V(\hat{X}_{nh})\|^2 + 2d(t - nh) \lesssim h^2 \beta \kappa d + hd \lesssim hd,$$

where the last inequality uses $h \lesssim 1/(\beta \kappa)$. $\square$

Proof of Theorem 3.4: We consider the continuous-time interpolation of the LMC iterates, denoted by $(\hat{X}_t)_{t \geq 0}$ and defined for $t \in [nh, (n+1)h)$ by

$$d\hat{X}_t = -\nabla V(\hat{X}_{nh}) dt + \sqrt{2} dB_t.$$

Let $(Y_t)_{t \geq 0}$ be the exact Langevin diffusion driven by the *same* Brownian motion (synchronous coupling), started at stationarity:

$$dY_t = -\nabla V(Y_t) dt + \sqrt{2} dB_t, \quad Y_0 \sim \pi,$$

so that $Y_t \sim \pi$ for all $t \geq 0$. We couple the initial conditions $(\hat{X}_0, Y_0)$ *optimally* for W_2 , so that $\mathbb{E} \|\hat{X}_0 - Y_0\|^2 = W_2^2(\mu_0, \pi)$.

Let $e_t := \hat{X}_t - Y_t$. The Brownian terms cancel and

$$de_t = -(\nabla V(\hat{X}_{nh}) - \nabla V(Y_t)) dt, \quad t \in [nh, (n+1)h).$$

Computing the time derivative of the squared distance and splitting,

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \|e_t\|^2 &= -\langle e_t, \nabla V(\hat{X}_{nh}) - \nabla V(Y_t) \rangle \\ &= -\langle \hat{X}_t - Y_t, \nabla V(\hat{X}_t) - \nabla V(Y_t) \rangle - \langle \hat{X}_t - Y_t, \nabla V(\hat{X}_{nh}) - \nabla V(\hat{X}_t) \rangle. \end{aligned}$$

By α -strong convexity the first term is at most $-\alpha \|e_t\|^2$. For the second, Young's inequality $\langle a, b \rangle \leq \frac{\alpha}{2} \|a\|^2 + \frac{1}{2\alpha} \|b\|^2$ and β -smoothness give

$$-\langle \hat{X}_t - Y_t, \nabla V(\hat{X}_{nh}) - \nabla V(\hat{X}_t) \rangle \leq \frac{\alpha}{2} \|e_t\|^2 + \frac{\beta^2}{2\alpha} \|\hat{X}_t - \hat{X}_{nh}\|^2.$$

Summing and multiplying by 2,

$$\frac{d}{dt} \|e_t\|^2 \leq -\alpha \|e_t\|^2 + \frac{\beta^2}{\alpha} \|\hat{X}_t - \hat{X}_{nh}\|^2.$$

The map $t \mapsto \|e_t\|^2$ is continuous and piecewise C^1 , so taking expectations and applying Grönwall's lemma in variation-of-constants form on $[0, Nh]$ yields

$$\mathbb{E} [\|e_{Nh}\|^2] \leq e^{-\alpha Nh} \mathbb{E} [\|e_0\|^2] + \frac{\beta^2}{\alpha} \int_0^{Nh} e^{-\alpha(Nh-t)} \mathbb{E} [\|\hat{X}_t - \hat{X}_{[t/h]h}\|^2] dt.$$

Injecting the local bound (3.6) of Lemma 3.5 and $\int_0^{Nh} e^{-\alpha(Nh-t)} dt \leq 1/\alpha$,

$$\mathbb{E} [\|e_{Nh}\|^2] \leq e^{-\alpha Nh} W_2^2(\mu_0, \pi) + \mathcal{O}\left(\frac{\beta^2 h d}{\alpha^2}\right).$$

Since $(\hat{X}_{Nh}, Y_{Nh})$ is a coupling of $(\hat{\mu}_{Nh}, \pi)$, the coupling bound $W_2^2(\hat{\mu}_{Nh}, \pi) \leq \mathbb{E}\|\hat{X}_{Nh} - Y_{Nh}\|^2$ and the subadditivity of the square root give (3.5).

For the complexity claim, set $h \asymp \varepsilon^2/(\beta\kappa d)$; then the bias term is $\mathcal{O}(\varepsilon/\sqrt{\alpha})$, the condition $\varepsilon \leq \sqrt{d}$ ensures $h \lesssim 1/(\beta\kappa)$, and the exponential term is at most $\varepsilon/\sqrt{\alpha}$ once $Nh \gtrsim \frac{1}{\alpha} \log(\alpha W_2^2(\mu_0, \pi)/\varepsilon^2)$, i.e. after $N = \mathcal{O}(\kappa^2 d \varepsilon^{-2} \log(\alpha W_2^2(\mu_0, \pi)/\varepsilon^2))$ iterations. $\square$

Remark: The assumption that the potential V is α -strongly convex implies in particular that the target distribution can't be multimodal. Hence, for multimodal distributions, this theorem does not give any guarantees of convergence. There is a similar result which relies on the weaker assumption that the potential V satisfies a Poincaré inequality, involving the χ^2 -divergence. This assumption allows for multimodal distribution at a cost of large Poincaré constant, which implies slow convergence. However, we decided to present the result involving the Wasserstein distance, as the χ^2 -divergence is highly sensitive to tail behaviors and is notoriously prone to exploding in practical settings. Indeed, recall that:

$$D_{\chi^2}(P \parallel Q) = \int \frac{p(x)^2}{q(x)} dx - 1.$$

This expression shows that the divergence becomes large whenever $q(x)$ is very small in regions where $p(x)$ has non-negligible mass. In particular, if there exists a region A such that $q(x) \approx 0$ while $p(x) > 0$, then the integrand $\frac{p(x)^2}{q(x)}$ becomes arbitrarily large, potentially leading to $D_{\chi^2}(P \parallel Q) = +\infty$. This situation typically arises when q underestimates the tails of p .

By virtue of the previous subsections and of the latest remark, it is clear that the LMC inherits the fundamental flaws of the continuous Langevin diffusion, namely the fact that if the distribution is multimodal, or the data is supported in a low dimensional subspace, the convergence will be slow. Moreover, comparing the convergence of the continuous-time Langevin diffusion (Theorems 2.41 and 2.44) with the LMC bound reveals a crucial difference: while the continuous dynamics can exhibit a dimension-free contraction rate, the discretization error in LMC scales with the ambient dimension d . This dependency stems from the variance of the injected Brownian increments at each discrete step. Consequently, in high-dimensional settings, mitigating this error requires a smaller step size h , which inherently slows down the convergence of the algorithm.

Experiments: Please note that complete experimental details are available at: <https://github.com/Laurentmb/Mathematical-Foundations-of-Diffusion-Based-Generative-Modeling/tree/main>. The presentation here is deliberately high-level to allow the reader to focus on the core concepts.

We provide an example where the Langevin procedure can produce high-quality samples. We consider here that we have access to the true distribution to be able to compare the samples produced by Langevin with ones produced by the true distribution, but note that this is usually not the case in practice. We then run the Langevin algorithm with both the analytical score and the score approximated using denoising score matching. The distribution we consider has density in $\mathbb{R}^2$:

$$p(x) \propto e^{-\frac{(\|x\| - R)^2}{2s^2}} * (1 + a \cos(\theta))$$

with the following values: $R, s, a = 1.5, 0.3, 0.6$. Here, θ comes from the polar representation of $x : x = (r \cos \theta, r \sin \theta)$.

Here, we initialize Langevin with the density: $x = (r, 0)$, $r \sim \mathcal{N}(R, s^2)$, which is an adversarial choice, as it puts mass only on the x axis. The role of the Langevin algorithm is then to recover the angular distribution. We obtain the following samples:

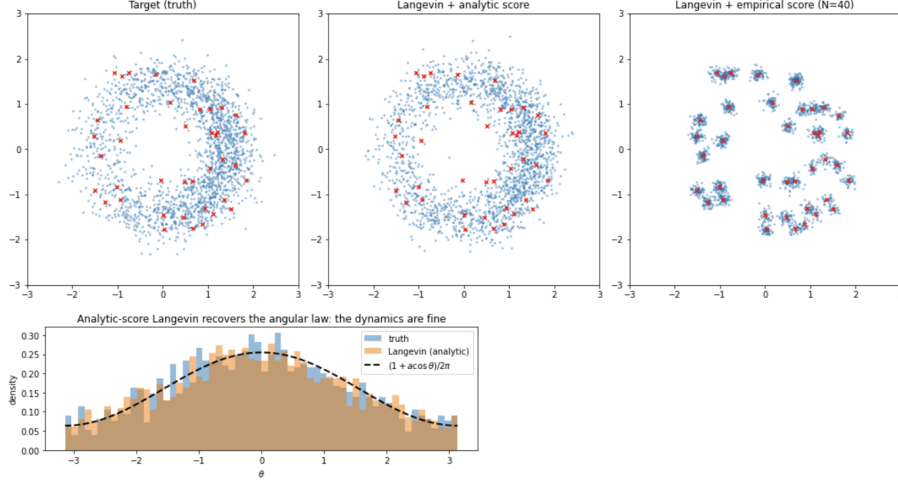


Figure 3.1: Comparison between samples obtained via Langevin with analytical score, Langevin with empirical score, and samples from the original data distribution

Remark: The red dots represent the training samples.

Langevin with the *analytical* score succeeds here: although the density varies along the ring, its support is connected and the density never vanishes, so there is no low-density barrier to cross.

Now, we must specify what "Langevin with the empirical score" means. Rather than training a network, we use the score of the kernel density estimate of the training set: each of the $N = 40$ data points is replaced by a Gaussian bump of width h , giving the smooth density $\hat{p}_h(x) = \frac{1}{N} \sum_i \mathcal{N}(x; x_i, h^2 I)$, whose score is available in closed form. This choice is not a simplification but a strengthening of the argument: Indeed, recall that the score of $\hat{p}_h$ is exactly the global minimizer of denoising score matching at noise level $\sigma = h$ on this dataset. Any failure is thus attributable to the estimation target itself, not to network capacity or training issues.

We see then that Langevin driven by the empirical score essentially samples only near the training points. Since the bandwidth ($h = 0.07$) is small compared to the typical spacing between the 40 points ($\approx 2\pi R/N \approx 0.24$), the density $\hat{p}_h$ consists of 40 isolated bumps rather than a ring. Increasing h does not resolve this: once h is comparable to the spacing the bumps merge, but one then samples a blurred version of the target. In this two-dimensional example the problem could also be fixed by adding data until the manifold is densely covered. However in high dimension the number of points required explodes, which is why the Langevin procedure fails under the manifold hypothesis.

We show now an example where even Langevin with analytical score fails. Langevin is initialized at $\mathcal{N}(0, I)$.

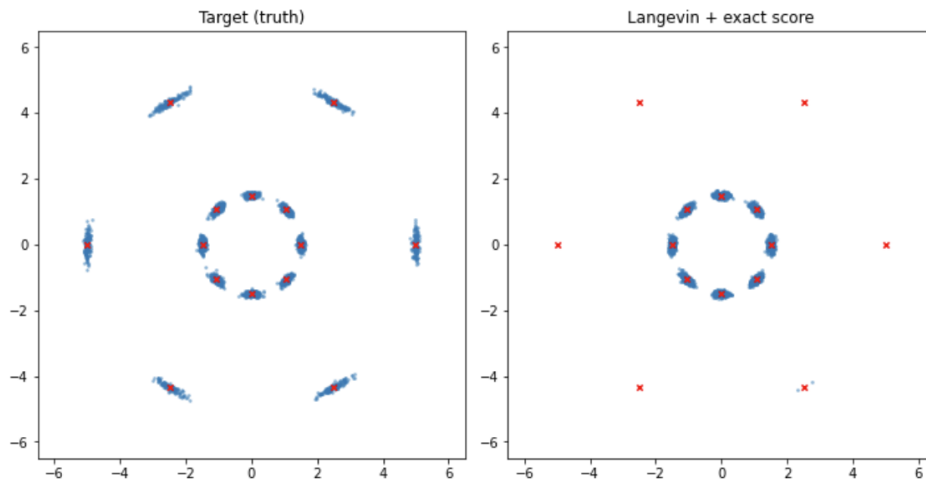


Figure 3.2: Comparison between samples obtained via Langevin with analytical score, and the truth

Remark: Note that here, the red dots represent the modes of the target distribution, and not the training samples anymore.

The target is multimodal with modes separated by wide regions of very low density. Hence the Poincaré constant is very large, and convergence is unattainable within the considered step budget, independently of the score, which is exact here for both methods.

The figure shows how severe this is: the outer ring is not mis-weighted but entirely missing. Two effects compound. First, the $\mathcal{N}(0, I)$ initialization essentially never seeds the outer ring, an effect unrelated to mixing, concerning only where the chains start. Second, a chain (one realization of the Markov chain defined by the Langevin update: a single trajectory $x_0, \dots, x_K$ obtained by iterating from one random initialization) that settles on the inner ring is very unlikely to leave it.

A more recent framework has been introduced to address the issues that the Langevin procedure was facing, known as diffusion models.

3.2 Classical Diffusion models

3.2.1 Forward process and Denoising Score Matching

For $t \in [0, T]$, we consider a generic SDE of the form

$$dX_t = f(X_t, t) dt + g(t) dW_t$$

with $X_0 \sim p_{\text{data}}$; we call it the *forward SDE*, and its terminal law p_T is (approximately) a tractable prior we can sample from.

This SDE transports the data towards an easy to sample from prior. The idea behind diffusion models is to do the opposite: transport the prior towards the data. To do so, we need to reverse the SDE. Then, if we can run the reverse SDE, starting at the prior distribution, we can sample from the data distribution. We define: $s := T - t \in [0, T]$ and the reverse process $Y_s := X_{T-s}$, so that s increases from 0 to T while the physical time $t = T - s$ decreases from T to 0. By extending what we did in the Time Reversal of SDEs section to a not-necessarily-positive diffusion coefficient (the proof is unchanged), Y solves the following SDE, driven forward in s :

$$dY_s = \left[-f(Y_s, T - s) + g^2(T - s) \nabla_x \log p_{T-s}(Y_s) \right] ds + g(T - s) d\widetilde{W}_s,$$

where $\widetilde{W}$ is a standard Brownian motion in the reversed clock s . By construction $Y_s \sim p_{T-s}$; in particular

$$Y_0 = X_T \sim p_T, \quad Y_T = X_0 \sim p_{\text{data}}.$$

Hence, if we can sample $Y_0 \sim p_T$ and approximate the score $\nabla_x \log p_{T-s}$ at each $s \in [0, T]$, we can integrate this SDE forward in s from 0 to T and obtain a sample $Y_T \sim p_{\text{data}}$. Since $T - s$ ranges over $[0, T]$ as s does, this amounts to approximating $\nabla_x \log p_t(X_t)$ for every $t \in [0, T]$.

To do so, we use the denoising score matching introduced in the previous section. More precisely, we want to solve:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{t \sim U(0, T)} \left[\lambda(t) \mathbb{E}_{x(0)} \mathbb{E}_{x(t)|x(0)} \left[\|s_{\theta}(x(t), t) - \nabla_x \log p_{0t}(x(t)|x(0))\|_2^2 \right] \right] \quad (3.7)$$

We emphasize here that X_0 is distributed according to the empirical data distribution. Indeed, as pointed out in the previous section, the true data distribution is unknown. The forward process is then initialized at the empirical data distribution, and $s_{\theta}(x(t), t)$ is a time-dependent score network aiming at approximating $\nabla_x \log p_t(x(t))$, the empirical score, at each time t ,

Remark:

Note that this is a scaled version of the DSM objective introduced in the previous section. Indeed, $x(0)$ is the noise-free sample, corresponding to x in the previous section, and $x(t)$ is the corrupted sample, corresponding to $\tilde{x}$ in the previous section. We then use the law of total expectation. That is, for any integrable random variables X and Y , with joint density $p(.,.)$ and any integrable function f , we have:

$$\mathbb{E}_{X, Y}[f(X, Y)] = \iint f(x, y) p(x, y) dx dy = \iint f(x, y) p(x) p(y|x) dx dy = \mathbb{E}_X \left[\mathbb{E}_{Y|X}[f(X, Y)] \right]$$

This allows us to write the DSM objective introduced in the previous section as:

$$\mathbb{E}_{x(0)} \mathbb{E}_{x(t)|x(0)} \left[\|s_{\theta}(x(t), t) - \nabla_x \log p_{0t}(x(t)|x(0))\|_2^2 \right]$$

However, because we are now considering a continuous diffusion process indexed by time $t \in [0, T]$, we must integrate this objective over the entire time horizon. Taking the expectation over $t \sim U(0, T)$ ensures that the score network learns to approximate the score field across all possible noise scales. Furthermore, a weighting function $\lambda(t)$ is introduced because the expected squared norm of the gradient varies drastically with t . For small t , the conditional distribution $p_{0t}(x(t)|x(0))$ is highly concentrated around $x(0)$ (approaching a Dirac mass), resulting in large gradients of the log-density. Conversely, as $t \rightarrow T$, the conditional distribution diffuses towards the smooth prior p_T , and the gradients become comparatively small. Consequently, without a proper weighting function, the optimization would disproportionately focus on small values of t . To compensate for this imbalance, $\lambda(t)$ is typically chosen to be inversely proportional to the expected squared norm of the true score:

$$\lambda(t) \propto \frac{1}{\mathbb{E}[\|\nabla_{x(t)} \log p_{0t}(x(t)|x(0))\|_2^2]}$$

To be able to compute this objective and update the weights of the score network in the training phase, we notice that we need to know the transition kernel $p_{0t}(x(t)|x(0))$. This motivates the choice of an affine drift f in the forward SDE. As a matter of fact, we can prove that with such a choice, the transition kernel will be gaussian.

Indeed, assume that $f(x, t) = a(t)x + b(t)$. Define the integrating factor $\Phi(t) = \exp\left(-\int_0^t a(s)ds\right)$ and let $Y_t = h(X_t, t) := \Phi(t)X_t$. Applying Itô's formula to Y_t , using that $\Phi'(t) = -a(t)\Phi(t)$, that the second derivative of h with respect to x vanishes, and that the process t is of finite variation, we get:

$$\begin{aligned} dY_t &= \Phi(t)dX_t + X_t d\Phi(t) \\ &= \Phi(t)\left[(a(t)X_t + b(t))dt + g(t)dW_t\right] - a(t)\Phi(t)X_t dt \\ &= \Phi(t)b(t)dt + \Phi(t)g(t)dW_t \end{aligned}$$

Integrating from 0 to t yields:

$$Y_t = X_0 + \int_0^t \Phi(s)b(s)ds + \int_0^t \Phi(s)g(s)dW_s \quad (3.8)$$

Multiplying by $\Phi(t)^{-1} = \exp\left(\int_0^t a(s)ds\right)$, we obtain the explicit solution for X_t :

$$X_t = \Phi(t)^{-1}X_0 + \int_0^t \frac{\Phi(s)}{\Phi(t)}b(s)ds + \int_0^t \frac{\Phi(s)}{\Phi(t)}g(s)dW_s \quad (3.9)$$

Conditioned on X_0 , the first two terms are purely deterministic. The third term is a stochastic integral of a deterministic function with respect to Brownian motion, which is a Gaussian random variable. Heuristically, this can be seen thanks to the fact that the integral is an infinite sum of increments of Brownian motions, which are independent centered gaussian random variables. Hence $\int_0^t \frac{\Phi(s)}{\Phi(t)}g(s)dW_s$ is a centered gaussian random variable, and its variance is $\int_0^t \left(\frac{\Phi(s)}{\Phi(t)}g(s)\right)^2 ds$ thanks to Theorem 2.16.

So finally, as a sum of deterministic variables and a Gaussian variable, $X_t|X_0$ is Gaussian.

Now that we know that choosing an affine drift is convenient, it remains to choose the affine function. As noted in [Son+21], there are two main choices for the forward process. These

2 choices actually reflect both discrete time generative models that emerged in the literature before the diffusion paradigm, namely Denoising Score Matching with Langevin Dynamics (introduced in [SE20]) and Denoising Diffusion Probabilistic Models (introduced in [HJA20]).

The first choice, corresponding to a continuous version of Denoising Score Matching with Langevin Dynamics, is :

$$dX_t = \sqrt{\frac{d}{dt}\sigma^2(t)}dW_t$$

Integrating both sides:

$$X_t = X_0 + \int_0^t \sqrt{\frac{d}{ds}\sigma^2(s)}dW_s$$

Observe that the stochastic integral has variance $\int_0^t \frac{d}{ds}\sigma^2(s)ds$, which increases with t . Hence the variance of X_t also increases with t , given that X_0 admits finite variance. This method is thus called the Variance Exploding method.

The second choice, corresponding to a continuous version of Denoising Diffusion Probabilistic Models, is the Ornstein–Uhlenbeck process:

$$dX_t = -\frac{1}{2}\beta(t)X_t dt + \sqrt{\beta(t)}dW_t$$

where β is a strictly positive continuous function.

We can actually solve this SDE. Indeed, let us define the integrating factor $I(t) = e^{\frac{1}{2}\int_0^t \beta(s)ds}$ and apply Itô's formula to the process $f(t, X_t) = I(t)X_t$:

$$\begin{aligned} d(I(t)X_t) &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial^2 x}d\langle X \rangle_t \\ &= \frac{1}{2}\beta(t)I(t)X_t dt + I(t)dX_t. \end{aligned}$$

Substituting the initial dynamics of dX_t :

$$\begin{aligned} d(I(t)X_t) &= \frac{1}{2}\beta(t)I(t)X_t dt + I(t)\left(-\frac{1}{2}\beta(t)X_t dt + \sqrt{\beta(t)}dW_t\right) \\ &= I(t)\sqrt{\beta(t)}dW_t. \end{aligned}$$

Integrating from 0 to t , with $I(0) = 1$:

$$I(t)X_t - X_0 = \int_0^t I(s)\sqrt{\beta(s)}dW_s.$$

Dividing by $I(t)$ and expanding its expression yields the explicit solution:

$$X_t = X_0 e^{-\frac{1}{2}\int_0^t \beta(s)ds} + \int_0^t e^{-\frac{1}{2}\int_s^t \beta(u)du} \sqrt{\beta(s)}dW_s. \quad (3.10)$$

From that, as the stochastic integral is independent of X_0 since increments of Brownian motion are independent of the past, we deduce that:

$$\text{Var}(X_t) = \text{Var}(X_0)e^{-\int_0^t \beta(s)ds} + \int_0^t e^{-\int_s^t \beta(u)du} \beta(s)ds$$

If now we set: $A(t) = \int_0^t \beta(s)ds$

We get:

$$\int_0^t e^{-\int_s^t \beta(u)du} \beta(s)ds = \int_0^t e^{-(A(t)-A(s))} \beta(s)ds = e^{-A(t)} \int_0^t e^{A(s)} \beta(s)ds$$

But note that by the fundamental theorem of calculus, $A'(s) = \beta(s)$.

Hence: $\int_0^t e^{A(s)} \beta(s)ds = e^{A(t)} - 1$

Finally:

$$\text{Var}(X_t) = 1 - e^{-A(t)} + e^{-A(t)} \text{Var}X_0$$

So if we choose X_0 such that $\text{Var}(X_0) = 1$, we get that $\text{Var}(X_t) = 1 \quad \forall t \geq 0$.

Limit distributions and Initialization of the backward process

Recall that in the Variance Exploding method:

$$X_t = X_0 + \int_0^t \sqrt{\frac{d}{ds} \sigma^2(s)} dW_s$$

Which leads to:

$$X_t|X_0 \sim N(X_0, [\sigma^2(t) - \sigma^2(0)]I_d)$$

Hence, no matter the initial distribution, the variance of X_t is increasing in t , so the process X does not admit any well-defined limit distribution. To initialize the backward process, one usually makes the assumption that $X_T \sim N(0, \sigma^2(T)I_d)$ (section 2.1 in [Son+21]).

Now, if we look at the variance preserving method:

$$X_t|X_0 \sim N\left(X_0 e^{-\frac{1}{2} \int_0^t \beta(s)ds}, (1 - e^{-\int_0^t \beta(s)ds})I_d\right)$$

Hence, if $\int_0^t \beta(s)ds$ diverges as $t \rightarrow \infty$ (which is the case for constant β for instance), we get that the conditional law $X_t|X_0$ converges to $N(0, I_d)$. Consequently, X_t also converges in law to $N(0, I_d)$ (by the law of total probability). This gives us that the limit distribution of X as $t \rightarrow \infty$ is $N(0, I_d)$.

Moreover, we can see that the standard normal distribution is also the stationary distribution of the process. Indeed:

$$X_t = X_0 e^{-\frac{1}{2} \int_0^t \beta(s) ds} + \int_0^t e^{-\frac{1}{2} \int_s^t \beta(u) du} \sqrt{\beta(s)} dW_s,$$

we know that conditionally on X_0 , X_t is a Gaussian random variable for any t . Moreover, if we assume $X_0 \sim N(0, I_d)$, the marginal distribution remains Gaussian, and we have:

$$\mathbb{E}[X_t] = e^{-\frac{1}{2} \int_0^t \beta(s) ds} \mathbb{E}[X_0] = 0$$

and

$$\text{Var}(X_t) = e^{-\int_0^t \beta(s) ds} \text{Var}[X_0] + (1 - e^{-\int_0^t \beta(s) ds}) I_d = I_d.$$

In practice, one initializes the backward distribution with $N(0, I_d)$, assuming that the distribution of $X_T = \hat{X}_0$ is close to the limit distribution, which is likely to be the case if T is large enough.

3.2.2 Solving the backward SDE

The forward SDE in this case is the variance-exploding SDE

$$dX_t = \sqrt{\frac{d}{dt} \sigma^2(t)} dW_t, \quad X_0 \sim p_{\text{data}},$$

so $f \equiv 0$, $g^2(t) = \frac{d}{dt} \sigma^2(t)$, and $p_t = p_{\text{data}} * \mathcal{N}(0, \sigma^2(t) I_d)$, assuming that $\sigma(0) = 0$.

Defining $s := T - t$, $Y_s := X_{T-s}$, the reverse process solves the SDE

$$dY_s = g^2(T-s) \nabla_x \log p_{T-s}(Y_s) ds + g(T-s) d\tilde{W}_s, \quad Y_0 = X_T \sim p_T,$$

where $\tilde{W}$ is a standard Brownian motion in s .

To solve this numerically, discretize $[0, T]$ into $\{t_0 = 0 < t_1 < \dots < t_N = T\}$ and traverse it from $t_N = T$ to $t_0 = 0$. The step from t_{i+1} to t_i has positive increment $\Delta s_i := t_{i+1} - t_i > 0$. Writing the sample at physical time t_i as X_{t_i} (i.e. Y_{T-t_i}) and approximating the marginal score at physical time t_{i+1} by the trained network $s_{\theta^*}(\cdot, t_{i+1}) \approx \nabla_x \log p_{t_{i+1}}$, the explicit Euler-Maruyama step reads

$$X_{t_i} = X_{t_{i+1}} + g^2(t_{i+1}) s_{\theta^*}(X_{t_{i+1}}, t_{i+1}) \Delta s_i + g(t_{i+1}) \sqrt{\Delta s_i} Z, \quad Z \sim \mathcal{N}(0, I_d).$$

For the variance-exploding SDE, $g^2(t_{i+1}) \Delta s_i = \frac{d}{dt} \sigma^2(t_{i+1}) (t_{i+1} - t_i) \approx \sigma_{i+1}^2 - \sigma_i^2$ (finite-difference of σ^2), where $\sigma_i^2 := \sigma^2(t_i)$ is increasing in i , and $g(t_{i+1}) \sqrt{\Delta s_i} = \sqrt{g^2(t_{i+1}) \Delta s_i} \approx \sqrt{\sigma_{i+1}^2 - \sigma_i^2}$. Hence

$$X_{t_i} = X_{t_{i+1}} + (\sigma_{i+1}^2 - \sigma_i^2) s_{\theta^*}(X_{t_{i+1}}, t_{i+1}) + \sqrt{\sigma_{i+1}^2 - \sigma_i^2} Z.$$

Predictor-corrector approach

An improvement is introduced in [Son+21]. The *predictor* is the reverse Euler-Maruyama step above, advancing by one grid increment (from physical time t_{i+1} to t_i). The *corrector* then refines X_{t_i} at the *fixed* noise level σ_i by running a few steps of Langevin Markov Chain Monte Carlo driven by the same score,

$$X_{t_i} \leftarrow X_{t_i} + \varepsilon_i s_{\theta^*}(X_{t_i}, t_i) + \sqrt{2\varepsilon_i} Z', \quad Z' \sim \mathcal{N}(0, I_d),$$

for a small step size $\varepsilon_i > 0$, so that X_{t_i} is brought closer to the marginal p_{t_i} before the next predictor step. More precisely, we use the following algorithm:

Algorithm 1: PC sampling (VE SDE)

```

 $\mathbf{x}_{t_N} \sim \mathcal{N}(\mathbf{0}, \sigma_{\max}^2 \mathbf{I})$ 
for  $i = N - 1$  to  $0$  do
    // Predictor (Euler-Maruyama)
     $\mathbf{x}'_{t_i} \leftarrow \mathbf{x}_{t_{i+1}} + (\sigma_{i+1}^2 - \sigma_i^2) \mathbf{s}_{\theta^*}(\mathbf{x}_{t_{i+1}}, t_{i+1})$ 
     $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
     $\mathbf{x}_{t_i} \leftarrow \mathbf{x}'_{t_i} + \sqrt{\sigma_{i+1}^2 - \sigma_i^2} \mathbf{z}$ 
    // Corrector (Langevin MCMC)
    for  $j = 1$  to  $M$  do
         $\mathbf{z}' \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
         $\mathbf{x}_{t_i} \leftarrow \mathbf{x}_{t_i} + \varepsilon_i \mathbf{s}_{\theta^*}(\mathbf{x}_{t_i}, t_i) + \sqrt{2\varepsilon_i} \mathbf{z}'$ 
return:  $\mathbf{x}_{t_0}$ 

```

Explanation of the algorithm: We start from the prior distribution. We predict the next point in the trajectory of the backward process, then we add M corrector steps. We iterate this procedure until we get to the last point of the backward process. The corrector works as follows: $\mathbf{s}_{\theta^*}(\mathbf{x}_{t_i}, t_i)$ indicates the direction in which the density increases the most, since it is the gradient of a monotone function of the density, and we know that the gradient of a function points towards the directions in which the function increases the most. Hence by adding $\varepsilon_i \mathbf{s}_{\theta^*}(\mathbf{x}_{t_i}, t_i)$, we move the sample in this direction. We then add the noise $\sqrt{2\varepsilon_i} \mathbf{z}'$ to avoid getting stuck in local maxima.

Thanks to the corrector approach, we can compensate for some of the discretization error induced by Euler-Maruyama. If we add only 1 corrector step at each timestep, we double the computation time. However, as noted in [Son+21], this is more efficient than simply doubling the number of steps in Euler-Maruyama.

Experiments: We provide examples where Diffusion models can produce high-quality samples where Langevin fail.

To do so, we take a mixture Gaussian distribution in $\mathbb{R}^2$, with well separated modes which have unequal weights. The modes are placed to form a ring. Note that as in the previous experiments, we have here access to the analytical score. However, this time we train a neural network to approximate the score for the Langevin procedure, as well as one for the diffusion model. We obtain the following results.

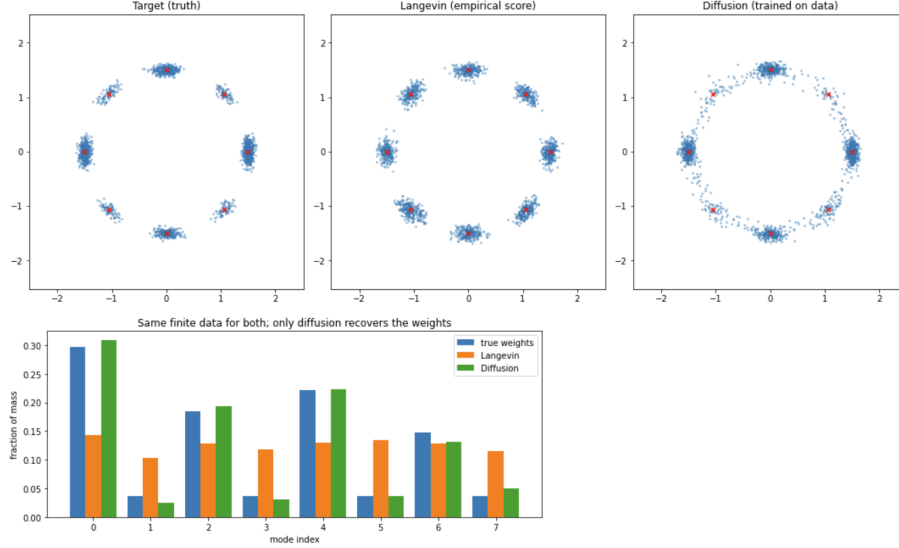


Figure 3.3: Langevin vs SDE: first case

Under the reported learned-score configurations, the reverse-SDE sampler produces mode frequencies closer to the target frequencies. However, since the methods use differently conditioned score networks and different sampling dynamics, this comparison does not isolate slow mixing as the sole cause.

Moreover, as we have access to the target distribution, we can compute some metrics to assess that the SDE procedure has indeed produced a distribution closer to the target distribution than the one of the Langevin procedure:

metric	Langevin	SGM
weight-TV (down)	0.3349	0.0288
mean loglik (up)	-0.1700	-5.3320
MMD^2 (down)	0.0044	0.0006
sliced-W2 (down)	0.0875	0.0140
precision (up)	0.9883	0.6680
recall (up)	0.9927	1.0000

Figure 3.4: Langevin vs SDE: performance metrics in the first case

The weight Total Variation is defined as:

$$\text{weight-TV} = \frac{1}{2} \sum_{k=1}^K |f_k - w_k|$$

where K is the number of modes in the target, w_k is the weight assigned to mode k by the target distribution, and f_k is the weight assigned to mode k by the distribution of the model, which is computed as: $f_k = \frac{1}{M} \sum_{j=1}^M \mathbf{1}[\arg \min_{k'} \|y_j - \mu_{k'}\| = k]$

In this metric, the SDE clearly outperforms the Langevin procedure. However, this metric is blind to points being sampled outside of the data manifold, as it assigns every point to a mode, even though such a point can be far from every mode. Moreover, it does not account for the dispersion of samples inside of the modes.

To circumvent this, we also observe the Sliced Wasserstein (sliced- W_2) metric. It aims at approximating the 2-Wasserstein distance (see Definition 2.21). As the 2-Wasserstein distance is a transport cost, it intuitively takes into account for the geometry of the samples, penalizing misplaced mass in proportion to how far it must be transported — so both misplaced points and wrong dispersion inside modes contribute. In that metric, the SDE method also largely outperforms the Langevin procedure.

Remark: Precision measures the proportion of generated samples that fall in the support of the target, and recall measures the fraction of real samples that are in the support of the generated distribution. However, the discriminative signal here lies in how mass is distributed over that support, which precision/recall do not measure — hence weight-TV and sliced- W_2 are the informative metrics.

Finally, we also provide the score network approximation for the diffusion model to see if the network has been able to correctly learn the score.

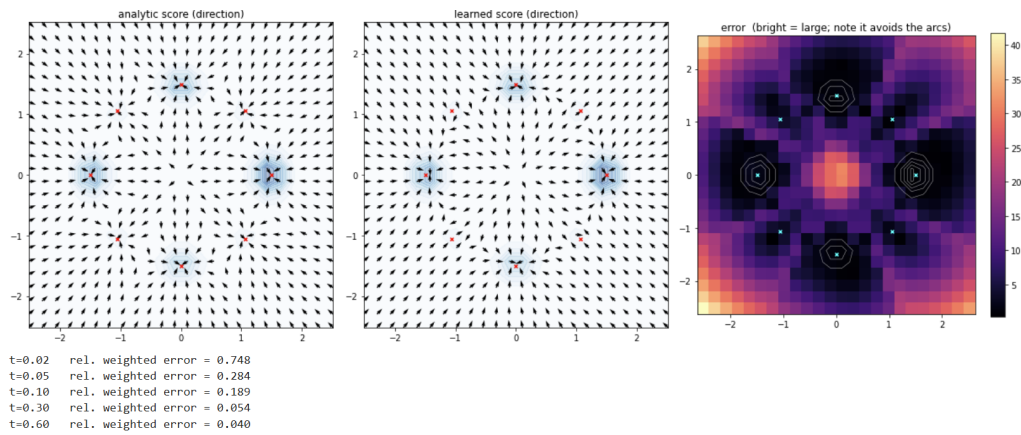


Figure 3.5: Score network approximation at $t = 0.05$

The figures show the approximation at time $t = 0.05$. One can then see that the approximation is good near the modes, but especially near the modes which have higher weights in the target distribution, as samples will naturally concentrate around those, improving the score estimation. The approximation is bad far from the modes, as the network does not have access to enough samples.

On another hand, as shown below the figures, the relative error decreases as t increases. This is because for small t , $\sigma_t \rightarrow 0$, the score $\nabla_x \log p_t$ then scales like $1/\sigma_t$ and becomes increasingly peaked, hence harder for any smooth network to represent. This is why we observe that the diffusion produces a distribution more spread than the target: the network underestimates the magnitude of the score near the manifold, leading to a more diffuse distribution, which we can confirm by looking at the precision metric.

Remark: Observe that we compare here the learned score with the analytical score, which is not the target of the network, as the latter is the empirical score. However, this can be justified for the two following reasons: firstly, for $t=0.05$, and if we have a large enough number of samples (which is the case for this example), the analytical score is not very different from the empirical score. Secondly, the score of the true distribution being the analytical score, we want to show that the network approximates it well to guarantee samples close to the true distribution, even though the minimizer of the loss used in training is actually the empirical score.

In the previous example, a meticulous reader could attribute the superior performance of the SDE based method to the fact that networks approximation are not the same for both methods (Langevin uses a mono-scale network, whereas diffusion uses multi-scale networks).

We show here an example where the SDE with exact analytical score outperforms the Langevin method with exact analytical score. Using analytic scores removes score-estimation and neural-network errors. The remaining comparison concerns the complete numerical sampling pipelines, including the annealing path used by the reverse-SDE, which the Langevin method does not have access to by nature.

In this example, the target distribution is the same as the one used for 3.2, and both methods are initialized at $N(0, I)$.

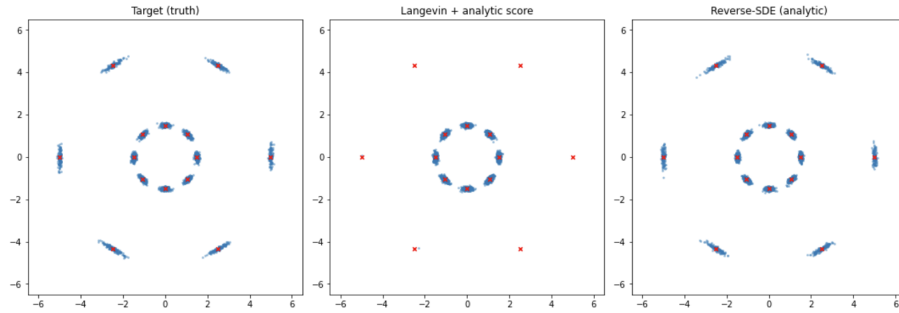


Figure 3.6: Langevin vs SDE: second case

The performance metrics in this case confirm the visual diagnostic: Diffusion still outperforms Langevin.

metric	Langevin	Reverse-SDE
weight-TV (down)	0.2721	0.0178
mean loglik (up)	-0.1732	-0.5460
MMD^2 (down)	0.0447	-0.0000
sliced-W2 (down)	1.4530	0.0410
precision (up)	0.9890	0.9890
recall (up)	0.8077	0.9910

Figure 3.7: Langevin vs SDE: performance metrics in the second case

3.2.3 Convergence of Diffusion models

One might naturally wonder how diffusion models overcome the two main bottlenecks of standard Langevin dynamics: the manifold hypothesis and slow mixing times on multi-modal distributions.

Diffusion models don't need Poincaré-inequality type assumptions

To deal with the fact that the performance of diffusion models do not suffer from multimodal distributions, we state the following result (Theorem 1 in [CDS24]).

We first define the forward process as:

$$d\vec{X}_t = -\vec{X}_t dt + \sqrt{2} dB_t, \vec{X}_0 \sim \mu^*$$

where μ^* is the data distribution.

We then need to define a discretization scheme, firstly introduced in [ZC23] (note however that the definition we give follows from the description of the method in section 2 of [CDS24]), which aims at increasing sampling speed:

Definition 3.6. Let $\tilde{p}_t := \vec{p}_t / \gamma^d$ with $\gamma^d = \mathcal{N}(0, I_d)$, and let $\tilde{s}_{\theta^*}(T - t_k, \cdot)$ approximate the scaled relative score $2\nabla \log \tilde{p}_{T-t_k}$. The backward process is

$$d\overleftarrow{X}_t = (-\overleftarrow{X}_t + 2\nabla \log \tilde{p}_{T-t}(\overleftarrow{X}_t)) dt + \sqrt{2} dB_t, \quad \overleftarrow{X}_0 \sim \gamma^d,$$

and, freezing the score on $[t_k, t_{k+1}]$ (step h) and solving the linear SDE exactly (which is possible as if we freeze the score, the backward process becomes simply an OU process, for which we have an analytical closed form), the EI update is

$$X_{t_{k+1}}^{\theta^*} = e^{-h} X_{t_k}^{\theta^*} + (1 - e^{-h}) \tilde{s}_{\theta^*}(T - t_k, X_{t_k}^{\theta^*}) + \sqrt{1 - e^{-2h}} Z_k, \quad Z_k \sim \mathcal{N}(0, I_d),$$

initialized at $X_0^{\theta^*} \sim \gamma^d$.

Then, we will need the two following assumptions.

Assumption H1. Let a sequence of step sizes $\{h_k\}_{k=1}^N$, $N \geq 1$, such that $\sum_{k=1}^N h_k = T$, and $(t_k)_{k \in \{0, \dots, N\}}$ defined by the recursion $t_{k+1} = t_k + h_{k+1}$. There exists $\varepsilon^2 > 0$ and $\theta^* \in \Theta$ such that

$$\frac{1}{T} \sum_{k=0}^{N-1} h_{k+1} \mathbb{E}_{p_{T-t_k}} \left[\left\| \tilde{s}_{\theta^*}(T - t_k, \vec{X}_{T-t_k}) - 2\nabla \log \tilde{p}_{T-t_k}(\vec{X}_{T-t_k}) \right\|^2 \right] \leq \varepsilon^2. \quad (3.11)$$

Finally, $\tilde{s}$ is the score network. We emphasize here that the expectation is taken with respect to the marginal $\tilde{p}_{T-t_k}$ for each $k \in \{0, \dots, N\}$. Hence, this is a bound over the approximation error of the true score, in contrast of score matching which is aiming at approximating the empirical score.

Assumption H2. μ^* is absolutely continuous with respect to the Gaussian measure γ_d and has finite relative Fisher information against γ_d , that is

$$I(\mu^* \mid \gamma_d) = \int_{\mathbb{R}^d} \left\| \nabla \log \frac{d\mu^*}{d\gamma_d}(x) \right\|^2 d\mu^*(x) < +\infty. \quad (3.12)$$

Theorem 3.7. Let $T \geq 1$, $h \leq 1$ and assume H1–H2. Consider the Exponential Integrator (EI) scheme $(X_t^{\theta^*})_{t \in [0, T]}$ with constant step size $h > 0$. Denoting for any $t \in [0, T]$ by $p_t^{\theta^*}$ the distribution of $X_t^{\theta^*}$, we have that

$$\text{KL}(\mu^* \mid p_T^{\theta^*}) \lesssim e^{-2T} \text{KL}(\mu^* \mid \gamma_d) + T\varepsilon^2 + hI(\mu^* \mid \gamma_d), \quad (3.13)$$

Moreover, the bound (3.13) also holds if we replace the term $e^{-2T} \text{KL}(\mu^* \mid \gamma_d)$ with $e^{-2T}(M_2^2 + d)$, where $M_2^2 = \int \|x\|^2 d\mu^*(x)$. In particular, choosing

$$T = \frac{1}{2} \log \left(\frac{M_2^2 + d}{\varepsilon^2} \right) \quad \text{and} \quad N = \frac{TI(\mu^* \mid \gamma_d)}{\varepsilon^2}$$

makes the approximation error $\tilde{\mathcal{O}}(\varepsilon^2)$, where the notation $\tilde{\mathcal{O}}$ indicates that logarithmic factors of d and ε have been dropped.

To have a better understanding of assumption H2, we state and prove the following lemma:

Lemma 3.8. *Let γ_d be the standard multivariate Gaussian distribution, and $\mu^\star$ be a probability measure on $\mathbb{R}^d$ admitting a density p with respect to the Lebesgue measure. Assume that:*

1. $\mu^\star$ has a finite second moment, i.e., $M_2^2 := \int_{\mathbb{R}^d} \|x\|^2 p(x) dx < \infty$.
2. p is weakly differentiable.
3. For all $i \in \{1, \dots, d\}$, $\lim_{|x_i| \rightarrow \infty} x_i p(x) = 0$ for almost every $x_{-i} \in \mathbb{R}^{d-1}$.
4. The relative Fisher information is finite: $I(\mu^\star | \gamma_d) < \infty$.

Then, the Fisher information $I(\mu^\star) := \int_{\mathbb{R}^d} \|\nabla \log p(x)\|^2 p(x) dx$ is finite, and:

$$I(\mu^\star | \gamma_d) = I(\mu^\star) + M_2^2 - 2d$$

Proof: Let p denote the density of $\mu^\star$. We first prove that $I(\mu^\star)$ is finite. By applying Minkowski's inequality in the space $L^2(\mu^\star)$ to the identity $\nabla \log p(x) = (\nabla \log p(x) + x) - x$, we obtain:

$$\sqrt{I(\mu^\star)} \leq \sqrt{\int_{\mathbb{R}^d} \|\nabla \log p(x) + x\|^2 p(x) dx} + \sqrt{\int_{\mathbb{R}^d} \|x\|^2 p(x) dx} = \sqrt{I(\mu^\star | \gamma_d)} + M_2 < \infty.$$

Now, recalling that $\nabla \log \gamma_d(x) = -x$ and using the identity $\nabla \log p(x)p(x) = \nabla p(x)$ (which holds almost everywhere by weak differentiability), we expand the relative Fisher information:

$$\begin{aligned} I(\mu^\star | \gamma_d) &= \int_{\mathbb{R}^d} \|\nabla \log p(x) - \nabla \log \gamma_d(x)\|^2 p(x) dx \\ &= \int_{\mathbb{R}^d} \|\nabla \log p(x) + x\|^2 p(x) dx \\ &= \int_{\mathbb{R}^d} \|\nabla \log p(x)\|^2 p(x) dx + \int_{\mathbb{R}^d} \|x\|^2 p(x) dx + 2 \int_{\mathbb{R}^d} \langle \nabla p(x), x \rangle dx \\ &= I(\mu^\star) + M_2^2 + 2 \int_{\mathbb{R}^d} \langle \nabla p(x), x \rangle dx. \end{aligned}$$

Note that the integrability of the cross-term is guaranteed by the Cauchy-Schwarz inequality applied in $\mathbb{R}^n$, and the Hölder inequality applied to the product $\|\nabla \log p(x)\| \|x\|$, since both $I(\mu^\star)$ and M_2^2 are finite. Indeed:

$$\begin{aligned} \int_{\mathbb{R}^d} |\langle \nabla \log p(x), x \rangle| p(x) dx &\leq \int_{\mathbb{R}^d} \|\nabla \log p(x)\| \|x\| p(x) dx \\ &\leq \left(\int_{\mathbb{R}^d} \|\nabla \log p(x)\|^2 p(x) dx \right)^{1/2} \left(\int_{\mathbb{R}^d} \|x\|^2 p(x) dx \right)^{1/2} \\ &= \sqrt{I(\mu^\star)} \cdot \sqrt{M_2^2} \end{aligned}$$

We can then apply integration by parts to the cross-term and get by Assumption 3:

$$2 \int_{\mathbb{R}^d} \langle \nabla p(x), x \rangle dx = 2 \sum_{i=1}^d \int_{\mathbb{R}^{d-1}} \left(\underbrace{[x_i p(x)]_{x_i \rightarrow -\infty}^{x_i \rightarrow +\infty}}_{=0} - \int_{\mathbb{R}} p(x) dx_i \right) dx_{-i} = -2 \sum_{i=1}^d \int_{\mathbb{R}^d} p(x) dx = -2d$$

Substituting this back gives $I(\mu^\star | \gamma_d) = I(\mu^\star) + M_2^2 - 2d$, which concludes the proof. $\square$

Hence, the assumption H2 basically amounts to having a score in L^2 and a data distribution which admits a second moment. This fully allows for multimodal distributions, in contrast to the necessary assumptions for the convergence of Langevin-based procedures.

By looking at the bound in (3.13), we observe that there is a tradeoff on the choice of T . If we choose T too small, the term $e^{-2T} \text{KL}(\mu^\star | \gamma_d)$ will be too large, and if we choose T too large, then the term $T\varepsilon^2$ will be too large.

The first term $e^{-2T} \text{KL}(\mu^\star | \gamma_d)$ might be intriguing at first sight. By looking at the proof, one can see it is actually $KL(\vec{p}_T^\star | \gamma_d)$ which appears in the proof. Proposition 5.5.1 in [BGL14] then proves that $KL(\vec{p}_T^\star | \gamma_d) \leq e^{-2T} \text{KL}(\mu^\star | \gamma_d)$. Hence, this term can be seen as the error due to the prior mismatch.

Recall that thanks to Assumption H1, we know that the weighted sum of score approximation error across the time grid is bounded by $T\varepsilon^2$. Hence the second term $T\varepsilon^2$ represents the accumulation of the score approximation error until time T .

The third term $h I(\mu^\star | \gamma^d)$ is the discretization error induced by the step size h . Let us explain why the relative Fisher information appears. The proof bounds the KL divergence by the integrated squared difference between the drift of the EI scheme and the true backward drift. After separating the score-approximation error, the contribution of time discretization is: $E := \sum_k \int_{kh}^{(k+1)h} \mathbb{E} \|Y_t - Y_{kh}\|^2 dt$, with $Y_t := 2\nabla \log \tilde{p}_{T-t}(\tilde{X}_t)$.

The stated bound then follows from two facts. First, $g(t) := \mathbb{E} \|Y_t\|^2$ has a particular meaning: since $\tilde{X}_t \sim \vec{p}_{T-t}$, one has $g(t) = 4 I(\mu^\star P_{T-t} | \gamma^d)$, so $g(T) = 4 I(\mu^\star | \gamma^d)$, and the authors prove then that g is increasing. Second, the SDE satisfied by Y (Proposition 2 in the paper) gives: $\mathbb{E} \|Y_t - Y_{kh}\|^2 \lesssim g((k+1)h) - g(kh)$. These increments actually telescope: $E \lesssim h(g(T) - g(0)) = 4h(I(\mu^\star | \gamma^d) - I(\mu^\star P_T | \gamma^d)) \leq 4h I(\mu^\star | \gamma^d)$.

The Fisher information thus enters not as a per-step regularity constant but as a "total variation budget", which controls the accumulated movement of the score between time 0 and time T . Because the per-interval bounds telescope, the discretization error is at most h times this fixed budget, regardless of where along the trajectory the score varies most. This makes the assumptions weaker, and more realistic, than uniform-in-time Lipschitz conditions: here the variation may concentrate near $t = 0$, as it does in practice (remember that the analytical score is very steep when t approaches 0), and the hypothesis would still remain valid.

Although this result shows that diffusion models allow for multimodal distributions, it does not allow for distributions satisfying the manifold hypothesis, as it requires distributions absolutely continuous with respect to the Gaussian distribution. The authors then propose this corollary:

Theorem 3.9 (Corollary 1). *Let $c \in (0, 1/2]$, $\delta \in (0, 1/2]$, $T \geq 1 + 2c$ and assume H1. Let $(P_t)_{t \geq 0}$ be the transition semigroup associated to the forward process. Assume only that $\mu^\star$ has finite second order moment. Set $L = d^{-1} I(\mu^\star P_\delta | \gamma_d)$ and assume that $L \geq 2$. Choose the constant and exponentially decreasing sequence of step-size, i.e., satisfying for $k < N$,*

$$h_{k+1} = c \min \{ \max \{ T - t_k, 1/L \}, 1 \}.$$

Denoting for any $t \in [0, T]$ by $p_t^{\theta^\star}$ the distribution of $X_t^{\theta^\star}$, we have that

$$\text{KL}(\mu^\star P_\delta | p_{T-\delta}^{\theta^\star}) \lesssim e^{-T} \{M_2^2 + d\} + C(T, \varepsilon) + c \left[(d + M_2^2) \left(\log(d + M_2^2) + \log(\delta^{-1}) + 1 \right) \right], \quad (3.14)$$

where $C(T, \varepsilon) = T\varepsilon^2$.

This result bounds the KL divergence between the noise-perturbed data distribution at time δ and the distribution of the sampled backward process at time $T - \delta$. What allows this to hold is that the noise-perturbed data distribution admits a density with respect to the Gaussian distribution for any noise level strictly above 0, with finite relative Fisher information, even though the data distribution might not have a density with respect to the Gaussian distribution. Observe also that the step-size sequence decreases as t_k approaches T : the steps shrink exactly where the score process accumulates most of its variation: near the data, where g increases most steeply, that is, where the discretization error per unit of time is largest.

While informative, this result deliberately stops short of the terminal time: the guarantee is stated against the smoothed target $\mu^* P_\delta$, not the actual target μ^* . To be able to decrypt the performance of diffusion models under the manifold hypothesis, we are going to proceed at the level of supports rather than metrics.

Diffusion models perform well under the manifold hypothesis

We first recall the manifold hypothesis. In words, the manifold hypothesis assumes that the data is not supported in the whole space $\mathbb{R}^d$, but rather on a lower-dimensional manifold. As a consequence, the assumption that $p_{\text{data}}(x) > 0 \quad \forall x \in \mathbb{R}^d$ (which was used in many of the results stated until now) is violated. Hence, p_{data} is not absolutely continuous with respect to the Lebesgue measure, and in particular, does not admit a density.

In practice, this assumption is realistic: for instance, in image generation, the ambient dimension d' is that of the pixel encoding, not of the content. Intrinsic-dimension estimators consistently return values orders of magnitude below d : [Pop+21] estimate the intrinsic dimension of ImageNet at between 26 and 43, against $224 \times 224 \times 3 = 150528$ pixels, and report similarly low values for MNIST and CIFAR-10.

The paper [Pid22] presents results about the ability of generalization of diffusion models, and the caveat of memorization. In its setting, generalization refers to the fact that diffusion models can produce samples that are not included in the training set, and similar in some sense as samples from the true (not empirical) data distribution. Mathematically, the author summarizes this by saying that the distribution of the generative model, p_{sample} , has the same support as the distribution of the data p_{data} .

Memorization then refers to the fact that p_{sample} shares its support with the empirical data distribution, denoted by $\hat{p}_{\text{data}}$. As the empirical data distribution is the uniform distribution over the training points, this would mean that the generative model is only able to generate samples from the training data.

Before giving the results that deal with these issues, we first summarize the machinery behind generative diffusion models. We denote by p_t the marginal at time t of the forward process when initialized at the data distribution. As mentioned before, in practice, the forward SDE is initialized at the empirical distribution and run until time T to train the score network. This network then actually approximates the empirical score, that is $\nabla \log \hat{p}_t$, where $\hat{p}_t$ is the marginal at time t of the forward process when initialized at the empirical data distribution. To generate new samples, we then run the backward process initialized at some prior distribution, which we assume to be close to p_T which is the first marginal of the backward process. Now, assume that the prior distribution matches exactly this first marginal, and that the score network approximates perfectly the empirical score at each time $t \in [\varepsilon, T]$, and that we stop the backward process as time $T - \varepsilon$. Then, p_{sample} would be exactly $\hat{p}_\varepsilon$, that is, a blurred version of the empirical data distribution. As this is valid for all ε , this means that the model essentially memorizes the training data. However, in practice, we notice that diffusion models can effectively generate samples that were not in the training set. Hence, this generalization

ability must come either from the prior mismatch, or the empirical score error made by the score network.

Remark: Although the original paper does not discuss this, one can wonder why the generalization ability of diffusion models can't come from discretization. In fact, discretization does introduce a gap: the discrete marginals $\hat{p}_{t_k}^\gamma$ differ from the continuous ones $\hat{p}_{t_k}$, so feeding the exact continuous score $\nabla \log \hat{p}_t$ into a discrete scheme does move p_{sample} away from $\hat{p}_0$. However, this gap is not generalization. Indeed, it vanishes as $\gamma \rightarrow 0$: if it were the source of generalization, finer solvers would push the model toward memorization, which is not observed in practice. Hence, we conjecture that generalization must come from the two other sources of error: the prior mismatch, and the approximation error the score network makes while approximating $\nabla \log \hat{p}_t$.

The authors then give two results, each dealing with one of these two aspects. In their work, the manifold hypothesis takes the following form:

Assumption 1: Denote by $\mathcal{M}$ the support of p_{data} . We assume that $\mathcal{M}$ is included in a ball of radius M .

Remark: This is actually an absence of strong assumptions on the data distribution, in the sense that this assumption fully allows for distributions supported on manifolds embedded in a higher-dimensional space.

In the following, the term forward SDE is used to talk about:

$$dX_t = \beta(X_t) dt + \sigma(t) dW_t, \quad X_0 \sim \pi_0, \quad (3.15)$$

where σ is positive, and β is linear.

The backward SDE then denotes the SDE satisfied by the corresponding reverse process:

$$dY_t = -\beta(Y_t) dt + \sigma(T-t)\sigma(T-t)^\top \nabla \log p_{T-t}(Y_t) dt + \sigma(T-t) dB_t, \quad Y_0 \sim q_0. \quad (3.16)$$

Note that the analytical score appears in the equation, in contrast to what is done in practice where we have a score network approximating the analytical score.

Lastly, p_{prior} denotes the distribution at which the backward process is initialized at the beginning of the generation phase, which is assumed to be different than p_T . We then have the following result (Theorem 1 in [Pid22]):

Theorem 3.10. *Denote by π_t the marginals of the forward SDE started in π_0 . Assume that Assumption 1 holds and that p_{prior} is absolutely continuous with respect to π_T . Then the following hold.*

- *Let Y_t be a solution to the backward SDE on $[0, T)$. The limit $Y_T := \lim_{t \rightarrow T} Y_t$ exists almost surely. We refer to its distribution as p_{sample} . The distribution p_{sample} is absolutely continuous with respect to π_0 . If π_T and p_{prior} are equivalent, then so are p_{sample} and π_0 .*
- *Furthermore,*

$$KL(p_{\text{sample}} \mid \pi_0) \leq KL(p_{\text{prior}} \mid \pi_T) \quad \text{and} \quad KL(\pi_0 \mid p_{\text{sample}}) \leq KL(\pi_T \mid p_{\text{prior}}). \quad (3.17)$$

Remark: The last point also holds in total variation.

Now, let us interpret this result. First of all, notice that as mentioned before, the forward and backward SDE considered here contain the analytical score and not the approximated one. This means that we consider here that there is no error due to the score network approximation. The first point of the theorem then tells us two different things. Remember that the forward and backward SDE admit a density with respect to Lebesgue measure at every time $t > 0$, regardless of the initial measure. Moreover, as the prior distribution is typically gaussian, the condition that π_T and p_{prior} are equivalent is satisfied in practice. If we take $\pi_0 = p_{\text{data}}$, then by the theorem, we get that p_{data} is equivalent to p_{sample} and the model generalizes. This means that if we do not have any errors due to score approximation and prior mismatch, diffusion models are in theory able to generalize.

Now, if we take $\pi_0 = \hat{p}_{\text{data}}$, which is what is done in practice, then we get that $\hat{p}_{\text{data}}$ is equivalent to p_{sample} and the model memorizes the training set. This means that the prior mismatch alone is not what allows generative diffusion models to generalize.

Furthermore, the theorem also indicates that the KL -divergence between the data distribution and the distribution we are sampling from is controlled by the KL -divergence induced by the prior mismatch, which is actually the same result as the one obtained in [CDS24] that we presented in the previous subsection, considering that in our setting, we do not have discretizations error and score-approximations error.

To study the effect of the score approximation error, the authors consider the following setting. They now define:

$$d\tilde{Y}_t = -\beta(\tilde{Y}_t) dt + \sigma(T-t)\sigma(T-t)^\top s(\tilde{Y}_t, T-t) dt + \sigma(T-t) dB_t$$

and the corresponding forward SDE, where s is the score approximation.

We then need the following assumptions:

$$\exists M > 0 \text{ such that } \|s(x, t) - \nabla_x \log \pi_t(x)\| \leq M \quad \forall (x, t) \in \mathbb{R}^d \times [0, T]$$

Furthermore, for each $S \in (0, T)$ and all x, y for which $\|x\|, \|y\| \leq N$ there is a constant $C_{S,N}$ such that:

$$s(x, t) \text{ is locally Lipschitz: } \|s(x, t) - s(y, t)\| \leq C_{S,N} \|x - y\| \quad \forall t \in [S, T]$$

We also assume that the reverse SDE $\tilde{Y}_t$ has a solution on $[0, T)$. For $t < T$, we define the Girsanov weights

$$Z_t = \exp \left(\int_0^t \sigma^\top(s(\tilde{Y}_s, s) - \nabla \log p_s(\tilde{Y}_s)) \cdot dB_s - \frac{1}{2} \int_0^t \|\sigma^\top(s(\tilde{Y}_s, s) - \nabla \log p_s(\tilde{Y}_s))\|^2 ds \right) \quad (3.18)$$

and assume that the Z_t are a uniformly integrable martingale.

Then we have the following result (Theorem 2 in [Pid22]):

Theorem 3.11. *Assume that the above assumptions and assumption 1 hold. Then,*

$\tilde{Y}_T := \lim_{t \rightarrow T} \tilde{Y}_t$ exists almost surely. Moreover, its distribution is equivalent to the distribution of Y_T .

Remark: The hypothesis of Theorem 3.11 do not concern the same objects as the assumptions of Theorem 3.10. That means that we can have an approximation error on the empirical score, and still be able to apply Theorem 3.10, as the latter is about another SDE than the one considered in Theorem 3.11.

We know that under the assumptions of Theorem 3.10, the distribution of Y_T is equivalent to p_{data} if the forward process is initialized at the data distribution. Hence, we deduce that under the assumptions of Theorem 3.11, which allow for prior mismatch and an error on the score approximation (and which are compatible with the assumptions of Theorem 3.10 by the latest remark), if the forward process is initialized at the true distribution, $\tilde{Y}_T$ is equivalent to p_{data} and the model generalizes. This is a good check that in theory, if we could start the prior process at the data distribution, the model would be able to generalize even with prior mismatch and score approximation error.

Now, we also know that under the assumptions of Theorem 3.10, the distribution of Y_T is equivalent to $\hat{p}_{\text{data}}$ when the forward process is initialized at the empirical data distribution. Hence, if the assumptions of Theorem 3.11 are also satisfied, the distribution of $\tilde{Y}_T$ is equivalent to $\hat{p}_{\text{data}}$, and the model memorizes the training set in the sense considered in the paper.

At this point, it is important to identify precisely which assumptions lead to this conclusion. Theorem 3.11 does not take boundedness of the score error as its fundamental change-of-measure assumption. Instead, it assumes that the approximate reverse SDE admits a solution on $[0, T]$ and that the associated Girsanov weights form a uniformly integrable martingale. Uniform boundedness of the score error is only a convenient sufficient condition for the latter property.

$$\theta_t = \sigma(T-t)^\top \left(\nabla \log p_{T-t}(\tilde{Y}_t) - s(\tilde{Y}_t, T-t) \right).$$

Novikov's condition reads

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T |\theta_t|^2 dt \right) \right] < \infty.$$

Whenever this condition holds, the corresponding Girsanov weights form a uniformly integrable martingale. In particular, provided that

$$\int_0^T \|\sigma(t)\|_{\text{op}}^2 dt < \infty,$$

a uniform bound

$$\sup_{(x,t) \in \mathbb{R}^d \times (0,T]} |s(x,t) - \nabla \log \pi_t(x)| < \infty$$

implies Novikov's condition and therefore verifies the Girsanov part of Assumption 2.

The way the paper handles this point is worth making explicit. The authors consider explicit drifts obtained by perturbing the empirical score by a constant error vector. Such a drift inherits the required local regularity from the empirical score, while its approximation error is uniformly bounded. Novikov's condition therefore holds and, under the remaining assumptions of Theorem 3.11, the conclusion is memorization. This behavior is also illustrated numerically in Figures 2b and 3b.

For the pretrained score network, by contrast, the numerical quantity appearing in Novikov's condition grows rapidly near the terminal time, while the generated samples do not collapse onto the training set. This numerical observation shows that Novikov's sufficient criterion can no longer be verified in that experiment. By itself, however, it does not prove that the Girsanov weights fail to be uniformly integrable.

We can nevertheless clarify the role of the regularity assumption. Finite networks built from affine layers and Lipschitz activations, such as ReLU or Swish, are locally Lipschitz by construction. The exact empirical score $\nabla \log \hat{p}_t$ is also smooth, and therefore locally Lipschitz, on compact space-time subsets with $t \in [S, T]$ and $S > 0$. Indeed, Gaussian smoothing ensures C^∞ regularity at every strictly positive time. For architectures involving additional operations such as normalization or attention, the corresponding local regularity should be checked separately. Moreover, local Lipschitz continuity alone does not establish global existence of the approximate reverse SDE, which remains a separate hypothesis.

Consequently, within the idealized continuous-time framework of the theorem, and provided that the remaining forward-process, prior-equivalence, existence and regularity assumptions hold, a model whose terminal distribution is not equivalent to the empirical data distribution must violate the uniformly integrable Girsanov-martingale assumption. Since a uniformly bounded score error would imply this assumption through Novikov’s condition, the score error cannot be uniformly bounded.

This is only a one-directional statement. An unbounded score error need not make Novikov’s condition fail, since it may become large only in regions that are rarely visited by the diffusion. More generally, unboundedness does not imply that the Girsanov weights fail to be uniformly integrable, and Theorem 3.11 provides no converse implication. Therefore, under the other assumptions of the theorem, the absence of a uniform bound on the empirical-score approximation error is a necessary, but not sufficient, condition for generalization.

Finally, this conclusion concerns the idealized continuous-time process run up to its singular endpoint and the support-based definition of generalization adopted in the paper. Practical samplers are discretized and are usually stopped at a strictly positive noise level. The argument should therefore be understood conditionally: it applies as long as the practical sampler is accurately represented by the limiting continuous-time model considered above.

Remark: The fact that this paper decides arbitrarily that "generalization" means that the support of p_{data} is the same as the support of p_{sample} can of course be criticized. Indeed, two distributions can share the same support and yet have very distinct shapes, consequently producing very different samples in practice. A more recent work by De Bortoli ([Bor23]) bounds the 1-Wasserstein distance between p_{data} and p_{sample} under the manifold hypothesis. Furthermore, the paper [Far+25] investigates a specific form of unbounded score-approximation error: smoothing the empirical score by a probability kernel. The structure of their contribution is threefold. First, as a modelling assumption, they posit that the inductive bias of a trained score network behaves as if the empirical score were smoothed by a probability kernel; the analysis does not train a network but studies this analytic proxy directly. Second, as their main theoretical result, they prove that such a smoothing is geometry-adaptive under mild conditions — approximating smoothing along the data manifold — with a quantitative bound between the smoothed score distribution and its manifold-adapted counterpart. Third, they leave as an open question whether real trained networks actually produce this kind of bias: if manifold-adaptive smoothing occurs in practice, it must emerge implicitly from architectural choices (convolutions, attention) and training dynamics, and characterising that link is deferred to future work.

3.3 Probability flow ODE

3.3.1 Another way of sampling

In the previous sections, we considered diffusion models, where we run a backward SDE to generate new samples. There is actually another way of sampling from the data distribution, starting from the SDE:

$$dX_t = f(X_t, t)dt + g(t)dW_t \text{ with } X_0 \sim p_{\text{data}} \quad (3.19)$$

Indeed, recall that in Section 2.4.3, we proved that there were infinitely many processes which had the same marginals as the reverse time process. One of these processes was the one with zero diffusion part, namely:

$$d\tilde{X}_\tau = \left[-f(\tilde{X}_\tau, T - \tau) + \frac{1}{2}g(T - \tau)^2 \nabla \log \tilde{p}_\tau(\tilde{X}_\tau) \right] d\tau. \quad (3.20)$$

where $\tau = T - t$. This equation is known as the probability flow ODE. If we index it by t , it becomes (as $d\tau = -dt$):

$$d\tilde{X}_{T-t} = \left[f(\tilde{X}_{T-t}, t) - \frac{1}{2}g(t)^2 \nabla \log \tilde{p}_{T-t}(\tilde{X}_{T-t}) \right] dt. \quad (3.21)$$

The process $\tilde{X}$ shares its marginals with the reverse-time process of 3.19. To generate samples from the data distribution, we integrate this Probability Flow ODE from $\tau = 0$ (corresponding to the noise prior at $t = T$) to $\tau = T$ (corresponding to the data distribution at $t = 0$).

3.3.2 Theoretical advantages of Probability Flow ODE over SDE-based Models

The main theoretical advantage of the Probability Flow ODE over SDE-based diffusion models is the ability to yield an exact likelihood formulation.

Indeed, if we consider the following deterministic forward process governed by an ODE:

$$dx(t) = f(x(t), t)dt \quad (3.22)$$

where f corresponds to the drift of the probability flow ODE associated with DDPM or SMLD, we have the following result (Theorem 1 in [Che+19]):

Theorem 3.12 (Instantaneous Change of Variables). *Let $x(t)$ be a finite continuous random variable with probability density $p(x(t))$ dependent on time. Let*

$$\frac{dx(t)}{dt} = f(x(t), t) \quad (3.23)$$

be a differential equation describing a continuous-in-time transformation of $x(t)$. Assuming that f is uniformly Lipschitz continuous in x and continuous in t , then the total derivative of the log-probability along the trajectory follows:

$$\frac{d}{dt} \log p_t(x(t)) = -\text{tr}(\nabla_x f(x(t), t)) \quad (3.24)$$

Integrating both sides of (3.24) between time $t = 0$ and $t = T$ yields:

$$\log p_0(x(0)) = \log p_T(x(T)) + \int_0^T \text{tr}(\nabla_x f(x(t), t)) dt \quad (3.25)$$

If x_0 is a sample from the target distribution p_{data} , the left-hand side represents the theoretical likelihood of observing x_0 under the distribution p_0 generated by the model.

Thanks to the deterministic nature of the ODE, this formula provides an exact theoretical expression for the likelihood. In practice, however, its computation is only exact up to numerical

errors. Starting from $x(0) = x_0$, we simulate the trajectory by running a numerical ODE solver. By evaluating the state $x(t)$ at each integration step, we can accumulate the trace of the Jacobian to compute the integral. Because exact trace computation scales poorly with dimensionality, efficient randomized estimators (such as the Hutchinson trace estimator) are typically used, as mentioned in Appendix D.2 of [Son+21]. Finally, since the terminal distribution p_T is chosen to be a simple known prior (e.g., a standard Gaussian), the term $\log p_T(x(T))$ can be evaluated analytically.

Remark on exactness and score interpretation: In theory, one could leverage this exact likelihood formulation to train the network directly via Maximum Likelihood Estimation (MLE), instead of relying on Denoising Score Matching (DSM). However, solving the ODE and approximating the trace of the Jacobian matrix at every integration step for each data point is computationally prohibitive during training. Furthermore, an arbitrary neural network yields a generic vector field, which is not guaranteed to be the gradient of a log-density. Therefore, while the ODE rigorously defines a probability flow, one should be careful not to strictly identify the learned neural vector field with an exact mathematical score.

Consequently, likelihood computation via the Probability Flow ODE is primarily used post-training. It serves as a rigorous (though numerically approximated) metric to evaluate the model and to compare it fairly against other generative frameworks, such as Continuous Normalizing Flows, the comparison being meaningful because both rely on the same instantaneous change-of-variables formula (see [Che+19]).

A further advantage of ODE-based methods comes from an inference perspective. Indeed, deterministic ODE solvers typically enable faster sample generation compared to SDE-based sampling (see Figure 3 in [Son+21]), if one is willing to accept a slightly larger numerical truncation error tolerance.

3.3.3 Practical caveats: Probability Flow ODE vs. Stochastic Sampling

While the Probability Flow ODE and the reverse-time SDE share the same theoretical marginals, their empirical performance often diverges during practical sampling. One of the primary reasons, highlighted in [Xu+23], is that deterministic ODE solvers accumulate score approximation errors. Consider an Euler step backward in time from t to $t - \Delta t$:

$$x_{t-\Delta t} = x_t - \left[f(x_t, t) - \frac{1}{2}g(t)^2 s_\theta(x_t, t) \right] \Delta t. \quad (3.26)$$

Because the trajectory is deterministic, an inaccurate prediction $s_\theta(x_t, t)$ pushes the sample off the data manifold, and this deviation becomes a flawed initial condition for the next step. Since the network outputs a smooth vector field, such local errors rarely cancel; they compound, driving the sample into out-of-distribution regions where the predictions degrade further. The deviation matters because the true score acts as a manifold-attracting force that diverges as $t \rightarrow 0$: approximating it imperfectly weakens or misdirects that attraction, and with no stochastic term to re-equilibrate, the errors accumulate.

It is important to specify which error is meant. The relevant quantity here is the gap to the *true* score, $\|s_\theta - \nabla_x \log p_t\|$, since the goal is to reproduce p_{data} , not its empirical counterpart. This is not the error monitored during training: the training loss measures the gap to the *empirical* score $\nabla_x \log \hat{p}_t$ (actually, as discussed earlier, it is the transition kernel that appears in the loss, but the minimizer of the loss remains the empirical score). The two must in fact behave differently: as discussed in the Subsection 3.2.3, the gap to the empirical score must remain unbounded as $t \rightarrow 0$ for the model to generate anything beyond the training set. The accumulation

mechanism above applies to either error, but only the gap to the true score explains generative performance; a small training loss at small t would signal memorization rather than quality.

Conversely, a discrete step of the SDE (e.g., using the Euler-Maruyama method) takes the following form:

$$x_{t-\Delta t} = x_t - \underbrace{\left[f(x_t, t) - \frac{1}{2}g(t)^2 s_\theta(x_t, t) \right] \Delta t}_{\text{probability-flow ODE step}} + \underbrace{\frac{1}{2}g(t)^2 s_\theta(x_t, t) \Delta t + g(t)\sqrt{\Delta t} z}_{\text{Langevin step at level } t, \epsilon=g(t)^2 \frac{\Delta t}{2}} \quad (3.27)$$

where $z \sim \mathcal{N}(0, I)$ is standard Gaussian noise. As noted in [Kar+22], this stochastic formulation introduces a built-in correction mechanism based on Langevin dynamics: the extra score term and the injected noise together form one step of Langevin dynamics at noise level t . Its stationary distribution, however, is not the true marginal p_t but the marginal implicitly defined by the network. At each level, the dynamics therefore actively re-equilibrate the sample distribution towards the law of the score-approximated reverse process at time t , contracting the deviations accumulated by the deterministic part. Hence, on the one hand, this correction removes accumulated integration error; but on the other hand, pulls toward the network’s own, possibly biased, version of the marginal.

Nevertheless, SDE-based methods are not flawless. First, one must be careful when comparing numerical discretization errors. It is tempting to assume that the Brownian term inherently forces larger errors due to the $\mathcal{O}(\sqrt{\Delta t})$ scale of its increments. However, this scale dictates the *strong* error (which measures pathwise accuracy). Under standard assumptions, the Euler-Maruyama method has a strong order of $1/2$ but a *weak* order of 1 (which measures distributional accuracy), as highlighted in [KP92]. Since generative modeling evaluates the quality of the final distribution rather than individual trajectories, the weak error is often the relevant metric. Therefore, the Brownian term does not strictly require smaller step sizes across the board; comparing ODE and SDE costs is meaningful only after specifying the exact solvers and the chosen accuracy criterion (strong vs. weak).

Second, the limit of the corrections highlighted before leads to the fact that the excess stochasticity in predictor-corrector approaches backfires: as Karras et al. observe empirically in [Kar+22], each corrector-style pass re-processes the sample at the same noise level, hence re-applies the network’s systematic bias, which accumulates with the number of passes. When the score approximation is accurate enough that its remaining error is dominated by a systematic bias rather than by unsystematic mistakes, excess stochasticity degrades results below the pure ODE: there is little accumulated integration error left to correct, while each additional stochastic pass re-applies the network’s bias.

This exposes a more nuanced core trade-off: ODEs more readily admit higher-order solvers (like Heun’s method, yielding better truncation errors for the same computational cost) but have no self-correction, whereas SDEs possess a self-correcting Langevin mechanism but are notoriously harder to upgrade to high-order numerical schemes. Karras et al. exploit exactly this tension, combining a high-order ODE solver for baseline accuracy with a calibrated dose of stochasticity for correction.

To formalize the intuitive trade-offs discussed above, the authors in [Xu+23] provide a rigorous theoretical framework. They define: $Q(r) = \Pr(a \geq r)$ for $a \sim \mathcal{N}(0, 1)$, and denote the value of x_r when initialized at $x_s = x$ as $\text{ODE}_\theta(x, s \rightarrow r)$. They then prove the following result:

Theorem 3.13. *Let $t_{\max}$ be the initial noise level. Let the initial random variables be $\bar{x}_{t_{\max}} =$*

$\bar{y}_{t_{\max}}$, and define:

$$\begin{aligned}\bar{x}_{t_{\min}} &= \text{ODE}_{\theta}(\bar{x}_{t_{\max}}, t_{\max} \rightarrow t_{\min}), \\ \bar{y}_{t_{\min}} &= \text{SDE}_{\theta}(\bar{y}_{t_{\max}}, t_{\max} \rightarrow t_{\min}).\end{aligned}$$

Let p_t denote the true population distribution at noise level t . Let $p_t^{\text{ODE}_{\theta}}$ and $p_t^{\text{SDE}_{\theta}}$ denote the distributions for x_t and y_t respectively.

Assume that for all x, y, s, t , the score function $s_{\theta}(x, t)$ satisfies:

$$\begin{aligned}\|ts_{\theta}(x, t) - ts_{\theta}(x, s)\| &\leq L_0|s - t|, \\ \|ts_{\theta}(x, t)\| &\leq L_1, \\ \|ts_{\theta}(x, t) - ts_{\theta}(y, t)\| &\leq L_2\|x - y\|,\end{aligned}$$

and the approximation error satisfies $\|ts_{\theta}(x, t) - t\nabla \log p_t(x)\| \leq \epsilon_{\text{approx}}$.

Assume in addition that for all $t \in [t_{\min}, t_{\max}]$, $\|x_t\| < B/2$ for any x_t in the support of p_t , $p_t^{\text{ODE}_{\theta}}$ or $p_t^{\text{SDE}_{\theta}}$. Then:

$$W_1(p_{t_{\min}}^{\text{ODE}_{\theta}}, p_{t_{\min}}) \leq B \cdot \text{TV}(p_{t_{\max}}^{\text{ODE}_{\theta}}, p_{t_{\max}}) + e^{L_2(t_{\max} - t_{\min})} \cdot (\delta(L_2L_1 + L_0) + \epsilon_{\text{approx}})(t_{\max} - t_{\min}) \quad (3.28)$$

$$\begin{aligned}W_1(p_{t_{\min}}^{\text{SDE}_{\theta}}, p_{t_{\min}}) &\leq B \cdot \left(1 - \lambda e^{-BL_1/t_{\min} - L_1^2 t_{\max}^2/t_{\min}^2}\right) \text{TV}(p_{t_{\max}}^{\text{SDE}_{\theta}}, p_{t_{\max}}) \\ &\quad + e^{2L_2(t_{\max} - t_{\min})} \left(\epsilon_{\text{approx}} + \delta L_0 + L_2 \left(\delta L_1 + \sqrt{2\delta dt_{\max}}\right)\right) (t_{\max} - t_{\min})\end{aligned} \quad (3.29)$$

where $\lambda := 2Q \left(\frac{B}{2\sqrt{t_{\max}^2 - t_{\min}^2}} \right)$.

δ represents the integration step size. It is typically close to 0, hence $\sqrt{\delta} \gg \delta$. Thanks to this result, we can see that if δ is large (for instance 0.1), the discretization error of the SDE, which scales as $\mathcal{O}(\sqrt{\delta})$, will be larger than that of the probability flow ODE, which scales as $\mathcal{O}(\delta)$. However, if δ is small enough, then the discretization error for both methods becomes negligible compared to the error term $\text{TV}(p_{t_{\max}}^{\text{ODE}_{\theta}}, p_{t_{\max}})$ or $\text{TV}(p_{t_{\max}}^{\text{SDE}_{\theta}}, p_{t_{\max}})$, which include both the prior mismatch and the accumulated score-approximation errors from time T to $t_{\max}$.

In this case, if $t_{\min}$ is sufficiently bounded away from 0, then the term $1 - \lambda e^{-BL_1/t_{\min} - L_1^2 t_{\max}^2/t_{\min}^2}$ is strictly less than 1. This multiplicative factor suggests a contraction mechanism where the SDE attenuates the accumulated TV error.

However, it is important to emphasize that comparing two upper bounds does not rigorously prove that SDE sampling is strictly more accurate than ODE sampling. It merely provides theoretical intuition supporting the empirical observation that SDEs are often more robust.

Remark: This theorem provides an interesting but idealized framework for analyzing the performance of ODE vs. SDE based methods, as its assumptions are fundamentally violated in standard practical settings. First, the bounded-support hypothesis ($\|x_t\| < B/2$) is not satisfied by ordinary Gaussian-smoothed marginals p_t , which have full support over $\mathbb{R}^d$. Second, the theorem only provides upper bounds on the Wasserstein distances, which can be loose as they involve exponential terms, and consequently fail to describe the exact realistic performance of these methods. Finally, the theorem relies on spatial global Lipschitz assumptions for the scaled score function $ts_{\theta}(x, t)$. While scaling by t is a theoretical artifice designed to mitigate the temporal explosion of the score as $t \rightarrow t_{\min}$, the assumption of spatial global Lipschitzness remains highly unrealistic. In practical settings where the target distribution is multimodal and

supported on a low-dimensional manifold, the score function makes sharp transitions between modes. Consequently, the spatial Lipschitz constant L_2 would be astronomical or undefined, further violating the assumptions of the theorem.

At the time they were introduced, diffusion models achieved State Of The Art performance over various benchmarks, especially for image generation (see Appendix H in [Son+21]). However, the main problem is that once the models have been trained; it is generating samples slower than pre-existing methods, such as GANs methods. Indeed, time T has to be taken large enough to be close to the prior distribution. Other models have since been developed to try to circumvent this issue. For instance, consistency models, introduced in [Son+23], can generate samples in 1 or 2 discretization steps. Latent diffusion models (from [Rom+22]) accelerate the sampling process by working in a compressed space. Another approach recently developed consists of solving the Schrödinger Bridge Problem to get a generative model.

3.4 Generative modeling with Schrödinger Bridge

3.4.1 Schrödinger Bridge as a generative model

Before stating the problem formally, it is worth explaining why solving a Schrödinger bridge yields a generative algorithm. A generative model is, in essence, a stochastic process that transports an easily sampled distribution p_{prior} onto the data distribution p_{data} . Score-based models obtain such a process indirectly: one prescribes a noising dynamics carrying p_{data} towards p_{prior} , and then reverses it in time. The Schrödinger bridge takes a different route and asks for the transport directly: among all path measures whose initial marginal is p_{data} and whose terminal marginal is p_{prior} , select the one closest to a given reference dynamics. Both endpoints are then matched exactly, by construction, and reading the solution backwards in time yields a process which starts exactly at the prior and ends exactly at the data distribution: that is, a generative model. However, note that "reading the solution backwards" implicitly assumes that the solution of the SBP is the law of a diffusion process whenever the reference is a diffusion process. A deep result in [Dai91] tells us that it is true. In addition, it also tells us that the diffusion coefficient of the solution of the SBP has to be the same that the one of the reference. Hence, finding the SBP solution amounts to change the drift of the reference, while keeping its diffusion part.

Recall the discretized version of the Dynamic Schrödinger Bridge problem introduced in Definition 2.33:

$$\pi^* = \arg \min_{\substack{\pi \in \mathcal{P}((\mathbb{R}^d)^{N+1}) \\ \pi_0 = \mu_0, \pi_N = \mu_T}} \text{KL}(\pi \parallel \mathcal{R})$$

where μ_0 and μ_T are probability measures on $\mathbb{R}^d$.

To adapt this optimization problem to generative modeling, the idea is to take the reference measure $\mathcal{R}$ to be the measure of the process which adds noise (the forward process); to take as μ_0 the data distribution, and μ_T the prior distribution. In the following, we assume that the forward process is the classical OU process.

In classical generative modeling, the forward process is imposed, and one has only $p_T \approx p_{\text{prior}}$, where p_T is the marginal at time T of the forward process. Since the backward process is initialized exactly at p_{prior} , this prior mismatch is a bias that can be mitigated only by enlarging T . In the Schrödinger bridge formulation the same dynamics are only a reference, while $\pi_N = p_{\text{prior}}$ becomes a hard constraint: the forward process is the minimal modification of the

Ornstein–Uhlenbeck dynamics reaching the prior *exactly* at time T . The bridge relaxes the forward process in order to tighten the terminal condition, which intuitively allows T to be taken substantially smaller.

If we define a time grid $0 = t_0 < \dots < t_N = T$, and denote by $p_{0:N}$ the law of the Euler–Maruyama discretization of the OU process $(X_{t_0}, \dots, X_{t_N})$, the Schrödinger Bridge Problem then becomes:

$$\pi^* = \underset{\substack{\pi \in \mathcal{P}_{N+1} \\ \pi_0 = p_{\text{data}}, \pi_N = p_{\text{prior}}}}{\operatorname{argmin}} \quad \operatorname{KL}(\pi \parallel p_{0:N})$$

where $\mathcal{P}_{N+1} := \mathcal{P}((\mathbb{R}^d)^{N+1})$.

By [Léo13], we know that under mild conditions, it admits the solution:

$$\pi^*(x_{0:N}) = \varphi^*(x_0) \left[\prod_{k=0}^{N-1} p(x_{k+1} \mid x_k) \right] \psi^*(x_N)$$

The solution therefore has an explicit structure: the reference process reweighted by two potentials, one at each endpoint. However, it is not computable: the potentials φ^* and ψ^* solve a coupled fixed-point system (the Schrödinger system) which admits no explicit solution in general. Hence to approximate this expression, an algorithm has been widely used in the literature.

3.4.2 Solving the Schrödinger Bridge Problem

Definition 3.14 (IPF). *The Iterative Proportional Fitting algorithm (IPF) is defined by the recursion:*

$$\begin{aligned} \pi^{2n+1} &= \underset{\pi \in \mathcal{P}_{N+1}, \pi_N = p_{\text{prior}}}{\operatorname{argmin}} \quad \operatorname{KL}(\pi \parallel \pi^{2n}) \\ \pi^{2n+2} &= \underset{\pi \in \mathcal{P}_{N+1}, \pi_0 = p_{\text{data}}}{\operatorname{argmin}} \quad \operatorname{KL}(\pi \parallel \pi^{2n+1}) \end{aligned} \tag{3.30}$$

started at $\pi^0 = p_{0:N}$.

We start by checking that under some conditions, the IPF algorithm is well defined. This is done by the authors, who give the following result (Proposition 32 in [Bor+23]):

Lemma 3.15. *Assume that there exists $\tilde{\pi} \in \mathcal{P}_{N+1}$ such that $\tilde{\pi}_0 = p_{\text{data}}$, $\tilde{\pi}_N = p_{\text{prior}}$ and $\operatorname{KL}(\tilde{\pi} \parallel \pi^0) < +\infty$. Then, the IPF is well defined, in the sense that π^n exists for each $n \in \mathbb{N}$.*

At each iteration, the IPF gives a probability measure which either has the data distribution as first marginal or the prior distribution as last marginal. Even though at all iterations, the probability measures given by the algorithm are defined over the same space, they do not have the same physical interpretations. Indeed, the measures defined at even iterations can be interpreted as laws of forward processes, starting from the data distribution, whereas the measures defined at odd iterations can be interpreted as laws of backward processes, starting from the prior distribution. One can then see the IPF as an alternation between forward and backward processes. If n is large enough, one can hope that π^n will satisfy both marginal constraints approximately: admitting exactly one of the two prescribed marginals and a close approximation of the other; while remaining, among such measures, the closest one to the reference.

There exists a classical method to obtain analytical forms for the iterations of the IPF, based on potential representations (proposition 27 in [Bor+23]):

Proposition 3.16. *Let $p_{0,N}$ be the joint law of the endpoints marginals of the original forward process considered. Assume that $\text{KL}(p_{\text{data}} \otimes p_{\text{prior}} \parallel p_{0,N}) < +\infty$. Then $(\pi^n)_{n \in \mathbb{N}}$ is well-defined and for any $n \in \mathbb{N}$ we have that π^{2n+1} and π^{2n+2} admit a density w.r.t. the Lebesgue measure, denoted q^n and p^{n+1} . In addition, we have for any $n \in \mathbb{N}$ and $x_{0:N} \in (\mathbb{R}^d)^{N+1}$:*

$$q^n(x_{0:N}) = \varphi_0^n(x_0) \prod_{k=0}^{N-1} p(x_{k+1} \mid x_k) \hat{\varphi}_N^n(x_N),$$

$$p^{n+1}(x_{0:N}) = \varphi_0^{n+1}(x_0) \prod_{k=0}^{N-1} p(x_{k+1} \mid x_k) \hat{\varphi}_N^n(x_N),$$

where for any $n \in \mathbb{N}$ we have for any $x_{0:N} \in (\mathbb{R}^d)^{N+1}$ and $k \in \{0, \dots, N-1\}$:

$$\begin{aligned} \hat{\varphi}_N^n(x_N) &= \frac{p_{\text{prior}}(x_N)}{\varphi_N^n(x_N)}, & \hat{\varphi}_k^n(x_k) &= \int_{\mathbb{R}^d} \hat{\varphi}_{k+1}^n(x_{k+1}) p(x_{k+1} \mid x_k) dx_{k+1}, \\ \varphi_0^{n+1}(x_0) &= \frac{p_{\text{data}}(x_0)}{\hat{\varphi}_0^n(x_0)}, & \varphi_{k+1}^{n+1}(x_{k+1}) &= \int_{\mathbb{R}^d} \varphi_k^{n+1}(x_k) p(x_{k+1} \mid x_k) dx_k, \end{aligned}$$

and $\varphi_0^0 = p_{\text{data}}$ and $\hat{\varphi}_N^{-1} = 1$.

Note that the two densities q^n and p^{n+1} share the same terminal potential $\hat{\varphi}_N^n$: passing from q^n to p^{n+1} amounts to projecting onto $\{\pi_0 = p_{\text{data}}\}$, which only updates the potential attached to x_0 .

This result shows that each of the iterations of the IPF admits a density which has the same shape as the structure of the SBP solution, that is a product of potentials evaluated at endpoints times the density of the reference. In other words, the IPF never leaves the parametric family in which the solution lies, and each half-step simply updates one of the two potentials. This is what makes the potential representation informative about the mechanics of the algorithm. However, it does not establish that the iterated potentials converge to the optimal ones φ^*, ψ^* . The convergence of the IPF iterations to the SBP solution is instead given by Proposition 5 in [Bor+23], whose proof does not rely on the potential representation stated above but on tools from information geometry:

Assumption 3.17. *Let p^0 denote the reference stochastic process. We define:*

- $p_{N|0}^0(x_N \mid x_0)$ as the conditional probability density (transition kernel) of the terminal state $X_N = x_N$ given the initial state $X_0 = x_0$ under the reference process.
- $p_N^0(x_N)$ as the marginal density of the reference process at the terminal time N . Assuming the process is initialized at the data distribution ($p_0^0 = p_{\text{data}}$), it is obtained by marginalizing over all possible initial states:

$$p_N^0(x_N) = \int_{\mathbb{R}^d} p_{N|0}^0(x_N \mid x_0) p_{\text{data}}(x_0) dx_0.$$

We assume then:

$p_N^0, p_{\text{prior}} > 0$, $|H(p_{\text{prior}})| < +\infty$, and

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} |\log p_{N|0}^0(x_N \mid x_0)| p_{\text{data}}(x_0) p_{\text{prior}}(x_N) dx_0 dx_N < +\infty.$$

Proposition 3.18. *Assume 3.17. Then there exists a solution $\pi^* \in \mathcal{P}_{N+1}$ to the SB problem and we have $\lim_{n \rightarrow +\infty} \|\pi^n - \pi^\infty\|_{TV} = 0$ with $\pi^\infty \in \mathcal{P}_{N+1}$. Let $h = p_{0,N}/(p_0^0 \otimes p_N^0)$, where p_0^0 and p_N^0 denote the marginals of the reference $p_{0:N}$ at times 0 and N , and assume that $h \in \mathcal{C}((\mathbb{R}^d)^2, (0, +\infty))$ and that there exist $\alpha_0, \alpha_N \in \mathcal{C}(\mathbb{R}^d, (0, +\infty))$ such that*

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} (|\log h(x_0, x_N)| + |\log \alpha_0(x_0)| + |\log \alpha_N(x_N)|) p_{\text{data}}(x_0) p_{\text{prior}}(x_N) dx_0 dx_N < +\infty,$$

with $h(x_0, x_N) \leq \alpha_0(x_0) \alpha_N(x_N)$. If $p_{0:N}$ is absolutely continuous w.r.t. π^∞ then $\pi^\infty = \pi^*$.

To solve the IPF, one could then follow the potentials representation defined above, and compute the potentials at each iteration. However, this becomes intractable in large dimensions, as each potential update is an integral over $\mathbb{R}^d$, composed N times. Indeed, they are defined as:

$$\begin{aligned} \Psi_k^n(x_k) &= \int_{\mathbb{R}^d} \Psi_{k+1}^n(x_{k+1}) p(x_{k+1} | x_k) dx_{k+1} \\ \hat{\Psi}_{k+1}^{n+1}(x_{k+1}) &= \int_{\mathbb{R}^d} \hat{\Psi}_k^{n+1}(x_k) p(x_{k+1} | x_k) dx_k \end{aligned}$$

The authors then propose a novel approach.

3.4.3 Diffusion Schrödinger Bridge

The authors prove an alternative representation of the IPF iterations, which does not rely on the potentials:

Proposition 3.19. *Assume that $\text{KL}(p_{\text{data}} \otimes p_{\text{prior}} \| \pi_{0,N}^0) < +\infty$. Then for any $n \in \mathbb{N}$, π^{2n} and π^{2n+1} admit positive densities w.r.t. the Lebesgue measure, denoted as p^n resp. q^n , and for any $x_{0:N} \in \mathbb{R}^{d^{N+1}}$, we have $p^0(x_{0:N}) = \pi^0$ and*

$$\begin{aligned} q^n(x_{0:N}) &= p_{\text{prior}}(x_N) \prod_{k=0}^{N-1} p_{k|k+1}^n(x_k | x_{k+1}), \\ p^{n+1}(x_{0:N}) &= p_{\text{data}}(x_0) \prod_{k=0}^{N-1} q_{k+1|k}^n(x_{k+1} | x_k). \end{aligned}$$

Let us now interpret this result. Recall that

$$\begin{aligned} \pi^{2n+1} &= \underset{\pi \in \mathcal{P}_{N+1}, \pi_N = p_{\text{prior}}}{\text{argmin}} \quad \text{KL}(\pi \| \pi^{2n}) \\ \pi^{2n+2} &= \underset{\pi \in \mathcal{P}_{N+1}, \pi_0 = p_{\text{data}}}{\text{argmin}} \quad \text{KL}(\pi \| \pi^{2n+1}) \end{aligned} \tag{3.31}$$

where $\pi_0 = p_{0:N}$.

Let us place ourselves at some even step $2n$ of the procedure, so that we already have $\pi^{2n} = p^n$ in hand. The proposition tells us that the next iterate $\pi^{2n+1} = q^n$ is obtained from p^n by keeping its reverse-time transitions and replacing only the terminal marginal by p_{prior} . This is the content of the backward decomposition

$$q^n(x_{0:N}) = p_{\text{prior}}(x_N) \prod_{k=0}^{N-1} p_{k|k+1}^n(x_k | x_{k+1}). \tag{3.32}$$

Because the density is written from the terminal time backward, starting at p_{prior} at step N and moving down to step 0, it is expressed through the reverse-time transitions $p_{k|k+1}^n$.

To give the intuition about why we have this result, recall that the definition of π^{2n+1} minimizes

$$\text{KL}(\pi \parallel \pi^{2n}) = \text{KL}(\pi_N \parallel p_N^n) + \mathbb{E}_{\pi_N}[\text{KL}(\pi_{0:N-1|N} \parallel p_{0:N-1|N}^n)].$$

while imposing $\pi_N = p_{\text{prior}}$. The constraint fixes the first term, and the second is minimized to zero by keeping the conditional law of $x_{0:N-1}$ given x_N . Hence, it follows by the product rule and the unicity of the backward transitions kernels that:

$$q_{k|k+1}^n(x_k \mid x_{k+1}) = p_{k|k+1}^n(x_k \mid x_{k+1}), \quad k = 0, \dots, N-1. \quad (3.33)$$

which then gives (3.32) by the product rule and the constraint over q_N^n .

Hence, q^n inherits from p^n exactly its reverse kernels and nothing more; changing the endpoint marginal and re-propagating those same kernels changes every marginal, and hence every forward kernel.

Note that the backward factorization of q^n is itself a sampling procedure: each factor $q_{k|k+1}^n$ is a conditional law, so if we had access to these reverse-time transitions, we would obtain a generative model by drawing successively

$$X_N \sim p_{\text{prior}} \longrightarrow X_{N-1} \sim q_{N-1|N}^n(\cdot \mid X_N) \longrightarrow \dots \longrightarrow X_0 \sim q_{0|1}^n(\cdot \mid X_1).$$

For n large enough, the resulting law of X_0 , namely the marginal q_0^n , is close to p_{data} .

Hence, the whole problem is to obtain these reverse-time transitions, which is the heart of the method of the authors.

The resulting difficulty. The process p^n was simulated forward, starting from p_{data} at step 0. What we can therefore sample from is the *forward* transition $p_{k+1|k}^n$, whereas (3.32) requires the *reverse* transition $p_{k|k+1}^n$. Hence, we need to apply the Bayes' rule to get:

$$p_{k|k+1}^n(x_k \mid x_{k+1}) = \frac{p_{k+1|k}^n(x_{k+1} \mid x_k) p_k^n(x_k)}{p_{k+1}^n(x_{k+1})}, \quad (3.34)$$

One then needs the marginals p_k^n and p_{k+1}^n , whose evaluation requires a high-dimensional integral over the whole trajectory, $p_k^n(x) = \int p_{k|k-1}^n(x \mid x') p_{k-1}^n(x') dx'$, and is therefore intractable in high dimension (e.g. $d = 3072$ in the original paper). To circumvent this, the authors proceed as follows.

Fix $n \in \mathbb{N}$ and $k \in \{0, \dots, N-1\}$, and write $\gamma := \gamma_{k+1}$ to lighten notation, where $\gamma_{k+1} > 0$ denotes the step size of the Euler-Maruyama discretization between times k and $k+1$, fixed before running the algorithm (so that $T = \sum_{k=1}^N \gamma_k$). Let f_k^n be the drift of the n -th forward process at the k -th step of the discretization. The authors then *assume* that the forward kernel is Gaussian with fixed covariance $2\gamma\mathbf{I}$:

$$p_{k+1|k}^n(x_{k+1} \mid x_k) = \mathcal{N}(x_{k+1}; F_k^n(x_k), 2\gamma\mathbf{I}), \quad F_k^n(x) = x + \gamma f_k^n(x). \quad (3.35)$$

At $n = 0$ this is not an assumption but a fact: the IPF is initialized at $\pi^0 = p_{0:N}$, the reference noising process, whose Euler-Maruyama discretization of $dX_t = f(X_t) dt + \sqrt{2} dB_t$ is exactly (3.35) with $f_k^0 = f$, i.e. $F_k^0(x) = x + \gamma f(x)$. In the Ornstein-Uhlenbeck case $f(x) = -\alpha x$ used

throughout, this is the usual noising kernel. For $n \geq 1$, however, $p_{k+1|k}^n$ is no longer given a priori but produced by the previous IPF half-step, and (3.35) becomes a genuine assumption, whose legitimacy is discussed in Remark 3.4.3 below. We further assume that the step $\|x_{k+1} - x_k\|$ is small, which holds for γ small since $x_{k+1} - x_k = O(\gamma) + O(\sqrt{\gamma})Z$ with $Z \sim \mathcal{N}(0, \mathbf{I})$.

Now, we derive an approximation for the reverse transition $q_{k|k+1}^n$. Take logarithms in (3.34) and expand each intractable term for small steps, using three approximations:

$$\underbrace{p_k^n \approx p_{k+1}^n}_{\text{(i) slow marginals}}, \quad \underbrace{\log p_{k+1}^n(x_k) \approx \log p_{k+1}^n(x_{k+1}) + \langle g, x_k - x_{k+1} \rangle}_{\text{(ii) first-order Taylor at } x_{k+1}}, \quad \underbrace{f_k^n(x_k) \approx f_k^n(x_{k+1})}_{\text{(iii) frozen drift}}, \quad (3.36)$$

where $g := \nabla \log p_{k+1}^n(x_{k+1})$.

Taking logarithms in (3.34),

$$\log q_{k|k+1}^n(x_k | x_{k+1}) = \underbrace{\log p_{k+1|k}^n(x_{k+1} | x_k)}_{(A)} + \underbrace{\log p_k^n(x_k)}_{(B)} - \underbrace{\log p_{k+1}^n(x_{k+1})}_{(C)}.$$

By (3.35),

$$(A) = -\frac{\|x_{k+1} - F_k^n(x_k)\|^2}{4\gamma} + \text{cst} = -\frac{\|x_{k+1} - x_k - \gamma f_k^n(x_k)\|^2}{4\gamma} + \text{cst}.$$

Using $f_k^n(x_k) \approx f_k^n(x_{k+1})$ and setting $m = x_{k+1} - \gamma f_k^n(x_{k+1})$ (constant in x_k),

$$(A) \approx -\frac{\|x_{k+1} - x_k - \gamma f_k^n(x_{k+1})\|^2}{4\gamma} + \text{cst} = -\frac{\|x_k - m\|^2}{4\gamma} + \text{cst}.$$

$$(B) = \log p_k^n(x_k) \approx \log p_{k+1}^n(x_k) \approx \underbrace{\log p_{k+1}^n(x_{k+1})}_{\text{cst}} + \langle g, x_k - x_{k+1} \rangle = \langle g, x_k \rangle + \text{cst}.$$

Since (C) is constant in x_k ,

$$\log q_{k|k+1}^n(x_k | x_{k+1}) \approx -\frac{\|x_k - m\|^2}{4\gamma} + \langle g, x_k \rangle + \text{cst}, \quad m = x_{k+1} - \gamma f_k^n(x_{k+1}).$$

Now expand the square and isolate the linear part:

$$-\frac{\|x_k - m\|^2}{4\gamma} + \langle g, x_k \rangle = -\frac{\|x_k\|^2}{4\gamma} + \frac{\langle m, x_k \rangle}{2\gamma} + \langle g, x_k \rangle + \text{cst}.$$

Factor $\frac{1}{2\gamma}$ out of the two linear terms:

$$\frac{\langle m, x_k \rangle}{2\gamma} + \langle g, x_k \rangle = \frac{\langle m + 2\gamma g, x_k \rangle}{2\gamma} := \frac{\langle m', x_k \rangle}{2\gamma}.$$

But the expansion of the square centered at m' gives:

$$-\frac{\|x_k - m'\|^2}{4\gamma} = -\frac{\|x_k\|^2}{4\gamma} + \frac{\langle m', x_k \rangle}{2\gamma} + \text{cst}.$$

As the quadratic term $-\|x_k\|^2/(4\gamma)$ and the linear term $\langle m', x_k \rangle/(2\gamma)$ coincide, the two expressions differ only by a constant:

$$-\frac{\|x_k - m\|^2}{4\gamma} + \langle g, x_k \rangle = -\frac{\|x_k - (m + 2\gamma g)\|^2}{4\gamma} + \text{cst}$$

Since the constant terms are uniquely determined by the constraint that the density must integrate to 1, $q_{k|k+1}^n$ is Gaussian, with covariance still $2\gamma\mathbf{I}$ and mean:

$$m + 2\gamma g = x_{k+1} + \gamma \underbrace{\left[-f_k^n(x_{k+1}) + 2\nabla \log p_{k+1}^n(x_{k+1}) \right]}_{:= b_{k+1}^n(x_{k+1})} := B_{k+1}^n(x_{k+1}), \quad (3.37)$$

that is,

$$\boxed{q_{k|k+1}^n(x_k | x_{k+1}) \approx \mathcal{N}(x_k; B_{k+1}^n(x_{k+1}), 2\gamma\mathbf{I}), \quad B_{k+1}^n(x) = x + \gamma b_{k+1}^n(x).} \quad (3.38)$$

Hence to get access to $q_{k|k+1}^n$, we just need to be able to compute B_{k+1}^n .

Remark. :[Discrete analogue of continuous time-reversal] This is the Euler–Maruyama counterpart of the fact that the time-reversal of $dX_t = f(X_t)dt + \sqrt{2}dB_t$ is again a diffusion, $dY_t = \{-f(Y_t) + 2\nabla \log p_{T-t}(Y_t)\}dt + \sqrt{2}dB_t$, with the *same* diffusion coefficient $\sqrt{2}$ and drift shifted by $2\nabla \log p_{T-t}$. Equation (3.38) is exactly this at the discrete level.

Consequence: the reverse mean as a regression As just said, the Gaussian form (3.38) reduces the approximate reverse kernel to a single function B_{k+1}^n , its mean. Let us note that while the mean of the exact reverse kernel $q_{k|k+1}^n$ is the true conditional expectation $\mathbb{E}[X_k | X_{k+1}]$, the function B_{k+1}^n defined in (3.37) is an approximation of it, coming from the Taylor expansion. However, we can find an exact regression formulation for this approximate mean B_{k+1}^n .

Recall that the forward kernel is

$$p_{k+1|k}^n(x_{k+1} | x_k) = \mathcal{N}(x_{k+1}; F_k^n(x_k), 2\gamma\mathbf{I}), \quad F_k^n(x) = x + \gamma f_k^n(x),$$

so that

$$\log p_{k+1|k}^n(x_{k+1} | x_k) = -\frac{\|x_{k+1} - F_k^n(x_k)\|^2}{4\gamma} + \text{cst.}$$

Differentiate the marginal

$$p_{k+1}^n(x_{k+1}) = \int_{\mathbb{R}^d} p_k^n(x_k) p_{k+1|k}^n(x_{k+1} | x_k) dx_k$$

with respect to x_{k+1} (only the kernel depends on x_{k+1}):

$$\nabla_{x_{k+1}} p_{k+1}^n(x_{k+1}) = \int_{\mathbb{R}^d} p_k^n(x_k) \nabla_{x_{k+1}} p_{k+1|k}^n(x_{k+1} | x_k) dx_k.$$

The gradient of the Gaussian kernel is

$$\nabla_{x_{k+1}} \log p_{k+1|k}^n(x_{k+1} | x_k) = -\frac{x_{k+1} - F_k^n(x_k)}{2\gamma} = \frac{F_k^n(x_k) - x_{k+1}}{2\gamma},$$

hence, using $\nabla p = p \nabla \log p$,

$$\nabla_{x_{k+1}} p_{k+1|k}^n(x_{k+1} | x_k) = p_{k+1|k}^n(x_{k+1} | x_k) \frac{F_k^n(x_k) - x_{k+1}}{2\gamma}.$$

Substituting back,

$$\nabla_{x_{k+1}} p_{k+1}^n(x_{k+1}) = \int_{\mathbb{R}^d} p_k^n(x_k) p_{k+1|k}^n(x_{k+1} | x_k) \frac{F_k^n(x_k) - x_{k+1}}{2\gamma} dx_k.$$

Dividing by $p_{k+1}^n(x_{k+1})$ gives the score:

$$\nabla \log p_{k+1}^n(x_{k+1}) = \int_{\mathbb{R}^d} \frac{p_k^n(x_k) p_{k+1|k}^n(x_{k+1} | x_k)}{p_{k+1}^n(x_{k+1})} \frac{F_k^n(x_k) - x_{k+1}}{2\gamma} dx_k,$$

and Bayes' rule identifies the weight as the reverse kernel,

$$\frac{p_k^n(x_k) p_{k+1|k}^n(x_{k+1} | x_k)}{p_{k+1}^n(x_{k+1})} = p_{k|k+1}^n(x_k | x_{k+1}),$$

so the integral gives an exact identity for the score as a conditional expectation:

$$\nabla \log p_{k+1}^n(x_{k+1}) = \mathbb{E} \left[\frac{F_k^n(X_k) - x_{k+1}}{2\gamma} \mid X_{k+1} = x_{k+1} \right], \quad (X_k, X_{k+1}) \sim p_{k,k+1}^n.$$

We can now substitute this exact identity back into the definition of our approximate mean B_{k+1}^n . By definition (3.37), $B_{k+1}^n(x_{k+1}) = x_{k+1} + \gamma(-f_k^n(x_{k+1}) + 2\nabla \log p_{k+1}^n(x_{k+1}))$. Substituting the score yields exactly:

$$B_{k+1}^n(x_{k+1}) = \mathbb{E}[x_{k+1} + F_k^n(X_k) - F_k^n(x_{k+1}) \mid X_{k+1} = x_{k+1}].$$

Note that if we were to apply the "frozen drift" approximation $f_k^n(X_k) \approx f_k^n(x_{k+1})$ within this expectation, we would recover the true conditional expectation $\mathbb{E}[X_k \mid X_{k+1} = x_{k+1}]$. Maintaining this exact unsimplified formulation for B_{k+1}^n , however, is what motivates the regression objective.

Indeed, since the conditional expectation is the L^2 -projection, $\mathbb{E}[Y \mid X = \cdot] = \operatorname{argmin}_{g \in L^2} \mathbb{E}\|g(X) - Y\|^2$, we get the following result:

Proposition 3.20. *With the Gaussian kernels (3.35) and (3.38), for any $n \in \mathbb{N}$ and $k \in \{0, \dots, N-1\}$,*

$$B_{k+1}^n = \operatorname{argmin}_{B \in L^2(\mathbb{R}^d, \mathbb{R}^d)} \mathbb{E}_{p_{k,k+1}^n} [\|B(X_{k+1}) - (X_{k+1} + F_k^n(X_k) - F_k^n(x_{k+1}))\|^2], \quad (3.39)$$

$$F_k^{n+1} = \operatorname{argmin}_{F \in L^2(\mathbb{R}^d, \mathbb{R}^d)} \mathbb{E}_{q_{k,k+1}^n} [\|F(X_k) - (X_k + B_{k+1}^n(X_{k+1}) - B_{k+1}^n(x_k))\|^2]. \quad (3.40)$$

Why this is the whole point. The objective (3.39) needs only pairs (X_k, X_{k+1}) sampled from p^n , obtained by simulating (3.35) forward from p_{data} with the known map F_k^n . The intractable marginals p_k^n, p_{k+1}^n that blocked Bayes (3.34) are never evaluated: they enter only implicitly through the law of the samples. Note that equation (3.40) is the mirror image for the other half-step: to build p^{n+1} one needs the forward transitions of q^n , sampled by simulating q^n backward from p_{prior} with the just-learned B_{k+1}^n . F_k^{n+1} is obtained from B_{k+1}^n , itself obtained from F_k^n . The starting point of the whole chain is F_k^0 , which is *not* learned: it is the drift of the initial forward process. The algorithm is thus initialized on a known object, and each half-step refines the previous one.

Remark (regression objective): By expressing B_{k+1}^n through $\nabla \log p_{k+1}^n$, we obtained the residual target $X_{k+1} + F_k^n(X_k) - F_k^n(x_{k+1})$. Because $B_{k+1}^n(x_{k+1})$ is exactly the conditional expectation of this residual target, it is naturally the minimizer of an L^2 objective. While one could theoretically use the naive target X_k (which directly targets the true conditional expectation $\mathbb{E}[X_k \mid X_{k+1}]$ rather than the Gaussian projection B_{k+1}^n), the residual target has a significantly better structure: it is of the form identity-plus-small-perturbation, matching the network parameterization $B_{k+1}^n(x) = x + \gamma b_{k+1}^n(x)$. The paper makes this choice explicit. It

presents in Proposition 28 the “bare-score” variant (regressing the score, equivalently X_k , more directly) and in Proposition 3 the residual variant, concluding in Appendix E.2.3 that the latter yields better results in practice and conjecturing that this is due to the residual structure of the networks used.

In practice B_k^n, F_k^n are replaced by neural networks $B_{\beta^n}(k, x) \approx B_k^n(x)$ and $F_{\alpha^n}(k, x) \approx F_k^n(x)$ trained on the empirical versions of (3.39)–(3.40), leading to the following algorithm:

Algorithm 2: Diffusion Schrödinger Bridge

```

for  $n \in \{0, \dots, L\}$  do
  while not converged do
    Sample  $\{X_k^j\}_{k,j=0}^{N,M}$ , where  $X_0^j \sim p_{\text{data}}$ , and  $X_{k+1}^j = F_{\alpha^n}(k, X_k^j) + \sqrt{2\gamma_{k+1}} Z_{k+1}^j$ ;
    Compute  $\hat{\ell}_n^b(\beta^n)$  approximating (3.39);
     $\beta^n \leftarrow \text{Gradient Step}(\hat{\ell}_n^b(\beta^n))$ ;
  while not converged do
    Sample  $\{X_k^j\}_{k,j=0}^{N,M}$ , where  $X_N^j \sim p_{\text{prior}}$ , and  $X_{k-1}^j = B_{\beta^n}(k, X_k^j) + \sqrt{2\gamma_k} \tilde{Z}_k^j$ ;
    Compute  $\hat{\ell}_{n+1}^f(\alpha^{n+1})$  approximating (3.40);
     $\alpha^{n+1} \leftarrow \text{Gradient Step}(\hat{\ell}_{n+1}^f(\alpha^{n+1}))$ ;

```

Result: (α^{L+1}, β^L)

Interpretation of the DSB Algorithm: Each half-step of the IPF is a KL projection which, by Proposition 3.19, amounts to time-reversing the process just built and restarting it from the other endpoint. The role of the algorithm is therefore to approximate these successive reversals, and by Proposition 3.20 each of them is the solution of an L^2 regression.

At each half-step, the expectation defining the target (over the joint law $p_{k,k+1}^n$ or $q_{k,k+1}^n$) is replaced by a Monte-Carlo estimate: we simulate M full trajectories $\{X_k^j\}_{k=0}^N$ of the relevant process, whose consecutive pairs (X_k^j, X_{k+1}^j) are samples from that joint law, and average the loss over them. Crucially, this never requires evaluating the intractable marginals, only sampling the process.

Concretely, the odd half-step (projection onto $\{\pi_N = p_{\text{prior}}\}$) is carried out by simulating forward from p_{data} using the current F_{α^n} , forming the empirical loss $\hat{\ell}_n^b$ approximating (3.39), and updating β^n by gradient descent; the even half-step (projection onto $\{\pi_0 = p_{\text{data}}\}$) is carried out by simulating backward from p_{prior} using the just-learned B_{β^n} , forming $\hat{\ell}_{n+1}^f$ approximating (3.40), and updating α^{n+1} . The alternation between forward and backward processes is thus exactly the alternation of the two IPF projections. Each network is trained from the previous one, along the chain $F_k^n \rightarrow B_{k+1}^n \rightarrow F_k^{n+1}$, starting from the reference drift $F_k^0(x) = x + \gamma f(x)$: the first backward network B_{β^0} therefore reverses the reference noising process, which is precisely classical score-based generative modeling, and each subsequent iteration refines it by tightening the marginal constraints. Iterating over $n = 0, \dots, L$ yields the final generative model, which samples $X_N \sim p_{\text{prior}}$ and runs the backward network B_{β^L} .

Note that this requires simulating M trajectories at each half-step, in order to get access to the pairs (X_k^j, X_{k+1}^j) used to train the network giving the drift of the next forward or backward process. Hence this method requires training $2L$ neural networks, whereas classical score-based generative modeling trains a single one (corresponding, as noted above, to the first IPF iteration only). The authors are fully conscious of this and present several techniques in the appendix to mitigate the issue.

Remark (validity and consequence of the starting hypothesis): The authors’ assumption on the forward kernels $p_{k+1|k}^n$ is therefore consistent with their scheme: the Gaussian form with fixed covariance $2\gamma\mathbf{I}$ is exact at $n = 0$ (the Euler–Maruyama reference kernel) and, following the iterations, is preserved at every n . This preservation, however, hinges crucially on the first-order Taylor expansion: truncating to first order keeps the exponent quadratic in x_k (a linear term added to a quadratic remains quadratic, with unchanged covariance), so the Taylor step is what *makes* the class stable, by projecting at each half-step the true non-Gaussian reversal onto the Gaussian family. The price is a departure from the exact IPF: its exact iterates are not Gaussian, hence an approximation error is induced at each half-step. This error is not controlled cumulatively, and it is precisely what prevents the convergence results for the exact IPF (which assume the projections are performed exactly) from applying to the authors’ algorithm. More recent works, like [Shi+23], propose a novel approach which aims at reducing this accumulated bias.

Chapter 4

Generative Modeling under Structural Constraints: Time Series and Manifolds

After their empirical success in generating pictures or videos, generative models have now been applied to a broad class of data. We study here two very different applications: financial time series, and protein backbones.

4.1 Generative models for time series

The Schrödinger Bridge approach turns out to be particularly well suited to generate what is known as time series.

Definition 4.1. *A time series indexed by $\mathcal{T}$ is the set of realizations $\{Y_t(w), t \in \mathcal{T}\}$ of a stochastic process Y defined over Ω , for some $w \in \Omega$.*

4.1.1 The importance of Time Series generation

Time series arise across nearly every quantitative discipline: meteorology records temperature and precipitation over time, medicine monitors physiological signals such as ECG, EEG or vital signs, and even a video can be regarded as a high-dimensional time series of images. In all these settings, being able not only to predict future values but to generate new, realistic trajectories is of considerable practical importance. In this work we deal with the generation problem: learning the distribution underlying a family of observed series so we can sample new ones. More specifically, we focus on financial time series.

Finance is a particularly compelling domain for generation, and for a structural reason: for any given market, history has produced only a single realized trajectory of prices. One cannot re-run the 2008 crisis under slightly different conditions, nor observe the alternative paths the market could have taken. As a result, it is impossible to assess from historical data alone the distribution of outcomes of a strategy or a portfolio, since one holds only a single draw from that distribution. A generative model that captures the law underlying the observed series addresses this limitation directly: it produces many plausible scenarios consistent with the statistical properties of the real market, effectively turning a single history into a population of credible alternatives.

This capability underpins several concrete applications. The foremost is stress-testing and backtesting, as done in [FJ23]: evaluating a trading or risk-management strategy on the single

historical path tends to overfit that path and to underestimate risk ([Bai+17]), whereas a set of synthetic yet realistic scenarios yields a whole distribution of possible outcomes. Generated series can also be used for data augmentation, enlarging the training set of a predictive model, for instance one forecasting whether a price will rise or fall, such as in [Alo+26].

4.1.2 Schrödinger Bridge for time series

Throughout this section, $\Omega = \{\omega \in C([0, T]; \mathbb{R}^d) : \omega(0) = 0\}$ denotes the canonical space, $X_t(\omega) := \omega(t)$ the coordinate process, $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ its natural filtration, and $\mathcal{F} = \mathcal{F}_T$. Under the Wiener measure $\mathbb{W}$, the process X is a Brownian motion.

If the index set is the finite time horizon $[0, T]$, note that we can interpret time series as trajectories of the stochastic process Y between 0 and T . As in practice, for most cases, we only have access to time series via their values at some discrete time grid $(t_1, \dots, t_N)$, we will consider in the following that time series are samples from $(\mathbb{R}^d)^N$ (assuming the data is d -dimensional).

The Schrödinger Bridge Problem for time series has been defined in [HHP23].

Definition 4.2. *Let $\mathbb{Q}$ be a reference measure on Ω . The Schrödinger Bridge Problem for time series is defined as follows:*

$$\mathbb{P}^* = \operatorname{argmin}\{KL(\mathbb{P}|\mathbb{Q}) : \mathbb{P} \in \mathcal{P}(\Omega), \pi_{\#}\mathbb{P} = \mu, \quad \text{where } \pi(\omega) = (X_{t_1}(\omega), \dots, X_{t_N}(\omega))\},$$

where μ is a probability measure on $(\mathbb{R}^d)^N$.

From the perspective of generative modeling, we have access to M i.i.d. samples from a distribution p_{data} (the population law, not the empirical law of the samples) on $(\mathbb{R}^d)^N$, and we then take $\mu = p_{\text{data}}$. We are demanding that the objective measure has finite dimensional distributions on the grid $\{t_1, \dots, t_N\}$ equal to p_{data} , which results in a constraint on a joint distribution of N random variables, instead of separate constraints for two marginals as in the usual Schrödinger Bridge Problem.

Remark: Note that only the constraint encodes fidelity to the data. No matter the reference measure, if the problem is analytically solved, the joint law p_{data} will be reproduced. There exist then infinite ways to interpolate between the marginals on the grid of this joint law. On a side-note, one can see that all of the samples making a training set need to be available as the same times $t_1, \dots, t_N$, which could be a limitation in practical use cases.

In the following, we assume that $\mathbb{Q} = \mathbb{W}^\sigma$, where $\mathbb{W}^\sigma$ is the scaled Wiener measure. That is, under $\mathbb{W}^\sigma$, $W = X/\sigma$ is a standard Brownian motion. We also define the filtration $\mathbb{F}^{\mathbb{W}^\sigma} = (\mathcal{F}_t^{\mathbb{W}^\sigma})_{t \geq 0}$ as follows.

Let $\mathcal{N}^{\mathbb{W}^\sigma}$ be the class of all $\mathbb{W}_\sigma$ -null sets, i.e. sets contained in some $A \in \mathcal{F}_T$ such that $\mathbb{W}_\sigma(A) = 0$. Formally, the completed filtration $\mathbb{F}^{\mathbb{W}^\sigma}$ is then defined as:

$$\mathcal{F}_t^{\mathbb{W}^\sigma} = \sigma(\mathcal{F}_t \cup \mathcal{N}^{\mathbb{W}^\sigma})$$

To solve the SBP for time series, we are not going to rely on an IPF-type algorithm as in the original case. Actually, in the time series case, we are able to compute an analytical closed form. For this, we need to do some computations involving stochastic calculus.

Let $\mathbb{P} \in \mathcal{P}(\Omega)$ such that $KL(\mathbb{P} | \mathbb{W}_\sigma) < \infty$. Then necessarily, $\mathbb{P} \ll \mathbb{W}_\sigma$.

Lemma 4.3 (Föllmer drift under Brownian scaling). *Let Ω be endowed with its Borel σ -field and its canonical filtration $(\mathcal{F}_t)_{t \in [0, T]}$, and let $X_t(\omega) := \omega_t$ be the coordinate process. Let $\mathbb{W}$ denote Wiener measure on Ω .*

Fix $\sigma > 0$, define

$$S_\sigma(\omega) := \sigma\omega,$$

and let

$$\mathbb{W}^\sigma := (S_\sigma)_\# \mathbb{W}.$$

Thus, under $\mathbb{W}^\sigma$, the coordinate process X is a Brownian motion with covariance matrix $\sigma^2 t I_d$ at time t .

Let $\mathbb{P}$ be a measure on the path space.

If $\mathbb{P} \ll \mathbb{W}^\sigma$, then there exists an $(\mathcal{F}_t)$ -progressively measurable process u such that, under $\mathbb{P}$,

$$B_t := X_t - \int_0^t u_s \, ds$$

is a Brownian motion with covariance parameter σ^2 , i.e. B/σ is a standard Brownian motion. Moreover,

$$\text{KL}(\mathbb{P} \mid \mathbb{W}^\sigma) = \frac{1}{2\sigma^2} \mathbb{E}_{\mathbb{P}} \left[\int_0^T |u_s|^2 \, ds \right].$$

The equality is understood in $[0, +\infty]$.

Proof: Let

$$g := S_\sigma^{-1}, \quad g(\omega) = \frac{\omega}{\sigma},$$

and define

$$\bar{\mathbb{P}} := g_\# \mathbb{P}.$$

Since

$$g_\# \mathbb{W}^\sigma = \mathbb{W},$$

the assumption $\mathbb{P} \ll \mathbb{W}^\sigma$ implies

$$\bar{\mathbb{P}} \ll \mathbb{W}.$$

Moreover, relative entropy is invariant under a measurable bijection, and therefore

$$\text{KL}(\mathbb{P} \mid \mathbb{W}^\sigma) = \text{KL}(g_\# \mathbb{P} \mid g_\# \mathbb{W}^\sigma) = \text{KL}(\bar{\mathbb{P}} \mid \mathbb{W}).$$

By Theorem 2 of [Leh11], there exists an adapted process $\bar{u}$ such that, under $\bar{\mathbb{P}}$,

$$\bar{B}_t(\eta) := \eta_t - \int_0^t \bar{u}_s(\eta) \, ds$$

is a standard Brownian motion and

$$\text{KL}(\bar{\mathbb{P}} \mid \mathbb{W}) = \frac{1}{2} \mathbb{E}_{\bar{\mathbb{P}}} \left[\int_0^T |\bar{u}_s|^2 \, ds \right].$$

Define, for $\omega \in \Omega$,

$$u_t(\omega) := \sigma \bar{u}_t(g\omega) = \sigma \bar{u}_t\left(\frac{\omega}{\sigma}\right).$$

Since g is non-anticipative and preserves the canonical filtration, u is progressively measurable. Furthermore,

$$\begin{aligned} \frac{1}{\sigma} \left(X_t(\omega) - \int_0^t u_s(\omega) ds \right) &= \frac{\omega_t}{\sigma} - \int_0^t \bar{u}_s \left(\frac{\omega}{\sigma} \right) ds \\ &= \bar{B}_t(g\omega). \end{aligned}$$

Because $g_{\#}\mathbb{P} = \bar{\mathbb{P}}$, the process $\bar{B} \circ g$ is a standard Brownian motion under $\mathbb{P}$. Consequently,

$$B_t := X_t - \int_0^t u_s ds$$

is a Brownian motion with covariance parameter σ^2 under $\mathbb{P}$.

Finally,

$$\begin{aligned} \mathbb{E}_{\mathbb{P}} \left[\int_0^T |u_s|^2 ds \right] &= \sigma^2 \mathbb{E}_{\mathbb{P}} \left[\int_0^T |\bar{u}_s(g\omega)|^2 ds \right] \\ &= \sigma^2 \mathbb{E}_{\bar{\mathbb{P}}} \left[\int_0^T |\bar{u}_s|^2 ds \right]. \end{aligned}$$

Combining this identity with the entropy identity above gives

$$\text{KL}(\mathbb{P} \mid \mathbb{W}^{\sigma}) = \frac{1}{2\sigma^2} \mathbb{E}_{\mathbb{P}} \left[\int_0^T |u_s|^2 ds \right].$$

□

From now on, we take $\sigma = 1$ for simplicity. Thanks to the proof, we know that there exists α adapted such that:

$$d\mathbf{X}_t = \alpha_t dt + d\mathbf{W}_t^{\mathbb{P}}$$

where $\mathbf{W}_t^{\mathbb{P}}$ is a $\mathbb{P}$ -Brownian motion. Define also:

$$J(\alpha) = \frac{1}{2} \mathbb{E}_{\mathbb{P}} \left[\int_0^T |\alpha_s|^2 ds \right]$$

Thanks to this result, we can now formulate the SBP as a stochastic control problem that becomes:

$$\boxed{\begin{aligned} &\min_{\alpha} J(\alpha) \\ \text{s.t. } &\begin{cases} d\mathbf{X}_t = \alpha_t dt + d\mathbf{W}_t^{\mathbb{P}} \\ \mathbf{X}_{t_1}, \dots, \mathbf{X}_{t_N} \sim \mu \\ \alpha \text{ is adapted with } J(\alpha) < \infty \end{cases} \end{aligned}} \quad (4.1)$$

We define:

•

$$V_{SBTS} := \inf_{\alpha \in \mathcal{A}_{\mathcal{T}}^{\mu}} J(\alpha),$$

where $\mathcal{A}_{\mathcal{T}}^{\mu}$ is the set of adapted (with respect to the canonical filtration) processes that satisfy the constraints of the minimization problem.

- $\mu_{\mathcal{T}}^{\mathbb{W}}$ as the law of the random vector $(W_{t_1}, \dots, W_{t_N})$ where W is the canonical process on Ω under the Wiener measure, and for ease of notations, we also identify it with its density with respect to Lebesgue measure.

Now, we are going to investigate why this problem admits a solution which has an analytical closed form.

Why is the solution explicit, in contrast to the original Schrödinger Bridge Problem?

The closed form is best understood through the chain rule for relative entropy under the grid projection $\pi(X) := (X_{t_1}, \dots, X_{t_N})$. For any $\mathbb{P} \ll \mathbb{W}$, disintegrating along the grid values y ,

$$\text{KL}(\mathbb{P} \parallel \mathbb{W}) = \underbrace{\text{KL}(\pi_{\#}\mathbb{P} \parallel \pi_{\#}\mathbb{W})}_{\text{grid term}} + \underbrace{\int \text{KL}(\mathbb{P}^y \parallel \mathbb{W}^y) (\pi_{\#}\mathbb{P})(dy)}_{\text{bridge term}}, \quad \mathbb{P}^y := \mathbb{P}(\cdot \mid \pi = y). \quad (4.2)$$

On the admissible set $\{\pi_{\#}\mathbb{P} = \mu\}$, the grid term is frozen to the constant $\text{KL}(\mu \parallel \mu_{\mathcal{T}}^{\mathbb{W}})$. Heuristically, the bridge term is minimized pointwise, with no coupling to solve (as we verify rigorously below) by $\mathbb{P}^y = \mathbb{W}^y$, i.e. Brownian bridges. This should be contrasted with the classical two-marginal SBP, where disintegrating along (X_0, X_T) leaves the endpoint term $\text{KL}(\pi \parallel \mathbb{Q}_{0,T})$ with π free over the couplings $\Pi(\mu_0, \mu_T)$: this residual entropic optimal transport problem has no closed form in general and is solved by Sinkhorn iterations. The difference is that the SBTS constraint fixes the full joint law of the observed coordinates, absorbing all the variability of the endpoint term, whereas two marginal constraints do not.

Now, we verify rigorously the above heuristic. Assume that

$$\text{KL}(\mu \mid \mu_{\mathcal{T}}^{\mathbb{W}}) < \infty.$$

In particular, $\mu \ll \mu_{\mathcal{T}}^{\mathbb{W}}$. Let

$$g := \frac{d\mu}{d\mu_{\mathcal{T}}^{\mathbb{W}}}.$$

Define a measure $\mathbb{P}^{\star}$ on Ω by

$$\boxed{\frac{d\mathbb{P}^{\star}}{d\mathbb{W}}(\omega) = g(\pi(\omega))}. \quad (4.3)$$

This measure is a probability measure, since

$$\begin{aligned} \mathbb{P}^{\star}(\Omega) &= \int_{\Omega} g(\pi(\omega)) \mathbb{W}(d\omega) \\ &= \int_{(\mathbb{R}^d)^N} g(y) \mu_{\mathcal{T}}^{\mathbb{W}}(dy) = 1. \end{aligned}$$

Moreover, for every Borel set $B \subseteq (\mathbb{R}^d)^N$,

$$\begin{aligned} (\pi_{\#}\mathbb{P}^{\star})(B) &= \mathbb{P}^{\star}(\pi^{-1}(B)) \\ &= \int_{\Omega} \mathbf{1}_B(\pi(\omega)) g(\pi(\omega)) \mathbb{W}(d\omega) \\ &= \int_B g(y) \mu_{\mathcal{T}}^{\mathbb{W}}(dy) \\ &= \mu(B). \end{aligned}$$

Therefore,

$$\pi_{\#}\mathbb{P}^{\star} = \mu,$$

so that $\mathbb{P}^*$ is admissible.

We also obtain

$$\begin{aligned} \text{KL}(\mathbb{P}^* \mid \mathbb{W}) &= \int_{\Omega} g(\pi(\omega)) \log g(\pi(\omega)) \mathbb{W}(d\omega) \\ &= \int_{(\mathbb{R}^d)^N} g(y) \log g(y) \mu_{\mathcal{T}}^{\mathbb{W}}(dy) \\ &= \text{KL}(\mu \mid \mu_{\mathcal{T}}^{\mathbb{W}}). \end{aligned}$$

The disintegration theorem (Theorem 5.3.1 in [AGS08]) then gives the existence of $((\mathbb{W})^y)_{y \in (\mathbb{R}^d)^N}$, a regular conditional distribution of $\mathbb{W}$ given $\pi = y$ such that

$$(\mathbb{W})^y(\pi^{-1}(\{y\})) = 1 \quad \text{for } \mu_{\mathcal{T}}^{\mathbb{W}}\text{-almost every } y.$$

Since $\mu \ll \mu_{\mathcal{T}}^{\mathbb{W}}$, this also holds μ -almost every y . Disintegrating (4.3) with respect to π yields, for every $A \in \mathcal{F}$,

$$\begin{aligned} \mathbb{P}^*(A) &= \int_{\Omega} \mathbf{1}_A(\omega) g(\pi(\omega)) \mathbb{W}(d\omega) \\ &= \int_{(\mathbb{R}^d)^N} g(y) (\mathbb{W})^y(A) \mu_{\mathcal{T}}^{\mathbb{W}}(dy) \\ &= \int_{(\mathbb{R}^d)^N} (\mathbb{W})^y(A) \mu(dy). \end{aligned}$$

Thus,

$$\mathbb{P}^* = \int_{(\mathbb{R}^d)^N} (\mathbb{W})^y \mu(dy), \tag{4.4}$$

and the conditional laws of $\mathbb{P}^*$ given the grid values satisfy

$$(\mathbb{P}^*)^y = (\mathbb{W})^y \quad \text{for } \mu\text{-almost every } y.$$

Now let $\mathbb{P}$ be any admissible probability measure, so that $\pi_{\#}\mathbb{P} = \mu$, and let $(\mathbb{P}^y)_{y \in (\mathbb{R}^d)^N}$ be a regular conditional distribution of $\mathbb{P}$ given $\pi = y$.

Now, by using Equation (4.2), since relative entropy is nonnegative,

$$\text{KL}(\mathbb{P} \mid \mathbb{W}) \geq \text{KL}(\mu \mid \mu_{\mathcal{T}}^{\mathbb{W}}) = \text{KL}(\mathbb{P}^* \mid \mathbb{W}).$$

Hence $\mathbb{P}^*$ is a minimizer. Equality holds if and only if

$$\mathbb{P}^y = (\mathbb{W})^y \quad \text{for } \mu\text{-almost every } y.$$

In that case, for every $A \in \mathcal{F}$,

$$\mathbb{P}(A) = \int_{(\mathbb{R}^d)^N} \mathbb{P}^y(A) \mu(dy) = \int_{(\mathbb{R}^d)^N} (\mathbb{W})^y(A) \mu(dy) = \mathbb{P}^*(A).$$

Consequently, $\mathbb{P} = \mathbb{P}^*$, which proves that $\mathbb{P}^*$ is the unique minimizer.

Hence, the measure $\mathbb{P}^*$ is the unique solution of the Schrödinger Bridge Problem for time series. However, from a generative modeling point of view, we want to be able to sample from the measure μ . Hence, we need to obtain a SDE whose solution is a process which has law $\mathbb{P}^*$ on the path space, but the measure $\mathbb{P}^*$ itself is not of much help for us. The main result of [HHP23] (Theorem 3.1) gives us such a SDE.

Theorem 4.4. Assume that $\text{KL}(\mu|\mu_{\mathcal{T}}^{\mathbb{W}}) < \infty$. The diffusion process $X_t = \int_0^t \alpha_s^* ds + W_t$, $0 \leq t \leq T$, with α^* defined as

$$\alpha_t^* = a^*(t, X_t; (X_{t_i})_{t_i \leq t}), \quad 0 \leq t < T,$$

with $a^*(t, x; \mathbf{x}_i)$, for $t \in [t_i, t_{i+1})$, $\mathbf{x}_i = (x_1, \dots, x_i) \in (\mathbb{R}^d)^i$, $x \in \mathbb{R}^d$, given by

$$a^*(t, x; \mathbf{x}_i) = \nabla_x \ln \mathbb{E}_{\mathbb{W}} \left[\frac{\mu}{\mu_{\mathcal{T}}^{\mathbb{W}}} (X_{t_1}, \dots, X_{t_N}) \mid \mathbf{X}_{t_i} = \mathbf{x}_i, X_t = x \right],$$

where we set $\mathbf{X}_{t_i} = (X_{t_1}, \dots, X_{t_i})$, induces a probability measure

$$\mathbb{P}^* = \frac{\mu}{\mu_{\mathcal{T}}^{\mathbb{W}}} (X_{t_1}, \dots, X_{t_N}) \mathbb{W},$$

which solves the Schrödinger bridge time series problem. Moreover, we have:

$$V_{SBTS} = \text{KL}(\mathbb{P}^*|\mathbb{W}) = \text{KL}(\mu|\mu_{\mathcal{T}}^{\mathbb{W}}).$$

Moreover, the optimal drift also takes the expression:

$$a^*(t, x; \mathbf{x}_i) = \frac{1}{t_{i+1} - t} \frac{\mathbb{E}_{\mu}[(X_{t_{i+1}} - x) F_i(t, X_{t_i}, x, X_{t_{i+1}}) \mid \mathbf{X}_{t_i} = \mathbf{x}_i]}{\mathbb{E}_{\mu}[F_i(t, X_{t_i}, x, X_{t_{i+1}}) \mid \mathbf{X}_{t_i} = \mathbf{x}_i]} \quad (4.5)$$

for $t \in [t_i, t_{i+1})$, $\mathbf{x}_i = (x_1, \dots, x_i) \in (\mathbb{R}^d)^i$, $x \in \mathbb{R}^d$, where

$$F_i(t, x_i, x, x_{i+1}) = \exp \left(-\frac{\|x_{i+1} - x\|^2}{2(t_{i+1} - t)} + \frac{\|x_{i+1} - x_i\|^2}{2(t_{i+1} - t_i)} \right).$$

Remark: The following fact will be used throughout the proof. Let $0 \leq t < t' \leq T$. Writing $X_{t'} = X_t + (X_{t'} - X_t)$, the increment $X_{t'} - X_t$ is, under $\mathbb{W}$, independent of $\mathcal{F}_t$ and distributed as $\mathcal{N}(0, (t' - t)I_d)$ (X is a Brownian motion under the Wiener measure). Consequently, the conditional law of $X_{t'}$ given $\mathcal{F}_t$ is $\mathcal{N}(X_t, (t' - t)I_d)$, with density $p_{t'-t}(X_t, \cdot)$; in particular it depends on $\mathcal{F}_t$ only through X_t . More generally, for $t \in [t_i, t_{i+1})$, iterating this one-step description shows that the conditional law of the vector $(X_{t_{i+1}}, \dots, X_{t_N})$ given $\mathcal{F}_t$ has density

$$(x_{i+1}, \dots, x_N) \longmapsto p_{t_{i+1}-t}(X_t, x_{i+1}) \prod_{j=i+1}^{N-1} p_{t_{j+1}-t_j}(x_j, x_{j+1}),$$

the product structure reflecting the independence of the successive increments. The same description applies to a standard Brownian motion B started at 0: the vector $(B_{t_{i+1}-t}, \dots, B_{t_N-t})$ has density $p_{t_{i+1}-t}(0, x_{i+1}) \prod_j p_{t_{j+1}-t_j}(x_j, x_{j+1})$.

The following lemma will also be used in the proof:

Lemma 4.5 (Independence (freezing) lemma; [Shr04]). Let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -algebra, U a $\mathcal{G}$ -measurable random vector, V a random vector independent of $\mathcal{G}$, and $f \geq 0$ measurable. Define the deterministic function $g(u) := \mathbb{E}[f(u, V)]$, where the expectation is taken with respect to the joint law of V . Then $\mathbb{E}[f(U, V) \mid \mathcal{G}] = g(U)$ a.s.

While the original proof provided by the authors is concise, we offer here a more explicit derivation with detailed intermediate steps. The structure of the proof is as follows: we construct a measure $\mathbb{P}^*$ on $(\Omega, \mathcal{F})$ and then exhibit a $\mathbb{P}^*$ -Brownian motion turning X into the announced diffusion. We write $\rho := \frac{\mu}{\mu_{\mathcal{T}}^{\mathbb{W}}}$, set $\mathbf{X}_{t_i} := (X_{t_1}, \dots, X_{t_i})$, and denote by $p_s(x, y) := (2\pi s)^{-d/2} e^{-\|y-x\|^2/2s}$ the Gaussian kernel. For $a, b \in \mathbb{R}^d$, $a \cdot b$ denotes the Euclidean inner product.

Proof: Step 1 (Construction of the density process of $\mathbb{P}^$)* Not that thanks to Paragraph 4.1.2, we know that the density of $\mathbb{P}^*$ satisfies:

$$Z_t := \mathbb{E}_{\mathbb{W}} \left[\frac{d\mathbb{P}^*}{d\mathbb{W}} \middle| \mathcal{F}_t \right] = \mathbb{E}_{\mathbb{W}} \left[\rho(X_{t_1}, \dots, X_{t_N}) \middle| \mathcal{F}_t \right] \quad 0 \leq t \leq T,$$

is a uniformly integrable $(\mathbb{F}, \mathbb{W})$ -martingale by the tower property, for which we choose the continuous version (every martingale of a Brownian filtration admits one by Itô's representation theorem). Note $Z_0 = 1$ ($\mathcal{F}_0$ is trivial, $X_0 = 0$) and $Z_T = \rho(X_{t_1}, \dots, X_{t_N})$ (as $\mathcal{F} = \mathcal{F}_T$).

Step 2 (Functional form of Z via the freezing lemma). Fix $i \in \{0, \dots, N-1\}$ and $t \in [t_i, t_{i+1})$. Write, for $j \geq i+1$, $X_{t_j} = X_t + V_j$ with $V_j := X_{t_j} - X_t$. The vector $V := (V_{i+1}, \dots, V_N)$ is a function of the increments of X after time t ; since X is a $\mathbb{W}$ -Brownian motion, V is independent of $\mathcal{F}_t$ and centered Gaussian with $\text{Cov}(V_j, V_k) = (t_{j \wedge k} - t)I_d$, i.e. V has the law of $(B_{t_{i+1}-t}, \dots, B_{t_N-t})$ for a standard Brownian motion B .

Applying Lemma 4.5 with $\mathcal{G} = \mathcal{F}_t$, $U = (\mathbf{X}_{t_i}, X_t)$, the independent vector V , and $f((\mathbf{x}_i, x), v) := \rho(\mathbf{x}_i, x + v_{i+1}, \dots, x + v_N)$, we get $Z_t = h_i(t, X_t; \mathbf{X}_{t_i})$, where

$$\begin{aligned} h_i(t, x; \mathbf{x}_i) &:= \mathbb{E}[\rho(\mathbf{x}_i, x + B_{t_{i+1}-t}, \dots, x + B_{t_N-t})] \\ &= \int_{(\mathbb{R}^d)^{N-i}} \rho(\mathbf{x}_i, x_{i+1}, \dots, x_N) p_{t_{i+1}-t}(x, x_{i+1}) \prod_{j=i+1}^{N-1} p_{t_{j+1}-t_j}(x_j, x_{j+1}) dx_{i+1} \cdots dx_N. \end{aligned} \quad (4.6)$$

where the second equality comes from the remark about the increments of Brownian motion.

Step 3 (One-step recursion, by Fubini–Tonelli). With the terminal convention $h_N(t_N, x; \mathbf{x}_{N-1}, x) := \rho(x_1, \dots, x_{N-1}, x)$, we claim that, for every $(t, x, \mathbf{x}_i)$,

$$h_i(t, x; \mathbf{x}_i) = \int_{\mathbb{R}^d} h_{i+1}(t_{i+1}, y; \mathbf{x}_i, y) p_{t_{i+1}-t}(x, y) dy. \quad (4.7)$$

Indeed, plugging the integral form (4.6) of h_{i+1} (with $x_{i+1} = y$) into the right-hand side and applying Fubini–Tonelli (the integrand is nonnegative) reassembles exactly the right-hand side of (4.6).

Step 4 (The Gaussian kernel solves the heat equation). From $\ln p_s(x, y) = -\frac{d}{2} \ln(2\pi s) - \frac{\|y-x\|^2}{2s}$ we compute

$$\partial_s p_s = p_s \left(\frac{\|y-x\|^2}{2s^2} - \frac{d}{2s} \right), \quad \nabla_x p_s = p_s \frac{y-x}{s}, \quad \Delta_x p_s = p_s \left(\frac{\|y-x\|^2}{s^2} - \frac{d}{s} \right),$$

whence $\partial_s p_s = \frac{1}{2} \Delta_x p_s$. Since in (4.7) the kernel appears with $s = t_{i+1} - t$ and $\partial_t = -\partial_s$, the kernel satisfies the backward equation $\partial_t p_{t_{i+1}-t}(x, y) + \frac{1}{2} \Delta_x p_{t_{i+1}-t}(x, y) = 0$.

Lemma 4.6 (Gaussian smoothing). *Let $g : \mathbb{R}^d \rightarrow [0, \infty)$ be measurable, and suppose there exist $a \in \mathbb{R}^d$ and $\sigma^2 > 0$ such that $M := \int g(y) e^{-\|y-a\|^2/2\sigma^2} dy < \infty$. Then $u(s, x) := \int g(y) p_s(x, y) dy$ is finite and of class C^∞ on $(0, \sigma^2) \times \mathbb{R}^d$, all derivatives may be taken under the integral sign, and $\partial_s u = \frac{1}{2} \Delta_x u$ there.*

Proof: Every (s, x) -partial derivative of $p_s(x, y)$ is of the form $q(\frac{y-x}{s}, \frac{1}{s}) p_s(x, y)$ with q a polynomial. Fix a compact $K = [s_0, s_1] \times \overline{B}_R$ with $0 < s_0 \leq s_1 < \sigma^2$. For $\varepsilon \in (0, 1)$, the inequality $\|y-x\|^2 \geq (1-\varepsilon)\|y-a\|^2 - C(\varepsilon, R, a)$, valid for $\|x\| \leq R$, gives $p_s(x, y) \leq C_1 e^{-(1-\varepsilon)\|y-a\|^2/2s_1}$ on K . Choosing ε small enough that $(1-\varepsilon)/s_1 > 1/\sigma^2$, the exponential absorbs any polynomial factor: for all $(s, x) \in K$,

$$\left| q\left(\frac{y-x}{s}, \frac{1}{s}\right) \right| p_s(x, y) \leq C_K e^{-\|y-a\|^2/2\sigma^2}.$$

Hence every differentiated integrand is dominated on K by $C_K g(y) e^{-\|y-a\|^2/2\sigma^2} \in L^1(dy)$, and iterated differentiation under the integral sign applies. The identity $\partial_s u = \frac{1}{2} \Delta_x u$ then follows from $\partial_s p_s = \frac{1}{2} \Delta_x p_s$ (Step 4) by linearity. $\square$

Step 5 (Regularity, positivity, and the PDE for h_i). These three properties are exactly the hypotheses needed in Step 6: $C^{1,2}$ is the minimal regularity required by Itô's formula, strict positivity legitimises $\nabla_x \ln h_i$, and the PDE will cancel the dt -term. All three follow from the recursion (4.7), in which the (t, x) -dependence sits entirely in the kernel, $g(y) := h_{i+1}(t_{i+1}, y; \mathbf{x}_i, y) \geq 0$ being fixed. Apply Lemma 4.6 with $\sigma^2 = t_{i+1} - t_i$, $a = x_i$, and $M = h_i(t_i, x_i; \mathbf{x}_i) = Z_{t_i} < \infty$ (finite for μ^W -a.e. prefix $\mathbf{x}_i$, which suffices since h_i is only ever evaluated at $\mathbf{X}_{t_i}$): since $s = t_{i+1} - t$ ranges over $(0, t_{i+1} - t_i)$ as t ranges over (t_i, t_{i+1}) , the function $h_i(\cdot, \cdot; \mathbf{x}_i)$ is C^∞ on $(t_i, t_{i+1}) \times \mathbb{R}^d$ and satisfies there

$$\partial_t h_i + \frac{1}{2} \Delta_x h_i = 0$$

No regularity at the endpoints $t \in \{t_i, t_{i+1}\}$ is needed: Itô's formula will be applied on compact subintervals $[t_i + \varepsilon, u] \subset (t_i, t_{i+1})$ and the endpoints recovered by the a.s. continuity of the martingale Z . Finally, strict positivity needs to be established only for the relevant prefixes. Since

$$Z_{t_i} = h_i(t_i, X_{t_i}; \mathbf{X}_{t_i}) = \mathbb{E}_{\mathbb{W}}[Z_T \mid \mathcal{F}_{t_i}],$$

we have

$$\begin{aligned} \mathbb{P}^*(Z_{t_i} = 0) &= \mathbb{E}_{\mathbb{W}}[Z_T \mathbf{1}_{\{Z_{t_i}=0\}}] \\ &= \mathbb{E}_{\mathbb{W}}[Z_{t_i} \mathbf{1}_{\{Z_{t_i}=0\}}] = 0. \end{aligned}$$

Moreover, $Z_{t_i} < +\infty$ $\mathbb{P}^*$ -almost surely. Therefore, for $\mathbb{P}^*$ -almost every realized prefix $\mathbf{X}_{t_i}$,

$$0 < h_i(t_i, X_{t_i}; \mathbf{X}_{t_i}) < +\infty.$$

For every such prefix, the non-negative function

$$y \longmapsto h_{i+1}(t_{i+1}, y; \mathbf{X}_{t_i}, y)$$

appearing in the recursion (4.7) is not Lebesgue-almost everywhere zero. Since the Gaussian kernel is strictly positive, it follows that

$$h_i(t, x; \mathbf{X}_{t_i}) > 0 \quad \text{for every } (t, x) \in (t_i, t_{i+1}) \times \mathbb{R}^d.$$

Consequently, $\nabla_x \log h_i$ is well defined along the canonical process, $dt \otimes \mathbb{P}^*$ -almost everywhere.

Step 6 (Itô's formula and the stochastic exponential). Fix $t_i < t_i + \varepsilon \leq u < t_{i+1}$. We apply Itô's formula on $[t_i + \varepsilon, u]$ to the deterministic function $(t, x) \mapsto h_i(t, x; \mathbf{x}_i)$, which is at least of class $C^{1,2}$ by Step 5, along the process (t, X_t) , with the parameter frozen at the $\mathcal{F}_{t_i}$ -measurable value $\mathbf{x}_i = \mathbf{X}_{t_i}$. Since $d\langle X^k, X^l \rangle_t = \delta_{kl} dt$ under $\mathbb{W}$,

$$dZ_t = \left(\partial_t h_i + \frac{1}{2} \Delta_x h_i \right)(t, X_t; \mathbf{X}_{t_i}) dt + \nabla_x h_i(t, X_t; \mathbf{X}_{t_i}) \cdot dX_t = \nabla_x h_i(t, X_t; \mathbf{X}_{t_i}) \cdot dX_t,$$

the dt -term vanishing by the backward heat equation. By the strict positivity of h_i we may write

$$dZ_t = Z_t \nabla_x \ln h_i(t, X_t; \mathbf{X}_{t_i}) \cdot dX_t = Z_t \alpha_t^* \cdot dX_t, \quad \alpha_t^* := \alpha^*(t, X_t; \mathbf{X}_{t_i}) = \nabla_x \ln h_i(t, X_t; \mathbf{X}_{t_i}).$$

Letting $u \uparrow t_{i+1}$, $\varepsilon \downarrow 0$, and using the a.s. continuity of Z together with the terminal condition $h_i(t_{i+1}, x; \mathbf{x}_i) = h_{i+1}(t_{i+1}, x; \mathbf{x}_i, x)$ from (4.7), the identity glues across the grid dates, so that on $[0, T]$, Z is the stochastic exponential

$$Z_t = \mathcal{E} \left(\int_0^t \alpha_s^* \cdot dX_s \right)_t = \exp \left(\int_0^t \alpha_s^* \cdot dX_s - \frac{1}{2} \int_0^t |\alpha_s^*|^2 ds \right), \quad \text{in particular } \frac{d\mathbb{P}^*}{d\mathbb{W}} = Z_T.$$

Step 7 (Canonical weak representation) The process Z constructed in Step 1 is the uniformly integrable density process of $\mathbb{P}^*$ with respect to $\mathbb{W}$. Moreover, Step 6 gives:

$$d\langle Z, X \rangle_t = Z_t \alpha_t^* dt$$

on each interval (t_i, t_{i+1}) . Since $Z_t > 0$, $dt \otimes \mathbb{P}^*$ -almost everywhere, Girsanov's theorem (see Theorem 2.14) yields that

$$W_t := X_t - \int_0^t \frac{1}{Z_s} d\langle Z, X \rangle_s = X_t - \int_0^t \alpha_s^* ds$$

is a Brownian motion under $\mathbb{P}^*$. Consequently, under $\mathbb{P}^*$, the canonical process satisfies

$$X_t = \int_0^t \alpha_s^* ds + W_t.$$

Thus, the canonical process under $\mathbb{P}^*$ provides a weak solution with the announced drift.

By Theorem 2 of [Leh11] applied to $\mathbb{P}^*$, there exists a Föllmer drift u^* such that

$$X_t = \int_0^t u_s^* ds + W_t$$

under $\mathbb{P}^*$, and

$$\frac{1}{2} \mathbb{E}_{\mathbb{P}^*} \left[\int_0^T |u_t^*|^2 dt \right] = \text{KL}(\mathbb{P}^* \mid \mathbb{W}).$$

On the other hand, the preceding steps show that

$$X_t = \int_0^t \alpha_s^* ds + \tilde{W}_t,$$

where $\tilde{W}$ is also a $\mathbb{P}^*$ -Brownian motion. By uniqueness of the semimartingale decomposition,

$$\alpha^* = u^* \quad dt \otimes \mathbb{P}^*\text{-almost everywhere.}$$

Consequently,

$$J(\alpha^*) = \frac{1}{2} \mathbb{E}_{\mathbb{P}^*} \left[\int_0^T |\alpha_t^*|^2 dt \right] = \text{KL}(\mathbb{P}^* \mid \mathbb{W}) = \text{KL}(\mu \mid \mu_{\mathcal{T}}^{\mathbb{W}}) < +\infty.$$

Therefore, α^* is an admissible finite-energy control. The grid constraint, the optimality and the uniqueness of $\mathbb{P}^*$ have already been established in Paragraph 4.1.2.

Step 8 (Equivalent expression for the drift). By (4.6) and $\rho = \mu / \mu_{\mathcal{T}}^W$,

$$h_i(t, x; \mathbf{x}_i) = \int_{(\mathbb{R}^d)^{N-i}} \frac{\mu(x_1, \dots, x_N)}{\mu_{\mathcal{T}}^W(x_1, \dots, x_N)} p_{t_{i+1}-t}(x, x_{i+1}) \prod_{j=i+1}^{N-1} p_{t_{j+1}-t_j}(x_j, x_{j+1}) dx_{i+1} \cdots dx_N. \quad (4.8)$$

Write $\mu_{\mathcal{T}}^W(x_1, \dots, x_N) = \prod_{k=0}^{N-1} (2\pi\Delta t_k)^{-d/2} \exp(-\frac{\|x_{k+1}-x_k\|^2}{2\Delta t_k})$, and note $p_{t_{j+1}-t_j}(x_j, x_{j+1}) = (2\pi\Delta t_j)^{-d/2} \exp(-\frac{\|x_{j+1}-x_j\|^2}{2\Delta t_j})$. Collecting the Gaussian normalisation constants into C , the integrand of (4.8) equals

$$C \mu(x_1, \dots, x_N) \underbrace{\exp\left(-\frac{\|x_{i+1}-x\|^2}{2(t_{i+1}-t)}\right)}_{\text{step } t \rightarrow t_{i+1}} \prod_{k=0}^{N-1} \exp\left(+\frac{\|x_{k+1}-x_k\|^2}{2\Delta t_k}\right) \prod_{j=i+1}^{N-1} \exp\left(-\frac{\|x_{j+1}-x_j\|^2}{2\Delta t_j}\right).$$

Three groups of exponential factors combine:

- (i) the *future* grid steps $k = j = i + 1, \dots, N - 1$ cancel exactly (the $\mathbb{W}$ -transition between two future grid dates coincides with the corresponding factor of $\mu_{\mathcal{T}}^W$);
- (ii) the *past* steps $k = 0, \dots, i - 1$ involve only $x_1, \dots, x_i$, hence are absorbed into C ;
- (iii) the *junction* step $k = i$ survives and combines with the $t \rightarrow t_{i+1}$ factor to give

$$\exp\left(-\frac{\|x_{i+1}-x\|^2}{2(t_{i+1}-t)} + \frac{\|x_{i+1}-x_i\|^2}{2(t_{i+1}-t_i)}\right) = F_i(t, x_i, x, x_{i+1}),$$

which does not cancel precisely because the conditioning is at the intermediate time t (position x), not at t_i .

Hence, with $\mu_i(x_1, \dots, x_i) := \int \mu(x_1, \dots, x_N) dx_{i+1} \cdots dx_N$ the marginal density of $(X_{t_1}, \dots, X_{t_i})$ under μ ,

$$h_i(t, x; \mathbf{x}_i) = C \int F_i(t, x_i, x, x_{i+1}) \mu(x_1, \dots, x_N) dx_{i+1} \cdots dx_N = C \mu_i(\mathbf{x}_i) \int F_i \frac{\mu(x_1, \dots, x_N)}{\mu_i(\mathbf{x}_i)} dx_{i+1} \cdots dx_N.$$

Since $\frac{\mu(x_1, \dots, x_N)}{\mu_i(\mathbf{x}_i)}$ is the conditional density of $(X_{t_{i+1}}, \dots, X_{t_N})$ given $\mathbf{X}_{t_i} = \mathbf{x}_i$ under μ , and absorbing $\mu_i(\mathbf{x}_i)$ (a function of $\mathbf{x}_i$ only) into C ,

$$h_i(t, x; \mathbf{x}_i) = C(t, \mathbf{x}_i) \mathbb{E}_{\mu}[F_i(t, x_i, x, X_{t_{i+1}}) \mid \mathbf{X}_{t_i} = \mathbf{x}_i]. \quad (4.9)$$

Step 9 (Differentiation). The factor $C(t, \mathbf{x}_i)$ in (4.9) does not depend on x , so it disappears under $\nabla_x \ln$:

$$a^*(t, x; \mathbf{x}_i) = \nabla_x \ln h_i = \frac{\nabla_x \mathbb{E}_{\mu}[F_i(t, x_i, x, X_{t_{i+1}}) \mid \mathbf{X}_{t_i} = \mathbf{x}_i]}{\mathbb{E}_{\mu}[F_i(t, x_i, x, X_{t_{i+1}}) \mid \mathbf{X}_{t_i} = \mathbf{x}_i]}.$$

Only the first term of the exponent of F_i depends on x , and

$$\nabla_x F_i(t, x_i, x, x_{i+1}) = \frac{x_{i+1} - x}{t_{i+1} - t} F_i(t, x_i, x, x_{i+1}).$$

Interchanging ∇_x and $\mathbb{E}_{\mu}$ (differentiation under the integral sign, justified as in Lemma 4.6) gives

$$\nabla_x \mathbb{E}_{\mu}[F_i \mid \mathbf{X}_{t_i} = \mathbf{x}_i] = \frac{1}{t_{i+1} - t} \mathbb{E}_{\mu}[(X_{t_{i+1}} - x) F_i(t, x_i, x, X_{t_{i+1}}) \mid \mathbf{X}_{t_i} = \mathbf{x}_i],$$

and dividing by the denominator yields (4.5). □

Remark (Choice of the reference measure): The proof justifies the choice of the Wiener measure as the reference. First, from a computational perspective, the Wiener measure provides explicit Gaussian transition kernels, which are essential to compute the terms appearing in the theorem. Second, while other processes like the Ornstein-Uhlenbeck process also offer explicit Gaussian transitions, the Wiener measure is typically preferred because it has strictly zero drift. Consequently, it imposes no a priori structural bias (such as mean-reversion) on the physical dynamics.

4.1.3 Practical challenges

Estimating the conditional density

The previous theoretical result suggests a practical way to sample from an approximation of the data distribution. By running an Euler-Maruyama discretization scheme at the grid times $t_1, \dots, t_N$, one can simulate approximate trajectories of the optimal diffusion, provided that the exact drift expression is replaced by a suitable estimator. One classical nonparametric method to do so is the Nadaraya-Watson estimator.

We estimate $m(x) = \mathbb{E}[Y \mid X = x]$. Writing $f(x, y)$ for the joint density of (X, Y) and f_X for its marginal, the definition of the conditional density gives:

$$m(x) = \int_{\mathbb{R}} y \frac{f(x, y)}{f_X(x)} dy$$

Given an i.i.d. sample $(x_i, y_i)_{i=1}^n$, plug in the density estimator $\hat{f}(x, y) = \frac{1}{nh^2} \sum_i K(\frac{x-x_i}{h})K(\frac{y-y_i}{h})$.

Using $\int \frac{1}{h} K(\frac{y-y_i}{h}) dy = 1$, one then obtains:

$$\hat{f}_X(x) = \frac{1}{nh} \sum_i K(\frac{x-x_i}{h}), \quad \int_{\mathbb{R}} y \hat{f}(x, y) dy = \frac{1}{nh} \sum_i y_i K(\frac{x-x_i}{h}).$$

Taking the ratio, we obtain:

$$\hat{m}(x) = \frac{\sum_{i=1}^n K(\frac{x-x_i}{h}) y_i}{\sum_{i=1}^n K(\frac{x-x_i}{h})} = \sum_{i=1}^n w_i(x) y_i, \quad w_i(x) = \frac{K(\frac{x-x_i}{h})}{\sum_j K(\frac{x-x_j}{h})},$$

a locally weighted average of the y_i with weights $w_i(x) \geq 0$, $\sum_i w_i(x) = 1$.

Defining $K_h(x) = \frac{1}{h^d} K(\frac{x}{h})$, we get:

$$\mathbb{E}[Y \mid X = x] \approx \frac{\sum_{i=1}^n K_h(\mathbf{x} - \mathbf{X}_i) \mathbf{Y}_i}{\sum_{i=1}^n K_h(\mathbf{x} - \mathbf{X}_i)} \quad (4.10)$$

In our case, this yields the following approximation for the drift:

$$a(t, x; \mathbf{x}_i) = \frac{1}{t_{i+1} - t} \frac{\sum_{m=1}^M (X_{t_{i+1}}^{(m)} - x) F_i(t, X_{t_i}^{(m)}, x, X_{t_{i+1}}^{(m)}) \tilde{K}_i^m}{\sum_{m=1}^M F_i(t, X_{t_i}^{(m)}, x, X_{t_{i+1}}^{(m)}) \tilde{K}_i^m}$$

where $\tilde{K}_i^m = \prod_{j=1}^i K_h(x_j - X_{t_j}^{(m)})$, for $t \in [t_i, t_{i+1}[$, $\mathbf{x}_i \in (\mathbb{R}^d)^i$, $i \in \{1, \dots, N-1\}$.

(4.11)

Let us now interpret this expression. For a given sample m , the vector $\frac{X_{t_{i+1}}^{(m)} - x}{t_{i+1} - t}$ is a velocity: it points from the current position x toward the sample's next value $X_{t_{i+1}}^{(m)}$, scaled by the time $t_{i+1} - t$ left to reach it. The drift $a(t, x; \mathbf{x}_i)$ is thus a weighted average of these velocities.

The factor $F_i(t, X_{t_i}^{(m)}, x, X_{t_{i+1}}^{(m)}) = \exp\left(-\frac{\|X_{t_{i+1}}^{(m)} - x\|^2}{2(t_{i+1} - t)} + \frac{\|X_{t_{i+1}}^{(m)} - X_{t_i}^{(m)}\|^2}{2(t_{i+1} - t_i)}\right)$ weights samples by the dynamics. Its first term is (proportional to) the Brownian transition density between

x at time t and $X_{t_{i+1}}^{(m)}$ at time t_{i+1} , and favors endpoints reachable from x by the Brownian motion within time $\sqrt{t_{i+1} - t}$. This must be balanced by the second term, since the true law of the paths is not Brownian: it divides the reference transition density of the sample's own increment $X_{t_i}^{(m)} \rightarrow X_{t_{i+1}}^{(m)}$, so that a genuinely large spread between the value at t and the value at t_{i+1} , typical under μ but improbable for a Brownian motion, is not penalized.

Finally, the kernel term $\tilde{K}_i^m = \prod_{j=1}^i K_h(x_j - X_{t_j}^{(m)})$ implements the conditioning on the past path. The event $\{\mathbf{X}_{t_i} = \mathbf{x}_i\} = \{X_{t_1} = x_1, \dots, X_{t_i} = x_i\}$ has zero probability, so it cannot be conditioned on directly: the kernel replaces it by the positive-probability event that each X_{t_j} lies within distance h of x_j . Without it, the average would run over all samples, including those whose past is not similar to the conditioning path $\mathbf{x}_i$. Thanks to the compact support of K_h , any sample departing from $\mathbf{x}_i$ by more than h at even a single date is eliminated, while the retained ones are weighted according to their proximity to $\mathbf{x}_i$.

Choice of the bandwidth h

Note that this method requires no training and relies on estimation only. However, it crucially depends on the choice of the bandwidth h . As noted by the authors of [Alo+25], this choice governs a bias-variance trade-off. A small h yields faithful conditioning (low bias) but few surviving samples, hence a noisy estimate (high variance), degenerating to 0/0 as i grows. A large h gives many samples and low variance, but blurred conditioning (high bias); this is the regime where nearly all samples are used. Crucially, using all samples is not what we want, even though they are all drawn from μ . The required drift is a conditional expectation given the constructed past, $\mathbb{E}_\mu[\cdot \mid \mathbf{X}_{t_i} = \mathbf{x}_i]$, whereas averaging over all samples estimates the unconditional $\mathbb{E}_\mu[\cdot]$. This comes from the fact that the constraint in the problem is on the joint law at times $(t_1, \dots, t_N)$. Dropping the conditioning yields the right marginals but the wrong joint law.

The paper [Alo+25] gives the following procedure regarding the choice of h :

Given a train set $\mathbf{X} = (\mathbf{X}_{t_1}^{(m)}, \dots, \mathbf{X}_{t_N}^{(m)})_{m=1, \dots, M}$, a validation set $\mathbf{Y} = (\mathbf{Y}_{t_1}^{(q)}, \dots, \mathbf{Y}_{t_N}^{(q)})_{q=1, \dots, Q}$, both from real data, and a list of bandwidths $\mathbf{H} = \{h_1, \dots, h_K\}$, we generate L realizations of $\hat{\mathbf{Y}}_{t_N}^{(q)}$ given the first real values of the series $(\mathbf{Y}_{t_1}^{(q)}, \dots, \mathbf{Y}_{t_{N-1}}^{(q)})$ for each q , using 4.11. Then, we choose $h^* \in \mathbf{H}$ such that it minimizes:

$$\text{MSE}_h = \frac{1}{Q} \sum_{q=1}^Q \left| \frac{1}{L} \sum_{l=1}^L \hat{\mathbf{Y}}_{t_N}^{(q,l)} - \mathbf{Y}_{t_N}^{(q)} \right|^2 \quad (4.12)$$

with $\hat{\mathbf{Y}}_{t_N}^{(q,l)}$ the l -th generated realizations of $\hat{\mathbf{Y}}_{t_N}^{(q)}$ given $(\mathbf{Y}_{t_1}^{(q)}, \dots, \mathbf{Y}_{t_{N-1}}^{(q)})$.

Using the average of L realizations instead of just one can be justified as follows. Conditionally on the observed history

$$\mathbf{Y}_{< t_N}^{(q)} := (\mathbf{Y}_{t_1}^{(q)}, \dots, \mathbf{Y}_{t_{N-1}}^{(q)}),$$

the L generated terminal values are independent realizations from the estimated conditional distribution associated with the bandwidth h . Consequently, their empirical average is a Monte Carlo approximation of the corresponding conditional mean:

$$\frac{1}{L} \sum_{l=1}^L \hat{\mathbf{Y}}_{t_N, h}^{(q,l)} \longrightarrow \mathbb{E}_h [\hat{\mathbf{Y}}_{t_N} \mid \mathbf{Y}_{< t_N}^{(q)}] \quad \text{as } L \rightarrow \infty.$$

Moreover, its conditional covariance is divided by L . Since the conditional mean is the optimal predictor under squared-error loss, averaging over several realizations reduces the Monte Carlo variance and makes the validation criterion more representative of the predictive performance associated with the bandwidth h .

Remark. While this ensemble MSE can be a defendable criterion for bandwidth selection, it does assess only point prediction quality, and not the similarity of the distribution obtained with the distribution of the original dataset. To evaluate the generative model, distributional metrics should be monitored alongside MSE. This is done later in the original paper, to verify the quality of the samples once the choice of h is already done.

Statistical Assumptions, Data Scarcity, and Evaluation Metrics

If we want to sample from the distribution p_{data} , which is the distribution that we assume our samples are sampled from, we need our samples to be identically distributed and independent, as these are the core assumptions of generative modeling.

We must precise what samples are used by the authors. If we take for instance the stocks dataset used in the paper (see Section 4.1 in [Alo+25]), it is made of the historical sample of Google daily historical data from 2004 to 2019. The authors hence only have one sample. To circumvent this issue, they divide this single path into overlapping sliding windows of same size. But then, for these windows to be identically distributed, one requires strongly stationary data, meaning that the joint law of the underlying process over each window is the same. However, stock prices are notoriously non-stationary. Hence, the authors transform them into log-returns prior to generation. Nonetheless, log-returns are generally admitted to be only approximately weakly stationary (possessing a time-invariant mean and autocovariance), see [Tsa10].

However, the authors pass over in silence the statistical nature of the M samples. Taking overlapping windows obviously introduces severe serial dependence, fundamentally violating the independent assumption.

One intuitive alternative to restore independence would be to compute the autocorrelation function (ACF) of the returns and space the windows apart by a lag x where the linear correlation vanishes. However, this approach fails in practice for two fundamental reasons. First, zero linear correlation ($\text{Cov}(X, Y) = 0$) does not imply independence. It is known (see for instance, [Con01]) that financial time series are governed by non-linear dependencies, most notably volatility clustering, hence the ACF for squared or absolute returns remains high even for spaced windows. Second, enforcing strictly non-overlapping windows reduces the effective sample size (for instance, extracting independent 24-day trajectories from a decade of daily historical data yields a mere ~ 100 samples). Feeding such a sparse dataset into a high-dimensional kernel estimator makes the method suffering the curse of dimensionality. Hence, the literature accepts the use of overlapping windows.

This dependency creates a critical generative flaw: Theoretically, feeding highly correlated path samples into the kernel estimator should restrict the generated measure to a narrow neighborhood around the single historical realization. However, the authors report highly competitive, state-of-the-art results, which can raise some questions. We conjecture that the answer lies in their choice of evaluation metrics. The authors rely on the Discriminative Score and Predictive Score. Because their model has memorized the dense, overlapping historical sequences, the generated synthetic paths can fool a discriminator network and achieve low prediction errors on the historical dynamics. These scores measure local sequence-to-sequence similarity, which masks the model’s conjectured inability to capture true stochastic diversity.

That being said, to make a rigorous statement of the heuristic discussed above, one should reproduce the results obtained by the authors, and define a metric that captures the fact that

the model essentially memorizes the training data. Only then, one could assert the above discussion. This is deferred to future work.

Method limitations

In addition to the practical challenges raised earlier, there are some foundational limitations to the method.

First, the performance is highly dependent of the choice of the bandwidth h .

Second, as the solution to the SBP for time series is a continuous process, it has a hard time modeling very abrupt variations, as noted by the authors about the Energy dataset (section 4.3).

Third, the framework imposes constant diffusion coefficient, which can be a limiting factor for performances on financial time-series, which are known to have non-constant volatility. It is directly observed by the authors in section 5.2.

Lastly, the method requires finite entropy with respect to Wiener measure.

Further work

The paper [Alo+26] deals with the last two points. It relies on another framework, which mixes Schrödinger Bridge (optimization on the drift only) and Bass martingale transport (optimization on the volatility only, see [BP23]) to get a method that jointly optimizes drift and volatility. In addition, the authors do not solely check that the model has produced realistic data, but give a concrete application. They aim at predicting the direction of the next return with a transformer based model. They then compare the performance of this model with a base training set, and the one of the model with a training set augmented with the time series produced by their method. The augmented training set indeed increases the out-of-sample accuracy (the proportion of correctly predicted signs) as well as the ROC AUC score (which represents the probability that a randomly chosen positive return is assigned a higher predicted probability than a randomly chosen negative return). However, as noted in their appendix C.2.5, when translating these predictions into a trading strategy, the resulting improvement in the Sharpe ratio cannot be regarded as statistically significant due to overlapping bootstrap confidence intervals.

The augmented training set increases the ROC AUC score from 0.497 to 0.521 (section 5.2.5 in [Alo+26]). That can look like a pretty small increase, however in quantitative finance, the signal-to-noise ratio is typically very low, hence a ROC slightly above 0.5 could constitute a valuable and exploitable statistical edge (see [CT08]). That being said, the authors assume no transaction costs, and we conjecture that with transaction costs; this edge would most likely cancel. Nonetheless, to be able to rigorously assert this, one must reproduce the work of the authors, and incorporate transaction costs.

In a more general way, we interpret the fact that the augmented data set provided such a small increase in the following way: for financial time series, generative modeling is better suited to risk management than to return prediction. Generative data augmentation amounts to learning a distribution and resampling from it, and the resulting directional benefit is inherently capped by the market's near-zero signal-to-noise ratio. Whatever the training objective, such a generator is optimized to reproduce the unconditional law of returns, such as volatility clustering, heavy tails or cross-sectional dependence, but this says little about direction. Indeed, the predictable signal lives in the conditional law, which in such a low-signal regime is only a small but not necessarily negligible perturbation of a coin flip. Matching the unconditional distribution therefore gives

no guarantee on the faint conditional signal (“alpha”) that prediction relies on, and a generator that excels on distributional diagnostics may in fact have captured only noise.

Direct conditional forecasting avoids this indirection by modeling $p(y_{t+1} \mid \mathcal{F}_t)$ explicitly, rather than fitting a separate predictor on synthetic paths; yet it meets the same ceiling: its value lies in probabilistic forecasts of the next-step distribution, not its direction. Moreover, non-stationarity increases the difficulty, since a conditional law estimated on past data need not transfer to the future one.

This distributional fidelity, by contrast, aligns naturally with risk-oriented applications: scenario generation, stress testing, and backtesting all exploit it, turning a single historical path into a sampling distribution of performance and helping to mitigate backtest overfitting.

4.2 Generative modeling for protein design

4.2.1 Motivation

The role of machine learning

Proteins are polymers made of amino acids (also referred as residues in the following) which fold in a certain three-dimensional structure, which enables them to perform a wide variety of biological and chemical functions. The ability to design new proteins is therefore important for the development of therapeutics, vaccines, molecular binders, enzymes and biomaterials.

Machine learning contributes to two complementary aspects of protein engineering: structure prediction and protein generation. Structure prediction, of which the Nobel prize winning AlphaFold ([Jum+21]) and its variants are the best-known example, aims to infer the three-dimensional structure of a protein from its amino-acid sequence. More precisely, the sequence strongly constrains the conformational ensemble of the protein and often determines a dominant native fold. Nevertheless, the correspondence is not strictly one-to-one: a protein may adopt several conformations, and many different amino-acid sequences can be compatible with the same fold.

Protein generation addresses another problem. Given a desired structural or functional specification, the goal is to generate protein structures and sequences that are likely to satisfy it. This problem is particularly difficult because the function of a protein depends not only on the global shape of the protein, but also on the precise three-dimensional arrangement of chemically active residues.

Each residue contains a common backbone, $N - C_\alpha - C$ atoms, and a side chain attached to the C_α atom. The physicochemical interactions between these side chains, together with the geometry of the backbone and the surrounding environment, largely determine how the protein folds and what function it performs. In many design problems, a small set of residues whose side chains must occupy a particular spatial arrangement is identified as a *functional motif*. The role of the remainder of the protein is then to form a stable scaffold that supports this motif in the desired geometry.

A common strategy is therefore to separate structure generation from sequence design. First, a model generates the protein backbone without explicitly generating all the side-chain atoms. A second model then searches for amino-acid sequences compatible with the generated backbone. This decomposition separates the continuous geometric search over three-dimensional structures from the combinatorial search over amino-acid sequences, making the overall design problem more tractable.

Diffusion models are particularly well suited to the backbone-generation step ([Wat+23]). Starting from randomly initialized residue frames, they iteratively remove noise until a protein-like backbone is obtained. Their stochastic nature makes it possible to generate many structurally diverse candidates for the same design problem. Moreover, the denoising process can be conditioned on additional information, such as a functional motif or a binding target. In motif-scaffolding applications, the functional motif is kept fixed or strongly constrained while the diffusion model constructs candidate backbones around it. The model thus aims to generate *designable* backbones, that is backbones for which at least one amino-acid sequence is expected to fold into the intended structure. Designability is not guaranteed by the diffusion model alone and must be assessed in subsequent steps.

We display the whole experimental pipeline in the figure below:

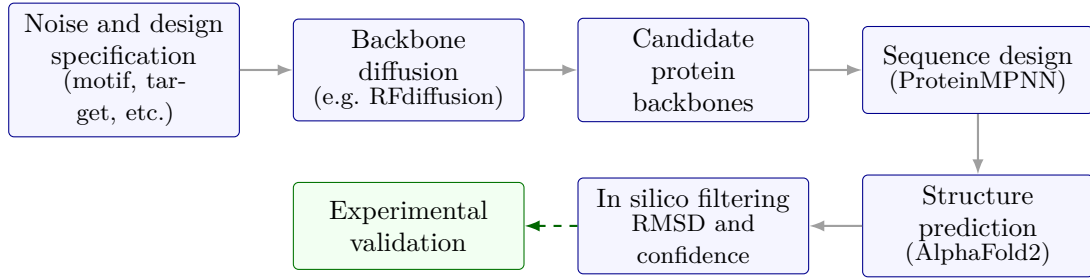


Figure 4.1: Typical computational protein-design pipeline. Multiple backbones can be generated from the same design specification, and multiple amino-acid sequences are generally sampled for each candidate backbone. The resulting designs are filtered computationally before experimental validation.

More precisely, the diffusion model first generates a collection of candidate backbones. For each backbone, ProteinMPNN (see [Dau+22]) samples several amino-acid sequences that are predicted to be compatible with its geometry. AlphaFold, or another structure-prediction model, is then used to predict the structure associated with each candidate sequence. The predicted structure is compared with the backbone produced by the diffusion model using a self-consistency backbone metric such as Root Mean Square Deviation (RMSD). This comparison is generally combined with confidence scores or predicted aligned error, as well as application-specific criteria concerning the functional motif or a binding interface.

A low RMSD and high prediction confidence indicate that the generated backbone and the proposed sequence are computationally self-consistent. They do not, however, prove that the sequence will fold into the desired structure or perform the intended function in the laboratory. Conversely, a large RMSD does not identify a unique source of failure: the generated backbone may be difficult to design, the sampled sequence may be incompatible with it, or the structure-prediction model may be inaccurate for that candidate. The final candidates must therefore be synthesized and tested experimentally.

The need for this computational filtering, hence these machine learning methods becomes clear from the size of the search space. With 20 standard amino-acid types, a protein containing 200 residues admits $20^{200} \approx 1.6 \times 10^{260}$ possible sequences. Exploring even a negligible fraction of this space experimentally is impossible. Machine learning does not search this space exhaustively; instead, backbone diffusion models, inverse-folding models such as ProteinMPNN and structure predictors such as AlphaFold concentrate the computational and experimental effort on a small set of plausible candidates. This reduction of an otherwise intractable search problem is the central motivation for using machine learning in protein design.

Geometrical constraints

In this part, we describe how proteins have been represented (see e.g. [Yim+23]) in the literature to make them well suited to generative modeling. As said before, proteins are chains of amino acids. Hence, we need to represent amino acids geometrically. As discussed before, amino acids contain 3 atoms $N - C_\alpha - C$ and a lateral chain, however the backbones consist only of the atoms $N - C_\alpha - C$. For chemical reasons, the length between each of the 3 atoms is fixed, as well as the angle $\widehat{NC_\alpha C}$. Hence each amino acid backbone can differ by the position of the C_α atom which lives in $\mathbb{R}^3$, and by rotation of its structure.

We can then represent backbones of amino acids in a reference frame, by defining a fictive triangle of reference representing the atoms $N - C_\alpha - C$, which has coordinates:

$$q_{C_\alpha} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad q_C = \begin{pmatrix} \ell_C \\ 0 \\ 0 \end{pmatrix}, \quad q_N = \begin{pmatrix} \ell_N \cos \theta \\ \ell_N \sin \theta \\ 0 \end{pmatrix},$$

where:

$$\ell_C = \|C - C_\alpha\|, \quad \ell_N = \|N - C_\alpha\|, \quad \theta = \widehat{NC_\alpha C}.$$

Thus:

- C_α is placed at the origin;
- the C_α - C bond is aligned with the x-axis;
- the C_α - N bond lies in the xy -plane, at an angle θ from the x-axis.

Any backbone will then be represented relatively with respect to this triangle of reference. More precisely, any backbone is a translation and a rotation of this triangle of reference.

Rotations of an oriented object about the origin in $\mathbb{R}^3$ can be represented by the special orthogonal group

$$SO(3) := \{M \in \mathbb{R}^{3 \times 3} : M^\top M = I_3, \det(M) = 1\}.$$

To understand this definition, first observe that the condition $M^\top M = I_3$ means that M preserves the Euclidean inner product. Indeed, for any $x, y \in \mathbb{R}^3$,

$$(Mx)^\top(My) = x^\top M^\top My = x^\top y.$$

Consequently, M preserves lengths and angles. Moreover, since M is linear, it fixes the origin. It therefore acts as a rigid transformation around the origin.

Equivalently, writing

$$M = \begin{pmatrix} | & | & | \\ m_1 & m_2 & m_3 \\ | & | & | \end{pmatrix},$$

the columns m_1, m_2, m_3 form an orthonormal basis of $\mathbb{R}^3$. Indeed, denoting by e_i the i -th canonical basis vector, we have

$$m_i^\top m_j = (Me_i)^\top(Me_j) = e_i^\top M^\top Me_j = e_i^\top e_j = \delta_{ij}.$$

Thus, M maps the canonical orthonormal frame (e_1, e_2, e_3) to the orthonormal frame (m_1, m_2, m_3) .

The additional condition $\det(M) = 1$ ensures that this new frame has the same orientation as the canonical frame. In other words, it is a right-handed frame satisfying

$$m_3 = m_1 \times m_2.$$

If instead $\det(M) = -1$, the transformation would reverse orientation and would contain a reflection. Therefore, $SO(3)$ consists precisely of the linear isometries of $\mathbb{R}^3$ that fix the origin and preserve orientation; these are exactly the rotations of $\mathbb{R}^3$.

This description also shows that $SO(3)$ has three degrees of freedom. The first column m_1 can be chosen anywhere on the unit sphere S^2 , which gives two degrees of freedom. Once m_1 has been fixed, the second column m_2 must be a unit vector in the plane orthogonal to m_1 . It therefore lies on a unit circle S^1 , giving one additional degree of freedom. Finally, the third column is uniquely determined by

$$m_3 = m_1 \times m_2.$$

Hence, an element of $SO(3)$ is determined by a total of

$$2 + 1 = 3$$

degrees of freedom.

Consequently, the pose of the backbone frame associated with one amino-acid residue can be represented by an element of $SE(3)$. More precisely, a residue frame is represented by

$$T_i = (R_i, t_i), \quad R_i \in SO(3), \quad t_i \in \mathbb{R}^3.$$

As a manifold, $SE(3)$ can be identified with $SO(3) \times \mathbb{R}^3$. However, its group structure is that of the semidirect product

$$SE(3) = SO(3) \ltimes \mathbb{R}^3,$$

with group law

$$(R, t)(R', t') = (RR', t + Rt').$$

A protein backbone containing N residues can therefore be represented by a collection of residue frames

$$\mathbf{T} = (T_1, \dots, T_N) \in SE(3)^N.$$

However, the intrinsic structure of a protein does not depend on its global position or orientation in space. Such a global rigid transformation $g \in SE(3)$ acts simultaneously on all residue frames through the diagonal action

$$g \cdot \mathbf{T} = (gT_1, \dots, gT_N).$$

In particular, if $g = (Q, a)$ and $T_i = (R_i, t_i)$, then

$$gT_i = (QR_i, a + Qt_i).$$

Two backbone configurations $\mathbf{T}, \mathbf{T}' \in SE(3)^N$ are therefore considered equivalent if they differ only by a global rigid transformation:

$$\mathbf{T} \sim \mathbf{T}' \iff \exists g \in SE(3) \text{ such that } \mathbf{T}' = g \cdot \mathbf{T}.$$

The corresponding quotient space is the set of equivalence classes

$$SE(3)^N / SE(3) = \{[\mathbf{T}] : \mathbf{T} \in SE(3)^N\},$$

where

$$[\mathbf{T}] = \{g \cdot \mathbf{T} : g \in SE(3)\}$$

is the orbit of $\mathbf{T}$ under the diagonal action of $SE(3)$.

4.2.2 Adapting the diffusion framework

In the following, $\mathbb{M}^n$ will denote the space of squared matrices of size $n \times n$.

Why the classical Euclidean framework does not apply. Let us focus for now on the case where we want to generate backbones of amino acids, that is elements of $SE(3)$. The manifold $SE(3) = SO(3) \times \mathbb{R}^3$ has dimension 6 but embeds as a submanifold of $\mathbb{M}^3 \times \mathbb{R}^3 \cong \mathbb{R}^{12}$. It is therefore tempting to run the classical diffusion framework in the ambient space $\mathbb{R}^{12}$ and to ignore the constraint. However, a forward process driven by a Brownian motion in $\mathbb{R}^{12}$ started at a point of $SE(3)$ satisfies $\mathbb{P}(B_t \in SE(3)) = 0 \quad \forall t > 0$, since $SE(3)$ has Lebesgue measure zero in $\mathbb{R}^{12}$. Hence, one needs a framework to define diffusion models on manifolds.

The Riemannian forward process. Such a framework is developed in [Bor+22]. The goal of Riemannian score-based generative modelling is to sample from a distribution $X(0) \sim p_0$ supported on a Riemannian manifold M . One first constructs an M -valued forward process $(X(t))_{t \geq 0}$ evolving from p_0 towards an invariant density $p_{\text{inv}}(x) \propto e^{-\Phi(x)}$ according to

$$dX(t) = -\frac{1}{2} \nabla \Phi(X(t)) dt + dB(t)_M, \quad X(0) \sim p_0, \quad (4.13)$$

where $B(t)_M$ is a Brownian motion on M and Φ is a potential.

Recall that the Langevin dynamics (4.13) admits $p_{\text{inv}} \propto e^{-\Phi}$ as its invariant density. Choosing the potential is therefore the same as choosing the prior.

In the following, we follow the approach of [Yim+23].

Brownian motion on a manifold. In this setting, the usual Euclidean definition through independent stationary increments is not valid anymore. Indeed, an increment $X_t - X_s$ requires a vector space structure, which a manifold need not have. However, we can still give the definition via the Martingale problem. It is given in Definition 2.72.

Specialising to $SE(3)$: the choice of a metric. Since Δ_M , hence the Brownian motion, depends on the Riemannian metric, defining a forward process on $SE(3)$ requires choosing an inner product on each tangent space. The authors of [Yim+23] make the following choice.

Proposition 4.7 (Metric on $SE(3)$). *For $T = (r, x) \in SE(3)$ and $(a, y), (a', y') \in \text{Tan}_T SE(3)$, define*

$$\langle (a, y), (a', y') \rangle_{\text{Tan}_T SE(3)} := \langle a, a' \rangle_{\text{Tan}_r SO(3)} + \langle y, y' \rangle_{\mathbb{R}^3}.$$

Then:

- (a) *for any $f \in C^\infty(SE(3))$ and $T = (r, x)$, $\nabla_T f(T) = [\nabla_r f(r, x), \nabla_x f(r, x)]$;*
- (b) *for any $f \in C^\infty(SE(3))$ and $T = (r, x)$, $\Delta_{SE(3)} f(T) = \Delta_{SO(3)} f(r, x) + \Delta_{\mathbb{R}^3} f(r, x)$;*
- (c) *for any $t > 0$, $B(t)_{SE(3)} = [B(t)_{SO(3)}, B(t)_{\mathbb{R}^3}]$ with $B(t)_{SO(3)}$ and $B(t)_{\mathbb{R}^3}$ independent.*

Here (a, y) denotes a *tangent vector*, not a point of $SE(3)$: one has $a \in \text{Tan}_r SO(3)$ and $y \in \text{Tan}_x \mathbb{R}^3 \cong \mathbb{R}^3$.

Remark. The three statements say that, for this choice of metric, the diffusion on $SE(3)$ is the product of the diffusions on $SO(3)$ and $\mathbb{R}^3$ at the level of geometry (a), analysis (b) and probability (c). In particular, point (a) and (c) will be particularly useful in the following.

The two factor metrics. It remains to choose an inner product in $SO(3)$, which is also a Riemannian manifold. It can be viewed as a submanifold of $\mathbb{M}^3 \cong \mathbb{R}^9$, then every tangent space is a linear subspace of $\mathbb{M}^3$, and we take the restriction of the ambient Frobenius product: for $r \in SO(3)$ and $u, v \in \text{Tan}_r SO(3)$,

$$\langle u, v \rangle_{\text{Tan}_r SO(3)} = \frac{1}{2} \text{tr}(u^\top v).$$

Choice of the invariant measure and the resulting forward process. It remains to choose the invariant measure, equivalently the potential Φ . The authors of [Yim+23] take

$$p_{\text{inv}}^{SE(3)}(T) \propto \mathcal{U}_{SO(3)}(r) \mathcal{N}(x; 0, \text{Id}_3),$$

where $\mathcal{U}_{SO(3)}$ is the density of the uniform distribution on $SO(3)$. This density is taken with respect to the Riemannian volume measure (see Definitions 2.54 and 2.62).

From $p_{\text{inv}}^{SE(3)} \propto e^{-\Phi}$ we obtain

$$\Phi(T) = -\log p_{\text{inv}}^{SE(3)}(T) + \text{cst} = \underbrace{-\log \mathcal{U}_{SO(3)}(r)}_{\text{constant in } r} + \frac{\|x\|^2}{2} + \text{cst}.$$

Applying Proposition 4.7(a), we get: $\nabla_{SE(3)}\Phi(r, x) = [0, x]$. Substituting this and Proposition 4.7(c) into (4.13) gives the forward process

$$dT(t) = \left[0, -\frac{1}{2}X(t) \right]dt + [dB(t)_{SO(3)}, dB(t)_{\mathbb{R}^3}]. \quad (4.14)$$

Remark (On the vanishing drift). The asymmetry in the drift of (4.14) has actually a clear interpretation. On $\mathbb{R}^3$ a drift is necessary: a Brownian motion diverges and admits no invariant probability measure; and the drift of the Ornstein–Uhlenbeck process is what produces a normalisable prior. On $SO(3)$ compactness implies that the volume measure is always invariant for Brownian motion.

Remark (What the separable prior buys). Proposition 4.7 alone does not decouple the dynamics. For a general potential Φ the same computation would give

$$\begin{cases} dR(t) = -\frac{1}{2}\nabla_r\Phi(R(t), X(t))dt + dB(t)_{SO(3)}, \\ dX(t) = -\frac{1}{2}\nabla_x\Phi(R(t), X(t))dt + dB(t)_{\mathbb{R}^3}, \end{cases}$$

where the noises are independent but the drifts couple the two components, so that neither marginal process is Markov on its own. What decouples the forward process is the additional modelling choice of a product invariant density, equivalently an additively separable potential. This leads to the fact that the transition density factorise as

$$q(T_t | T_0) = q_{SO(3)}(r_t | r_0) q_{\mathbb{R}^3}(x_t | x_0),$$

which is what allows the score matching loss to split into a sum of two terms.

Reverse-time process and denoising score matching

The reverse-time process then satisfies the following SDE:

$$d\overleftarrow{T}^{(s)} = \left[\left(0, \frac{1}{2}\overleftarrow{X}^{(s)} \right) + \nabla_T \log p_{T_F-s}(T)|_{T=\overleftarrow{T}^{(s)}} \right] ds + d\tilde{B}_{SE(3)}^{(s)}.$$

Here, ∇_T denotes the Riemannian gradient with respect to the variable $T \in SE(3)$. In particular, the score is a tangent vector belonging to $T_T SE(3)$ (see Definition 2.68).

Just as in the Euclidean case, the marginal score $\nabla_T \log p_t(T)$ is unavailable in practice. It is approximated by a score network

$$s_\theta(t, T) \in T_T SE(3),$$

trained by minimizing the denoising score matching loss

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \text{Unif}([0, T_F])} \mathbb{E}_{T^{(0)} \sim p_0} \mathbb{E}_{T^{(t)} \sim p_{t|0}(\cdot | T^{(0)})} \left[\lambda(t) \left\| s_\theta(t, T^{(t)}) - \nabla_T \log p_{t|0}(T | T^{(0)}) \right\|_{T_{T^{(t)}} SE(3)}^2 \right]. \quad (4.15)$$

Using a norm here is valid as recall that any tangent space of a Riemmanian manifold is an Euclidean space, hence carries a norm.

Just as in the $\mathbb{R}^d$ -case, for a score network expressive enough, any population minimizer $\theta^* \in \arg \min_{\theta} \mathcal{L}(\theta)$ satisfies

$$s_{\theta^*}(t, T) = \nabla_T \log p_t(T)$$

for p_t -almost every $T \in SE(3)$.

Now, we need to have access to $\nabla_T \log p_{t|0} \left(T \mid T^{(0)} \right) \Big|_{T=T^{(t)}}$ in order to compute this loss. Note that conditioned on $T^{(0)} = (r_0, x_0)$, $R^{(t)}$ and $X^{(t)}$ are independent, as $R^{(t)}$ depends only on r_0 and $B_{SO(3)}$, $X^{(t)}$ depends only on x_0 and $B_{\mathbb{R}^3}$, and the two Brownian motion are independent. Equivalently:

$$p_{t|0}(r, x | r_0, x_0) = p_{t|0}(r | r_0, x_0) p_{t|0}(x | r_0, x_0) = p_{t|0}(r | r_0) p_{t|0}(x | x_0)$$

Taking logarithms yields:

$$\log(p_{t|0}(r, x | r_0, x_0)) = \log(p_{t|0}(r | r_0)) + \log(p_{t|0}(x | x_0))$$

Now, by point (a) in Proposition 4.7, we have:

$$\nabla_T \log p_{t|0} \left(T \mid T^{(0)} \right) \Big|_{T=T^{(t)}} = [\nabla_R \log p_{t|0} \left(R, X \mid R^{(0)}, X^{(0)} \right) \Big|_{R=R^{(t)}}; \nabla_X \log p_{t|0} \left(R, X \mid R^{(0)}, X^{(0)} \right) \Big|_{X=X^{(t)}}]$$

Now using the fact derived above, we get:

$$\nabla_T \log p_{t|0} \left(T \mid T^{(0)} \right) \Big|_{T=T^{(t)}} = [\nabla_R \log p_{t|0} \left(R \mid R^{(0)} \right) \Big|_{R=R^{(t)}}; \nabla_X \log p_{t|0} \left(X \mid X^{(0)} \right) \Big|_{X=X^{(t)}}]$$

since the other term in both sums are constant in R and X respectively and hence vanish by taking the gradient.

These two quantities can then be computed independently.

For the translation part, we are exactly in the case of the OU process in $\mathbb{R}^3$.

For the rotation part, however, we need to derive the law $p_{t|0} \left(R \mid R^{(0)} \right)$. The authors of [Yim+23] then prove the following result:

Proposition 4.8. *For any $t > 0$, $\omega \in (0, \pi)$, and $r(0), r(t) \in SO(3)$ we have that $p_{t|0}(r(t) \mid r(0)) = \text{IGSO}(3)(r(t); r(0), t)$ given by $\text{IGSO}(3)(r(t); r(0), t) = f(\omega(r(0)^\top r(t)), t)$, where $\omega(r)$ is the rotation angle in radians for any $r \in SO(3)$ — its length in the axis-angle representation — and*

$$f(\omega, t) = \sum_{\ell \in \mathbb{N}} \frac{(2\ell + 1) e^{-\ell(\ell+1)t/2} \sin \left(\left(\ell + \frac{1}{2} \right) \omega \right)}{\sin(\omega/2)}.$$

Note that $r(0)^T r(t)$ is the relative rotation between $r(0)$ and $r(t)$. Indeed:

$$r(0)(r(0)^T r(t)) = r(t)$$

In particular, $r(0)^T r(t)$ is an element of $SO(3)$, as $SO(3)$ is a group and $r(t)^{-1} = r(t)^T$.

Hence, this result amounts to say that the transition density between two rotations $r(0)$ and $r(t)$ depends only on the angle of the relative rotation between them. It is the analog of the Brownian motion in the Euclidean case, where the transition density depends only on the distance $\|x(t) - x(0)\|$ for $x(0)$ and $x(t) \in \mathbb{R}^d$.

Moreover, this formula gives intuition about the behavior of this transition density for large times.

For large times t , since $\left| \frac{\sin((\ell + \frac{1}{2})\omega)}{\sin(\omega/2)} \right| \leq 2\ell + 1$, we have $\sum_{\ell=0}^{\infty} (2\ell + 1)^2 e^{-\ell(\ell+1)t/2} < \infty$ and we can apply Dominated Convergence Theorem to get:

$$\lim_{t \rightarrow \infty} f(\omega, t) = \sum_{\ell=0}^{\infty} \lim_{t \rightarrow \infty} (2\ell + 1) e^{-\ell(\ell+1)t/2} \frac{\sin\left(\left(\ell + \frac{1}{2}\right)\omega\right)}{\sin(\omega/2)}$$

only the term $\ell = 0$ survives in the sum, so that:

$$\lim_{t \rightarrow \infty} f(\omega, t) = 1$$

In other words, at the limit, the transition density becomes uniform, meaning that the forward process forgets its initial distribution, just as in the OU case where the transition density becomes Gaussian independent of the initial distribution.

Doing some computations using the axis-angle representation of $SO(3)$, one can then prove:

Proposition 4.9. *For $t > 0$, $\omega \in (0, \pi)$, and $r(0), r(t) \in SO(3)$, we have*

$$\nabla_{r(t)} \log p_{t|0}(r(t) | r(0)) = r(t) \frac{\text{Log}(r(0, t))}{\omega^{(t)}} \frac{\partial_{\omega} f(\omega^{(t)}, t)}{f(\omega^{(t)}, t)},$$

where

$$r(0, t) = (r(0))^{\top} r(t)$$

is the relative rotation,

$$\omega^{(t)} = \omega(r(0, t))$$

is its rotation angle, and Log denotes the matrix principal logarithm, which is defined as the inverse of the matrix exponential on $SO(3)$ for $\omega \in (0, \pi)$.

If we are able to compute f , which is done in practice by truncating the series and computing the resulting finite sum, and to compute the matrix logarithm which appears in the score expression, we are then able to run the backward process on $SO(3)$. As we already knew how to run the backward process on $\mathbb{R}^3$, and the 2 processes can be run independently, we are able to run the backward process on $SE(3)$. That is, to get a generative model for backbones of amino acids.

4.2.3 From generating amino acids to generating proteins

That being said, we want to model protein backbones, not single amino acids backbones. That is, to generate proteins which consist of N amino acids, we need to define a generative model on $SE(3)^N$. Moreover, as said in the motivation section, proteins are invariant under rotations and translations of their whole structure. To deal with this matter, one idea could be to define a diffusion framework directly on the quotient space $SE(3)^N/SE(3)$. Indeed, this space remains a Riemannian manifold and is isomorphic to $SE(3)^{N-1}$ via the isomorphism (an isomorphism between two manifolds is a diffeomorphism, see Definition 2.51):

$$\begin{aligned}\Phi : SE(3)^N/SE(3) &\longrightarrow SE(3)^{N-1} \\ (T_1, \dots, T_N) &\longmapsto (T_1^{-1}T_2, \dots, T_1^{-1}T_N).\end{aligned}$$

This map is well-defined since for any $g \in SE(3)$:

$$(gT_1)^{-1}(gT_i) = T_1^{-1}g^{-1}gT_i = T_1^{-1}T_i.$$

Its inverse is given by:

$$(S_2, \dots, S_N) \longmapsto [I, S_2, \dots, S_N].$$

Hence, we can fully define forward and backward processes which take values in this quotient space using the framework for general Riemannian manifolds defined above. However, this quotient does not inherit a quotient-group structure from $SE(3)^N$, because $\{(g, \dots, g), g \in SE(3)\}$ is not a normal subgroup of $SE(3)^N$. Consequently, the closed form of the transition law of the Brownian motion in $SO(3)$, which is obtained thanks to the fact that $SO(3)$ is a compact Lie group (Definition 2.58), is not available anymore.

A few solutions can be thought about. The first one is to use Implicit Score Matching, defined for manifolds in [Bor+22], as it not needs the transition density. However, it would require techniques such as the Hutchinson estimator (also extended to the manifold framework in the same article) if N is large. Another solution would be to follow the approach used in [KL26], where the authors consider to use PINN to approximate the heat kernel of diffusions under some quotient spaces. However, their approach focuses for now on quotienting by discrete spaces or finite groups.

Lastly, and it is the path chosen by the authors of [Yim+23], instead of seeing the target measure as a measure on $SE(3)^N/SE(3)$, one can view the target measure as a measure on $SE(3)^N$ invariant under the action of $SE(3)$. That is, one wants to design a measure μ on $SE(3)^N$ such that for any $T_0 \in SE(3)$, and measurable $A \subset SE(3)^N$,

$$\mu(A) = \mu(\{T_0 \cdot T \mid T \in A\}),$$

where for any $T = (T_1, \dots, T_N)$, $T_0 \cdot T = (T_0T_1, \dots, T_0T_N)$.

However: there is no probability measure on $\mathbb{R}^{3N}$ which is invariant under the action of $\mathbb{R}^3$.

We prove this fact. We start by taking $N = 1$ and suppose that μ is a translation-invariant probability measure on $\mathbb{R}^3$, namely

$$\mu(A + a) = \mu(A)$$

for every Borel set $A \subseteq \mathbb{R}^3$ and every $a \in \mathbb{R}^3$.

Let:

$$Q = [0, 1]^3.$$

The sets $(Q + k)_{k \in \mathbb{Z}^3}$ are pairwise disjoint and satisfy

$$\mathbb{R}^3 = \bigsqcup_{k \in \mathbb{Z}^3} (Q + k).$$

By translation invariance,

$$\mu(Q + k) = \mu(Q)$$

for every $k \in \mathbb{Z}^3$. Therefore,

$$1 = \mu(\mathbb{R}^3) = \sum_{k \in \mathbb{Z}^3} \mu(Q).$$

If $\mu(Q) = 0$, the right-hand side is zero, whereas if $\mu(Q) > 0$, it is infinite. Both cases contradict $\mu(\mathbb{R}^3) = 1$. Hence, no translation-invariant probability measure exists on $\mathbb{R}^3$.

We now consider the general case. The diagonal action of $\mathbb{R}^3$ on $\mathbb{R}^{3N}$ is defined by

$$a \cdot (x_1, \dots, x_N) = (x_1 + a, \dots, x_N + a).$$

Suppose that μ is an invariant probability measure on $\mathbb{R}^{3N}$. Define the center-of-mass map

$$C : \mathbb{R}^{3N} \longrightarrow \mathbb{R}^3, \quad C(x_1, \dots, x_N) = \frac{1}{N} \sum_{i=1}^N x_i.$$

This map is equivariant under translations:

$$C(a \cdot X) = C(X) + a.$$

Let $\nu = C_{\#}\mu$ be the pushforward of μ by C . Then ν is a probability measure on $\mathbb{R}^3$. Moreover, for every Borel set $A \subseteq \mathbb{R}^3$ and every $a \in \mathbb{R}^3$,

$$\begin{aligned} \nu(A + a) &= \mu\left(C^{-1}(A + a)\right) \\ &= \mu\left(a \cdot C^{-1}(A)\right) \\ &= \mu\left(C^{-1}(A)\right) \\ &= \nu(A). \end{aligned}$$

Thus, ν would be a translation-invariant probability measure on $\mathbb{R}^3$, contradicting the first part. Hence, no probability measure on $\mathbb{R}^{3N}$ is invariant under diagonal translations.

Consequently, no probability measure in $SE(3)^N$ is invariant under the action of $SE(3)$.

The solution: defining measures on the centered translations That being said, one can still get a measure (but not a *probability* measure) on $SE(3)^N$ which is invariant under the action of $SE(3)$. Before stating the result, we are going to provide a characterization for invariant measures.

Let (S, Σ) be a measurable space endowed with a measurable left action of a group G . For every $g \in G$, define

$$\mathcal{L}_g : S \longrightarrow S, \quad \mathcal{L}_g(x) := g \cdot x.$$

For $A \in \Sigma$, we write

$$g \cdot A := \mathcal{L}_g(A) = \{g \cdot x : x \in A\}.$$

Since $\mathcal{L}_g^{-1} = \mathcal{L}_{g^{-1}}$, the definition of the pushforward measure gives

$$\begin{aligned} ((\mathcal{L}_g)_\# \mu)(A) &= \mu(\mathcal{L}_g^{-1}(A)) \\ &= \mu(\mathcal{L}_{g^{-1}}(A)) \\ &= \mu(g^{-1} \cdot A). \end{aligned}$$

Suppose first that μ is G -invariant, namely

$$\mu(g \cdot A) = \mu(A) \quad \text{for every } g \in G \text{ and } A \in \Sigma.$$

Since $g^{-1} \in G$, it follows that

$$((\mathcal{L}_g)_\# \mu)(A) = \mu(g^{-1} \cdot A) = \mu(A).$$

Therefore,

$$(\mathcal{L}_g)_\# \mu = \mu \quad \text{for every } g \in G.$$

Conversely, suppose that

$$(\mathcal{L}_g)_\# \mu = \mu \quad \text{for every } g \in G.$$

Applying this identity to g^{-1} , we obtain, for every $A \in \Sigma$,

$$\begin{aligned} \mu(g \cdot A) &= \mu(\mathcal{L}_{g^{-1}}^{-1}(A)) \\ &= ((\mathcal{L}_{g^{-1}})_\# \mu)(A) \\ &= \mu(A). \end{aligned}$$

Hence μ is G -invariant.

Consequently,

$$\boxed{\mu \text{ is } G\text{-invariant} \iff (\mathcal{L}_g)_\# \mu = \mu \text{ for every } g \in G.}$$

The authors of [Yim+23] then state the following result (Proposition G.3).

Proposition 4.10 (Construction of invariant measures). *Let*

$$\mathcal{C}_N := \left\{ [(r_1, z_1), \dots, (r_N, z_N)] \in SE(3)^N : \sum_{i=1}^N z_i = 0 \right\}.$$

This is the centered configuration space, denoted by $SE(3)_0^N$ in the original paper.

For every $R \in SO(3)$, define the diagonal rotational action $\rho_R : \mathcal{C}_N \rightarrow \mathcal{C}_N$ by

$$\rho_R([(r_1, z_1), \dots, (r_N, z_N)]) = [(Rr_1, Rz_1), \dots, (Rr_N, Rz_N)].$$

Let η be an $SO(3)$ -invariant probability measure on $\mathcal{C}_N$, equivalently

$$(\rho_R)_\# \eta = \eta \quad \text{for every } R \in SO(3).$$

Let λ_3 denote the Lebesgue measure on $\mathbb{R}^3$, and define

$$\Psi : \mathbb{R}^3 \times \mathcal{C}_N \longrightarrow SE(3)^N$$

by

$$\Psi(\bar{x}, [(r_1, z_1), \dots, (r_N, z_N)]) = [(r_1, z_1 + \bar{x}), \dots, (r_N, z_N + \bar{x})].$$

Define the measure ν on $SE(3)^N$ by

$$\nu := \Psi_\#(\lambda_3 \otimes \eta).$$

Equivalently, for every $B \in \mathcal{B}(SE(3)^N)$,

$$\nu(B) = \int_{\mathbb{R}^3} \int_{\mathcal{C}_N} \mathbf{1}_B(\Psi(\bar{x}, T_0)) \eta(dT_0) \lambda_3(d\bar{x}).$$

Then ν is invariant under the diagonal action of $SE(3)$ on $SE(3)^N$. More precisely, for $g \in SE(3)$, let

$$L_g([T_1, \dots, T_N]) := [gT_1, \dots, gT_N].$$

Then

$$(L_g)_\# \nu = \nu \quad \text{for every } g \in SE(3).$$

The proof is not given by the authors. We give it here for the sake of completeness.

Proof: Let $g = (R, a) \in SE(3)$, where $R \in SO(3)$ and $a \in \mathbb{R}^3$. Recall that the group law on $SE(3)$ is

$$(R, a)(r, x) = (Rr, a + Rx).$$

Let

$$T_0 = [(r_1, z_1), \dots, (r_N, z_N)] \in \mathcal{C}_N.$$

By the definitions of L_g , Ψ , and ρ_R , we have

$$\begin{aligned} L_g(\Psi(\bar{x}, T_0)) &= L_g([(r_1, z_1 + \bar{x}), \dots, (r_N, z_N + \bar{x})]) \\ &= [(Rr_1, a + Rz_1 + R\bar{x}), \dots, (Rr_N, a + Rz_N + R\bar{x})] \\ &= \Psi(a + R\bar{x}, \rho_R(T_0)). \end{aligned}$$

Let $f : SE(3)^N \rightarrow [0, +\infty]$ be measurable. By the defining property of the pushforward measure,

$$\begin{aligned} \int_{SE(3)^N} f(T) (L_g)_\# \nu(dT) &= \int_{SE(3)^N} f(L_g(T)) \nu(dT) \\ &= \int_{\mathbb{R}^3} \int_{\mathcal{C}_N} f(L_g(\Psi(\bar{x}, T_0))) \eta(dT_0) \lambda_3(d\bar{x}) \\ &= \int_{\mathbb{R}^3} \int_{\mathcal{C}_N} f(\Psi(a + R\bar{x}, \rho_R(T_0))) \eta(dT_0) \lambda_3(d\bar{x}). \end{aligned}$$

Since η is $SO(3)$ -invariant, we have

$$(\rho_R)_\# \eta = \eta.$$

We then obtain:

$$\begin{aligned} \int_{SE(3)^N} f(T) (L_g)_\# \nu(dT) &= \int_{\mathbb{R}^3} \int_{\mathcal{C}_N} f(\Psi(a + R\bar{x}, \rho_R(T_0))) \eta(dT_0) \lambda_3(d\bar{x}) \\ &= \int_{\mathbb{R}^3} \int_{\mathcal{C}_N} f(\Psi(a + R\bar{x}, T'_0)) (\rho_R)_\# \eta(dT'_0) \lambda_3(d\bar{x}) \\ &= \int_{\mathbb{R}^3} \int_{\mathcal{C}_N} f(\Psi(a + R\bar{x}, T'_0)) \eta(dT'_0) \lambda_3(d\bar{x}), \end{aligned}$$

where the second equality follows from the change of variable $T'_0 = \rho_R(T_0)$ and the last equality follows from $(\rho_R)_\# \eta = \eta$.

Now set

$$\bar{x}' = a + R\bar{x}.$$

Because $R \in SO(3)$, we have $\det(R) = 1$. Hence Lebesgue measure is invariant under the affine isometry $\bar{x} \mapsto a + R\bar{x}$, so that

$$\lambda_3(d\bar{x}') = \lambda_3(d\bar{x}).$$

Consequently,

$$\begin{aligned} \int_{SE(3)^N} f(T) (L_g)_\# \nu(dT) &= \int_{\mathbb{R}^3} \int_{\mathcal{C}_N} f(\Psi(\bar{x}', T'_0)) \eta(dT'_0) \lambda_3(d\bar{x}') \\ &= \int_{SE(3)^N} f(T) \nu(dT). \end{aligned}$$

This equality holds for every nonnegative measurable function f . In particular, taking $f = \mathbf{1}_B$ for an arbitrary $B \in \mathcal{B}(SE(3)^N)$ gives

$$(L_g)_\# \nu(B) = \nu(B).$$

Therefore,

$$(L_g)_\# \nu = \nu \quad \text{for every } g \in SE(3),$$

which proves that ν is $SE(3)$ -invariant. $\square$

Furthermore, the converse is also true: every measure on $SE(3)^N$ which is invariant under the action of $SE(3)$ is of this form. Indeed:

Proposition 4.11. *Let μ be a measure on $SE(3)^N$, which is $SE(3)$ -invariant and can be written as a countable sum of finite measures, each with compact support. Writing $T := [T_1, \dots, T_N]$, assume that for every $f \in C_c^\infty(SE(3)^N)$ the map*

$$x \mapsto \int_{SE(3)^N} f(T) dK(x, T)$$

is continuous, and that $K(x, SE(3)^N) < +\infty$ for every $x \in \mathbb{R}^3$. Then, there exist η an $SO(3)$ -invariant probability measure on $SE(3)_0^N$ and $\bar{\mu}$ proportional to the Lebesgue measure on $\mathbb{R}^3$ such that

$$d\mu([(r_1, x_1), \dots, (r_N, x_N)]) = d\bar{\mu}\left(\frac{1}{N} \sum_{i=1}^N x_i\right) \times d\eta\left(\left[\left(r_1, x_1 - \frac{1}{N} \sum_{i=1}^N x_i\right), \dots, \left(r_N, x_N - \frac{1}{N} \sum_{i=1}^N x_i\right)\right]\right).$$

The proof is this time given by the authors and can be found in appendix G.

Remark: In the first result, we only use the two following facts about $\mathcal{C}_N$:

- ρ_R must map $\mathcal{C}_N$ into $\mathcal{C}_N$; otherwise, $(\rho_R)_\# \eta = \eta$ is ill-defined;
- $\mathcal{C}_N$ must be a Borel set, ensuring that η and the restriction of Ψ are well-defined.

Hence, the proposition remains true for any set which satisfies these two points. However, the reciprocal statement relies on the fact that with this choice for $\mathcal{C}_N$, Ψ is a bijection, which leads to the result.

The practical consequence:

Thanks to this result, it is sufficient to construct an $SO(3)$ -invariant probability measure on the centered configuration space $SE(3)_0^N$. In practice, samples from this probability measure are centered representatives of protein structures. This does not restrict the intrinsic protein structures being modeled: as discussed earlier, two configurations that differ only by a global translation represent the same physical structure.

As a consequence, our goal is now to define a diffusion framework on $SE(3)_0^N$, in order to get a generative model that gives a probability measure on $SE(3)_0^N$, which is $SO(3)$ -invariant.

Generative modeling on $SE(3)_0^N$

Now, we are ready to define generative modeling on $SE(3)_0^N$ as done in [Yim+23]. We first need to extend the generative modeling framework from $SE(3)$ to $SE(3)^N$.

The forward process is defined as the concatenation of the individual forward processes:

$$dT(t) = \left[0, -\frac{1}{2}X(t) \right] dt + \left[dB^{\text{SO}(3)^N}(t), dB^{\mathbb{R}^{3N}}(t) \right]$$

where the Brownian motion components are independent. Hence, conditionally to $T^{(0)}$ (that is, the protein backbone), the noise added to each residue is independent. Equivalently:

$$p_{t|0}(T(t) \mid T(0)) = \prod_{i=1}^N p_{t|0}(T_i(t) \mid T_i(0)).$$

Here, the manifold $SE(3)^N$ is endowed with the metric:

$$\langle (v_1, \dots, v_N), (w_1, \dots, w_N) \rangle_{SE(3)^N} = \sum_{i=1}^N \langle v_i, w_i \rangle_{SE(3)}.$$

The Laplacian then becomes:

$$\Delta_{SE(3)^N} = \sum_{i=1}^N \Delta_{SE(3),i}$$

Now, we need to define the forward process on $SE(3)_0^N$. To do so, we just need to remove the center of mass of the $\mathbb{R}^{3N}$ part. The authors do so by defining a projection matrix, thanks to the following definition:

Definition 4.12 (Kronecker Product). *Let A be an $m \times n$ matrix and B be a $p \times q$ matrix. The Kronecker product $A \otimes B$ is the $mp \times nq$ block matrix defined as:*

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}$$

Let $P = \left(I_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top\right) \otimes I_3$. For $X = (X_1, \dots, X_N) \in \mathbb{R}^{3N}$, it satisfies:

$$(PX)_i = X_i - \frac{1}{N} \sum_{j=1}^N X_j.$$

The forward process is then defined by:

$$dT(t) = \left[0, -\frac{1}{2}PX(t)\right] dt + \left[dB^{\text{SO}(3)^N}(t), P dB^{\mathbb{R}^{3N}}(t)\right] \quad (4.16)$$

If $T^0 = (R^0, X^{(0)}) \in SE(3)_0^N$, then $T^{(t)} = (R^{(t)}, X^{(t)}) \in SE(3)_0^N$. Indeed, the SDE for the i -th translation component reads

$$dX_i(t) = -\frac{1}{2} \left(X_i(t) - \bar{X}(t)\right) dt + dB_i(t) - \frac{1}{N} \sum_{j=1}^N dB_j(t).$$

Summing over $i \in \{1, \dots, N\}$ yields

$$d\left(\sum_{i=1}^N X_i(t)\right) = -\frac{1}{2} \left[\sum_{i=1}^N X_i(t) - N\bar{X}(t)\right] dt + \sum_{i=1}^N dB_i(t) - \frac{N}{N} \sum_{j=1}^N dB_j(t).$$

Thus,

$$d\left(\sum_{i=1}^N X_i(t)\right) = 0.$$

Therefore, the sum is invariant over time.

Note then that the forward process needs to start from a distribution over $SE(3)_0^N$. However, proteins in the Protein Data Bank (the dataset used by the authors) are elements of $SE(3)^N$. Hence the authors employ the following procedure: given $T^{\text{raw}} = [(r_1, x_1^{\text{raw}}), \dots, (r_N, x_N^{\text{raw}})]$, they compute the center of mass

$$\bar{x}^{\text{raw}} = \frac{1}{N} \sum_{i=1}^N x_i^{\text{raw}},$$

and then update each translation component as

$$x_i(0) = x_i^{\text{raw}} - \bar{x}^{\text{raw}}.$$

Remark: While the components of the Brownian motion on $\mathbb{R}^{3N}$ were independent before projection, this is not the case anymore after projection. Indeed, we have to introduce correlation between them to make sure that the sum of the components of $X^{(t)}$ remain null for each t . More precisely: Let $B = (B_1, \dots, B_N)$ be a standard Brownian motion in $\mathbb{R}^{3N}$, where the B_i are independent Brownian motions in $\mathbb{R}^3$. For $a, b \in \{1, 2, 3\}$,

$$d\langle B_i^a, B_j^b \rangle_t = \delta_{ij} \delta_{ab} dt.$$

Now, define $\tilde{B} = PB$. Since

$$\tilde{B}_i^a = \sum_{k=1}^N \left(\delta_{ik} - \frac{1}{N}\right) B_k^a,$$

bilinearity of the quadratic covariation gives

$$\begin{aligned}
 d\langle \tilde{B}_i^a, \tilde{B}_j^b \rangle_t &= \sum_{k,\ell=1}^N \left(\delta_{ik} - \frac{1}{N} \right) \left(\delta_{j\ell} - \frac{1}{N} \right) d\langle B_k^a, B_\ell^b \rangle_t \\
 &= \sum_{k=1}^N \left(\delta_{ik} - \frac{1}{N} \right) \left(\delta_{jk} - \frac{1}{N} \right) \delta_{ab} dt \\
 &= \sum_{k=1}^N \left(\delta_{ik}\delta_{jk} - \frac{\delta_{ik}}{N} - \frac{\delta_{jk}}{N} + \frac{1}{N^2} \right) \delta_{ab} dt \\
 &= \left(\delta_{ij} - \frac{1}{N} \right) \delta_{ab} dt,
 \end{aligned}$$

Equivalently,

$$d\langle \tilde{B}_i, \tilde{B}_j \rangle_t = \left(\delta_{ij} - \frac{1}{N} \right) I_3 dt.$$

For $i \neq j$, the covariation of the projected noises is then negative:

$$d\langle \tilde{B}_i, \tilde{B}_j \rangle_t = -\frac{1}{N} I_3 dt.$$

This negative correlation is induced by the centering constraint: a noise displacement of one residue must be compensated collectively by the other residues so that

$$\sum_{i=1}^N d\tilde{B}_i^{(t)} = 0.$$

Invariant distribution of the forward process: We now find the invariant distribution for this forward process. Note that:

$$\mathcal{C}_N = SO(3)^N \times V_N.$$

where $V_N := \text{Im}(P)$. Recall that, on a Riemannian manifold $\mathcal{M}$, the Langevin diffusion

$$dY^{(t)} = -\frac{1}{2} \nabla U(Y^{(t)}) dt + dB_{\mathcal{M}}^{(t)}$$

has invariant measure

$$\pi(dy) = \frac{1}{Z} e^{-U(y)} \text{vol}_{\mathcal{M}}(dy),$$

provided that the normalizing constant Z is finite.

In the case of the forward process:

$$dT^{(t)} = \left[0, -\frac{1}{2} P X^{(t)} \right] dt + \left[dB_{SO(3)^N}^{(t)}, P dB_{\mathbb{R}^{3N}}^{(t)} \right].$$

The rotational invariant measure is the normalized Riemannian volume on $SO(3)^N$, since the potential is zero in the rotational variables. Namely:

$$\pi_R = \mathcal{U}(SO(3))^{\otimes N}.$$

For the translational component, the invariant density on V_N is proportional to

$$\exp\left(-\frac{1}{2} \|x\|^2\right).$$

It can be identified with

$$\pi_X = P_{\#} \left(\mathcal{N}(0, I_3)^{\otimes N} \right) = P_{\#} \mathcal{N}(0, I_{3N}).$$

Indeed, by definition of the pushforward, for every Borel set $A \subseteq V_N$,

$$(P_{\#} \mathcal{N}(0, I_{3N})) (A) = \mathcal{N}(0, I_{3N}) \left(P^{-1}(A) \right).$$

Finally, the rotational and translational components of the forward process are decoupled and driven by independent Brownian motions. Hence their invariant measures combine as a product probability measure:

$$\pi_{\text{inv}} = \mathcal{U}(SO(3))^{\otimes N} \otimes P_{\#} \left(\mathcal{N}(0, I_3)^{\otimes N} \right)$$

We will need the following fact in subsection 4.2.4:

Lemma 4.13. *The measure π_{inv} is $SO(3)$ -invariant.*

Proof: Write $\gamma := \mathcal{N}(0, I_3)^{\otimes N} = \mathcal{N}(0, I_{3N})$ and $\Phi_Q(R, X) = ((QR_i)_i, \hat{Q}X)$. Since Φ_Q acts on the two factors separately, the pushforward of the product measure is the product of the pushforwards, and it suffices to treat each factor.

For the rotational factor, $\mathcal{U}(SO(3))$ is the normalised Haar measure of $SO(3)$, hence $(L_Q)_{\#} \mathcal{U}(SO(3)) = \mathcal{U}(SO(3))$ by left invariance; the same then holds componentwise for $\mathcal{U}(SO(3))^{\otimes N}$.

For the translational factor, $\hat{Q} = I_N \otimes Q$ is orthogonal, so $\hat{Q}_{\#} \gamma = \gamma$, and $P\hat{Q} = \hat{Q}P$. Therefore

$$\hat{Q}_{\#} (P_{\#} \gamma) = (\hat{Q} \circ P)_{\#} \gamma = (P \circ \hat{Q})_{\#} \gamma = P_{\#} (\hat{Q}_{\#} \gamma) = P_{\#} \gamma.$$

Hence $(\Phi_Q)_{\#} \pi_{\text{inv}} = \pi_{\text{inv}}$ for every $Q \in SO(3)$. $\square$

Backward process: The backward process then satisfies the following coupled SDE:

$$d\overleftarrow{R}(t) = \nabla_r \log p_{T_F-t}(\overleftarrow{T}(t)) dt + dB^{\text{SO}(3)^N}(t), \quad (4.17)$$

$$d\overleftarrow{X}(t) = P \left\{ \frac{1}{2} \overleftarrow{X}(t) + \nabla_x \log p_{T_F-t}(\overleftarrow{T}(t)) \right\} dt + P dB^{\mathbb{R}^{3N}}(t). \quad (4.18)$$

Finally, we need to check that the distribution of this backward process at time T_F is indeed $SO(3)$ -invariant, so that we can use Proposition 4.10 to couple it with the Lebesgue measure in order to obtain an invariant measure on $SE(3)^N$. However, the way the original article handles this is problematic.

4.2.4 Building an invariant diffusion under the $SO(3)$ action

We give here the proposition and the corollary used in the original paper, and we explain why the corollary does not follow from the proposition.

Proposition 4.14 (*G*-invariance and SDEs, 3.6). *Let G be a Lie group and H a subgroup of G . If (a) $X(0) \sim p_0$ for an H -invariant distribution p_0 and (b) $dX(t) = b(t, X(t))dt + \Sigma^{1/2}dB(t)$ for bounded, H -equivariant coefficients b and Σ satisfying $b \circ L_h = dL_h(b)$ and $\Sigma dL_h(\cdot) = dL_h(\Sigma \cdot)$, and where $B(t)$ is a Brownian motion associated with a left-invariant metric. Then for every $t \geq 0$:*

(a) *the distribution p_t of $X(t)$ is H -invariant, and*

(b) its score $\nabla_{X(t)} \log p_t(X(t))$ is H -equivariant.

Corollary [3.7] Suppose $\{T(t)\}_{t \geq 0}$ has $SO(3)$ invariant initial distribution p_0 and evolves according to the forward process. Then for every $t \in (0, T_F)$, $\nabla \log p_{T_F-t}(\overleftarrow{T}(t))$ is $SO(3)$ equivariant, and the distribution of $(\overleftarrow{R}(t), \overleftarrow{X}(t))$ implied by the backward process is $SO(3)$ -invariant.

Proposition 3.6 of the original article considers a Lie group G and a subgroup $H \subseteq G$ acting on G by left multiplication. Under the assumption that the initial distribution is H -invariant, the proposition asserts that all marginal distributions of the forward process are H -invariant and that their scores are H -equivariant. In the application leading to Corollary 3.7, the authors implicitly take

$$G = SE(3)_0^N, \quad H = SO(3),$$

where $SO(3)$ is identified with its diagonal copy

$$H_{\text{diag}} := \{((Q, 0), \dots, (Q, 0)) : Q \in SO(3)\} \subseteq SE(3)^N.$$

In other words, there exists a morphism Φ_{diag} such that: $\Phi_{\text{diag}} : SO(3) \rightarrow SE(3)_N^0$, $\Phi_{\text{diag}}(Q) = ((Q, 0), \dots, (Q, 0)) \quad \forall Q \in SO(3)$.

The authors assert that “this defines a subgroup of $SE(3)^N$ denoted $SE(3)_0^N$... Note that $SE(3)_0^N$ is still a Lie group and $SO(3)$ is a subgroup of $SE(3)_0^N$.” That assertion is false.

$SE(3)_0^N$ is not stable under the componentwise law The group law inherited from the standard semidirect-product law on $SE(3)$ is

$$((R_i, x_i))_{i=1}^N ((S_i, y_i))_{i=1}^N = ((R_i S_i, x_i + R_i y_i))_{i=1}^N,$$

and the constraint $\sum_{i=1}^N x_i = 0$ is not preserved by it. Concretely, take $N = 2$ and let $R_z(\pi)$ denote the rotation by π about the third axis. Set

$$g = ((I, 0), (R_z(\pi), 0)), \quad h = ((I, e_1), (I, -e_1)).$$

Both lie in $SE(3)_0^2$, yet

$$gh = ((I, e_1), (R_z(\pi), -R_z(\pi)e_1)) = ((I, e_1), (R_z(\pi), e_1)), \quad \sum_{i=1}^2 x_i = 2e_1 \neq 0.$$

The set is not stable under inversion either: since $(r, x)^{-1} = (r^\top, -r^\top x)$, the inverse of $((I, e_1), (R_z(\pi), -e_1)) \in SE(3)_0^2$ has translation sum $-2e_1 \neq 0$ (However, for $N = 1$ one has $SE(3)_0^1 = SO(3) \times \{0\}$, which is a subgroup; the obstruction requires $N \geq 2$).

Consequently $SE(3)_0^N$ is not a Lie group for the law used in the article and Proposition 3.6 cannot be applied literally with $G = SE(3)_0^N$.

Why the most natural alternative group laws do not resolve the issue

(i) A different law on $SE(3)_0^N$. As a manifold, $SE(3)_0^N \cong SO(3)^N \times V_N$, which does carry Lie group structures, e.g. the direct product law $((R_i), X) * ((S_i), Y) = ((R_i S_i), X + Y)$. But for that law, if we consider the diagonal action, which is defined as the left multiplication by Φ_{diag} , then we obtain: $Q \cdot (R, X) = \Phi_{\text{diag}}(Q) \cdot (R, X) = (QR, X + 0) = (QR, X)$: it leaves the translations untouched, which is not the global rotation action we need. We can't apply

Proposition 3.6 with another action of $SO(3)$, as invariance and equivariance are defined with respect to this left diagonal action.

(ii) *Working in the ambient group $G = SE(3)^N$.* Here, G is a Lie group and H_{diag} is indeed a subgroup for the required law. But the law of the process 4.16 is carried by the submanifold $SE(3)_0^N$, hence admits no density with respect to the Haar measure of $SE(3)^N$, whereas the proof of Proposition 3.6 is written throughout for such a density p_t .

The correct fix is therefore to regard $SE(3)_0^N$ as a Riemannian manifold equipped with the action of $SO(3)$ by global rotations, and to extend Proposition 3.6 from subgroups acting by left multiplication on Lie groups to Lie groups acting isometrically on Riemannian manifolds. Here, H_{diag} is a Lie subgroup of the ambient group $SE(3)^N$, endowed with its usual componentwise semidirect-product law. Although the centered space $SE(3)_0^N$ is not itself a subgroup of $SE(3)^N$, ambient left multiplication by H_{diag} preserves $SE(3)_0^N$ and restricts to the desired smooth action

$$\rho_Q(R, X) = (QR, QX).$$

Consequently, an extension of Proposition 3.6 to isometric Lie-group actions on Riemannian manifolds provides the appropriate framework for the setting considered in the paper.

Second issue: unbounded coefficients Proposition 3.6 is stated and proved for bounded coefficients. However, the forward drift

$$b(R, X) = \left(0, -\frac{1}{2}PX\right)$$

is unbounded. The article states, without proof, that the result extends to unbounded coefficients under suitable growth assumptions. The proposition below therefore replaces boundedness by the hypothesis of a well-posed, non-explosive martingale problem, which is verified directly in the application.

Extension of the original proposition

We first isolate the transformation rule for the gradient.

Lemma 4.15 (Gradient under an isometry). *Let (M, g) be a Riemannian manifold and $\Phi : M \rightarrow M$ an isometry. Then for every $f \in C^1(M)$ and every $x \in M$,*

$$d\Phi_x(\nabla(f \circ \Phi)(x)) = \nabla f(\Phi(x)).$$

Proof: For every $u \in T_x M$, by using the definition of Riemannian gradient twice and using the chain rule:

$$\langle \nabla(f \circ \Phi)(x), u \rangle_x = d(f \circ \Phi)_x(u) = df_{\Phi(x)}(d\Phi_x u) = \langle \nabla f(\Phi(x)), d\Phi_x u \rangle_{\Phi(x)}.$$

Since Φ is an isometry, the left-hand side also equals $\langle d\Phi_x \nabla(f \circ \Phi)(x), d\Phi_x u \rangle_{\Phi(x)}$. Hence:

$$\langle d\Phi_x \nabla(f \circ \Phi)(x) - \nabla f(\Phi(x)), d\Phi_x u \rangle_{\Phi(x)} = 0$$

But then, as Φ is a diffeomorphism, $d\Phi_x : T_x M \rightarrow T_{\Phi(x)} M$ is also an diffeomorphism hence is surjective, and taking u such that $d\Phi_x u = d\Phi_x \nabla(f \circ \Phi)(x) - \nabla f(\Phi(x))$, we get the result. $\square$

Proposition 4.16 (Invariant diffusions under isometric group actions). *Let (M, g) be a Riemannian manifold and let H be a Lie group acting smoothly on M by Riemannian isometries;*

for $h \in H$ write $\Phi_h(x) = h \cdot x$. Let $(X^{(t)})_{t \in [0, T]}$ be a time-inhomogeneous diffusion on M with generator

$$\mathcal{L}_t f = \langle b_t, \nabla f \rangle + \frac{1}{2} \operatorname{div}(\Sigma_t \nabla f), \quad f \in C_c^\infty(M),$$

where $(t, x) \mapsto b_t(x)$ is a measurable, locally bounded family of vector fields and Σ_t a measurable family of smooth, symmetric, positive semidefinite fields of endomorphisms of TM , the tangent bundle of M . Assume that for every $h \in H$, $x \in M$, $v \in T_x M$ and $t \in [0, T]$,

$$b_t(h \cdot x) = h \cdot b_t(x), \quad \Sigma_t(h \cdot x)(h \cdot v) = h \cdot (\Sigma_t(x)v),$$

where $h \cdot v$ for $v \in T_x M$ for some $x \in M$ denotes the action induced by $d\Phi_h$ on tangent vectors. Assume moreover that the martingale problem associated with $(\mathcal{L}_t)_{t \in [0, T]}$ is non-explosive (any solution takes values in M for any time $t \in [0, T]$), and well posed, i.e. for any initial distribution, there exists a unique solution to the martingale problem.

Then, if the law μ_0 of $X^{(0)}$ is H -invariant, then the law μ_t of $X^{(t)}$ is H -invariant for every $t \in [0, T]$:

$$(\Phi_h)_\# \mu_t = \mu_t, \quad h \in H.$$

If in addition μ_t admits a positive C^1 density p_t with respect to the Riemannian volume measure, then its score is H -equivariant:

$$\nabla \log p_t(h \cdot x) = h \cdot \nabla \log p_t(x).$$

Proof: We are first going to prove that: $\mathcal{L}_t f(h \cdot x) = \mathcal{L}_t(f \circ \Phi_h)(x)$. For $f \circ \Phi_h$ to be a correct test function for the martingale problem, we need that $f \circ \Phi_h \in C_c^\infty(M)$. First, note $f \circ \Phi_h \in C^\infty(M)$ since Φ_h is smooth. Moreover, note that:

$\Phi_h^{-1}(\operatorname{supp}(f))$ is compact as Φ_h^{-1} is continuous and $\operatorname{supp}(f)$ is compact; hence it is closed as $\mathcal{M}$ is Hausdorff. Furthermore, $f(\Phi_h(x)) \neq 0 \implies \Phi_h(x) \in \operatorname{supp}(f)$ hence $\operatorname{supp}(f \circ \Phi_h) \subset \Phi_h^{-1}(\operatorname{supp}(f))$. As $\operatorname{supp}(f \circ \Phi_h)$ is closed and included in a compact set, it is compact.

Hence, $f \circ \Phi_h \in C_c^\infty(M)$.

To prove the equality for the first term of the generator, using that $d\Phi_h$ is a linear isometry, and then Lemma 4.15 and the assumption about the H -equivariance of b_t , we get:

$$\langle b_t(x), \nabla(f \circ \Phi_h)(x) \rangle_x = \langle h \cdot b_t(x), h \cdot \nabla(f \circ \Phi_h)(x) \rangle_{h \cdot x} = \langle b_t(h \cdot x), \nabla f(h \cdot x) \rangle_{h \cdot x}.$$

For the second term, set

$$Y(x) := \Sigma_t(x) \nabla(f \circ \Phi_h)(x), \quad Z(x) := \Sigma_t(x) (\nabla f(x)).$$

Using the H -equivariance of Σ_t and then Lemma 4.15 gives:

$$d\Phi_h(Y(x)) = h \cdot (\Sigma_t(x) \nabla(f \circ \Phi_h)(x)) = \Sigma_t(h \cdot x) (h \cdot \nabla(f \circ \Phi_h)(x)) = \Sigma_t(h \cdot x) (\nabla f(h \cdot x)) = Z(\Phi_h(x)),$$

But then recall that for a diffeomorphism Φ and X a vector field, we can define: $\Phi_* X(x) := d\Phi|_{\Phi^{-1}(x)} X(\Phi^{-1}(x))$, which gives for $y = \Phi(x)$: $\Phi_* X(\Phi(y)) = d\Phi|_y X(y)$. Hence: $Z = (\Phi_h)_* Y$. Moreover, Φ_h is an isometry, and hence by Lemma 2.63, it preserves the Riemannian volume measure and for any volume-preserving diffeomorphism φ , one has that $\operatorname{div}(\varphi_* Y) \circ \varphi = \operatorname{div} Y$. Hence

$$\operatorname{div}(\Sigma_t \nabla(f \circ \Phi_h))(x) = \operatorname{div}(\Sigma_t \nabla f)(h \cdot x),$$

and therefore

$$\mathcal{L}_t(f \circ \Phi_h) = (\mathcal{L}_t f) \circ \Phi_h.$$

Now, fix $h \in H$ and set $Y^{(t)} := \Phi_h(X^{(t)})$. Since X solves the martingale problem for $(\mathcal{L}_t)$, for every $f \in C_c^\infty(M)$, as $f \circ \Phi_h \in C_c^\infty(M)$, the process

$$M_f^{(t)} := (f \circ \Phi_h)(X^{(t)}) - (f \circ \Phi_h)(X^{(0)}) - \int_0^t \mathcal{L}_s(f \circ \Phi_h)(X^{(s)}) \, ds$$

is a martingale for the filtration of X . But since $\mathcal{L}_t(f \circ \Phi_h) = (\mathcal{L}_t f) \circ \Phi_h$, we can write it as:

$$M_f^{(t)} = f(\Phi_h X^{(t)}) - f(\Phi_h X^{(0)}) - \int_0^t \mathcal{L}_s(f)(\Phi_h X^{(s)}) \, ds$$

and Y is adapted to the same filtration. Hence, $\Phi_h(X)$ and X solve the same martingale problem.

Since μ_0 is H -invariant, $\Phi_h^{(0)} = h \cdot X^{(0)} \stackrel{\text{law}}{=} X^{(0)}$. Hence, as the martingale problem is assumed to be well-posed, if two processes share the same initial law and solve the same martingale problem, they have the same law on the path space $\mathcal{C}([0, T], M)$. In particular: $(\Phi_h)_\# \mu_t = \mu_t$ for every $t \in [0, T]$.

Now, we prove the equivariance of the score. Suppose that

$$\mu_t(dx) = p_t(x) \, \text{vol}(dx),$$

where $p_t > 0$ is of class C^1 . Fix $h \in H$. For every nonnegative measurable function $\psi : M \rightarrow \mathbb{R}$, the definition of the pushforward measure gives

$$\begin{aligned} \int_M \psi(y) (\Phi_h)_\# \mu_t(dy) &= \int_M \psi(\Phi_h(x)) \mu_t(dx) \\ &= \int_M \psi(\Phi_h(x)) p_t(x) \, \text{vol}(dx). \end{aligned}$$

Remembering that Φ_h preserves the Riemannian volume measure and performing the change of variables $y = \Phi_h(x)$:

$$\int_M \psi(y) (\Phi_h)_\# \mu_t(dy) = \int_M \psi(y) p_t(\Phi_h^{-1}(y)) \, \text{vol}(dy).$$

Since this equality holds for every nonnegative measurable function ψ , we conclude that

$$(\Phi_h)_\# \mu_t(dy) = p_t(\Phi_h^{-1}(y)) \, \text{vol}(dy),$$

or equivalently,

$$\frac{d(\Phi_h)_\# \mu_t}{d \, \text{vol}} = p_t \circ \Phi_h^{-1} \quad \text{vol-a.e.}$$

If μ_t is H -invariant, then

$$(\Phi_h)_\# \mu_t = \mu_t.$$

and we get:

$$p_t \circ \Phi_h^{-1} = p_t \quad \text{vol-a.e.}$$

Since the invariance holds for every $h \in H$, replacing h by h^{-1} gives

$$p_t \circ \Phi_h = p_t \quad \text{vol-a.e.}$$

Both functions are continuous, and the Riemannian volume measure has full support on M . Hence the equality holds everywhere:

$$p_t \circ \Phi_h = p_t \quad \text{on } M.$$

Applying Lemma 4.15 to $\log p_t \in C^1(M)$ and using $\log p_t \circ \Phi_h = \log p_t$,

$$\nabla \log p_t(h \cdot x) = h \cdot \nabla (\log p_t \circ \Phi_h)(x) = h \cdot \nabla \log p_t(x). \quad \square$$

Application to the forward process on $SE(3)_0^N$

In this paragraph, we define the objects used throughout, and we check that we can safely apply Proposition 4.16.

Geometry and the $SO(3)$ action

Let

$$V_N := \left\{ X = (x_1, \dots, x_N) \in \mathbb{R}^{3N} : \sum_{i=1}^N x_i = 0 \right\}, \quad \mathcal{M}_N := SE(3)_0^N = SO(3)^N \times V_N,$$

Remember that we defined the matrix P as the orthogonal projection of $\mathbb{R}^{3N}$ onto V_N ; in particular $\|P\|_{\text{op}} \leq 1$ and $P|_{V_N} = \text{Id}$.

We equip $\mathcal{M}_N$ with the product metric: at $T = (R, X)$, a tangent vector is (A, v) with $A_i \in T_{R_i}SO(3)$ and $v \in V_N$, and

$$\langle (A, v), (A', v') \rangle_T := \sum_{i=1}^N \frac{1}{2} \text{Tr}(A_i A_i'^\top) + \langle v, v' \rangle_{\mathbb{R}^{3N}}.$$

The global rotation action is:

$$\Phi_Q(R, X) := ((QR_i)_{i=1}^N, \hat{Q}X).$$

It preserves $\mathcal{M}_N$, since $\sum_i Qx_i = Q \sum_i x_i = 0$. It is the restriction to $\mathcal{M}_N$ of left multiplication by $((Q, 0), \dots, (Q, 0)) \in SE(3)^N$, and it is smooth in (Q, R, X) . If we embed the submanifold $\mathcal{M}_N$ into the ambient vector space $(\mathbb{R}^{3 \times 3})^N \times \mathbb{R}^{3N}$, we get that Φ_Q is linear (linearity demands vector space structure, which $SO(3)^N$ does not have). Then the tangent spaces of this ambient vector space are isomorphic to the vector space itself, and since Φ_Q is linear, the differential of this linear map is simply the map itself. Restricting this differential back to the tangent space of the submanifold yields:

$$d\Phi_Q(A, v) = (QA, \hat{Q}v), \quad QA := (QA_1, \dots, QA_N),$$

which defines then the action of $SO(3)$ on the tangent space by: $Q \cdot (A, v) := (QA, \hat{Q}v)$, $QA := (QA_1, \dots, QA_N)$,

The action is isometric.

We compute:

$$\begin{aligned} \langle d\Phi_Q|_T(A, v), d\Phi_Q|_T(A', v') \rangle_{\Phi_Q(T)} &= \langle (QA, \hat{Q}v), (QA', \hat{Q}v') \rangle_{\Phi_Q(T)} \\ &= \sum_{i=1}^N \frac{1}{2} \text{Tr}((QA_i)(QA_i')^\top) + \langle \hat{Q}v, \hat{Q}v' \rangle_{\mathbb{R}^{3N}} \\ &= \sum_{i=1}^N \frac{1}{2} \text{Tr}(QA_i A_i'^\top Q^\top) + \langle \hat{Q}v, \hat{Q}v' \rangle_{\mathbb{R}^{3N}} \\ &= \sum_{i=1}^N \frac{1}{2} \text{Tr}(A_i A_i'^\top) + \langle v, v' \rangle_{\mathbb{R}^{3N}} \\ &= \langle (A, v), (A', v') \rangle_T. \end{aligned}$$

where we used the cyclicity of the trace and the fact that $\hat{Q}$ is orthogonal. Hence, Φ_Q preserves the Riemannian metric and is therefore a Riemannian isometry (see Definition 2.56).

Identifying the generator Since $X \in V_N$ and $P|_{V_N} = \text{Id}_{V_N}$, the intrinsic drift on

$$\mathcal{M}_N = SO(3)^N \times V_N$$

is

$$b(R, X) = \left(0, -\frac{1}{2}X\right).$$

Moreover, $P dB_{\mathbb{R}^{3N}}$ is a standard Brownian increment on V_N for the induced Euclidean metric. Indeed, let $\tilde{B}_t := PB_{\mathbb{R}^{3N}_t}$. Since P is deterministic and constant,

$$d\tilde{B}_t = P dB_t.$$

Moreover, $\tilde{B}$ is null at 0, takes values in V_N , and is a continuous local martingale. For every $u, v \in V_N$,

$$d[\langle u, \tilde{B} \rangle, \langle v, \tilde{B} \rangle]_t = \langle P^\top u, P^\top v \rangle dt = \langle u, v \rangle dt,$$

because $P = P^\top$ and $P|_{V_N} = I_{V_N}$. Therefore, by Lévy's characterization (Theorem 2.17), $\tilde{B}$ is a standard Brownian motion on V_N equipped with its induced Euclidean metric.

Together with the Brownian motion on $SO(3)^N$, this shows that the intrinsic covariance is

$$\Sigma_T = \text{Id}_{T_T \mathcal{M}_N}.$$

Therefore, the generator of the forward process is

$$\mathcal{L}f = \langle b, \nabla f \rangle + \frac{1}{2} \text{div}(\Sigma \nabla f) = \langle b, \nabla f \rangle + \frac{1}{2} \Delta_{\mathcal{M}_N} f.$$

Well-posedness In this paragraph, we check that the martingale problem associated to $\mathcal{L}$ is well posed (i.e., for any distribution μ_0 , there is a unique solution to the martingale problem which initial distribution is μ_0).

First, we need this definition (Definition 1.3.1 in [Hsu02]).

Definition 4.17 (Diffusion Process and Diffusion Measure). *Let M be a smooth manifold, and define $W(M) := C([0, \infty), M_\Delta)$.*

- (i) *An $\mathbb{F}$ -adapted stochastic process $X : \Omega \rightarrow W(M)$ defined on a filtered probability space $(\Omega, \mathbb{F}, P)$ is called a diffusion process generated by L (or simply an L -diffusion) if X is an M -valued $\mathbb{F}$ -semimartingale up to $\zeta(X)$ (the explosion time of X , see Definition 2.71) and*

$$M^f(X)_t = f(X_t) - f(X_0) - \int_0^t Lf(X_s) ds, \quad 0 \leq t < \zeta(X),$$

is a local $\mathbb{F}$ -martingale for all $f \in C^\infty(M)$.

- (ii) *A probability measure μ on the standard filtered path space $(W(M), \mathcal{B}(W(M)))^\mathbb{F}$ is called a diffusion measure generated by L (or simply an L -diffusion measure) if*

$$M^f(\omega)_t = f(\omega_t) - f(\omega_0) - \int_0^t Lf(\omega_s) ds, \quad 0 \leq t < \zeta(\omega),$$

is a local $\mathcal{B}(W(M))^\mathbb{F}$ -martingale for all $f \in C^\infty(M)$.

We are going to rely on the two following Theorems (1.3.4 and 1.3.6 in [Hsu02]):

Theorem 4.18. *Let L be a smooth second-order elliptic operator on a differentiable manifold M and μ_0 a probability measure on M . Then there exists an L -diffusion measure with initial distribution μ_0 .*

Theorem 4.19. *An $\mathcal{L}$ -diffusion measure with a given initial distribution is unique.*

Combining these two results shows that, for every initial distribution μ_0 , there exists a unique $\mathcal{L}$ -diffusion measure with initial distribution μ_0 . Equivalently, every $\mathcal{L}$ -diffusion process with initial distribution μ_0 has the same law on the path space. We would like to conclude that this yields that if our generator $\mathcal{L}$ satisfies these hypotheses, then the martingale problem associated is well-posed. This correspondence is true, but not immediate.

Claim: Let $\mathcal{L}$ be a second-order smooth operator without zero-order term. Then, a process X is a solution to the martingale problem associated to $\mathcal{L}$ as defined in 2.71 if and only if it is a $\mathcal{L}$ -diffusion process as defined in 4.17.

Proof: Let X be a $\mathcal{L}$ -diffusion process, denote by ζ its explosion time. Let $f \in \mathcal{C}_c^\infty(M)$. Note that $\text{supp}(\mathcal{L}f) \subset \text{supp}(f)$, hence $\mathcal{L}f$ also has compact support. This yields that for any $t \geq 0$: $|M^f(X)_t| = |f(X_t) - f(X_0) - \int_0^t \mathcal{L}f(X_s) ds| \leq 2\|f\|_\infty + t\|\mathcal{L}f\|_\infty$. Hence, $M^f(X)$ is a local martingale bounded on every compact time interval by a deterministic constant. Then, there exists a localizing sequence $(\tau_n)_{n \in \mathbb{N}}$ such that for every $n \in \mathbb{N}$:

$$\mathbb{E}[M^f(X)_{t \wedge \tau_n} | \mathcal{F}_s] = M^f(X)_{s \wedge \tau_n}$$

.

But as $t \wedge \tau_n \leq t$ and that on $[0, t]$, $M^f(X)$ is bounded, we can apply dominated convergence to let $n \rightarrow \infty$ and get, as $\tau_n \rightarrow \zeta$:

$$\mathbb{E}[M^f(X)_{t \wedge \zeta} | \mathcal{F}_s] = M^f(X)_{s \wedge \zeta}$$

.

That is, $M^f(X)^\zeta$ is a true martingale (adaptability is clear and integrability follows from the fact that f and $\mathcal{L}f$ have compact support). But as f is compactly supported, $f(\Delta) = \mathcal{L}f(\Delta) = 0$, so that $M^f(X)$ is constant after ζ . That is, $M^f(X) = M^f(X)^\zeta$ and $M^f(X)$ is a true martingale.

Conversely, let X be a solution to the martingale problem associated to $\mathcal{L}$, and $f \in \mathcal{C}^\infty(M)$. Choose a compact exhaustion (meaning that every K_i in the following sequence is compact).

$$K_1 \subset \text{int}(K_2) \subset \dots, \quad \bigcup_{n=1}^{\infty} K_n = M,$$

This is always possible on a manifold which admits a countable basis (which is included in our definition of manifold) thank to Prop A.60 in [LL12]. Now, applying Lemma 2.78 to the pair $K_{n+1} \subset \text{int}(K_{n+2})$ yields $\chi_n \in \mathcal{C}_c^\infty(M)$ with $0 \leq \chi_n \leq 1$, $\chi_n \equiv 1$ on K_{n+1} and $\text{supp } \chi_n \subset \text{int}(K_{n+2})$. In particular $\chi_n \equiv 1$ on the open set $\text{int}(K_{n+1}) \supseteq K_n$, so all derivatives of χ_n vanish on K_n . Set

$$f_n := \chi_n f \in \mathcal{C}_c^\infty(M)$$

since $\text{supp}(\chi_n) \subset K_{n+2}$ which is compact, and $\text{supp}(\chi_n)$ is closed. Define the stopping times

$$\tau_n := \inf\{t \geq 0 : X_t \notin K_n\}.$$

Since χ_n is identically equal to 1 on a neighborhood of K_n , all its derivatives vanish there. Hence, on K_n ,

$$f_n = f, \quad Lf_n = Lf.$$

Consequently, for every $t \geq 0$,

$$M^f(X)_{t \wedge \tau_n} = M^{f_n}(X)_{t \wedge \tau_n} \quad (4.19)$$

as for $t \leq \tau_n$, $X_t \in K_n$. Since $f_n \in C_c^\infty(M)$, the process $(M^{f_n}(X))_{t \geq 0}$ is a martingale as X solves the martingale problem associated to $\mathcal{L}$. Hence, by the stopping theorem, the stopped process $(M^{f_n}(X))_{t \geq 0}$ is also a martingale, for any $n \in \mathbb{N}$. Therefore, by (4.19), $(M^{f_n}(X))_{t \geq 0}$ is a martingale for every n . Moreover, $\tau_n \uparrow \zeta$, where ζ denotes the explosion time of X . It follows that $(M^f(X))_{t < \zeta}$ is a local martingale. $\square$

Now, it remains to prove that our generator $\mathcal{L}$ which satisfies: $\mathcal{L}f = \langle b, \nabla f \rangle + \frac{1}{2} \operatorname{div}(\Sigma \nabla f) = \langle b, \nabla f \rangle + \frac{1}{2} \Delta_{\mathcal{M}_N} f$ is elliptic and smooth. For the smooth part, b and Σ are both smooth. For the elliptic part, this is direct by Lemma 2.77, as $\Sigma_T = \operatorname{Id}_{T_T \mathcal{M}_N}$ is positive definite.

Consequently, by Theorems 4.18 and 4.19, the martingale problem associated to $\mathcal{L}$ is well-posed.

Non-explosion The two components decouple:

$$dR^{(t)} = dB_{SO(3)^N}^{(t)}, \quad dX^{(t)} = -\frac{1}{2}X^{(t)} dt + dB_{V_N}^{(t)},$$

where $B_{SO(3)^N}$ and B_{V_N} are independent. The first equation is the Brownian motion on the compact manifold $SO(3)^N$. As the generator does not contain any zero-order term, we can apply Proposition 2.73 to get that the martingale problem is non-explosive. The second is the Ornstein–Uhlenbeck process on the vector space V_N ; it admits the unique strong solution $X^{(t)} = e^{-t/2}X^{(0)} + \int_0^t e^{-(t-s)/2} dB_{V_N}^{(s)}$, which does not explode, as it has continuous trajectories hence is bounded on every compact interval.

Then, for any initial law μ_0 , we define: $Z_t = (R_t, X_t)$, $Z_0 = (R_0, X_0)$, where R_t satisfies: $dR^{(t)} = dB_{SO(3)^N}^{(t)}$ and X_t satisfies: $dX^{(t)} = -\frac{1}{2}X^{(t)} dt + dB_{V_N}^{(t)}$. If we denote by ζ_Z the explosion time of Z , ζ_R , ζ_X the explosion times of the rotation and translation part respectively, we have:

$$\zeta_Z = \zeta_R \wedge \zeta_X = \infty \quad a.s.$$

As we know that the generator of Z is $\mathcal{L} = \langle b, \nabla f \rangle + \frac{1}{2} \Delta_{\mathcal{M}_N} f$, we get that Z is a solution of the martingale problem that is non-explosive.

But for the martingale problem to be non-explosive, it is required that any solution is non-explosive.

Hence, let $\tilde{Z}$ be another solution to the martingale problem with initial distribution μ_0 . As we proved earlier that the martingale problem is well posed, we have that $\tilde{Z}$ and Z share the same distribution on the path space. In particular: $\mathbb{P}(\tilde{\zeta} \leq T) = \mathbb{P}(\zeta \leq T) = 0 \quad \forall T \geq 0$. Hence: $\mathbb{P}(\tilde{\zeta} < \infty) = \mathbb{P}(\cup_{n=1}^\infty \{\tilde{\zeta} \leq n\}) \leq \sum_{n=1}^\infty \mathbb{P}(\tilde{\zeta} \leq n) = 0$.

Hence, as μ_0 was arbitrary, we get that for any initial distribution, any solution to the martingale problem is non-explosive. That is, the martingale problem is non-explosive.

Equivariance of the coefficients The matrices P and $\hat{Q}$ commute:

$$P\hat{Q} = \left(I_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top\right) \otimes Q = \hat{Q}P,$$

using $(A \otimes B)(C \otimes D) = AC \otimes BD$. Hence, for every $Q \in SO(3)$,

$$b(\Phi_Q(R, X)) = (0, -\frac{1}{2}P\hat{Q}X) = (0, -\frac{1}{2}\hat{Q}PX) = d\Phi_Q(0, -\frac{1}{2}PX) = d\Phi_Q b(R, X),$$

so the drift is $SO(3)$ -equivariant.

For the diffusion coefficient, note that:

$$\begin{aligned}
 \Sigma_{\Phi_Q(T)} \circ d\Phi_Q|_T &= \text{Id}_{T_{\Phi_Q(T)}\mathcal{M}_N} \circ d\Phi_Q|_T \\
 &= d\Phi_Q|_T \\
 &= d\Phi_Q|_T \circ \text{Id}_{T_T\mathcal{M}_N} \\
 &= d\Phi_Q|_T \circ \Sigma_T.
 \end{aligned}$$

Hence the diffusion coefficient is also equivariant.

The density is indeed C^1 for $t > 0$ By the product structure, the transition kernel of the forward process factorizes as the heat kernel of $SO(3)^N$ times a non-degenerate Gaussian kernel on V_N . The former is smooth and strictly positive for every $t > 0$; the latter is smooth and strictly positive. Hence for every $t > 0$ the law μ_t admits a smooth, strictly positive density p_t with respect to the Riemannian volume measure of $\mathcal{M}_N$, so the second conclusion of Proposition 4.16 applies on $(0, T_F]$.

The data distribution can be made $SO(3)$ -invariant Proposition 4.16 explicitly requires the initial distribution p_0 to be $SO(3)$ -invariant; applied to the forward process, p_0 is the data distribution. One could think this holds structurally, since the physical process generating protein structures is isotropic. The empirical distribution obtained from the Protein Data Bank, however, does not satisfy it: deposited coordinate frames inherit systematic conventions from the crystallographic unit cell or the reconstruction pipeline, so orientations are not Haar-distributed. The authors do not discuss this (the same property is needed for the application of their Proposition 3.6 to the forward process). The hypothesis can be restored by applying a random rotation to every training sample, which replaces p_0 by its Haar average $\bar{p}_0$; since symmetrisation commutes with the forward SDE (its coefficients being equivariant), the resulting model is exactly SE(3)-Diff for $\bar{p}_0$. Note that this is a repair of the argument rather than a change in training: under exact equivariance the objective is already orbit-averaged, so augmentation leaves the learned model unchanged.

Conclusion Finally, we can conclude:

- for every $t \in [0, T_F]$, the law μ_t of the forward process is $SO(3)$ -invariant;
- for every $t \in (0, T_F]$, the score $\nabla \log p_t$ is $SO(3)$ -equivariant.

Remark: One might wonder why we highlight that the score of the reverse process is $SO(3)$ -equivariant in the proposition. Indeed, the goal of the proposition is to get that the marginals of the reverse process are $SO(3)$ -invariant, which is immediate since the marginals of the forward process are $SO(3)$ -invariant, a fact that we obtained without having to consider that the score of the reverse process is $SO(3)$ -equivariant. However, once we consider the reverse process with the approximated score, it does not share its marginals with the forward process anymore, hence this identification breaks. This is the focus of the next paragraph.

Getting $SO(3)$ -invariance in practice: the approximate reverse process

In practice, the exact score is unavailable and is replaced by a neural network trained by denoising score matching, as discussed above. The approximate reverse process is the diffusion

on $\mathcal{M}_N$ with generator

$$\mathcal{L}_t^\theta f = \langle b_\theta(t, \cdot), \nabla f \rangle + \frac{1}{2} \Delta_{\mathcal{M}_N} f, \quad f \in C_c^\infty(\mathcal{M}_N),$$

that is, with drift b_θ and diffusion coefficient $\Sigma_T^\theta = \text{Id}_{T_T \mathcal{M}_N}$, where

$$b_\theta(t, T) = \left(s_\theta^R(T_F - t, T), P \left(\frac{1}{2} X + s_\theta^X(T_F - t, T) \right) \right), \quad T = (R, X).$$

Assume that the score network is $SO(3)$ -equivariant, namely

$$s_\theta^R(t, \Phi_Q(T)) = \left(Q s_{\theta,i}^R(t, T) \right)_{i=1}^N$$

and

$$s_\theta^X(t, \Phi_Q(T)) = \widehat{Q} s_\theta^X(t, T)$$

Remark. Enforcing $SO(3)$ -equivariance restricts the architecture of the score network. However, since the exact score of every marginal (excluding the marginal at time T_F) of the reverse process is itself $SO(3)$ -equivariant, this constraint does not exclude the exact score from the admissible function class. Thus, assuming sufficient expressive power within the class of equivariant networks, it introduces no approximation bias.

For the translational component, using $P\widehat{Q} = \widehat{Q}P$, we obtain

$$\begin{aligned} b_{\theta,X}(t, \Phi_Q(T)) &= P \left(\frac{1}{2} \widehat{Q} X + s_\theta^X(T_F - t, \Phi_Q(T)) \right) \\ &= P \widehat{Q} \left(\frac{1}{2} X + s_\theta^X(T_F - t, T) \right) \\ &= \widehat{Q} P \left(\frac{1}{2} X + s_\theta^X(T_F - t, T) \right) \\ &= \widehat{Q} b_{\theta,X}(t, T). \end{aligned}$$

The rotational component is equivariant by assumption. Consequently,

$$b_\theta(t, \Phi_Q(T)) = d\Phi_Q(T) b_\theta(t, T). \quad (4.20)$$

In practice, this approximated reverse process is initialized at the stationary distribution of the forward process, that is:

$$\nu_0 = \mathcal{U}(SO(3))^{\otimes N} \otimes P_\# \left(\mathcal{N}(0, I_3)^{\otimes N} \right),$$

which is $SO(3)$ -invariant.

Moreover, time reversal does not affect the diffusion coefficient: $\Sigma_T^\theta = \Sigma_T = \text{Id}_{T_T \mathcal{M}_N}$, which was shown above to be $SO(3)$ -equivariant.

Let $\tau < T_F$ and assume that the martingale problem associated with the approximate reverse process is non-explosive and well posed on $[0, \tau]$. All the hypotheses of Proposition 4.16 are then satisfied. Therefore, for every $Q \in SO(3)$ and $t \in [0, \tau]$,

$$(\Phi_Q)_\# \text{Law} \left(\overleftarrow{T}_\theta^{(t)} \right) = \text{Law} \left(\overleftarrow{T}_\theta^{(t)} \right).$$

Equivalently,

$$\boxed{Q \cdot \overleftarrow{T}_\theta^{(t)} \stackrel{\text{law}}{=} \overleftarrow{T}_\theta^{(t)}}.$$

Since $\tau < T_F$ is arbitrary, the marginal distribution of the approximate reverse process is $SO(3)$ -invariant for every $t \in [0, T_F]$.

Remark. To prove that the martingale problem is well-posed and non-explosive, we can't follow the same procedure as in the forward process case. Indeed, the results we used are only valid for generators that are time independent. One possibility would be to define the manifold $\widehat{\mathcal{M}}_N = J \times \mathcal{M}_N$, where J is an open subset of $[0, T_F - \varepsilon]$. This would yield a time-independent generator, and we could then use the same approach as for the forward process. However, this would require smooth networks, which is not likely in practice. As a consequence, the direct application of Proposition 4.16 to the frame of the paper [Yim+23] is only complete for the theoretical case where we have access to the analytical score. The application for the score-approximated reverse process is deferred to future work.

In this section, we showed mathematically how to build a generative model that can generate protein backbones, based on the paper [Yim+23]. The authors then dedicate the rest of the paper to practical implementation, which is out of the scope of this work.

The following figure summarizes the key mathematical steps of the procedure:

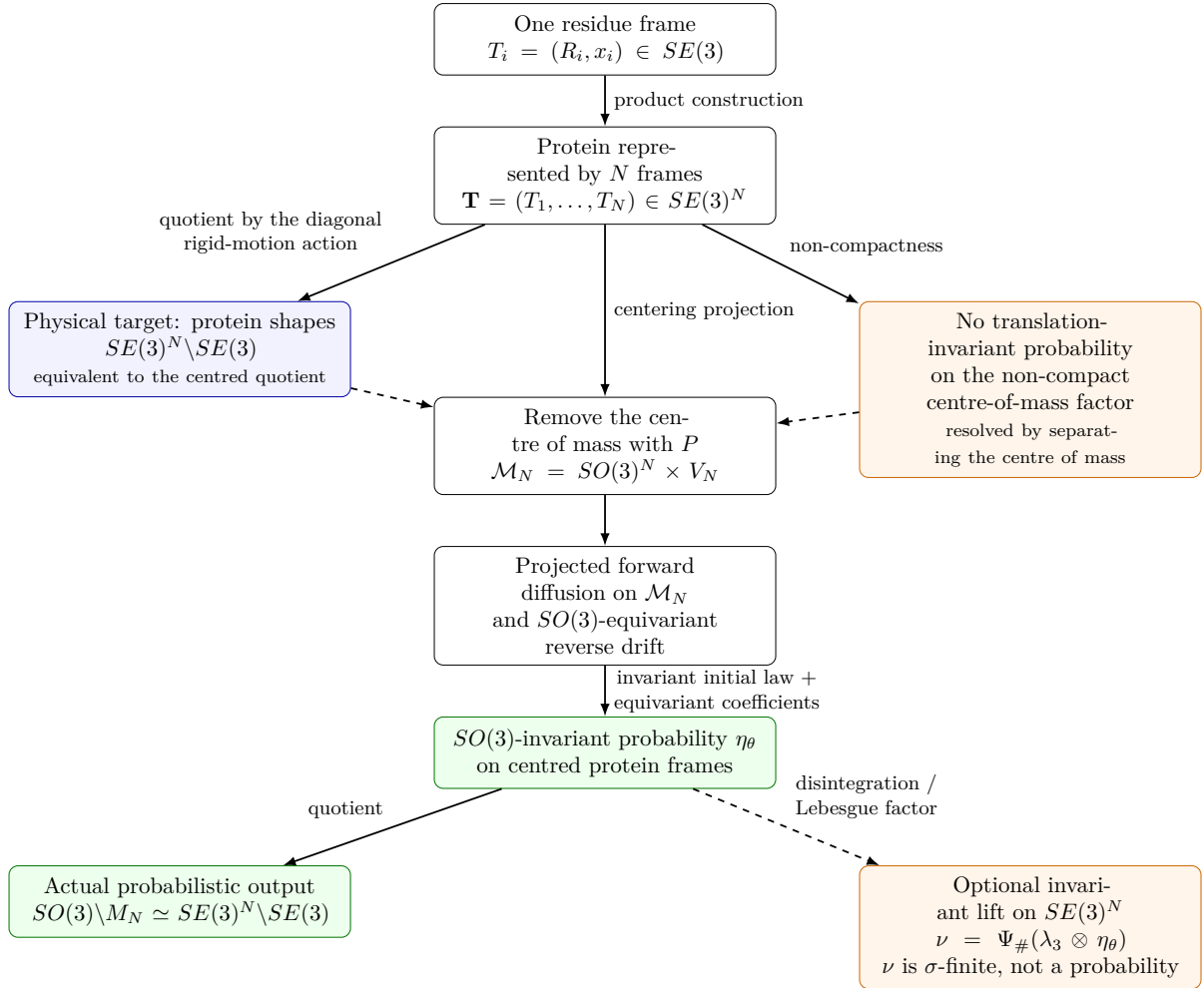


Figure 4.2: Logical path from a diffusion on one rigid frame to a generative distribution on protein shapes. The probability model lives naturally on centred frames (or on their rotation quotient); the translation-invariant lift to $SE(3)^N$ has infinite total mass.

Chapter 5

Conclusion

This thesis examined the mathematical foundations of diffusion-based generative modeling and their adaptation to temporally and geometrically structured data.

The mathematical foundations developed in the first part of the thesis reveal several complementary interpretations of generative modeling. Stochastic analysis describes how a reference diffusion can be modified through an adapted drift, while time-reversal results identify the score as the central quantity governing the backward dynamics. Optimal transport and Schrödinger bridges provide a variational interpretation in which generation is formulated as the construction of a path measure satisfying prescribed distributional constraints while remaining close, in relative entropy, to a reference process. Differential geometry becomes necessary when the support of the data is a non-Euclidean Riemannian manifold.

For classical diffusion models, under suitable assumptions, the exact reverse-time SDE recovers the forward marginals in reverse order. This population-level identity does not, however, imply that the diffusion model exactly reproduces the data distribution. Score-estimation error, prior mismatch, time truncation, and numerical discretization all contribute to the final sampling error. Similarly, although the probability-flow ODE and the reverse SDE share the same one-time marginals in the exact setting, they are not pathwise equivalent and may behave differently after approximation and discretization.

The manifold hypothesis raises a fundamental difficulty for the standard convergence theory of diffusion models: if the data distribution is supported on a lower-dimensional subset of the ambient space, then it does not admit a Lebesgue density at the terminal time. Consequently, convergence guarantees formulated in terms of relative entropy cannot generally be applied directly to the original data distribution. Forward diffusion resolves this singularity at every strictly positive time, but the resulting estimates then concern a smoothed distribution rather than the singular target itself. The results studied in the thesis provide a complementary, qualitative perspective. They establish that generalization beyond the training samples requires the approximation error with respect to the empirical score to be unbounded. This provides a necessary, but not sufficient, condition for support-level generalization: an unbounded approximation error does not by itself imply that the model generalizes. This conclusion is qualitative. It neither controls the distance between the generated and population distributions nor provides convergence rates depending on the intrinsic dimension of the data manifold. The study of such dimension-adaptive quantitative guarantees lies beyond the scope of the present thesis.

The Schrödinger bridge perspective arises in two distinct settings. In the classical unconstrained $\mathbb{R}^d$ framework, it primarily aims at reducing prior mismatch and allowing a shorter time horizon.

In the time-series setting, the Schrödinger bridge approach allows to transform the generative problem into a stochastic control problem, whose minimizer admits an explicit expression: the target distribution on the grid, interpolated by Brownian bridges between consecutive grid points.

The protein-backbone application illustrates a different type of structural difficulty. Protein frames involve rotations and translations, and their physically meaningful distribution must be invariant under global rigid transformations. Since no translation-invariant probability measure exists on the translation component, the probability model naturally lives on a centered configuration space, or equivalently on an appropriate quotient. A direct application of the group-theoretic proposition used in the source paper is not possible because the centered space is not a subgroup of $\text{SE}(3)^N$ under the componentwise semidirect-product law. The appropriate formulation is instead to regard it as a Riemannian manifold carrying an isometric $\text{SO}(3)$ -action. Proposition 4.16 formalizes this approach at the level of a diffusion generator: if the generator commutes with the group action and the corresponding martingale problem is well posed and non-explosive, an invariant initial distribution remains invariant.

The contributions of this thesis consist of a unified mathematical presentation of several diffusion-based frameworks, detailed reconstructions of important arguments, the identification of assumptions and distinctions that are sometimes left implicit, and corrected formulations for the two structured applications. The geometric treatment of the centered protein space and the corresponding invariance proposition constitute the principal original mathematical contribution, while the time-series analysis provides a critical conjecture about the ability of the considered model to generalize.

Several questions remain open. For the time-series model, a complete analysis should formulate a precise and empirically testable criterion or metric distinguishing memorization from distributional generalization. For manifold-valued diffusion models, future work should establish well-posedness and non-explosion of the martingale problem for time-dependent reverse drifts represented by non-smooth neural networks.

More broadly, the analysis suggests that progress in diffusion generative modeling requires the theoretical and algorithmic levels to be studied together. Theoretical tools like stochastic time reversal, entropy minimization, and geometric invariance provide powerful principles, but their conclusions depend on the exact approximation and discretization regime under consideration. Keeping these objects distinct is essential both for rigorous mathematical analysis and for understanding what the implemented generative model actually learns.

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